

MATHEMATICS FOR THE

Working Programmer

A complete rebuild of the mathematics you were never properly taught — from what a number is, through calculus, linear algebra, probability and statistics, to the derivation of backpropagation — written for someone who already knows how to think precisely, because they write code.

Intuition first. Proofs shown. Exercises worked.

No step skipped.

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How to Read This Book

What this is

This is a rebuild, not a refresher.

The assumption behind it is specific: you are a working programmer. You think precisely for a living. You can hold an abstraction in your head, you can follow a chain of reasoning ten steps long, and you have written code that had to be **correct**, not merely plausible. What you do not have is the mathematics, because high school was a while ago and high school was not very good at this anyway.

That combination is unusual and it changes what the right book looks like. Most “math for programmers” books assume you remember calculus and want a quick tour. Most “start from scratch” books assume you are sixteen and need to be coaxed. Neither is you. You need to start from genuinely low ground — we will define what a number is — but you can be moved fast, and you can be told the truth about why things work instead of being handed rules to obey.

So this is the deal the book makes:

- **Nothing is assumed.** Not logarithms, not trigonometry, not what a function is. Chapter 1 starts at the counting numbers.
- **Nothing is hand-waved either.** When a result matters, you get a proof. Not a gesture at a proof — an actual argument you can check line by line.
- **The intuition comes first, always.** Before any definition there is a paragraph telling you what picture to have in your head and what problem the definition was invented to solve. Definitions in mathematics are not arbitrary. Every one of them is somebody else’s solution to a difficulty, and you will be told what the difficulty was.
- **You already know how to compute.** So the book leans on that. Algorithms appear as pseudocode. Numerical behaviour — overflow, cancellation, conditioning — is treated as first-class mathematics, because for you it is.

What is in it, and why

The six parts are ordered by dependency, not by tradition.

Part I — Rebuilding the Ground. Numbers, algebra, functions, exponentials and logarithms, trigonometry, complex numbers. Then proof technique. This part exists because every later part will silently use it, and a shaky base here shows up three hundred pages later as “I can follow each step but I have no idea what is happening.”

The proof chapter is the hinge of the whole book. You already know induction from reasoning about recursion; that is a much better starting point than most people have. We build outward from it.

Part II — Discrete Mathematics. Sets, relations, counting, graphs, recurrences and asymptotics, number theory. This is the mathematics already inside your code whether or not you have named it. It is also the gentlest place to practise proof, because the objects are finite and concrete.

Part III — Calculus. Limits, derivatives, integrals, series, and then multivariable calculus. The destination is the gradient. Everything in machine learning that “learns” is doing one thing: computing a gradient and stepping against it. You cannot understand that from the outside.

Part IV — Linear Algebra. If you only had time for one part, it would be this one. Matrices are not tables of numbers; they are functions, and the whole subject becomes easy the moment that clicks. We go all the way to the singular value decomposition and to matrix calculus, which is the notation ML papers are actually written in.

Part V — Probability and Statistics. From the axioms through to maximum likelihood, Bayesian inference, hypothesis testing, and information theory. Cross-entropy loss is not a magic formula somebody chose; it falls out of information theory in about four lines, and you will see those four lines.

Part VI — Computation, Geometry, and Learning. Floating point and numerical stability, root-finding and quadrature, optimization, the geometry of graphics and simulation, and then a complete from-scratch derivation of backpropagation and of the mathematics inside a transformer. The appendices contain worked solutions to every exercise, a curated list of free and openly-licensed mathematics books, a notation reference, and a study plan that puts the whole thing in order.

How to actually use it

The idea: reading mathematics is not reading.

Prose is designed to be read at a steady speed. Mathematics is not. A single line can take twenty minutes and that is normal, not a failure. The correct reading speed for a definition is **slow enough to construct an example of it and an example of something that just fails it**. If you cannot do both, you have not read it yet.

Concretely:

1. **Read the intuition box, then look away and say it back.** If you cannot, read it again. This costs thirty seconds and saves an hour.
2. **When you meet a definition, build two things immediately:** one object that satisfies it, and one that nearly does but does not. The near-miss is where the understanding lives. A definition without a near-miss is a definition you have memorised, not understood.
3. **Read proofs twice.** First pass, ask only: is each step true? Second pass, ask: why would anyone have thought of this? The second question is the one that transfers to problems you have not seen.
4. **Do the exercises.** This is not a moral instruction, it is a mechanical one. Mathematics is a motor skill in a way that reading obscures. The solutions in Appendix A are fully worked, but a solution read before a genuine attempt teaches you close to nothing — you get the pleasant feeling of understanding without acquiring any of it. Give each problem a real fifteen minutes first.
5. **Skipping is allowed; going backwards is mandatory.** If Part III is hard going, the fault is almost always in Part I. Go back. That is not a defeat, it is the only debugging technique that works here.

On the proofs

A proof is not a ritual and it is not there to convince a grader. A proof is the only technology anybody has ever found for being **sure**.

You already believe this, in a different vocabulary. You do not trust a function because you ran it once; you trust it because you can argue about what it does on every input — the invariant holds on entry, the loop body preserves it, therefore it holds on exit. That is a proof. You have been writing them for years without the notation.

What this book adds is the notation, the standard moves (contradiction, contraposition, construction, pigeonhole, induction in its stronger forms), and the habit of noticing which move a situation is asking for.

Trap: the two failure modes.

Beginners fail in exactly two ways, and they are opposites.

The first is **hand-waving**: writing something that sounds like an argument but has a hole where the hard part should be. The word “clearly” is where this usually hides.

The second is **ritual**: writing symbols in the shapes that proofs tend to have, without any of them meaning anything. This one is more common among people who have been graded a lot.

The cure for both is the same. After every line, ask: **what exactly am I now entitled to assume, and what am I still trying to establish?** If you cannot answer both, stop and fix it before writing the next line.

A note on notation

Mathematical notation is a terrible user interface with a very good excuse: it was optimised for writing by hand, at speed, by people who already know what it means. Much of what makes a page look impenetrable is not depth, it is compression.

This book introduces every symbol at the moment it is first needed and never before, and Appendix C collects all of them in one table you can keep open. When something looks unreadable the problem is nearly always that one symbol is unfamiliar — not that the idea is hard. Look it up and try the line again.

The honest promise

Working through this book will not make you a mathematician. It will make you someone who can open a machine learning paper, a graphics textbook, or an algorithms monograph and **read it** — slowly, with effort, but without bouncing off. It will make the mathematics in your own work visible to you instead of invisible.

That is a real change and it does not take a decade. It takes a few hundred hours of the specific kind of attention described above.

Start at the beginning. The beginning is genuinely the beginning.

PART I

Rebuilding the Ground

Everything below calculus, done properly this time. Numbers, algebra as a rewriting system, functions, exponentials, trigonometry, complex numbers — and then the single most important skill in the book: how to read and write a proof.

CHAPTER 1.

Numbers

1.1. Why start here

You might reasonably expect a book aimed at a programmer to open somewhere more interesting than “what is a number”. Here is why it does not.

Every hard thing later in this book is a statement about numbers that is not obvious. “Every bounded increasing sequence converges.” “A continuous function on a closed interval attains its maximum.” “The eigenvalues of a real symmetric matrix are real.” Each of those is **false** if you change which numbers you are talking about. The first two are false over the rationals. The third is only interesting because it is false in general over the complexes.

So the number systems are not throat-clearing. They are the place where the later theorems get their teeth. And there is a second reason, closer to home: the numbers in your computer are **not** the numbers in this book, and knowing exactly where the two diverge is the difference between code that works and code that silently produces garbage at scale.

The idea: each number system is a bug fix.

Do not think of $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ as a hierarchy of increasing fanciness. Think of it as a changelog.

Each system is the previous one, patched, because somebody tried to do a perfectly reasonable operation and the answer did not exist. Subtraction broke $\mathbb{N}$. Division broke $\mathbb{Z}$. Taking limits broke $\mathbb{Q}$. Square roots of negatives broke $\mathbb{R}$. Each time, the fix was to add exactly the missing answers and nothing else.

When you see it that way, the definitions stop being arbitrary. They are patch notes.

1.2. The natural numbers

Definition 1.1 (natural numbers).

The **natural numbers** are

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

Some authors start at 1. This book starts at 0, for the same reason your arrays do.

What actually characterises $\mathbb{N}$ is not the list — lists trail off into “...”, which is not a definition — but the **rule for generating it**. There is a starting element 0, and there is a successor operation $s(n) = n + 1$, and every natural number is reached from 0 by applying s some finite number of times.

That is a recursive data type, and you have declared it before:

The naturals, as you would declare them

```
type Nat =  
  | Zero  
  | Succ of Nat
```


The mathematical name for the fact that this generates **everything** — that there are no stray naturals floating around that 0 and s never reach — is the **induction axiom**, and it is the reason proof by induction works. We will use it constantly from Chapter 7 onwards. It is worth noticing now that induction is not a clever trick somebody found; it is a restatement of what $\mathbb{N}$ is.

Definition 1.2 (closure).

A set S is **closed** under an operation if applying that operation to members of S always lands you back inside S .

Closure is the organising idea of this chapter. $\mathbb{N}$ is closed under addition and multiplication: add or multiply two naturals, get a natural. It is **not** closed under subtraction: $3 - 5$ is not a natural number.

That is the first bug.

1.3. The integers

Definition 1.3 (integers).

The **integers** are

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\},$$

the naturals together with a negative $-n$ for each n , where $-n$ is **defined** as the unique solution to $n + x = 0$.

The letter is $\mathbb{Z}$ from the German **Zahlen**, numbers. Note the definition carefully: we did not say “negative numbers are numbers less than zero” and leave it at that. We said $-n$ is **the thing you add to n to get 0**. That is a functional specification, and it is what makes negatives usable.

$\mathbb{Z}$ is closed under $+$, $-$, and $\times$. A set with those three operations behaving sensibly is called a **ring**, and $\mathbb{Z}$ is the ring everyone learns first.

$\mathbb{Z}$ is not closed under division: $1/2$ is not an integer.

That is the second bug.

Trap: your integers are not these integers.

A 64-bit signed integer in your language of choice is not an element of $\mathbb{Z}$. It is an element of $\mathbb{Z}/2^{64}\mathbb{Z}$ — the integers modulo 2^{64} — which is a genuinely different mathematical object, and one we will study properly in Chapter 12.

The practical difference: in $\mathbb{Z}$, if $a > 0$ and $b > 0$ then $a + b > a$. In 64-bit arithmetic that is false, and the counterexample is called overflow. Every binary-search bug of the form $(lo + hi) / 2$ is this theorem failing.

1.4. The rationals

Definition 1.4 (rational numbers).

The **rational numbers** are

$$\mathbb{Q} = \left\{ \frac{p}{q} : p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0 \right\},$$

where p/q and r/s are regarded as **the same number** exactly when $ps = qr$.

That last clause is the whole content of the definition and it is worth pausing on. A rational number is not a pair of integers. It is an **equivalence class** of pairs of integers: $1/2$, $2/4$ and $(-3)/(-6)$ are three names for one object.

This is a pattern you know. It is the difference between a value and its representation, and between structural and reference equality. Writing `Fraction(1,2) == Fraction(2,4)` requires you to **define** what equality means for the type, and cross-multiplication is that definition.

Example 1.5 (why cross-multiplication is the right rule).

We want p/q to mean “the number x with $qx = p$ ”. If $qx = p$ and $sx = r$ describe the same x , multiply the first by s and the second by q : $qsx = ps$ and $qsx = qr$. Hence $ps = qr$.

So the rule is not a convention. It is forced by what a fraction is supposed to mean.

$\mathbb{Q}$ is closed under $+$, $-$, $\times$, and $\div$ by anything nonzero. A set with all four is a **field**, and fields are the natural home of linear algebra (Part IV).

So $\mathbb{Q}$ has all four arithmetic operations. Why is it not enough?

1.5. The gap in the rationals

Here is the first genuine theorem in this book, and it is one of the oldest in mathematics. Read the proof slowly; it is the model for a great deal of what follows.

Theorem 1.6 (irrationality of the square root of 2).

There is no rational number x with $x^2 = 2$.

Proof. Suppose, for contradiction, that there is one. Then we can write it as $x = p/q$ with $p, q \in \mathbb{Z}$, $q \neq 0$, and — this is the move that makes the proof work — with p and q having **no common factor**. We may always assume this, because if they had a common factor we could cancel it and rename.

From $x^2 = 2$ we get

$$\frac{p^2}{q^2} = 2, \quad \text{so} \quad p^2 = 2q^2.$$

So p^2 is even. Now, if p were odd, say $p = 2k + 1$, then $p^2 = 4k^2 + 4k + 1$, which is odd. So p cannot be odd: p is even. Write $p = 2m$.

Substituting: $(2m)^2 = 2q^2$, that is $4m^2 = 2q^2$, that is $q^2 = 2m^2$.

By the identical argument, q is even.

But now p and q are both even, so they share the common factor 2 — contradicting the assumption that they had none.

The assumption that a rational square root of 2 exists leads to a contradiction, so no such rational exists. ■

The idea: what just happened.

Three techniques appeared, and you will use all three for the rest of your life.

Proof by contradiction. Assume the thing you want to disprove, derive something impossible, conclude the assumption was wrong.

Choosing a good representative. “In lowest terms” was not decoration; the contradiction was manufactured entirely out of it. Picking the right representative of an equivalence class is a recurring, powerful move.

Descent. Notice the shape: from a solution we built a **smaller** solution ($p/2, q/2$ also works). An infinite chain of strictly smaller positive integers is impossible. That is the same well-foundedness argument you use to prove a recursion terminates.

So the rationals have holes. The diagonal of a unit square has a length, and that length is not a rational number. Worse, this is not a rare defect: we will see in Chapter 17 that in a precise sense **almost every** number is irrational.

1.6. The real numbers

The fix is $\mathbb{R}$: fill in the holes. Making that precise took mathematicians about two thousand years, and there are two standard constructions (Dedekind cuts and Cauchy sequences). We will not build $\mathbb{R}$ from scratch — that is a real analysis course — but we must state the property that distinguishes it, because every theorem in calculus rests on it.

Definition 1.7 (upper bound and supremum).

Let $S \subseteq \mathbb{R}$ be nonempty. A number b is an **upper bound** for S if $x \leq b$ for every $x \in S$.

A number b is the **least upper bound**, or **supremum**, written $\sup S$, if it is an upper bound and no smaller number is an upper bound.

Theorem 1.8 (completeness of the reals).

Every nonempty subset of $\mathbb{R}$ that has an upper bound has a least upper bound in $\mathbb{R}$.

This is an axiom rather than something we prove — it is part of what “the real numbers” means. Its importance is best seen by watching it fail.

Example 1.9 (completeness fails over the rationals).

Let $S = \{x \in \mathbb{Q} : x^2 < 2\}$. Inside $\mathbb{Q}$, S is nonempty and bounded above (by 2, say). But it has no least upper bound **in $\mathbb{Q}$** : for any rational upper bound you propose, there is a smaller rational one, because the only candidate for the least is $\sqrt{2}$ and that is not available.

Concretely, this is why the bisection root-finder of Chapter 33 works. Bisection on $f(x) = x^2 - 2$ produces a sequence of ever-tighter intervals whose endpoints are rational. Over $\mathbb{Q}$ the process converges to nothing at all. Over $\mathbb{R}$ it converges, and completeness is precisely the guarantee that it does.

The idea: completeness in one sentence.

Completeness says: **if a sequence of numbers is closing in on something, there is something there for it to close in on.** Every convergence theorem in calculus is a cash-out of that single promise.

1.6.1. Decimal expansions

Every real number has a decimal expansion, and rationals are exactly the ones whose expansion eventually repeats.

Proposition 1.10.

A real number is rational if and only if its decimal expansion is eventually periodic.

Proof. ($\Rightarrow$) Compute p/q by long division. At each step the state of the algorithm is the current remainder, which is one of the q values $0, 1, \dots, q-1$. After at most q steps a remainder must repeat — there are only finitely many states — and from that point the digits repeat with the same period. This is the pigeonhole principle (Chapter 9), and it is also just the observation that a finite-state machine must eventually cycle.

($\Leftarrow$) Suppose the expansion repeats with period k from some point on. Let x be the number and shift the decimal point: $10^m x$ and $10^{m+k} x$ have **identical** tails. Subtract, and the infinite tails cancel exactly, leaving

$$(10^{m+k} - 10^m)x = \text{some integer},$$

so x is that integer divided by the integer $10^{m+k} - 10^m$, hence rational. ■

Example 1.11 (the 0.999... business).

Apply the second half of that proof to $x = 0.\bar{9}$. Here $m = 0$, $k = 1$, so $10x - x = 9$, giving $9x = 9$ and $x = 1$.

This is not a trick or an approximation. $0.\bar{9}$ and 1 are two decimal strings naming the same real number, the same way **1.0** and **1.00** name the same value. The map from decimal strings to reals is surjective but not injective, and this is the only way it fails to be injective.

1.7. How many numbers are there?

Here is a result that has no business being as surprising as it is, and it is directly relevant to computation.

Definition 1.12 (countable).

A set is **countable** if its elements can be listed in a sequence $a_1, a_2, a_3, \dots$ such that every element appears exactly once. Equivalently, it can be put in one-to-one correspondence with $\mathbb{N}$.

$\mathbb{Z}$ is countable: list it as $0, 1, -1, 2, -2, 3, -3, \dots$. Nothing is missed, nothing is repeated.

Theorem 1.13 (the rationals are countable).

$\mathbb{Q}$ is countable.

Proof. Every positive rational in lowest terms is a pair (p, q) of positive integers with no common factor. Order these pairs by $p + q$ (finitely many pairs for each value of the sum), and within each group by p . This enumerates every positive rational exactly once:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{1}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \dots$$

Then interleave with the negatives and 0 as we did for $\mathbb{Z}$. ■

Theorem 1.14 (the reals are not countable).

$\mathbb{R}$ is not countable.

Proof. It suffices to show the interval $(0, 1)$ is not countable. Suppose it were, and let

$$a_1, a_2, a_3, \dots$$

be a complete list, each written as a decimal $a_n = 0.d_{n1}d_{n2}d_{n3}\dots$ (to avoid the $0.\bar{9}$ ambiguity above, always choose the expansion that does not end in all nines).

Now construct a new number $b = 0.b_1b_2b_3\dots$ by the rule

$$b_n = \begin{cases} 5 & \text{if } d_{nn} \neq 5 \\ 4 & \text{if } d_{nn} = 5 \end{cases}$$

Then $b \in (0, 1)$, and b differs from a_n in its n -th digit, for every n . So b is not on the list. But the list was supposed to contain every number in $(0, 1)$ — contradiction. ■

The idea: why a programmer should care about the diagonal.

That proof is the ancestor of the halting problem, of Gödel's incompleteness theorems, and of Russell's paradox. The shape is always the same: you are handed an enumeration that claims to be complete, and you build the one object that is deliberately different from the n -th entry in the n -th place.

There is also a blunt computational corollary. There are only countably many finite programs (a program is a finite string over a finite alphabet, and those are countable). There are uncountably many real numbers. Therefore **almost every real number is not computable** — no program produces its digits. The numbers you can actually name — π , e , $\sqrt{2}$, every number in every paper ever written — form a set of measure zero inside the reals.

1.8. Order, absolute value, intervals

Definition 1.15 (absolute value).

For $x \in \mathbb{R}$,

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

It is the distance from x to 0; and $|x - y|$ is the distance between x and y .

Reading $|x - y|$ as **distance** rather than as a case-split is the single most useful reflex in analysis. Almost every ε in Chapter 13 is a statement about distances.

Theorem 1.16 (triangle inequality).

For all $x, y \in \mathbb{R}$:

$$|x + y| \leq |x| + |y|.$$

Proof. For any real t we have $-|t| \leq t \leq |t|$, directly from the definition. Applying this to x and to y and adding the two chains:

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

A number u satisfying $-c \leq u \leq c$ with $c \geq 0$ satisfies $|u| \leq c$: if $u \geq 0$ then $|u| = u \leq c$, and if $u < 0$ then $|u| = -u \leq c$ from the left inequality. Apply that with $u = x + y$ and $c = |x| + |y|$. ■

The name comes from the geometric version: the direct route from A to C is no longer than going via B . In the form $|x - z| \leq |x - y| + |y - z|$ that reading is literal, and this is the version you will use.

Definition 1.17 (intervals).

For $a < b$:

$$(a, b) = \{x : a < x < b\}, \quad [a, b] = \{x : a \leq x \leq b\},$$

and the half-open $[a, b)$, $(a, b]$ analogously. A square bracket includes the endpoint, a round bracket excludes it. We also allow $(a, \infty) = \{x : x > a\}$ and similar; ∞ is never an endpoint that gets included, because it is not a number.

The distinction between $[a, b]$ and (a, b) looks pedantic and is not. The Extreme Value Theorem (Chapter 13) is true on $[0, 1]$ and false on $(0, 1)$: the function $f(x) = x$ has no maximum on the open interval, because for any candidate there is a larger x still inside. Half-open intervals are also exactly the convention behind Python slicing and C++ iterator ranges, for the same reason: $[a, b)$ concatenates cleanly with $[b, c)$.

1.9. The numbers in your machine

Everything above is mathematics. Here is what your CPU actually has.

Definition 1.18 (floating point, informally).

An IEEE-754 double is a number of the form

$$\pm m \times 2^e$$

where the **significand** m has 53 bits of precision and the exponent e is an integer in a bounded range. There are finitely many of them — about 2^{64} — and they are not evenly spaced: they are dense near zero and sparse far from it.

Three consequences, each of which is a mathematical statement failing:

Floats are not closed under arithmetic. The sum of two doubles is generally not a double, so it is rounded. Therefore addition is **not associative**: $(a + b) + c$ and $a + (b + c)$ can differ. Every parallel reduction in your codebase is nondeterministic for this reason.

Floats are not dense. Between any two distinct reals there is another real. Between two adjacent doubles there is nothing. Near 1, the gap is about 2.2×10^{-16} (this is **machine epsilon**); near 10^{16} , the gap exceeds 1, so incrementing does nothing at all.

Floats are not the rationals either. 0.1 is a perfectly good rational and is **not** representable, because $1/10$ in binary is a repeating expansion — exactly the phenomenon of the proposition above, in base 2. Hence the famous $0.1 + 0.2 \neq 0.3$.

Trap: the practical rule.

Never test floating-point values with `==`. Test $|x - y| < \varepsilon$ for a tolerance ε chosen with reference to the **magnitude** of x and y , not an absolute constant — because the spacing scales with magnitude. Chapter 33 makes this quantitative and gives the correct relative-error form.

None of this makes the mathematics useless; it makes it **necessary**. You cannot reason about the error of a computation without first knowing what the exact answer would have been.

1.10. Exercises

- 1.1 Prove that $\sqrt{3}$ is irrational. Then explain exactly where the same argument breaks down if you try to run it on $\sqrt{4}$ — it must break down somewhere, since $\sqrt{4} = 2$ is rational.
- 1.2 Show that the sum of a rational and an irrational number is always irrational. Then show, by giving examples, that the sum of two irrationals can be either rational or irrational.
- 1.3 Convert $0.\overline{142857}$ to a fraction using the shift-and-subtract technique from the proposition on decimal expansions.
- 1.4 Prove the **reverse triangle inequality**:

$$||x| - |y|| \leq |x - y|.$$

Hint: write $x = (x - y) + y$ and apply the ordinary triangle inequality.

- 1.5 Is the set of all finite-length strings over the alphabet $\{a, b\}$ countable? Is the set of all **infinite** strings over $\{a, b\}$ countable? Justify both answers.
- 1.6 In IEEE-754 double precision, find a concrete triple a, b, c for which $(a + b) + c \neq a + (b + c)$, and explain in terms of rounding why your example works. You may reason about it on paper; you do not need to run it.
- 1.7 Let $S = \{1 - \frac{1}{n} : n \in \mathbb{N}, n \geq 1\}$. Write down $\sup S$. Is $\sup S$ an element of S ? Explain what this shows about the difference between a supremum and a maximum.

CHAPTER 2.

Algebra as a Rewriting System

2.1. What algebra actually is

School presents algebra as a bag of procedures: “move it to the other side and flip the sign”, “cross-multiply”, “FOIL”. Learned that way it is unmemorable and, worse, unextendable — the moment an expression does not match a procedure you have seen, you are stuck.

There is a much better way to see it, and it is one you already have.

The idea: algebra is term rewriting.

An algebraic expression is a **syntax tree**. Algebra is a collection of **rewrite rules** that transform one tree into another tree denoting the same value. Solving an equation is a **search**: apply rewrites until the tree has the shape $x = \langle \text{something with no } x \text{ in it} \rangle$.

That is all of it. “Moving to the other side” is not a rule; it is a derived shortcut for “apply the same operation to both sides, then simplify”. If you keep the real rules in mind you can rederive every shortcut and you will never be stuck at an unfamiliar shape.

The rules are these. They are not theorems — for the real numbers they are axioms, the defining properties of a **field**.

Definition 2.1 (field axioms).

For all $a, b, c \in \mathbb{R}$:

- | | |
|--|------------------|
| (F1) $a + b = b + a$ and $ab = ba$ | (commutativity) |
| (F2) $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$ | (associativity) |
| (F3) $a + 0 = a$ and $a \cdot 1 = a$ | (identities) |
| (F4) $a + (-a) = 0$, and if $a \neq 0$, $a \cdot a^{-1} = 1$ | (inverses) |
| (F5) $a(b + c) = ab + ac$ | (distributivity) |

Every manipulation you will ever perform on a real-number expression is a finite composition of these five, plus substitution of equals for equals.

Example 2.2 (why a minus times a minus is a plus).

This is usually asserted. It is provable, and only from the axioms above.

First, $a \cdot 0 = 0$ for every a . Why: $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$ by (F3) and (F5). Adding $-(a \cdot 0)$ to both sides gives $0 = a \cdot 0$.

Next, $(-a)b = -(ab)$. Why: $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$ by (F5) and the previous paragraph. So $(-a)b$ is **the** additive inverse of ab , which is what $-(ab)$ means.

Finally, $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, using the previous line twice and the fact that $-(-x) = x$.

Nothing was assumed except the axioms. A negative times a negative is a positive **because it has to be** — any other convention would break distributivity.

2.2. Solving equations: which steps are safe

Here is the part that school genuinely gets wrong, and it costs people marks and, later, correctness bugs.

An equation is a **predicate**: a statement that is true for some values of x and false for others. Its **solution set** is the set of x making it true. When you rewrite an equation you are transforming one predicate into another, and the only thing that matters is what happens to the solution set.

Definition 2.3 (equivalent, and one-way, transformations).

A transformation of an equation is **equivalent** (reversible) if the new equation has exactly the same solution set.

It is **one-way** if the new solution set may be **larger** — in which case you have introduced possible **extraneous solutions** and must check each candidate — or **smaller**, in which case you have **lost** solutions, which is worse because nothing warns you.

Operation	Safe?	Why / what to watch
Add or subtract anything from both sides	safe	Always reversible.
Multiply both sides by a nonzero constant	safe	Reversible: divide back.
Multiply both sides by an expression	danger	If the expression can be 0, you gain roots.
Divide both sides by an expression	danger	If it can be 0, you lose roots.
Square both sides	danger	Gains roots: $x = -2$ becomes $x^2 = 4$.
Take square root of both sides	danger	Must write $\pm$, or you lose one.
Apply a strictly increasing function	safe	E.g. exp, or log on positives.
Cancel a common factor	danger	Same as dividing. State the case where it is zero.

Example 2.4 (losing a root by cancelling).

Solve $x^2 = 5x$.

The tempting move is to divide both sides by x , giving $x = 5$. That is **wrong**: it silently assumed $x \neq 0$, and $x = 0$ is a genuine solution.

The correct move is to never divide by something that might vanish. Instead bring everything to one side and factor:

$$x^2 - 5x = 0 \Rightarrow x(x - 5) = 0 \Rightarrow x = 0 \text{ or } x = 5.$$

The step from a product being zero to a factor being zero is legitimate and has a name.

Theorem 2.5 (zero product property).

In $\mathbb{R}$ (indeed in any field), if $ab = 0$ then $a = 0$ or $b = 0$.

Proof. Suppose $ab = 0$ and $a \neq 0$. Then a^{-1} exists, and

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

■

Trap: this fails in the integers modulo n .

In arithmetic mod 12, $3 \times 4 = 0$ but neither factor is zero. Such elements are called **zero divisors**, and their presence is exactly why $\mathbb{Z}/12\mathbb{Z}$ is not a field. Since your machine integers live in $\mathbb{Z}/2^{64}\mathbb{Z}$, factoring-based reasoning about them is not automatically valid. Chapter 12 sorts out when it is.

Example 2.6 (gaining a root by squaring).

Solve $\sqrt{x+6} = x$.

Square both sides: $x+6 = x^2$, so $x^2 - x - 6 = 0$, so $(x-3)(x+2) = 0$, so $x = 3$ or $x = -2$.

Now **check**, because squaring was one-way. $x = 3$: $\sqrt{9} = 3$. Good. $x = -2$: $\sqrt{4} = 2 \neq -2$. Rejected.

The extraneous root appeared because squaring cannot distinguish 2 from -2 ; it is not injective.

Any time you apply a non-injective function to both sides, you must check.

2.3. Expanding and factoring

Expanding is mechanical: apply distributivity until no products of sums remain. Factoring is the reverse and is genuinely a search, which is why it is harder. These identities are the ones worth having in memory, because they are the patterns the search looks for.

$$a^2 - b^2 = (a - b)(a + b)$$

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$$

The last one contains the others and is worth understanding rather than memorising.

Proposition 2.7 (the general difference of powers).

For every integer $n \geq 1$,

$$a^n - b^n = (a - b) \sum_{k=0}^{n-1} a^{n-1-k} b^k.$$

Proof. Expand the product. Distributing a over the sum gives $\sum_{k=0}^{n-1} a^{n-k} b^k$, and distributing $-b$ gives $-\sum_{k=0}^{n-1} a^{n-1-k} b^{k+1}$. Reindex the second sum with $j = k + 1$, so it runs $j = 1$ to n and equals $-\sum_{j=1}^n a^{n-j} b^j$.

Now subtract. The two sums have identical terms $a^{n-k} b^k$ for $k = 1, \dots, n-1$, which cancel in pairs. What survives is the $k = 0$ term of the first sum, a^n , and the $j = n$ term of the second, $-b^n$. ■

The idea: telescoping.

That proof is a **telescoping sum**: almost everything cancels with its neighbour and only the two ends survive. Telescoping is one of the three or four genuinely reusable summation techniques; you will see it again for series (Chapter 17) and for the fundamental theorem of calculus (Chapter 16), which is, in essence, one enormous telescoping sum.

Setting $b = 1$ gives the geometric-sum identity, which you will use constantly for complexity analysis:

$$1 + a + a^2 + \cdots + a^{n-1} = \frac{a^n - 1}{a - 1} \quad (a \neq 1).$$

Example 2.8 (why doubling arrays is amortised $O(1)$).

Growing an array by doubling, from size 1 to size $n = 2^k$, copies

$$1 + 2 + 4 + \cdots + 2^{k-1} = \frac{2^k - 1}{2 - 1} = n - 1$$

elements in total. Spread over n insertions that is under one copy each. The geometric sum is the entire content of the amortised analysis.

2.4. Quadratics, properly

Theorem 2.9 (the quadratic formula).

Let $a \neq 0$. The solutions of $ax^2 + bx + c = 0$ are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Proof. This is **completing the square**, and the technique matters more than the formula.

Divide by a (legal, $a \neq 0$):

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

We want to recognise $x^2 + \frac{b}{a}x$ as the start of a perfect square. Since $(x + t)^2 = x^2 + 2tx + t^2$, matching the linear terms forces $2t = \frac{b}{a}$, so $t = \frac{b}{2a}$. Add and subtract t^2 :

$$\underbrace{x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}}_{\left(x + \frac{b}{2a}\right)^2} - \frac{b^2}{4a^2} + \frac{c}{a} = 0.$$

Rearranging,

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}.$$

Now take square roots — remembering the $\pm$, since this step is one-way:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a},$$

and subtract $\frac{b}{2a}$. ■

Definition 2.10 (discriminant).

$\Delta = b^2 - 4ac$. If $\Delta > 0$ there are two distinct real roots; if $\Delta = 0$ one repeated real root; if $\Delta < 0$ no real roots (but two complex ones — Chapter 6).

Completing the square is not a one-off trick for quadratics. It reappears as: the derivation of the Gaussian integral (Chapter 27), the reason the normal distribution has the density it does, the standard proof that a positive definite quadratic form has a unique minimiser (Chapter 34), and the algebraic core of the Cholesky decomposition. Learn the move, not the formula.

Trap: the quadratic formula is numerically unstable.

If $b^2 \gg 4ac$ then $\sqrt{b^2 - 4ac} \approx |b|$, and one of the two numerator terms is a difference of nearly equal numbers — **catastrophic cancellation**. You lose most of your significant digits.

The fix uses the fact that the product of the roots is c/a . Compute the well-conditioned root

$$x_1 = \frac{-b - \operatorname{sgn}(b)\sqrt{b^2 - 4ac}}{2a}$$

(both numerator terms have the same sign, so no cancellation), then get the other from $x_2 = c/(ax_1)$. Chapter 33 explains the general principle.

2.5. Polynomials

Definition 2.11 (polynomial).

A **polynomial** in x is an expression

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

with the a_i constants and $a_n \neq 0$; n is the **degree**. A **root** of p is a value r with $p(r) = 0$.

Theorem 2.12 (factor theorem).

r is a root of p if and only if $(x - r)$ divides $p(x)$ exactly.

Proof. ($\Leftarrow$) If $p(x) = (x - r)q(x)$ then $p(r) = 0 \cdot q(r) = 0$.

($\Rightarrow$) Divide p by $(x - r)$ with remainder. Because the divisor has degree 1, the remainder has degree 0 — it is a constant c :

$$p(x) = (x - r)q(x) + c.$$

Substituting $x = r$ gives $p(r) = c$. So if $p(r) = 0$ then $c = 0$ and the division is exact. ■

Corollary 2.13.

A polynomial of degree n has at most n roots.

Proof. Induction on n . Degree 0: a nonzero constant has no roots. Suppose the claim holds for degree $n - 1$, and let p have degree n . If p has no root we are done. Otherwise let r be a root; by the factor

theorem $p(x) = (x - r)q(x)$ with $\deg q = n - 1$. Any root $s \neq r$ of p satisfies $(s - r)q(s) = 0$ with $s - r \neq 0$, so $q(s) = 0$ by the zero product property. By hypothesis q has at most $n - 1$ roots, so p has at most n . ■

Example 2.14 (this corollary is a real tool).

Two polynomials of degree at most n that agree at $n + 1$ distinct points are **identical**. (Subtract them; the difference has degree at most n and more than n roots, so it is the zero polynomial.)

That single fact underwrites polynomial interpolation, Reed–Solomon error correction, Shamir secret sharing, and the Schwartz–Zippel lemma behind randomised polynomial identity testing.

2.5.1. Vieta and coefficient relations

If $p(x) = x^2 + bx + c$ has roots r_1, r_2 then $p(x) = (x - r_1)(x - r_2) = x^2 - (r_1 + r_2)x + r_1r_2$, so

$$r_1 + r_2 = -b, \quad r_1r_2 = c.$$

In general, for a monic degree- n polynomial, the coefficient of x^{n-1} is minus the sum of the roots and the constant term is $(-1)^n$ times their product. These will resurface in Chapter 23: the trace of a matrix is the sum of its eigenvalues and the determinant is their product, and that is **this identity**, applied to the characteristic polynomial.

2.6. Rational expressions and partial fractions

A **rational function** is a quotient $p(x)/q(x)$ of polynomials. Adding them requires a common denominator; the operation you will need far more often is the reverse.

Definition 2.15 (partial fraction decomposition).

Given p/q with $\deg p < \deg q$ and q factored into distinct linear factors $q(x) = (x - r_1) \cdots (x - r_n)$, there are unique constants A_i with

$$\frac{p(x)}{q(x)} = \frac{A_1}{x - r_1} + \frac{A_2}{x - r_2} + \cdots + \frac{A_n}{x - r_n}.$$

Example 2.16 (doing one).

Decompose

$$\frac{3x + 1}{x^2 - x - 2}$$

.

Factor the bottom: $x^2 - x - 2 = (x - 2)(x + 1)$. Write

$$\frac{3x + 1}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1}.$$

Multiply through by $(x - 2)(x + 1)$:

$$3x + 1 = A(x + 1) + B(x - 2).$$

This must hold for **all** x , so we may choose convenient values. Put $x = 2$: $7 = 3A$, so $A = 7/3$. Put $x = -1$: $-2 = -3B$, so $B = 2/3$. Hence

$$\frac{3x+1}{x^2-x-2} = \frac{7/3}{x-2} + \frac{2/3}{x+1}.$$

The reason to care: Chapter 16 will show that $\int (dx)/(x-r) = \ln|x-r|$, so decomposing a rational function turns an impossible-looking integral into a sum of logarithms. The same decomposition is how generating functions are solved for closed-form recurrences (Chapter 11).

2.7. Inequalities

Inequalities obey the field axioms with one crucial exception.

Definition 2.17 (order axioms).

For all $a, b, c \in \mathbb{R}$:

- (O1) Exactly one of $a < b$, $a = b$, $a > b$ holds.
- (O2) If $a < b$ and $b < c$ then $a < c$.
- (O3) If $a < b$ then $a + c < b + c$.
- (O4) If $a < b$ and $c > 0$ then $ac < bc$.

Proposition 2.18 (multiplying by a negative flips the inequality).

If $a < b$ and $c < 0$ then $ac > bc$.

Proof. $c < 0$ means $-c > 0$ (add $-c$ to both sides of $c < 0$ using (O3)). By (O4), $a(-c) < b(-c)$, i.e. $-ac < -bc$. Adding $ac + bc$ to both sides gives $bc < ac$. ■

That is the exception, and it is where nearly all inequality errors come from. The other common trap:

Trap: never cross-multiply an inequality blindly.

From $\frac{1}{x} < 2$ you may **not** conclude $1 < 2x$. That step multiplies by x , whose sign is unknown. If $x > 0$ it is valid and gives $x > 1/2$; if $x < 0$ the inequality flips and $1/x$ is negative, hence automatically less than 2, so every negative x works. The true solution set is $(-\infty, 0) \cup (1/2, \infty)$ — and the naive answer misses half of it.

The reliable method is **sign analysis**: move everything to one side, factor, find where each factor changes sign, and tabulate.

Theorem 2.19 (AM–GM for two terms).

For $a, b \geq 0$:

$$\frac{a+b}{2} \geq \sqrt{ab},$$

with equality iff $a = b$.

Proof. $(\sqrt{a} - \sqrt{b})^2 \geq 0$ because squares of reals are nonnegative. Expanding: $a - 2\sqrt{ab} + b \geq 0$, so $a + b \geq 2\sqrt{ab}$. Equality holds exactly when $\sqrt{a} - \sqrt{b} = 0$, i.e. $a = b$. ■

The “prove it is a square” move is the most common way to establish an inequality, and it generalises: in Chapter 24 the Cauchy–Schwarz inequality is proved by observing that a certain quadratic is a square and therefore has non-positive discriminant.

2.8. Summation notation

Definition 2.20 (sigma notation).

$$\sum_{i=m}^n f(i) = f(m) + f(m+1) + \cdots + f(n).$$

If $n < m$ the sum is **empty** and equals 0 by convention. The analogous product $\prod$ has empty value 1.

This is a **for** loop with an accumulator, and the conventions match the ones you would choose. The empty-sum convention exists so that $\sum_{i=m}^n + \sum_{i=n+1}^p = \sum_{i=m}^p$ holds without special cases — exactly the reason your slice indices are half-open.

Three identities, each proved once and used forever:

Theorem 2.21 (the three standard sums).

For $n \geq 1$ and $r \neq 1$:

$$\begin{aligned}\sum_{i=1}^n i &= \frac{n(n+1)}{2} \\ \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6} \\ \sum_{i=0}^{n-1} r^i &= \frac{r^n - 1}{r - 1}\end{aligned}$$

Proof. **First.** Write the sum forwards and backwards and add termwise:

$$\begin{aligned}S &= 1 + 2 + \cdots + n \\ S &= n + (n-1) + \cdots + 1\end{aligned}$$

Each of the n columns sums to $n+1$, so $2S = n(n+1)$.

Second. Telescope $\sum_{i=1}^n [(i+1)^3 - i^3]$. On the one hand it collapses to $(n+1)^3 - 1$. On the other, expanding the bracket gives $3i^2 + 3i + 1$, so the sum is $3 \sum i^2 + 3 \sum i + n$. Equate:

$$3 \sum i^2 = (n+1)^3 - 1 - 3 \frac{n(n+1)}{2} - n,$$

and simplifying the right side gives $n(n+1)(2n+1)/2$. Divide by 3.

Third. This is the difference-of-powers proposition with $a = r$, $b = 1$. ■

The idea: the perturbation trick.

The second proof is worth internalising as a general method. To find $\sum i^k$, telescope $\sum [(i+1)^{k+1} - i^{k+1}]$: the left side collapses, the right side expands into $\sum i^k$ plus lower-order sums you already

know. That gives every power sum by recursion, and it is the discrete ancestor of integration by parts.

2.9. Exercises

2.1 Using only the field axioms (F1)–(F5), prove that the additive identity is unique: if $a + z = a$ for every a , then $z = 0$.

2.2 Solve $\sqrt{2x+3} = x$, being explicit about which step is one-way and which candidate roots must be checked and why.

2.3 Complete the square on $2x^2 - 12x + 7$ to write it in the form $a(x-h)^2 + k$. Read off the minimum value of the expression and the x at which it occurs, without using calculus.

2.4 Prove that $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$ — that is, the sum of the first n cubes is the square of the sum of the first n integers. Use the perturbation trick.

2.5 Decompose

$$\frac{5x-4}{x^2-x-6}$$

into partial fractions.

2.6 Solve the inequality

$$\frac{x-1}{x+2} \geq 1$$

by sign analysis. State the solution set as a union of intervals, and say explicitly what goes wrong if you instead multiply both sides by $x+2$.

2.7 Prove that for all real x and y , $x^2 + y^2 \geq 2xy$, and identify when equality holds. Then deduce AM–GM for two nonnegative numbers from it by a substitution.

2.8 A polynomial p of degree 3 satisfies $p(0) = p(1) = p(2) = p(3) = 0$. What is p ? Justify your answer with the corollary to the factor theorem.

CHAPTER 3.

Functions

3.1. The definition, and why it is so fussy

You already have a working notion of “function”: something you call with an argument and get a result. That intuition is right, and mathematics makes it precise in a way that is worth absorbing, because the precision buys you things later.

Definition 3.1 (function).

A **function** f from a set A to a set B , written $f : A \rightarrow B$, assigns to **each** element $a \in A$ **exactly one** element $f(a) \in B$.

A is the **domain**, B is the **codomain**, and

$$\text{range}(f) = \{f(a) : a \in A\} \subseteq B$$

is the **range** (or image): the set of values actually attained.

Three clauses in there do real work.

“**Each**”. f must be defined on every element of A . A “function” that throws on some inputs is not a function on that domain — it is a function on a smaller domain. This is why mathematicians fuss about domains: $f(x) = \frac{1}{x}$ is not a function $\mathbb{R} \rightarrow \mathbb{R}$; it is a function $\mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$.

“**Exactly one**”. No multi-valued results. This is why $\sqrt{x}$ is **defined** to be the nonnegative root: if it returned both, it would not be a function. It is why we write $x = \pm\sqrt{c}$ rather than pretending the square root is two-valued.

Codomain versus range. The codomain is declared; the range is discovered. This is exactly the difference between a function’s declared return type and the set of values it can actually produce. `fn f(x: i32) -> i32 { x * x }` has codomain `i32` and range “the perfect squares”. The distinction is not pedantry; whether a function is **onto** depends entirely on which codomain you declared.

The idea: a function is a lookup table you may not be able to store.

The set-theoretic definition of a function is literally its table of pairs: f is the set $\{(a, f(a)) : a \in A\}$. That set is called the **graph**.

So a function is not a formula. A formula is one way to **specify** a function, the way source code is one way to specify a behaviour. Two different formulas can define the same function (they are then equal, as functions), and plenty of functions have no formula at all.

Trap: f and $f(x)$ are different things.

f is the function. $f(x)$ is the **value** of the function at x — a number, not a function. Writing “the function $f(x) = x^2$ ” is a universal and harmless abuse of notation, but when you get to Chapter 26 and start differentiating with respect to functions, the distinction stops being cosmetic. In programming terms: `f` versus `f(x)`.

3.2. Composition

Definition 3.2 (composition).

Given $f : A \rightarrow B$ and $g : B \rightarrow C$, their **composition** $g \circ f : A \rightarrow C$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

Note the order: $g \circ f$ means “ f first”. This reads backwards and everyone finds it irritating; it is that way so that the notation $g(f(x))$ matches.

Proposition 3.3 (composition is associative).

$(h \circ g) \circ f = h \circ (g \circ f)$, whenever the compositions are defined.

Proof. Two functions are equal iff they have the same domain and codomain and agree at every point. Both sides go $A \rightarrow D$. And for any $x \in A$,

$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))),$$

$$(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))).$$

They agree everywhere, so they are equal. ■

That proof looks trivial and is included on purpose: it is the template for every “these two functions are the same” argument, and it is the reason function composition forms a **monoid**, which is why you can pipeline transformations in any grouping you like.

3.3. Injective, surjective, bijective

These three words describe how a function can fail to be invertible, and they are worth getting exactly right because they recur in linear algebra as statements about matrices.

Definition 3.4 (injective, surjective, bijective).

$f : A \rightarrow B$ is

- **injective** (one-to-one) if $f(a_1) = f(a_2)$ implies $a_1 = a_2$. Equivalently: distinct inputs give distinct outputs; no collisions.
- **surjective** (onto) if for every $b \in B$ there is some $a \in A$ with $f(a) = b$. Equivalently: the range is the whole codomain.
- **bijective** if both.

Example 3.5 (the four cases, concretely).

- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$. Not injective ($f(2) = f(-2)$); not surjective (nothing maps to -1).
- $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$. Both. Bijective.
- $f : \mathbb{R} \rightarrow [0, \infty), f(x) = x^2$. Surjective now, still not injective.
- $f : [0, \infty) \rightarrow \mathbb{R}, f(x) = x^2$. Injective now, not surjective.

Same formula, four different functions, because the domain and codomain are part of the function. This is exactly the point of declaring them.

Theorem 3.6 (invertibility).

$f : A \rightarrow B$ has a two-sided inverse $f^{-1} : B \rightarrow A$ (meaning $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$) if and only if f is bijective.

Proof. ($\Rightarrow$) Suppose f^{-1} exists. If $f(a_1) = f(a_2)$, apply f^{-1} to both sides: $a_1 = a_2$, so f is injective. Given $b \in B$, the element $a = f^{-1}(b)$ satisfies $f(a) = b$, so f is surjective.

($\Leftarrow$) Suppose f is bijective. For each $b \in B$, surjectivity says there is at least one a with $f(a) = b$, and injectivity says there is at most one. So there is exactly one; define $f^{-1}(b)$ to be it. That definition is legitimate precisely because “exactly one” holds, and the two composition identities are immediate. ■

The idea: the same three words in linear algebra.

Keep these. In Part IV, for a matrix A viewed as a function $x \mapsto Ax$:

- injective $\leftrightarrow$ the null space is $\{0\}$ $\leftrightarrow$ columns are linearly independent;
- surjective $\leftrightarrow$ the columns span the whole target space;
- bijective $\leftrightarrow A$ is invertible $\leftrightarrow \det A \neq 0$.

Linear algebra is largely the study of what these three conditions look like when the function happens to be linear.

3.4. Graphs and transformations

The **graph** of $f : \mathbb{R} \rightarrow \mathbb{R}$ is the set of points $(x, f(x))$ in the plane. Reading a graph fluently is a genuine skill and it is mostly about knowing how five modifications move it.

Expression	Effect on the graph	Watch out
$f(x) + c$	shift up by c	outside the function: vertical
$f(x + c)$	shift left by c	inside: horizontal, and backwards
$af(x)$	stretch vertically by a	$a < 0$ also flips over the x -axis
$f(ax)$	compress horizontally by a	inside: backwards again
$f(-x)$	reflect in the y -axis	
$-f(x)$	reflect in the x -axis	

The idea: why inside-the-function transformations run backwards.

This trips up everyone once. Consider $g(x) = f(x - 3)$. To find g 's value at $x = 10$, you ask f for its value at 7. So whatever f was doing at 7 now happens at 10: the picture has moved **right**, even though the formula says minus.

The rule: transformations applied to the **output** do what they say; transformations applied to the **input** do the opposite, because you are changing where you have to stand to see the same value.

Definition 3.7 (even and odd).

f is **even** if $f(-x) = f(x)$ for all x (symmetric about the y -axis), and **odd** if $f(-x) = -f(x)$ (symmetric under a 180° rotation about the origin).

$\cos$ and x^2 are even; $\sin$ and x^3 are odd; e^x is neither. Every function on $\mathbb{R}$ splits uniquely into an even part and an odd part:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}.$$

This decomposition is the baby version of a Fourier transform, and the reason $\int_{-a}^a$ of an odd function is always 0.

3.5. Monotonicity and boundedness

Definition 3.8 (monotone).

f is **increasing** on an interval if $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$, and **strictly increasing** if the conclusion is $f(x_1) < f(x_2)$. Decreasing is the mirror image. **Monotone** means one or the other.

Proposition 3.9.

A strictly monotone function is injective.

Proof. Let f be strictly increasing and suppose $a_1 \neq a_2$. Without loss of generality $a_1 < a_2$ (otherwise swap names). Then $f(a_1) < f(a_2)$, so in particular $f(a_1) \neq f(a_2)$. The decreasing case is identical with the inequality reversed. ■

This is the everyday test for invertibility, and it is why “apply a strictly increasing function to both sides” was listed as a **safe** operation on equations in Chapter 2: strictly increasing means injective means the step is reversible.

Definition 3.10 (bounded).

f is **bounded above** on S if there is M with $f(x) \leq M$ for all $x \in S$; bounded below similarly; **bounded** if both, equivalently if $|f(x)| \leq M$ for some M .

3.6. The functions you must know on sight

Polynomials. Defined in Chapter 2. Continuous and smooth everywhere. Degree n behaviour at large $|x|$ is dominated by the leading term $a_n x^n$; this is the entire content of “big-O of a polynomial is its leading term”.

Rational functions $p(x)/q(x)$. Undefined where $q = 0$. Near such a point the graph typically shoots off to $\pm\infty$ — a **vertical asymptote**. As $|x| \rightarrow \infty$, the behaviour is governed by comparing $\deg p$ and $\deg q$:

$$\begin{aligned} \deg p < \deg q &\Rightarrow y \rightarrow 0, \\ \deg p = \deg q &\Rightarrow y \rightarrow a_n/b_m \text{ (horizontal asymptote),} \\ \deg p > \deg q &\Rightarrow |y| \rightarrow \infty. \end{aligned}$$

Piecewise functions, including $|x|$, $\lfloor x \rfloor$, $\lceil x \rceil$ and the sign function. These are where continuity and differentiability go to die, and therefore where the interesting counterexamples live. $|x|$ is continuous everywhere but not differentiable at 0 — which is exactly why ReLU networks need a subgradient convention (Chapter 36).

The exponential and logarithm, Chapter 4. The trigonometric functions, Chapter 5.

3.7. Sequences are functions too

Definition 3.11 (sequence).

A **sequence** of real numbers is a function $a : \mathbb{N} \rightarrow \mathbb{R}$. We write a_n rather than $a(n)$, and denote the whole sequence by $(a_n)_{n=0}^{\infty}$ or just (a_n) .

That is not a reframing for its own sake. Once a sequence is a function, all the vocabulary transfers: a sequence can be increasing, bounded, injective. And the central theorem of Chapter 13 — every bounded monotone sequence converges — becomes a statement about functions on $\mathbb{N}$, provable directly from the completeness axiom of Chapter 1.

Example 3.12 (a recurrence is a recursively defined function).

The Fibonacci sequence $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ is a function $\mathbb{N} \rightarrow \mathbb{N}$ specified by recursion rather than by a formula. That it **is** a well-defined function — that the recursion determines exactly one value at each n — is itself a theorem, and its proof is strong induction (Chapter 7). Chapter 11 finds the closed form.

3.8. Functions as values

One place where your instincts are **ahead** of a standard maths course.

In mathematics, functions are objects. You can form sets of them, add them, compose them, and — crucially — feed them to other functions. The set of all functions $A \rightarrow B$ is written B^A , and that notation is not a joke: if $|A| = m$ and $|B| = n$ then there are exactly n^m such functions (Chapter 9).

An **operator** is a function whose input is a function. You know several:

- differentiation, $D(f) = f'$, takes a function and returns a function (Chapter 14);
- definite integration, $I(f) = \int_a^b f$, takes a function and returns a number (Chapter 16);
- in linear algebra, a matrix **is** a function, and matrix multiplication **is** composition (Chapter 20).

The idea: the punchline of Part IV, stated early.

Here is the whole of linear algebra in one sentence, so it is in your head before you get there.

The set of functions $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is unmanageably huge. Restrict to those satisfying $f(x + y) = f(x) + f(y)$ and $f(cx) = cf(x)$ — the **linear** ones — and something remarkable happens: every such function is completely determined by what it does to n basis vectors, so it can be written down as an $m \times n$ grid of numbers. That grid is a matrix.

Matrices are not tables of data that happen to have a weird multiplication rule. They are a **finite encoding of a function**, and the multiplication rule is composition. Everything else in Part IV follows from that.

3.9. Exercises

- 3.1** For each function, state whether it is injective, surjective, both, or neither, **as declared**:
(a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 2n$; (b) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = \lfloor n/2 \rfloor$; (c) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - x$; (d) $f : \mathbb{R} \rightarrow (0, \infty)$, $f(x) = e^x$ (take the usual properties of e^x for granted).
- 3.2** Prove: if $g \circ f$ is injective, then f is injective. Then give an example showing g need not be.
- 3.3** Let $f(x) = 2x + 3$ and $g(x) = x^2$. Compute $f \circ g$ and $g \circ f$ and confirm they differ. Find every x at which they agree.
- 3.4** Describe, in words and in order, the sequence of transformations taking the graph of $y = f(x)$ to the graph of $y = -3f(2x - 4) + 1$. Be careful about the order of the horizontal ones.
- 3.5** Decompose $f(x) = e^x$ into its even and odd parts using the formula in the text. (The two pieces have names: they are $\cosh x$ and $\sinh x$.)
- 3.6** Let A be a set with m elements and B a set with n elements. Explain carefully why there are exactly n^m functions $A \rightarrow B$, and determine how many of them are injective (your answer will be a product).
- 3.7** Find the domain, any vertical asymptotes, and the behaviour as $x \rightarrow \pm\infty$ of

$$f(x) = \frac{2x^2 - 3}{x^2 - 4}.$$

CHAPTER 4.

Exponentials and Logarithms

4.1. Why this chapter matters more than it looks

Exponentials and logarithms are the two functions a programmer meets most often and understands least. Complexity classes, information content, entropy, learning rates, sigmoid and softmax, log-likelihood, decibels, floating-point exponents, log-scale plots: all of it is this chapter.

The good news is that essentially everything follows from **one** rule, applied carefully.

4.2. Powers, built up from nothing

We define a^n in stages, and at each stage the definition is **forced** by insisting one law keeps holding.

Definition 4.1 (positive integer powers).

For $a \in \mathbb{R}$ and $n \in \mathbb{N}$ with $n \geq 1$:

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ copies}}.$$

From this, immediately:

$$a^m a^n = a^{m+n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n.$$

The first is the important one; call it **the law**. Everything below is the answer to “what must a^x mean, if the law is to keep working?”

Zero exponent. The law demands $a^0 a^n = a^{0+n} = a^n$. Dividing by a^n (assuming $a \neq 0$) gives $a^0 = 1$. Not a convention — a consequence.

Negative exponents. The law demands $a^{-n} a^n = a^0 = 1$, so

$$a^{-n} = \frac{1}{a^n}.$$

Fractional exponents. The law demands $(a^{1/n})^n = a^{n/n} = a^1 = a$, so $a^{1/n}$ must be an n -th root of a . For $a > 0$ there is exactly one positive such root, and we take it. Then

$$a^{p/q} = (a^{1/q})^p.$$

Irrational exponents. Now we are stuck: there is no way to write $2^{\sqrt{2}}$ as repeated anything. The definition is by **continuity**: $\sqrt{2}$ is a limit of rationals $1, 1.4, 1.41, 1.414, \dots$, and we define $2^{\sqrt{2}}$ to be the limit of $2^1, 2^{1.4}, 2^{1.41}, \dots$. That this limit exists, and does not depend on which sequence of rationals you chose, is a genuine theorem requiring the completeness of $\mathbb{R}$ from Chapter 1.

The idea: the pattern to notice.

This is how mathematics extends a definition, every single time: you have an operation that works on a small domain and a **law** it satisfies; you extend the domain in the unique way that preserves the law.

It is exactly how you would extend an API without breaking callers. The same move appears again for $0! = 1$ (Chapter 9), for $x^0 = I$ on matrices (Chapter 20), and for the gamma function extending the factorial to non-integers.

Trap: negative bases break everything.

$(-8)^{1/3}$ looks like it should be -2 , and over the reals that is a defensible convention. But then $(-8)^{2/6}$ ought to equal $((-8)^2)^{1/6} = 64^{1/6} = 2$, a different answer for the same exponent written differently.

The law and negative bases are incompatible. So a^x for general real x is defined **only for $a > 0$** . Every identity in this chapter carries that hypothesis, silently.

4.3. The exponential function

Fix $a > 0$, $a \neq 1$. The function $f(x) = a^x$ is defined for all real x , is strictly monotone (increasing if $a > 1$, decreasing if $a < 1$), is everywhere positive, and has range $(0, \infty)$.

Being strictly monotone, it is injective (Chapter 3); onto $(0, \infty)$, it is surjective. So it is a bijection $\mathbb{R} \rightarrow (0, \infty)$ and therefore has an inverse. That inverse is the logarithm.

4.3.1. Where e comes from

Among all the bases you could pick, one is special, and the reason is worth seeing now even though the tools only arrive in Chapter 14.

Consider the growth rate of a^x — how fast the value changes per unit of x . For any base,

$$\frac{a^{x+h} - a^x}{h} = a^x \cdot \frac{a^h - 1}{h}.$$

The growth rate is **proportional to the current value**: that is the defining feature of exponential growth, and it is why compound interest, population models, and radioactive decay all produce exponentials.

The constant of proportionality is $\lim_{h \rightarrow 0} (a^h - 1)/h$, which depends on the base. For $a = 2$ it is about 0.693; for $a = 3$, about 1.099. Somewhere in between there is a base making the constant exactly 1.

Definition 4.2 (Euler's number).

e is the unique base for which $\lim_{h \rightarrow 0} (e^h - 1)/h = 1$. Equivalently, and more usefully for computation,

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!} \approx 2.71828182845905.$$

The consequence, proved properly in Chapter 14, is

$$\frac{d}{dx} e^x = e^x.$$

e^x is the function that is its own derivative. That single property is why e appears everywhere: any time a quantity changes at a rate proportional to itself — and that is a great many quantities — the solution is an exponential in base e .

Example 4.3 (the compound interest reading).

Invest 1 unit at 100% annual interest, compounded n times a year. Each period multiplies by $(1 + 1/n)$, and there are n periods, so you finish with $(1 + 1/n)^n$. Compounding more often helps, but with sharply diminishing returns: $n = 1$ gives 2, $n = 12$ gives 2.613, $n = 365$ gives 2.7146. The limit — continuous compounding — is e .

4.4. Logarithms

Definition 4.4 (logarithm).

For $a > 0$, $a \neq 1$, and $x > 0$, $\log_a x$ is the unique real number y with $a^y = x$. That is, $\log_a$ is the inverse of $x \mapsto a^x$:

$$a^{\log_a x} = x \quad \text{and} \quad \log_a(a^y) = y.$$

Standard abbreviations: $\ln x = \log_e x$, $\lg x = \log_2 x$, and (in most mathematics, and in this book) an unadorned log means $\ln$. In computer science an unadorned log usually means $\lg$ — and inside a big-O it does not matter, for a reason we will see in a moment.

The idea: a logarithm is a question.

$\log_a x$ asks: **to what power must I raise a to get x ?**

Read it that way and the values become obvious rather than memorised. $\log_2 8 = 3$ because $2^3 = 8$. $\log_2(1/4) = -2$ because $2^{-2} = 1/4$. $\log_a 1 = 0$ for every a , because anything to the zero is one.

The second reading, equally useful: $\lg x$ is **the number of bits you need to write x down**, and $\log_{10} x$ is the number of decimal digits. A logarithm counts digits. That is why it grows so slowly, and why an $O(\log n)$ algorithm is effectively free.

Every logarithm law is an exponent law wearing a disguise. Here they are, with the disguise removed.

Theorem 4.5 (the logarithm laws).

For $x, y > 0$ and any real r :

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^r) = r \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a} \quad (\text{change of base})$$

Proof. **Product.** Let $u = \log_a x$ and $v = \log_a y$, so $a^u = x$ and $a^v = y$. Then $xy = a^u a^v = a^{u+v}$ by the law. Taking $\log_a$ of both ends, $\log_a(xy) = u + v$.

Quotient. Identical, with $a^u/a^v = a^{u-v}$.

Power. $x^r = (a^u)^r = a^{ur}$, so $\log_a(x^r) = ur = r \log_a x$.

Change of base. Start from $a^{\log_a x} = x$ and apply $\log_b$ to both sides. The power law gives $(\log_a x)(\log_b a) = \log_b x$. Divide by $\log_b a$, which is nonzero since $a \neq 1$. ■

Example 4.6 (why the base does not matter inside big-O).

Change of base says $\log_2 n = (\ln n)/(\ln 2) = 1.4427... \times \ln n$. The two logarithms differ by a **constant factor**, and big-O discards constant factors. So $O(\log_2 n) = O(\ln n) = O(\log_{10} n)$, and writing “ $O(\log n)$ ” without a base is not sloppiness — it is the observation that the base is not part of the information.

Contrast 2^n and 3^n : those differ by a factor of $(3/2)^n$, which is not constant. Bases matter in exponents and do not matter in logarithms.

4.5. Growth rates: the hierarchy you must have memorised

For large x , and for any constants $\varepsilon > 0$, $k > 0$, $b > 1$:

$$\log x \ll x^\varepsilon \ll x^k \ll b^x \ll x! \ll x^x$$

where $f \ll g$ means $f(x)/g(x) \rightarrow 0$. Each of these gaps is enormous and each one gets proved in Chapter 15 once we have L'Hôpital's rule.

The two facts that surprise people:

Logarithms beat every power of x , no matter how small the power. $\log x$ grows more slowly than $x^{0.001}$. Eventually. (The crossover is astronomically far out, which is why this is surprising.)

Every exponential beats every polynomial. 1.0001^x eventually dwarfs x^{1000} . This is the entire reason exponential-time algorithms are considered intractable and polynomial ones are not.

Example 4.7 (Stirling, and why it is everywhere).

$n!$ sits awkwardly between b^n and n^n . **Stirling's approximation** pins it down:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad \text{so} \quad \ln(n!) \approx n \ln n - n.$$

The logarithmic form is the one you will actually use. It is why comparison-based sorting needs $\Omega(n \log n)$ comparisons: there are $n!$ possible orderings, a comparison tree of depth d distinguishes at most 2^d of them, so $2^d \geq n!$, giving $d \geq \lg(n!) \approx n \lg n - n/\ln 2$.

The entire lower bound is one application of the logarithm laws to Stirling's formula.

4.6. The functions built out of exp and log

4.6.1. The logistic (sigmoid) function

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

It maps $\mathbb{R}$ bijectively onto $(0, 1)$, is strictly increasing, and satisfies $\sigma(-x) = 1 - \sigma(x)$. Its inverse is the **logit** or **log-odds**:

$$\sigma^{-1}(p) = \ln\left(\frac{p}{1-p}\right).$$

Proof. Set $p = 1/(1 + e^{-x})$ and solve for x . Then $1 + e^{-x} = 1/p$, so $e^{-x} = (1-p)/p$, so $-x = \ln((1-p)/p)$, so $x = \ln(p/(1-p))$. ■

That is why logistic regression is called that: it models the **log-odds** as a linear function, and σ is the map back to probabilities. It is not an arbitrary squashing function; it is the inverse of the natural parameter of the Bernoulli distribution, which Chapter 31 makes precise.

4.6.2. Softmax

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

The output is a probability vector: every entry is positive and they sum to 1. Note that adding a constant c to every z_i changes nothing — the e^c factors cancel top and bottom. That invariance is not a curiosity; it is the fix for the following disaster.

Trap: naive softmax overflows, and log-sum-exp is the cure.

e^{1000} overflows a double, which caps out near e^{709} . Meanwhile e^{-1000} underflows to exactly zero. So computing softmax by the formula above fails on perfectly ordinary logits.

Use the shift invariance: subtract $m = \max_j z_j$ from every entry first. Then the largest exponent is $e^0 = 1$, nothing overflows, and the smallest terms underflow harmlessly to zero.

The same trick for the **log** of a sum of exponentials — which is what you need for log-likelihoods — is called **log-sum-exp**:

$$\ln \sum_j e^{z_j} = m + \ln \sum_j e^{z_j - m}, \quad m = \max_j z_j.$$

Every serious numerical library implements this. Now you know why it exists, and it is one line of algebra.

4.6.3. The hyperbolic functions

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

These are the even and odd parts of e^x from Chapter 3. They satisfy $\cosh^2 x - \sinh^2 x = 1$ — a hyperbola, hence the name, against the circle $\cos^2 + \sin^2 = 1$ of Chapter 5. And $\tanh x = 2\sigma(2x) - 1$, so it is the sigmoid rescaled to $(-1, 1)$, which is the whole reason it was the default activation function for twenty years.

4.7. Working in log space

A practical technique that is really a change of coordinates.

Products of many small probabilities underflow: 0.001^{300} is zero in a double. Sums of logarithms do not. So the standard move throughout machine learning and statistics is to work with $\ln p$ instead of p :

$$\ln \prod_i p_i = \sum_i \ln p_i.$$

This converts an underflowing product into a well-behaved sum, converts multiplication into addition (cheaper), and turns the likelihood function into the **log-likelihood**, whose maximiser is the same because $\ln$ is strictly increasing — a fact you now know is exactly the “safe transformation” of Chapter 2.

Example 4.8 (reading a log-log plot).

If $y = Cx^k$ then $\ln y = \ln C + k \ln x$. So on axes where both variables are plotted logarithmically, a **power law appears as a straight line with slope k** .

If instead $y = Cb^x$ then $\ln y = \ln C + x \ln b$: on a **semi-log plot** (log on y only) an **exponential** appears as a straight line.

So a glance at which kind of plot straightens your data tells you which model it obeys. This is the most useful five seconds of mathematics in performance engineering.

4.8. Exercises

- 4.1 Without a calculator, evaluate: (a) $\log_2 32$, (b) $\log_9 3$, (c) $\log_5(1/125)$, (d) $\log_a(a^7)$, (e) $2^{\log_2 11}$.
- 4.2 Prove the change-of-base formula in the specific form $\log_2 n = \ln n / \ln 2$, starting from the definition of the logarithm and using only the power law.
- 4.3 Solve for x : (a) $3^{2x-1} = 27$, (b) $\ln(x) + \ln(x-3) = \ln(10)$ (careful — check for extraneous roots), (c) $e^{2x} - 5e^x + 6 = 0$ (hint: substitute $u = e^x$).
- 4.4 Show that $\sigma(-x) = 1 - \sigma(x)$ for the logistic function, and use it to explain why a binary classifier only needs to output one number rather than two.
- 4.5 Show directly from the definition that softmax is invariant under adding a constant to every input, and then verify that log-sum-exp with the max shift gives the same value as the naive formula, in exact arithmetic.
- 4.6 A merge sort on n items does about $n \lg n$ comparisons. A quadratic sort does about $n^2/2$. At what n do they cross over? Solve as far as you can by hand and then describe how you would finish it numerically.
- 4.7 You measure a routine at $n = 1000$ taking 2 ms and at $n = 4000$ taking 16 ms. Assuming the cost is Cn^k , find k using logarithms. Then say what you would have concluded if the second measurement were 8 ms instead.
- 4.8 Prove that $\tanh x = 2\sigma(2x) - 1$ directly from the definitions.

CHAPTER 5.

Trigonometry

5.1. Forget the triangles

School trigonometry is taught with right triangles: SOH-CAH-TOA, opposite over hypotenuse, and so on. That definition is historically first and pedagogically last, because it only makes sense for angles between 0 and 90°, and every interesting use of trigonometry involves angles outside that range or no angles at all.

The right definition is the **unit circle**, and it makes everything else fall out.

Definition 5.1 (sine and cosine).

Let $\theta \in \mathbb{R}$. Start at the point $(1, 0)$ and travel counterclockwise along the unit circle $x^2 + y^2 = 1$ for an arc length of θ (going clockwise if $\theta < 0$). You arrive at a point. Define

$$(\cos \theta, \sin \theta) = \text{that point.}$$

That is the whole definition. Everything below is a consequence.

The idea: the one picture to keep.

A point going round and round a circle at constant speed. Its x -coordinate is $\cos$, its y -coordinate is $\sin$.

Immediately obvious from that picture, with no algebra:

- $\cos^2 \theta + \sin^2 \theta = 1$, because the point is on the unit circle. That is Pythagoras and it is now a triviality.
- Both functions live in $[-1, 1]$.
- Both repeat with period 2π , because that is how far round the circle is.
- $\cos$ is even and $\sin$ is odd: going $-\theta$ reflects the point in the x -axis, which negates y and leaves x alone.
- $\sin(\theta + \pi/2) = \cos \theta$: a quarter turn swaps the roles.

Every “identity” you were made to memorise is a statement about that rotating point.

5.2. Radians, and why degrees are wrong

The definition above measured angle as **arc length on the unit circle**. That unit is the **radian**. A full turn is the circumference, 2π ; a half turn is π ; a right angle is $\pi/2$.

Degrees are an arbitrary division of the circle into 360 parts, inherited from Babylonian astronomy. Nothing in mathematics prefers them, and using them breaks calculus.

Trap: degrees break the derivative.

In radians, $\frac{d}{d\theta} \sin \theta = \cos \theta$. In degrees, it is

$$\frac{d}{d\theta} \sin(\theta^\circ) = \left(\frac{\pi}{180}\right) \cos(\theta^\circ),$$

and that stray $\pi/180$ then infects every formula downstream.

The deep reason is the small-angle limit

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1,$$

which is **true only in radians** — and which is itself just the statement that for a small arc, arc length and chord length agree. Radians are the unit that makes that ratio 1, and calculus is built on it.

Your language's `sin()` takes radians. Assume it does, always.

Useful conversions to have automatic: $\pi/6 = 30^\circ$, $\pi/4 = 45^\circ$, $\pi/3 = 60^\circ$, $\pi/2 = 90^\circ$, $\pi = 180^\circ$.

5.3. The special values

These come from two triangles and are worth knowing cold.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin \theta$	0	1/2	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	1/2	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undef.

The idea: how to never memorise this table.

Write the sine row as

$$\frac{\sqrt{0}}{2}, \quad \frac{\sqrt{1}}{2}, \quad \frac{\sqrt{2}}{2}, \quad \frac{\sqrt{3}}{2}, \quad \frac{\sqrt{4}}{2}$$

— the numerators are just $\sqrt{0}, \sqrt{1}, \sqrt{2}, \sqrt{3}, \sqrt{4}$ in order. The cosine row is the same list backwards, because $\cos \theta = \sin(\pi/2 - \theta)$.

That is the entire table, reconstructible in five seconds.

5.4. The other four functions

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}.$$

$\tan \theta$ is the **slope** of the line from the origin through the point at angle θ — which is why it appears whenever you convert between a direction and a gradient. It has period π , not 2π , because opposite directions have the same slope, and it blows up at $\theta = \pi/2$ where the line is vertical.

5.5. The identities that matter

Theorem 5.2 (Pythagorean identity).

$$\cos^2 \theta + \sin^2 \theta = 1.$$

Proof. The point $(\cos \theta, \sin \theta)$ lies on the unit circle $x^2 + y^2 = 1$ by definition. That is the identity. ■

Dividing through by $\cos^2 \theta$ gives $1 + \tan^2 \theta = \sec^2 \theta$, and dividing by $\sin^2 \theta$ gives $\cot^2 \theta + 1 = \csc^2 \theta$. Do not memorise those two; derive them when needed, it takes four seconds.

Theorem 5.3 (angle addition).

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

These are the workhorses. Everything else — double angle, half angle, product-to-sum — is a corollary. Here is a proof that also explains **why** they are true, which the usual geometric proof does not.

Proof. Recall from Chapter 3 that a rotation of the plane by angle θ is a linear map, and (as we will verify in Chapter 20) its matrix is

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

because it must send $(1, 0) \mapsto (\cos \theta, \sin \theta)$ and $(0, 1) \mapsto (-\sin \theta, \cos \theta)$ — read straight off the unit circle.

Now, rotating by β and then by α is the same as rotating by $\alpha + \beta$. In matrix terms, $R_\alpha R_\beta = R_{\alpha+\beta}$. Multiply out the left side:

$$\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} = \begin{pmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & \cos \alpha \cos \beta - \sin \alpha \sin \beta \end{pmatrix}.$$

Setting this equal to $R_{\alpha+\beta}$ and comparing the top-left and bottom-left entries gives exactly the two identities. ■

The idea: what the angle addition formulas really are.

They are the statement that **rotations compose**. That is the whole content.

This is also why they reappear as the multiplication rule for complex numbers in polar form (Chapter 6), as the rotation matrices of computer graphics (Chapter 35), and as the reason the Fourier transform turns convolution into multiplication. One fact, four costumes.

Setting $\beta = \alpha$ gives the **double angle** formulas:

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha$$

The last two forms come from substituting the Pythagorean identity, and they rearrange into the **half angle** formulas

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}, \quad \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2},$$

which are how you integrate $\sin^2$ and $\cos^2$ in Chapter 16.

5.6. Waves: amplitude, frequency, phase

Definition 5.4 (sinusoid).

A **sinusoid** is a function

$$f(t) = A \sin(\omega t + \varphi)$$

with **amplitude** A (peak height), **angular frequency** ω (radians per unit time; the period is $2\pi/\omega$), and **phase** φ (where in the cycle it starts).

Theorem 5.5 (sum of two sinusoids of the same frequency).

For any a, b ,

$$a \cos(\omega t) + b \sin(\omega t) = R \cos(\omega t - \varphi)$$

where $R = \sqrt{a^2 + b^2}$ and $\varphi = \text{atan2}(b, a)$.

Proof. Expand the right side with the angle addition formula:

$$R \cos(\omega t - \varphi) = R \cos \varphi \cos(\omega t) + R \sin \varphi \sin(\omega t).$$

Matching coefficients requires $R \cos \varphi = a$ and $R \sin \varphi = b$. Squaring and adding: $R^2(\cos^2 \varphi + \sin^2 \varphi) = a^2 + b^2$, so $R = \sqrt{a^2 + b^2}$. Dividing: $\tan \varphi = b/a$, with the quadrant determined by the signs of a and b — which is exactly what `atan2` computes. ■

That theorem is the reason Fourier analysis works with sines and cosines rather than needing an amplitude-and-phase representation everywhere: any phase shift is a linear combination of a fixed sine and a fixed cosine. It is also, geometrically, the conversion between Cartesian (a, b) and polar (R, φ) coordinates — the same conversion, seen twice.

5.7. Inverse trigonometric functions

None of `sin`, `cos`, `tan` is injective — they repeat forever — so none has an inverse on its full domain. The fix, as always, is to restrict the domain until the function is injective, then invert.

Function	Restricted domain	Range of inverse	Notes
<code>arcsin</code>	$[-\pi/2, \pi/2]$	$[-\pi/2, \pi/2]$	input in $[-1, 1]$
<code>arccos</code>	$[0, \pi]$	$[0, \pi]$	input in $[-1, 1]$
<code>arctan</code>	$(-\pi/2, \pi/2)$	$(-\pi/2, \pi/2)$	input any real

Trap: `arctan` loses half the plane; use `atan2`.

Given a point (x, y) , you want its angle. Writing $\theta = \arctan(y/x)$ is wrong and it is wrong in production code all the time.

Two failures. First, $x = 0$ divides by zero. Second, and worse, the ratio y/x is identical for $(1, 1)$ and $(-1, -1)$, so `arctan` cannot tell the first quadrant from the third; it always returns an answer in $(-\pi/2, \pi/2)$, silently collapsing two quadrants onto two others.

The correct function is the two-argument `atan2(y, x)`, which inspects the signs of both arguments and returns a full-range angle in $(-\pi, \pi]$. Use it. Always. Note the argument order: y first.

5.8. Laws of sines and cosines

For a triangle with sides a, b, c and opposite angles A, B, C :

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad (\text{law of sines})$$

$$c^2 = a^2 + b^2 - 2ab \cos C \quad (\text{law of cosines})$$

The law of cosines is Pythagoras with a correction term, and it reduces to Pythagoras exactly when $C = \pi/2$, since $\cos(\pi/2) = 0$.

The idea: the law of cosines is the dot product.

In Chapter 19 we define the dot product of two vectors and prove

$$u \cdot v = \|u\| \|v\| \cos \theta.$$

Put u and v as two sides of a triangle from a common vertex; the third side is $v - u$. Then

$$\|v - u\|^2 = (v - u) \cdot (v - u) = \|v\|^2 - 2u \cdot v + \|u\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\| \|v\| \cos \theta.$$

That **is** the law of cosines, in one line. The classical geometric proof is a page long; the vector proof is three symbols wide. This is a fair preview of what Part IV does to geometry generally.

5.9. Where you will actually meet this

Rotations. Chapter 35. The matrix R_θ above is the single most-used object in computer graphics, robotics, and physics simulation.

Periodic signals and Fourier analysis. Any periodic function decomposes into a sum of sinusoids. That is the basis of audio processing, JPEG, MP3, and the convolution theorem.

Complex numbers. Chapter 6, where $e^{i\theta} = \cos \theta + i \sin \theta$ compresses this entire chapter into one line.

Distances on a sphere. The haversine formula for great-circle distance is trigonometry, and it is what every mapping API is doing.

Attention. In a transformer, rotary position embeddings (RoPE) encode token position by **rotating** pairs of coordinates by an angle proportional to position. The reason relative position falls out cleanly is the angle addition formula: rotating by θ_m then by $-\theta_n$ leaves a rotation by θ_{m-n} , which depends only on the difference.

5.10. Exercises

- 5.1** Using only the unit circle definition, evaluate $\sin(7\pi/6)$, $\cos(3\pi/4)$, and $\tan(5\pi/3)$. State which quadrant each angle is in and use that to check the sign of your answer.
- 5.2** Prove that $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ from the addition formula, using only the fact that $\sin$ is odd and $\cos$ is even.
- 5.3** Express $3 \cos t + 4 \sin t$ in the form $R \cos(t - \varphi)$. Give R exactly and φ as an `atan2` expression.
- 5.4** Derive $\tan(\alpha + \beta) = (\tan \alpha + \tan \beta) / (1 - \tan \alpha \tan \beta)$ from the sine and cosine addition formulas. State the condition under which your derivation is valid.
- 5.5** Show that $\sin^4 \theta$ can be written as a linear combination of 1 , $\cos 2\theta$, and $\cos 4\theta$, by applying the half-angle formula twice. (This is precisely the manipulation needed to integrate $\sin^4$.)

- 5.6** A point is at $(x, y) = (-3, -4)$. Compute its distance from the origin and its angle, giving the angle in $(-\pi, \pi]$. Explain in one sentence what $\arctan(y/x)$ would have returned instead and why that is wrong.
- 5.7** Prove the law of sines. Hint: drop a perpendicular from one vertex to the opposite side and compute its length two different ways.

CHAPTER 6.

Complex Numbers

6.1. The last bug fix

Chapter 1 described the number systems as a changelog: each one patches an operation the previous one could not perform. One patch remains.

Over $\mathbb{R}$, the equation $x^2 + 1 = 0$ has no solution, because squares are never negative. More generally, Chapter 2's discriminant told us a quadratic with $\Delta < 0$ has no real roots. That is unsatisfying in the same way that $3 - 5$ having no natural-number answer was unsatisfying, and the fix is the same: add the missing answer and see what else you are forced to add with it.

Definition 6.1 (the imaginary unit).

i is a symbol satisfying $i^2 = -1$.

Trap: i is not 'imaginary' in any useful sense.

The name is a seventeenth-century insult that stuck. There is nothing less real about i than about -1 (also once considered absurd) or $\sqrt{2}$ (which got a man killed, allegedly).

A complex number is an ordered pair of reals with a particular multiplication rule. That is a completely concrete object — you can store one in a struct — and it models rotation in the plane exactly. Nothing mystical is happening.

Definition 6.2 (complex numbers).

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\},$$

with $a + bi = c + di$ meaning $a = c$ and $b = d$. We call $a = \operatorname{Re}(z)$ the **real part** and $b = \operatorname{Im}(z)$ the **imaginary part** (note: $\operatorname{Im}(z)$ is a real number, not an imaginary one).

Arithmetic is what you get by treating i as a variable and reducing $i^2 \rightarrow -1$:

$$\begin{aligned}(a + bi) + (c + di) &= (a + c) + (b + d)i \\ (a + bi)(c + di) &= (ac - bd) + (ad + bc)i.\end{aligned}$$

Multiplication deserves a look: expanding gives $ac + adi + bci + bdi^2$, and the last term is $-bd$ because $i^2 = -1$. That single substitution is the entire difference from ordinary polynomial arithmetic.

Proposition 6.3.

$\mathbb{C}$ is a field: it is closed under $+$, $-$, $\times$, and division by any nonzero element.

Proof. Closure under $+$, $-$, $\times$ is visible in the formulas. For division, the question is whether $1/(a + bi)$ can be written in the form $c + di$. Multiply top and bottom by $a - bi$:

$$\frac{1}{a + bi} = \frac{a - bi}{(a + bi)(a - bi)} = \frac{a - bi}{a^2 + b^2} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

The denominator $a^2 + b^2$ is a real number, and it is nonzero exactly when $a + bi \neq 0$. So the inverse exists and has the required form. ■

Definition 6.4 (conjugate and modulus).

The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$. The **modulus** is $|z| = \sqrt{a^2 + b^2}$.

The identity used in that proof, $z\bar{z} = |z|^2$, is the workhorse. It is how you divide, how you rationalise, and how you convert a complex condition into a real one.

Useful and easily checked: $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z} \cdot \bar{w}$, $|zw| = |z||w|$, and z is real iff $z = \bar{z}$.

6.2. The complex plane

Plot $z = a + bi$ at the point (a, b) . Now:

- Addition is **vector addition**. Identical to Chapter 19's picture.
- $|z|$ is the distance from the origin.
- $\bar{z}$ is reflection in the real axis.
- And multiplication is... what, geometrically?

That question has a beautiful answer, and it is the reason complex numbers are useful rather than merely consistent.

Definition 6.5 (polar form).

Any nonzero z can be written

$$z = r(\cos \theta + i \sin \theta)$$

where $r = |z| > 0$ and $\theta = \arg(z)$ is the angle from the positive real axis, computed as `atan2(b, a)` (Chapter 5). θ is determined only up to adding multiples of 2π .

Theorem 6.6 (multiplication is rotation and scaling).

If $z = r(\cos \alpha + i \sin \alpha)$ and $w = s(\cos \beta + i \sin \beta)$ then

$$zw = rs(\cos(\alpha + \beta) + i \sin(\alpha + \beta)).$$

That is: **moduli multiply, arguments add**.

Proof. Multiply out:

$$zw = rs[(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta)].$$

The two bracketed expressions are exactly $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$ by the angle addition formulas of Chapter 5. ■

The idea: this is the whole point.

Multiplying by a complex number is a rotation combined with a scaling.

Multiplying by i (modulus 1, argument $\pi/2$) rotates by a quarter turn. Do it twice and you have rotated by half a turn, which is multiplication by -1 . So $i^2 = -1$ is not an arbitrary stipulation; it is the statement that **two quarter turns make a half turn**.

Once you see it, the entire theory of complex numbers becomes the theory of plane rotations, and the algebra and the geometry stop being separate subjects.

6.3. Euler's formula

Theorem 6.7 (Euler's formula).

For all real θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

The proof needs power series and appears in Chapter 17. But the **reason** it is true can be stated now, and it is the more valuable half.

The idea: why $e^{i\theta}$ has to be a rotation.

Chapter 4 said e^x is characterised by growing at a rate equal to its own value. Formally, $f(t) = e^{kt}$ is the solution of $f' = kf$ with $f(0) = 1$.

Now take $k = i$. The equation $f' = if$ says: the velocity of the point $f(t)$ is its position, rotated a quarter turn.

A velocity always perpendicular to the position vector is precisely **circular motion at constant speed**. Starting at $f(0) = 1$ on the unit circle, moving always perpendicular at unit speed, after time θ you have travelled arc length θ — and by the definition of Chapter 5, you are at $(\cos \theta, \sin \theta)$.

So $e^{i\theta} = \cos \theta + i \sin \theta$ is not a coincidence between unrelated functions. It is the statement that **the exponential of an imaginary number is a rotation**, and it holds because perpendicular velocity means circular motion.

Setting $\theta = \pi$: $e^{i\pi} = \cos \pi + i \sin \pi = -1$, so

$$e^{i\pi} + 1 = 0,$$

which is famous and is really just “half a turn takes you to -1 .”

Euler's formula collapses Chapter 5 into algebra. Polar form becomes $z = re^{i\theta}$, and then the multiplication theorem is nothing but the exponent law:

$$(re^{i\alpha})(se^{i\beta}) = rse^{i(\alpha+\beta)}.$$

The angle addition formulas **are** $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$, with real and imaginary parts read off separately. Two identities anyone would struggle to memorise, replaced by one they already know.

Theorem 6.8 (De Moivre's theorem).

For every integer n ,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Proof. $(e^{i\theta})^n = e^{in\theta}$ by the exponent law. Apply Euler's formula to both sides. ■

Example 6.9 (deriving a triple-angle formula in ten seconds).

Take $n = 3$ and expand the left side of De Moivre with the binomial theorem, writing $c = \cos \theta$, $s = \sin \theta$:

$$(c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = (c^3 - 3cs^2) + i(3c^2s - s^3).$$

Equating real parts with $\cos 3\theta$:

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta = 4 \cos^3 \theta - 3 \cos \theta,$$

using $s^2 = 1 - c^2$. The imaginary parts give $\sin 3\theta$ free of charge.

Doing this with trigonometric identities alone is genuinely unpleasant. This is what complex numbers are **for**: they make trigonometric manipulation into routine algebra.

6.4. Roots of unity

Definition 6.10 (**n -th roots of unity**).

The solutions of $z^n = 1$ in $\mathbb{C}$. There are exactly n of them:

$$\omega_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

Proof. Write $z = re^{i\theta}$. Then $z^n = r^n e^{in\theta}$, and this equals $1 = 1 \cdot e^{i \cdot 0}$ iff $r^n = 1$ and $n\theta$ is a multiple of 2π . Since $r > 0$ is real, $r = 1$; and $\theta = 2\pi k/n$ for integer k . Values of k differing by n give the same z , so there are exactly n distinct roots. ■

Geometrically these are n equally spaced points on the unit circle, one of them at 1 — the vertices of a regular n -gon.

Example 6.11 (**the roots of unity are the FFT**).

The discrete Fourier transform of a length- n signal is

$$X_k = \sum_{j=0}^{n-1} x_j \omega_n^{-jk}, \quad \omega_n = e^{2\pi i/n}.$$

The Fast Fourier Transform is fast because of one algebraic property of these roots: $\omega_n^2 = \omega_{n/2}$. Squaring an n -th root of unity gives an $(n/2)$ -th root of unity, so the length- n problem splits into two length- $n/2$ problems over the same kind of object. That is the divide-and-conquer, and it is the reason the algorithm is $O(n \log n)$ instead of $O(n^2)$.

The whole of the FFT is this chapter plus a recursion.

Proposition 6.12.

The n -th roots of unity sum to zero, for $n \geq 2$.

Proof. They are $1, \omega, \omega^2, \dots, \omega^{n-1}$ for $\omega = e^{2\pi i/n}$, a geometric series. By the formula in Chapter 2,

$$\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1 - 1}{\omega - 1} = 0,$$

legitimate since $\omega \neq 1$ when $n \geq 2$.

Geometrically: n equally spaced unit vectors are symmetric about the origin, so they cancel. ■

6.5. The fundamental theorem of algebra

Theorem 6.13 (fundamental theorem of algebra).

Every non-constant polynomial with complex coefficients has at least one complex root. Consequently, a polynomial of degree n factors completely:

$$p(z) = a_n(z - r_1)(z - r_2) \cdots (z - r_n)$$

for some complex $r_1, \dots, r_n$, not necessarily distinct.

The proof requires complex analysis and is beyond this book; the **consequence** is what you need.

The idea: why this closes the changelog.

$\mathbb{N}$ was not closed under subtraction. $\mathbb{Z}$ was not closed under division. $\mathbb{Q}$ was not closed under limits. $\mathbb{R}$ was not closed under taking roots.

$\mathbb{C}$ is closed under **everything algebraic**. You cannot escape it by solving polynomial equations, no matter how you try. That property is called being **algebraically closed**, and it is why $\mathbb{C}$ is where the number-system changelog ends.

Corollary 6.14 (complex roots of real polynomials come in pairs).

If p has real coefficients and $p(z) = 0$, then $p(\bar{z}) = 0$.

Proof. Conjugation preserves sums and products, and fixes real numbers. So

$$0 = \bar{0} = \overline{p(z)} = \overline{\sum a_k z^k} = \sum \overline{a_k} \bar{z}^k = \sum a_k \bar{z}^k = p(\bar{z}),$$

where $\overline{a_k} = a_k$ because the coefficients are real. ■

That is why a cubic with real coefficients always has at least one real root: the three roots either are all real, or one is real and the other two are a conjugate pair. Non-real roots cannot appear alone.

6.6. Where complex numbers show up in your work

Eigenvalues. A real matrix can have complex eigenvalues (Chapter 23), and when it does they come in conjugate pairs and encode a **rotation** in an invariant plane. A rotation matrix in $\mathbb{R}^2$ has eigenvalues $e^{\pm i\theta}$ — the angle of rotation is sitting right there in the eigenvalue.

Signal processing. Every frequency-domain method is complex arithmetic. A complex number carries amplitude (modulus) and phase (argument) in one object, which is exactly what a sinusoid needs.

Stability analysis. A linear recurrence or differential equation is stable iff certain roots lie inside the unit circle, or in the left half-plane. That is a statement about complex numbers, and it is what a control engineer means by “the poles”.

Quaternions. Chapter 35. The 3D rotation representation used in every game engine is a four-dimensional cousin of $\mathbb{C}$, built for the same reason: to make rotation into multiplication.

6.7. Exercises

6.1 Compute, in the form $a + bi$: (a) $(3 + 2i)(1 - 4i)$, (b) $(2 + i)/(3 - i)$, (c) i^{2027} , (d) $|3 - 4i|$.

- 6.2** Write $z = -1 + i$ in polar form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$, and then use De Moivre to compute z^8 .
- 6.3** Find all three solutions of $z^3 = 8$, in both polar and $a + bi$ form. Sketch where they sit in the plane.
- 6.4** Prove that $|zw| = |z||w|$ two ways: once by direct computation with $z = a + bi$, $w = c + di$, and once in one line using polar form.
- 6.5** Use Euler's formula to prove the identity

$$\cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

by writing each cosine as $(e^{ix} + e^{-ix})/2$ and multiplying out.

- 6.6** Let $\omega = e^{2\pi i/5}$. Compute $1 + \omega + \omega^2 + \omega^3 + \omega^4$ and explain the answer both algebraically and geometrically.
- 6.7** Show that the set of complex numbers with $|z| = 1$ is closed under multiplication and under taking inverses. (This makes it a **group** — the circle group — and it is the same group as 2D rotations.)

CHAPTER 7.

Logic and Proof

7.1. The most important chapter in the book

Everything from here on is proved. If you can read a proof, the rest of this book is a long but straightforward walk. If you cannot, every page will feel like being shown conclusions by someone who will not say why.

You have an advantage. You already reason formally — about invariants, about termination, about whether a refactor preserves behaviour. What follows is mostly **naming** things you do, plus a handful of moves that have no obvious programming analogue.

7.2. Propositions and connectives

Definition 7.1 (proposition).

A **proposition** is a statement that is definitely true or definitely false.

“7 is prime” is a proposition (true). “ $x > 3$ ” is not — it depends on x ; it is a **predicate**, a proposition-valued function. “This sentence is false” is not a proposition at all.

Definition 7.2 (connectives).

For propositions P and Q :

P	Q	$P \Rightarrow Q$	others
T	T	T	$\neg P$: negation, true iff P is false
T	F	F	$P \wedge Q$: true iff both
F	T	T	$P \vee Q$: true iff at least one (inclusive)
F	F	T	$P \Leftrightarrow Q$: true iff same truth value

Two of these need comment.

Or is inclusive. $P \vee Q$ is true when both are. Mathematical English never means exclusive-or unless it says “exactly one of”. This differs from ordinary speech and matches every programming language you use.

Implication is the strange one. $P \Rightarrow Q$ is false in exactly one case: P true and Q false. In particular, when P is false the implication is true regardless of Q .

The idea: why a false premise makes an implication true.

Read $P \Rightarrow Q$ as a **promise**: “if P happens, I guarantee Q .” The promise is broken only if P happens and Q does not. If P never happens, the promise was never tested, so it was not broken — and an unbroken promise counts as kept.

“If n is an integer with $n^2 = 2$, then I am the Pope” is **true**. The hypothesis is never satisfied, so nothing is ever claimed. This is called **vacuous truth** and it is not a loophole; it is what makes universally quantified statements about empty sets behave.

The programmer's version: `all([])` returns `True` and `any([])` returns `False`, for exactly this reason. Every element of the empty list satisfies every predicate, vacuously.

Definition 7.3 (converse, inverse, contrapositive).

Given $P \Rightarrow Q$:

- the **converse** is $Q \Rightarrow P$;
- the **inverse** is $\neg P \Rightarrow \neg Q$;
- the **contrapositive** is $\neg Q \Rightarrow \neg P$.

Theorem 7.4 (contraposition).

$P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are logically equivalent: they have the same truth value in every case.

Proof. Compare truth tables. $P \Rightarrow Q$ is false exactly when P is true and Q is false. $\neg Q \Rightarrow \neg P$ is false exactly when $\neg Q$ is true and $\neg P$ is false — that is, when Q is false and P is true. Same case. Since they are false in exactly the same situation, they are true in exactly the same situations. ■

Trap: the converse is a different statement.

Confusing a statement with its converse is the single most common logical error, in mathematics and in life.

“If it is a square, it is a rectangle” is true. Its converse is false. The contrapositive — “if it is not a rectangle, it is not a square” — is true, and it always is, because it is the same statement.

Note the inverse is the contrapositive of the converse, so **inverse and converse stand or fall together**, and neither follows from the original.

7.3. Quantifiers

Definition 7.5 (quantifiers).

$\forall x \in S : P(x)$ — “for all x in S , $P(x)$ holds.”

$\exists x \in S : P(x)$ — “there exists x in S with $P(x)$.”

Theorem 7.6 (negating quantifiers).

$\neg(\forall x : P(x))$ is equivalent to $\exists x : \neg P(x)$

$\neg(\exists x : P(x))$ is equivalent to $\forall x : \neg P(x)$

The rule in words: **push the negation inward and flip every quantifier**. To deny “all swans are white” you do not assert “all swans are non-white”; you exhibit one black swan.

Example 7.7 (negating something with three quantifiers).

Continuity of f at a (Chapter 13) is

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x : |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

To negate: flip each quantifier in turn, left to right, and negate the innermost statement. The negation of $A \Rightarrow B$ is $A \wedge \neg B$ (check the truth table). So:

$$\exists \varepsilon > 0 \forall \delta > 0 \exists x : |x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon.$$

In words: there is a target tolerance ε that you can never hit, however small you make δ — because for each δ someone can find a nearby x whose image is still far away.

Being able to do this mechanically is worth a great deal. Most “I do not understand this definition” is really “I cannot state its negation.”

Trap: the order of quantifiers changes the meaning entirely.

$$\forall x \exists y : y > x \quad \text{versus} \quad \exists y \forall x : y > x.$$

Over the reals the first is **true** — given any x , take $y = x + 1$; the y is allowed to depend on the x . The second is **false**: it claims a single y larger than every real number.

The rule: **inner variables may depend on outer ones; outer ones may not depend on inner ones**. This is the difference between “every request gets some response” and “there is one response that serves every request”, and it is the difference between pointwise and uniform continuity, and between pointwise and uniform convergence, both of which cause real trouble in analysis.

7.4. The proof techniques

7.4.1. Direct proof

Assume the hypothesis, derive the conclusion by valid steps.

Proposition 7.8.

If n is odd then n^2 is odd.

Proof. n odd means $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is $2m + 1$ with $m = 2k^2 + 2k$ an integer. So n^2 is odd. ■

Note the shape: **unpack the definition** of the hypothesis into an equation, manipulate, **repack into the definition** of the conclusion. A large fraction of all direct proofs are exactly that, and when you are stuck the first move is always “write out what the words mean”.

7.4.2. Proof by contraposition

Prove $\neg Q \Rightarrow \neg P$ instead. Use this when the negations are easier to work with than the originals — typically when P or Q contains “not”, or an existence claim.

Proposition 7.9.

If n^2 is even then n is even.

Proof. We prove the contrapositive: if n is odd then n^2 is odd. That is the previous proposition. ■

A direct attack would start from “ $n^2 = 2k$ ” and try to extract information about n , which is awkward. Contraposition made it free. This is exactly the step used inside the irrationality proof of Chapter 1.

7.4.3. Proof by contradiction

Assume the statement is false, derive something impossible.

Proposition 7.10.

There are infinitely many primes.

Proof. Suppose not: suppose there are finitely many, say $p_1, p_2, \dots, p_n$ is the complete list.

Consider $N = p_1 p_2 \cdots p_n + 1$.

$N > 1$, so it has a prime factor p (every integer greater than 1 does — see Chapter 12). That p must be on our list, since the list is complete. So p divides the product $p_1 \cdots p_n$. It also divides N . Therefore it divides their difference, which is 1.

No prime divides 1. Contradiction. So the list cannot have been complete, and there are infinitely many primes. ■

Trap: contradiction is over-used.

Beginners reach for contradiction reflexively, and it often adds nothing: they assume $\neg Q$, never use it, prove Q directly, and then announce a contradiction. That is a direct proof wearing a costume.

Use contradiction when the assumption $\neg Q$ gives you **something concrete to work with** — as above, where “the list is finite and complete” is a real handle. If you never use the negated assumption, delete it.

7.4.4. Proof by cases

Split into exhaustive possibilities and handle each. The only requirement is that the cases genuinely cover everything — say so explicitly.

Proposition 7.11.

For every integer n , $n^2 + n$ is even.

Proof. Every integer is even or odd; these cases are exhaustive.

Case 1: $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, even.

Case 2: $n = 2k + 1$. Then $n^2 + n = (4k^2 + 4k + 1) + (2k + 1) = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$, even.

Both cases give an even number, so the claim holds for all n . ■

7.4.5. Existence proofs, constructive and not

To prove $\exists x : P(x)$, the honest way is to exhibit an x and verify $P(x)$. That is a **constructive** proof, and it doubles as an algorithm.

A **non-constructive** proof shows something must exist without producing it. These feel like cheating and are not.

Proposition 7.12.

There exist irrational a, b with a^b rational.

Proof. Consider $s = \sqrt{2}^{\sqrt{2}}$. Either s is rational or it is not.

If s is rational, take $a = b = \sqrt{2}$ and we are done.

If s is irrational, take $a = s$ and $b = \sqrt{2}$; then

$$a^b = \left(\sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2,$$

which is rational, and we are done.

Either way such a, b exist. ■

We have proved existence without knowing which case holds — the proof never tells us which pair works. A constructivist rejects this argument; everyone else accepts it. The programmer’s reading: this is a proof that a value exists with no way to compute it, which is precisely why constructive logic and type theory care about the distinction.

7.4.6. Uniqueness

To prove “there is exactly one x with $P(x)$ ”, do two things: exhibit one (existence), then assume x_1 and x_2 both work and show $x_1 = x_2$.

Proposition 7.13.

A nonzero real a has exactly one multiplicative inverse.

Proof. **Existence:** $1/a$ works.

Uniqueness: suppose $ab = 1$ and $ac = 1$. Then

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c.$$

7.4.7. Disproof by counterexample

To disprove $\forall x : P(x)$, exhibit a single x with $\neg P(x)$. One is enough, and no amount of confirming instances substitutes for checking.

Example 7.14.

“Every $n^2 + n + 41$ is prime for $n \in \mathbb{N}$.” True for $n = 0, 1, \dots, 39$ — forty consecutive confirmations. False at $n = 40$, where the value is 41^2 .

Testing is not proving. You knew that.

7.5. Induction

Now the technique you already half-own.

Theorem 7.15 (principle of mathematical induction).

Let $P(n)$ be a predicate on $\mathbb{N}$. If

- (i) $P(0)$ holds (**base case**), and
 - (ii) for every k , $P(k) \Rightarrow P(k+1)$ (**inductive step**),
- then $P(n)$ holds for every $n \in \mathbb{N}$.

This is not a trick; it is a restatement of what $\mathbb{N}$ is (Chapter 1). Every natural number is reachable from 0 by finitely many successor steps, so if truth survives each step and holds at the start, it holds everywhere.

Example 7.16 (a first induction, in full).

Claim: $\sum_{i=1}^n i = n(n+1)/2$ for all $n \geq 1$.

Base case ($n = 1$). The left side is 1; the right side is $1 \cdot 2/2 = 1$. They agree.

Inductive step. Assume the claim for some $k \geq 1$; that is, assume

$$\sum_{i=1}^k i = \frac{k(k+1)}{2}. \quad (\star)$$

We must prove it for $k+1$. Compute:

$$\sum_{i=1}^{k+1} i = \underbrace{\sum_{i=1}^k i}_{\text{use } (\star)} + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1) \left(\frac{k}{2} + 1 \right) = \frac{(k+1)(k+2)}{2},$$

which is the claimed formula with $n = k+1$.

By induction, the claim holds for every $n \geq 1$.

The idea: the shape of every induction, and where the work is.

The inductive step always looks like this: take the expression for $k+1$, **carve out the part that looks like the case k** , replace it using the hypothesis, and simplify.

The single most common failure is not carving anything out — writing down the $k+1$ case and staring at it. If you have not used the inductive hypothesis, you have not done an induction.

The second most common failure is proving $P(k) \Rightarrow P(k+1)$ correctly and forgetting the base case. Then you have a chain with nothing attached to the ground. “ $n = n+1$ for all n ” has a perfectly valid inductive step.

7.5.1. Strong induction

Sometimes $P(k)$ alone is not enough and you need every earlier case.

Theorem 7.17 (strong induction).

If for every n , $[P(m) \text{ holds for all } m < n] \text{ implies } P(n)$, then $P(n)$ holds for all n .

Note there is no separate base case: at $n = 0$ the hypothesis is vacuously true (nothing is less than 0), so the step must prove $P(0)$ outright. In practice you handle the small cases explicitly anyway.

Proposition 7.18.

Every integer $n \geq 2$ is a product of primes.

Proof. Strong induction on n . Let $n \geq 2$ and suppose every integer m with $2 \leq m < n$ is a product of primes.

If n is prime, it is a product of one prime. Done.

Otherwise n is composite: $n = ab$ with $2 \leq a, b < n$. By the inductive hypothesis both a and b are products of primes, so n is the concatenation of those two products. ■

Ordinary induction fails here: knowing about $n - 1$ tells you nothing about the factors of n . This is the same reason your recursion on a tree calls itself on **both** subtrees rather than on “the previous node”.

7.5.2. Well-ordering, and why it is the same thing

Theorem 7.19 (well-ordering principle).

Every nonempty subset of $\mathbb{N}$ has a least element.

Well-ordering, ordinary induction, and strong induction are all equivalent — each can be proved from either of the others. Well-ordering is the most convenient form for “minimal counterexample” arguments:

Proposition 7.20.

Every integer $n \geq 2$ has a prime divisor.

Proof. Suppose not. Then the set S of integers $n \geq 2$ with no prime divisor is nonempty, so by well-ordering it has a least element n_0 .

n_0 cannot be prime (it would divide itself). So $n_0 = ab$ with $2 \leq a < n_0$. Since n_0 is the **least** member of S , $a \notin S$, so a has a prime divisor p . But $p \mid a$ and $a \mid n_0$, so $p \mid n_0$ — contradicting $n_0 \in S$. ■

The idea: well-ordering is termination.

You have used this principle every time you argued that a loop terminates. You exhibit a quantity that is a natural number and strictly decreases each iteration; since there is no infinite strictly decreasing sequence of naturals, the loop must stop. That is well-ordering, and the quantity has a name: a **variant**, or **ranking function**.

Likewise, the descent argument in Chapter 1’s irrationality proof was well-ordering: no infinite chain of ever-smaller positive integers.

7.5.3. Structural induction

Induction is not really about numbers; it is about anything built up from constructors by finitely many steps. To prove a property of every element of a recursively defined type, prove it for the base constructors and prove each recursive constructor preserves it.

Example 7.21 (structural induction on a binary tree).

Claim: a binary tree with n internal nodes has $n + 1$ leaves.

Base case: a tree that is a single leaf has $n = 0$ internal nodes and $1 = 0 + 1$ leaf.

Step: a tree formed as an internal node with subtrees L and R . Suppose L has ℓ internal nodes and $\ell + 1$ leaves, and R has r internal nodes and $r + 1$ leaves. The combined tree has $\ell + r + 1$

internal nodes (the two subtrees plus the new root) and $(\ell + 1) + (r + 1) = (\ell + r + 1) + 1$ leaves. Which is the claim.

Note this is **strong** induction in disguise: we assumed the result for two smaller trees, not for “the previous tree”.

7.5.4. Loop invariants are induction

Making the correspondence explicit, because it is the bridge from what you know to what this chapter is teaching.

```

Sum of an array
function sum(A[0..n-1]):
    s := 0
    i := 0
    while i < n:
        s := s + A[i]
        i := i + 1
    return s

```

Invariant: at the top of each iteration, $s = \sum_{j=0}^{i-1} A[j]$.

Initialisation (base case). Before the loop, $i = 0$ and $s = 0$, and the empty sum is 0. Holds.

Maintenance (inductive step). Suppose it holds with values s, i . The body sets $s' = s + A[i]$ and $i' = i + 1$. Then

$$s' = \sum_{j=0}^{i-1} A[j] + A[i] = \sum_{j=0}^{i'-1} A[j].$$

Holds again.

Termination. The loop exits with $i = n$, so $s = \sum_{j=0}^{n-1} A[j]$, which is what we wanted.

That is induction on the iteration count, verbatim. When you write a loop invariant you are writing $P(k)$; when you argue the body preserves it you are proving $P(k) \Rightarrow P(k + 1)$.

7.6. A checklist for reading and writing proofs

Reading:

1. What exactly is being claimed? Write the statement in symbols with all quantifiers explicit.
2. What is assumed and what must be shown?
3. For each line: which earlier fact or definition licenses it?
4. Where is the hypothesis actually used? If a hypothesis is never used, either the theorem is stronger than stated or the proof is wrong.

Writing:

1. State what you are proving and what technique you are using. “We proceed by induction on n .” “We prove the contrapositive.”
2. Unpack every definition into a formula before manipulating.
3. Say where each assumption is discharged.

4. Never write “clearly”, “obviously”, or “it is easy to see”. Those words mark the exact place where proofs are wrong. If it really is immediate, the sentence explaining why is short — write it.
5. End by observing that you have arrived at the claim.

7.7. Exercises

- 7.1** Write the contrapositive, the converse, and the inverse of: “If a graph is a tree then it has no cycles.” State which of the four statements are true.
- 7.2** Negate, pushing the negation all the way in:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - L| < \varepsilon.$$

Then say in plain English what your negation asserts.

- 7.3** Prove by contraposition: if $n^3 + 5$ is odd, then n is even.
- 7.4** Prove by induction that $n! > 2^n$ for all integers $n \geq 4$. Note carefully where the hypothesis $n \geq 4$ is needed.
- 7.5** Prove by induction that $n^3 - n$ is divisible by 6 for every $n \in \mathbb{N}$.
- 7.6** Prove by strong induction that every integer $n \geq 8$ can be written as $3a + 5b$ with a, b nonnegative integers. (You will need several base cases — work out how many and why.)
- 7.7** Find the flaw in this “proof” that all horses are the same colour.

Claim: in any set of n horses, all have the same colour. **Base:** $n = 1$, trivially true. **Step:** given a set of $k + 1$ horses, remove one to get k horses, which by hypothesis are all the same colour. Put it back and remove a different one; again all k are the same colour. So all $k + 1$ agree.

Identify the exact value of k at which the step fails, and explain why.

- 7.8** Write a loop invariant for the following, and use it to prove the routine returns the maximum of a nonempty array.

```
function max(A[0..n-1]):
    m := A[0]
    i := 1
    while i < n:
        if A[i] > m: m := A[i]
        i := i + 1
    return m
```


PART II

Discrete Mathematics

The mathematics that is already secretly inside your code: sets and relations, counting, graphs, recurrences and asymptotics, and the number theory that runs every hash table and every key exchange.

CHAPTER 8.

Sets, Relations, and Structure

8.1. Sets

Definition 8.1 (set).

A **set** is an unordered collection of distinct objects, called its **elements**. We write $x \in S$ for “ x is an element of S ” and $x \notin S$ otherwise.

Two sets are equal exactly when they have the same elements:

$$S = T \quad \text{iff} \quad \forall x : (x \in S \Leftrightarrow x \in T).$$

Unordered and distinct. $\{1, 2, 3\} = \{3, 1, 2\} = \{1, 1, 2, 3\}$ — all the same set. This is a **HashSet**, not a **Vec**, and equality is by contents.

Ways to write a set down:

- **Roster:** $\{2, 3, 5, 7\}$.
- **Set-builder:** $\{x \in \mathbb{Z} : x^2 < 10\}$, read “the x in $\mathbb{Z}$ such that...”. Some authors use $|$ instead of $:$.
- **Image:** $\{x^2 : x \in \mathbb{N}\}$, the set of all values of an expression.

The empty set is $\emptyset = \{\}$. Note $\{\emptyset\}$ is **not** empty — it is a set containing one element, which happens to be the empty set. The distinction between a box and an empty box is exactly the distinction between **None** and **Some(None)**.

Definition 8.2 (subset).

$S \subseteq T$ means every element of S is in T . If additionally $S \neq T$ we write $S \subsetneq T$ and say **proper subset**.

The idea: how to prove two sets are equal.

Almost always by **double inclusion**: show $S \subseteq T$ and $T \subseteq S$.

And to show $S \subseteq T$, the ritual is fixed: **let $x \in S$ be arbitrary; ... ; therefore $x \in T$** . If you find yourself stuck on a set identity, write that first line down and the rest usually follows.

8.1.1. Operations

$$S \cup T = \{x : x \in S \vee x \in T\} \quad (\text{union})$$

$$S \cap T = \{x : x \in S \wedge x \in T\} \quad (\text{intersection})$$

$$S \setminus T = \{x : x \in S \wedge x \notin T\} \quad (\text{difference})$$

$$\overline{S} = U \setminus S \quad (\text{complement, relative to a universe } U)$$

$$S \times T = \{(s, t) : s \in S, t \in T\} \quad (\text{Cartesian product})$$

Note $S \times T$ is a set of **ordered pairs**, and ordered pairs are ordered: $(1, 2) \neq (2, 1)$. The Cartesian product is where tuples come from, and $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ is where Part IV lives.

Theorem 8.3 (De Morgan's laws).

$$\overline{S \cup T} = \overline{S} \cap \overline{T}, \quad \overline{S \cap T} = \overline{S} \cup \overline{T}.$$

Proof. We prove the first by double inclusion.

($\subseteq$) Let $x \in \overline{S \cup T}$. Then $x \notin S \cup T$, so it is not the case that $(x \in S \text{ or } x \in T)$. By the logical De Morgan law of Chapter 7, $x \notin S$ and $x \notin T$. Hence $x \in \overline{S}$ and $x \in \overline{T}$, i.e. $x \in \overline{S} \cap \overline{T}$.

($\supseteq$) Every step above is an “iff”, so reading them upwards gives the reverse inclusion.

The second law follows by applying the first to $\overline{S}$ and $\overline{T}$ and complementing. ■

You know this law as `!(a || b) == !a && !b`. It is the same theorem; sets and booleans are two faces of the same algebra (a **Boolean algebra**), with $\cup$ as **or**, $\cap$ as **and**, and complement as **not**.

Definition 8.4 (power set).

$\mathcal{P}(S)$ is the set of all subsets of S .

Proposition 8.5.

If $|S| = n$ then $|\mathcal{P}(S)| = 2^n$.

Proof. A subset is determined by a yes/no decision for each of the n elements: in or out. That is a function from S to $\{0, 1\}$, and there are 2^n of them (Chapter 3).

Concretely, subsets of an n -element set correspond exactly to n -bit bitmasks — which is how you should implement them when $n \leq 64$. ■

Cantor’s theorem says $|\mathcal{P}(S)| > |S|$ for **every** set, finite or not, and its proof is the diagonal argument of Chapter 1 in general form. So there is no largest infinity.

8.2. Relations

Definition 8.6 (relation).

A **binary relation** from A to B is a subset $R \subseteq A \times B$. We write aRb for $(a, b) \in R$. A relation **on** A is a relation from A to A .

That is deliberately minimal: a relation is just a set of pairs, i.e. a table of which things are related to which. Equality, $<$, “divides”, “is reachable from”, and every foreign key in your database are relations.

Definition 8.7 (properties a relation on A may have).

- **Reflexive:** aRa for all a .
- **Symmetric:** aRb implies bRa .
- **Antisymmetric:** aRb and bRa together imply $a = b$.
- **Transitive:** aRb and bRc imply aRc .

Two combinations of these matter enormously.

8.2.1. Equivalence relations

Definition 8.8 (equivalence relation).

Reflexive, symmetric, and transitive. Written $\sim$. The **equivalence class** of a is

$$[a] = \{x \in A : x \sim a\}.$$

Theorem 8.9 (equivalence classes partition the set).

Let $\sim$ be an equivalence relation on A . Then the equivalence classes are nonempty, they cover A , and any two of them are either identical or disjoint.

Proof. **Nonempty and covering:** $a \in [a]$ by reflexivity, so every element is in its own class.

Identical or disjoint: suppose $[a] \cap [b] \neq \emptyset$; take c in both. Then $c \sim a$ and $c \sim b$. Let $x \in [a]$ be arbitrary, so $x \sim a$. By symmetry $a \sim c$, and by transitivity (twice) $x \sim c$ then $x \sim b$. So $x \in [b]$, giving $[a] \subseteq [b]$. By the same argument with a and b swapped, $[b] \subseteq [a]$. Hence $[a] = [b]$. ■

The idea: equivalence relations are 'equal for my purposes'.

An equivalence relation is a coarser notion of sameness than identity. It says: **these objects differ, but not in any way I care about right now.**

You have implemented this repeatedly:

- Case-insensitive string comparison. The classes are $\{\text{Cat}, \text{CAT}, \text{cat}, \dots\}$.
- `equals()` and `hashCode()` in Java — and the contract between them is exactly “equal objects land in the same class”.
- Modular arithmetic: $a \sim b$ iff $n \mid (a - b)$. The classes are the residues mod n (Chapter 12).
- Fractions: $(p, q) \sim (r, s)$ iff $ps = qr$. The classes are the rational numbers (Chapter 1).

The general move — **pass to the quotient** — is: build the set whose elements are the classes, and now the things you did not care about are literally gone. That is what $\mathbb{Q}$ and $\mathbb{Z}/n\mathbb{Z}$ are.

8.2.2. Partial and total orders

Definition 8.10 (partial order).

Reflexive, **antisymmetric**, and transitive. Written $\leq$. A set with a partial order is a **poset**.

If additionally any two elements are comparable ($a \leq b$ or $b \leq a$ for all a, b), it is a **total order**.

$\leq$ on $\mathbb{R}$ is total. Subset inclusion on $\mathcal{P}(S)$ is partial and not total: $\{1\}$ and $\{2\}$ are incomparable. Divisibility on $\mathbb{N}$ is partial: 3 and 5 are incomparable.

Example 8.11 (topological sort is exactly this).

A dependency graph defines a partial order: task $a \leq b$ if a must finish before b . Building the project requires a **total** order compatible with it — a linear sequence in which to run the tasks.

The theorem that this is always possible for a finite poset is called **linear extension**, and the algorithm that produces it is topological sort. A cycle in the dependency graph means antisymmetry fails, so you never had a partial order, and the sort correctly reports failure.

8.3. Functions, revisited

Chapter 3 defined a function informally. Now we can say it exactly.

Definition 8.12 (function, formally).

A function $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ such that for every $a \in A$ there is **exactly one** $b \in B$ with $(a, b) \in f$.

So a function is a special kind of relation — one that is total (defined everywhere) and single-valued. The graph of Chapter 3 was not a picture of the function; it **was** the function.

Definition 8.13 (image and preimage).

For $f : A \rightarrow B$, $S \subseteq A$, $T \subseteq B$:

$$f(S) = \{f(x) : x \in S\}, \quad f^{-1}(T) = \{x \in A : f(x) \in T\}.$$

Trap: f^{-1} is overloaded and the two meanings differ.

$f^{-1}(T)$ — the **preimage** of a set — always exists, for any function whatsoever. $f^{-1}(b)$ — the **inverse function** applied to a point — exists only when f is bijective.

Notation being reused for two different things is a real hazard here. The preimage of a set is the more general notion; think of it as a database query (“all rows whose **f** column lies in T ”), which always makes sense even when the mapping is many-to-one.

Preimages behave better than images. For any f :

$$f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T), \quad f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

But images only satisfy $f(S \cap T) \subseteq f(S) \cap f(T)$, and the inclusion can be strict — take $f(x) = x^2$, $S = \{1\}$, $T = \{-1\}$: the left side is empty, the right side is $\{1\}$. Equality for all S, T holds precisely when f is injective.

8.4. Cardinality

Definition 8.14 (same size).

$|A| = |B|$ means there exists a bijection $A \rightarrow B$.

For finite sets this is the ordinary notion. For infinite sets it produces the results of Chapter 1: $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$, but $|\mathbb{R}| > |\mathbb{N}|$.

Theorem 8.15 (inclusion–exclusion, two and three sets).

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Proof. For the first: adding $|A|$ and $|B|$ counts every element of $A \cup B$ once, except those in both, which are counted twice. Subtracting $|A \cap B|$ corrects each of those by one.

For the second, track an element in exactly k of the three sets. It is counted $\binom{k}{1}$ times by the singles, $\binom{k}{2}$ by the pairs, and $\binom{k}{3}$ by the triple, for a net count of $\binom{k}{1} - \binom{k}{2} + \binom{k}{3}$. This is 1 for $k = 1, 2, 3$

(check: $1; 2 - 1 = 1; 3 - 3 + 1 = 1$) and 0 for $k = 0$. So each element of the union is counted exactly once. ■

The general form, for n sets, alternates signs over all $2^n - 1$ nonempty intersections. Chapter 9 uses it to count derangements.

8.5. Algebraic structures, briefly

You will meet these names. They are worth five minutes now so they are not frightening later.

Definition 8.16 (a small zoo).

A set G with an operation $\star$ is a:

- **Monoid** if $\star$ is associative and there is an identity element.
- **Group** if additionally every element has an inverse.
- **Abelian group** if additionally $\star$ is commutative.

A set with two operations $(+, \times)$ is a:

- **Ring** if it is an abelian group under $+$, a monoid under $\times$, and $\times$ distributes over $+$.
- **Field** if additionally every nonzero element has a multiplicative inverse.

Example 8.17 (you already use monoids).

Every **reduce/fold** with an identity is a monoid: $(+, 0)$ on numbers, $(*, 1)$, $(\text{concat}, "")$ on strings, $(\text{max}, -\text{infinity})$, $(\text{union}, \{\})$.

Associativity is exactly the condition that lets you split the reduction across threads and combine the partial results in any grouping. The identity is what you seed an empty partition with. When a library requires your combiner to be “associative with an identity”, it is requiring a monoid, and the failure mode when you lie about it is nondeterministic output.

Chapter 1’s warning about floating-point addition now has a crisp statement: $(\text{float}, +, 0.0)$ is **not** a monoid, because $+$ is not associative on floats. That is why parallel float reductions are not reproducible.

Where the others show up: $\mathbb{Z}$ is a ring; $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields; invertible $n \times n$ matrices form a group under multiplication (non-abelian — Chapter 20); rotations form a group; $\mathbb{Z}/n\mathbb{Z}$ is a ring, and a field exactly when n is prime (Chapter 12).

8.6. Exercises

- 8.1** Prove by double inclusion that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
- 8.2** List all elements of $\mathcal{P}(\{a, b, c\})$ and verify there are 2^3 of them. Then describe the natural bijection between these subsets and the integers 0 through 7.
- 8.3** For each relation on $\mathbb{Z}$, determine which of reflexive, symmetric, antisymmetric, transitive it has: (a) aRb iff $a \leq b$; (b) aRb iff $a \mid b$; (c) aRb iff $a + b$ is even; (d) aRb iff $a \neq b$.
- 8.4** Show that “has the same remainder mod 5” is an equivalence relation on $\mathbb{Z}$, and describe its equivalence classes explicitly.
- 8.5** Give a function f and sets S, T for which $f(S \cap T)$ is a proper subset of $f(S) \cap f(T)$. Then prove that if f is injective, equality always holds.

- 8.6** In a class of 100 students, 60 take mathematics, 45 take physics, and 20 take both. How many take neither? Use inclusion–exclusion and state what universe you are complementing against.
- 8.7** Is $(\mathbb{R}, -)$ (the reals under subtraction) a monoid? Check each axiom and say precisely which one fails, if any.
- 8.8** Prove that a relation which is symmetric and transitive, but not reflexive, is possible — give an example — and explain why the “obvious” argument that symmetry plus transitivity implies reflexivity is wrong.

CHAPTER 9.

Counting

9.1. Why counting is hard

Counting sounds like the easiest topic in mathematics and it is one of the most error-prone, because the errors are silent. A wrong count looks exactly like a right count. There is no type checker.

The discipline that saves you is always the same: **say precisely what you are counting, and give a rule that pairs each thing you want to count with exactly one of your count's terms.** When a combinatorial argument goes wrong it is almost always because the pairing double-counts or misses.

9.2. The two basic rules

Theorem 9.1 (sum rule and product rule).

If A and B are **disjoint** finite sets,

$$|A \cup B| = |A| + |B|.$$

For any finite sets,

$$|A \times B| = |A| \cdot |B|.$$

Restated as decisions — which is how you will use them:

- If a thing can be built in one of several **mutually exclusive** ways, add the counts.
- If building a thing requires several **independent** choices in sequence, multiply the counts.

The word “disjoint” and the word “independent” are the whole game.

Example 9.2 (counting passwords).

How many strings of length exactly 8 over a 62-character alphabet, with at least one digit (10 of the characters are digits)?

Direct counting means splitting by how many digits appear — eight disjoint cases, all awkward. Instead count the complement:

$$62^8 - 52^8,$$

total strings minus those with no digit at all. This is **complementary counting**, and it is the first thing to try when the phrase “at least one” appears.

9.3. Permutations

Definition 9.3 (factorial and falling factorial).

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1, \quad 0! = 1.$$

The number of ordered selections of k items from n distinct items, without repetition, is

$$P(n, k) = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

$n!$ counts the orderings of n distinct objects: n choices for the first position, $n - 1$ for the second (one is used up), and so on. That is the product rule applied n times.

The idea: why $0! = 1$.

Not a convention pulled from nowhere. Two reasons, both forcing it.

Combinatorial: $n!$ counts arrangements of n objects. There is exactly one way to arrange nothing — the empty arrangement. So $0! = 1$.

Algebraic: we want $n! = n \cdot (n - 1)!$ to hold for all $n \geq 1$. At $n = 1$: $1 = 1 \cdot 0!$, forcing $0! = 1$.

Same pattern as $a^0 = 1$ in Chapter 4 and the empty sum being 0: the edge-case value is whatever makes the general law hold without exceptions. Getting these right is why your recursive base cases work.

9.4. Combinations

Definition 9.4 (binomial coefficient).

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is the number of k -element **subsets** of an n -element set. Read “ n choose k ”. It is 0 when $k < 0$ or $k > n$.

Proof. Count ordered selections two ways. There are $P(n, k) = n!/(n - k)!$ of them. On the other hand, every ordered selection is obtained by first choosing **which** k elements (some number C of ways) and then ordering them ($k!$ ways). By the product rule $P(n, k) = C \cdot k!$, so $C = n!/(k!(n - k)!)$. ■

The idea: the difference between permutation and combination.

One question: **does order matter?**

Podium finishes in a race: order matters, use $P(n, k)$. A hand of cards: order does not matter, use $\binom{n}{k}$. A committee: combination. A committee with named roles: permutation.

If you can state your objects concretely enough to answer that one question, you have solved most counting problems.

Identities worth knowing, each with a **counting** proof rather than an algebraic one — counting proofs are shorter and they explain.

Proposition 9.5 (symmetry).

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof. Choosing the k elements to include is the same act as choosing the $n - k$ to exclude. The two sides count the same objects via that bijection. ■

Proposition 9.6 (Pascal's rule).

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Fix a particular element x of the n -set. Every k -subset either contains x or does not — disjoint, exhaustive.

Those containing x : choose the remaining $k - 1$ from the other $n - 1$ elements, so $\binom{n-1}{k-1}$ of them.

Those not containing x : choose all k from the other $n - 1$, so $\binom{n-1}{k}$.

Add by the sum rule. ■

That is Pascal's triangle, and it is also the recurrence you would write for a memoised `choose(n,k)` — with the counting proof serving as the correctness argument.

Theorem 9.7 (binomial theorem).

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. Expand $(x + y)(x + y) \cdots (x + y)$ with n factors by distributing completely. Every term of the expansion is formed by picking either x or y from each factor and multiplying. A term ends up as $x^k y^{n-k}$ precisely when you picked x from exactly k of the n factors — and the number of ways to choose **which** k factors is $\binom{n}{k}$.

Collecting like terms gives the stated sum. ■

Setting $x = y = 1$ gives $\sum_k \binom{n}{k} = 2^n$: the total number of subsets, agreeing with Chapter 8. Setting $x = -1$, $y = 1$ gives $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \geq 1$: a set has equally many even-sized and odd-sized subsets.

9.5. Counting with repetition

Four cases, and the whole subject of elementary counting is knowing which one you are in.

	Order matters	Order does not
Repetition allowed	n^k	$\binom{n+k-1}{k}$
No repetition	$\frac{n!}{(n-k)!}$	$\binom{n}{k}$

The top-left is the product rule: k independent choices from n options each — strings of length k over an alphabet of size n .

The top-right is the only non-obvious one.

Theorem 9.8 (stars and bars).

The number of ways to choose k items from n types, with repetition allowed and order irrelevant — equivalently, the number of nonnegative integer solutions of $x_1 + x_2 + \cdots + x_n = k$ — is

$$\binom{n+k-1}{k}.$$

Proof. Represent a selection as a row of k stars separated into n groups by $n - 1$ bars. For example with $n = 4$ types and $k = 6$ items,

$$\star\star || \star\star\star | \star$$

means two of type 1, none of type 2, three of type 3, one of type 4.

Every such arrangement of k stars and $n - 1$ bars corresponds to exactly one selection, and every selection to exactly one arrangement. So we need to count the arrangements: there are $k + n - 1$ positions, and we choose which k of them hold stars. That is $\binom{n+k-1}{k}$. ■

The idea: why stars and bars is worth remembering.

It is the answer to “how many ways to distribute k identical things into n distinct bins”, which is an astonishingly common question in disguise: how many monomials of degree k in n variables; how many ways to split a budget into integer amounts; how many multisets of size k ; how many compositions of an integer.

The proof technique — **find a bijection to something you already know how to count** — is the single most powerful move in combinatorics.

9.6. Pigeonhole

Theorem 9.9 (pigeonhole principle).

If n items are placed into m boxes and $n > m$, some box contains at least two items.

More generally, some box contains at least $\lceil n/m \rceil$ items.

Proof. Suppose every box contained at most $\lceil n/m \rceil - 1$ items. Then the total is at most $m(\lceil n/m \rceil - 1) < m \cdot (n/m + 1) - m = n$, a contradiction, using $\lceil x \rceil < x + 1$. ■

Trivially obvious and startlingly powerful, because it produces existence proofs out of nothing.

Example 9.10 (three pigeonhole arguments you have already relied on).

Hash collisions are unavoidable. Any hash function from a domain of size $> 2^{64}$ into 64-bit outputs has collisions. No cleverness helps; there are more inputs than outputs. This is why “collision-free” hashing means “collisions are hard to **find**”, never “there are none”.

Lossless compression cannot always compress. A lossless compressor is injective. There are 2^n strings of length n and only $2^n - 1$ strings strictly shorter, so some length- n string does not shrink. Every compression scheme expands some inputs.

Every rational has a repeating decimal. Chapter 1’s proof: long division by q has at most q possible remainders, so a remainder must repeat within q steps. In general, a deterministic machine with m states run for more than m steps must revisit a state — which is also how cycle detection works.

9.7. Inclusion–exclusion, in general

Theorem 9.11 (inclusion–exclusion).

For finite sets $A_1, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

Proof. Take an element lying in exactly $k \geq 1$ of the sets. On the left it is counted once. On the right, it is counted by every subset S of those k sets, contributing $(-1)^{|S|+1}$ for each. So its total contribution is

$$\sum_{j=1}^k (-1)^{j+1} \binom{k}{j} = - \sum_{j=1}^k (-1)^j \binom{k}{j} = - \left[(1-1)^k - \binom{k}{0} \right] = 1,$$

using the binomial theorem with $x = -1$, $y = 1$. Elements in none of the sets contribute 0 to both sides. ■

Example 9.12 (derangements: nobody gets their own hat).

How many permutations of n items leave **no** item in its original position?

Let A_i be the set of permutations fixing item i . We want $n! - |\bigcup A_i|$. An intersection of j of the A_i fixes j specified items and permutes the rest freely: $(n-j)!$ permutations. There are $\binom{n}{j}$ such intersections. So

$$D_n = \sum_{j=0}^n (-1)^j \binom{n}{j} (n-j)! = n! \sum_{j=0}^n \frac{(-1)^j}{j!}.$$

The sum converges to e^{-1} (Chapter 17), so $D_n \approx n!/e$: **about 36.8% of all permutations are derangements**, essentially independently of n . Shuffle a deck against another deck and the chance no card lines up is about $1/e$, whether the decks have 5 cards or 5 million.

9.8. Counting proofs of identities

Definition 9.13 (double counting).

Count one set in two different ways; the two answers must be equal. This proves an identity, and the proof explains **why** the identity is true rather than merely verifying it.

Proposition 9.14 (Vandermonde's identity).

$$\binom{m+n}{k} = \sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}.$$

Proof. Count k -member committees drawn from a group of m women and n men.

Ignoring the split: $\binom{m+n}{k}$.

Splitting by how many women are chosen: if exactly j women, there are $\binom{m}{j}$ ways to pick them and $\binom{n}{k-j}$ ways to pick the remaining $k-j$ men. Sum over j , since the cases are disjoint and exhaustive. ■

Proposition 9.15 (hockey stick).

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}.$$

Proof. Count $(r+1)$ -subsets of $\{1, \dots, n+1\}$, classified by their **largest** element. If the largest is $i+1$, the remaining r elements come from $\{1, \dots, i\}$, giving $\binom{i}{r}$ subsets. Summing over the possible largest elements $i+1 = r+1, \dots, n+1$ gives the left side; the right side counts the same subsets without the classification. ■

9.9. Basic probability, foreshadowed

Counting and probability are the same subject when outcomes are equally likely.

Definition 9.16.

If a finite sample space Ω has all outcomes equally likely, then for an event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

That is the whole bridge; Part V generalises it to the case where outcomes are not equally likely.

Example 9.17 (the birthday problem).

With k people and 365 equally likely birthdays, the probability that all are distinct is

$$\frac{365 \cdot 364 \cdots (365 - k + 1)}{365^k} = \frac{P(365, k)}{365^k}.$$

At $k = 23$ this drops below $1/2$. The reason it is so small is that there are $\binom{23}{2} = 253$ **pairs**, not 23 people — collisions are about pairs, and pairs grow quadratically.

Same mathematics: a hash function with N possible outputs starts producing collisions after about $\sqrt{N}$ insertions, not N . For a 64-bit hash that is $2^{32} \approx 4$ billion — which is why 64-bit hashes are not safe against accidental collision at large scale, and why content-addressed stores use 256 bits.

9.10. Exercises

- 9.1** How many 5-card hands from a standard 52-card deck contain exactly two pairs (two cards of one rank, two of another, one of a third)? Show your decision sequence explicitly and say where you are careful not to double-count the two pairs.
- 9.2** How many strings of length 10 over $\{A, B, C\}$ contain at least one A ? Use complementary counting.
- 9.3** Give a double-counting proof of $\sum_{k=0}^n \binom{n}{k} = 2^n$ by counting the subsets of an n -set in two ways.
- 9.4** How many nonnegative integer solutions does $x_1 + x_2 + x_3 + x_4 = 15$ have? How many **positive** integer solutions? (For the second, substitute $y_i = x_i - 1$.)
- 9.5** Prove that among any $n+1$ integers chosen from $\{1, 2, \dots, 2n\}$, some one of them divides another. Hint: write each as $2^a m$ with m odd, and pigeonhole on m .
- 9.6** Use inclusion–exclusion to count the integers from 1 to 1000 divisible by none of 2, 3, or 5.

- 9.7** Prove $\binom{n}{k} = \binom{n}{k} \binom{n-1}{k-1}$ by a double-counting argument (count pairs consisting of a k -subset together with a distinguished element of it).
- 9.8** A hash table has m buckets and receives n keys uniformly at random. Write an exact expression for the probability of no collision, then use the approximation $1 - x \approx e^{-x}$ to show that collisions become likely once n is around $\sqrt{m}$.

CHAPTER 10.

Graphs

10.1. The most reusable abstraction in computing

A graph is a set of things and a set of connections between them. That is almost content-free as a definition, and that is exactly why it is useful: an enormous range of problems turn out to be graph problems once you name the right things and the right connections.

This chapter is the mathematics. The algorithms you probably already know; the point here is the theory that tells you when they work and why.

Definition 10.1 (graph).

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a set of **edges**.

In an **undirected** graph an edge is an unordered pair $\{u, v\}$; in a **directed** graph (a **digraph**) it is an ordered pair (u, v) .

Unless stated otherwise we assume **simple** graphs: no self-loops $\{v, v\}$ and no repeated edges. A graph allowing repeats is a **multigraph**.

Note that a directed graph is exactly a **relation** on V (Chapter 8), and an undirected graph is exactly a **symmetric** relation. Everything you learned about relations applies: reflexive means every vertex has a self-loop, transitive means every path of length two is shortcut by an edge.

Definition 10.2 (degree).

In an undirected graph, $\deg(v)$ is the number of edges incident to v . In a digraph we distinguish $\deg^-(v)$ (in-degree) and $\deg^+(v)$ (out-degree).

Theorem 10.3 (handshake lemma).

For any undirected graph,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. Count the pairs (v, e) where v is an endpoint of e , in two ways. Grouping by vertex gives $\sum_v \deg(v)$. Grouping by edge gives $2|E|$, since each edge has exactly two endpoints. ■

Corollary 10.4.

In any graph, the number of vertices of odd degree is even.

Proof. The total degree is even by the lemma. The even-degree vertices contribute an even amount, so the odd-degree vertices must contribute an even amount too — and a sum of odd numbers is even only if there is an even number of them. ■

That is a double-counting proof, exactly the technique of Chapter 9, and it is the prototype for graph theory generally: **count something two ways**.

10.2. Paths, cycles, connectivity

Definition 10.5 (walk, path, cycle).

A **walk** is a sequence of vertices with consecutive ones adjacent. A **path** is a walk with no repeated vertex. A **cycle** is a walk of length at least 3 that starts and ends at the same vertex, with no other repeats.

G is **connected** if there is a path between every pair of vertices. A **connected component** is a maximal connected subgraph.

Proposition 10.6.

“There is a path from u to v ” is an equivalence relation on the vertices of an undirected graph, and its equivalence classes are the connected components.

Proof. Reflexive: the length-zero path from u to itself. Symmetric: reverse the path. Transitive: concatenate the two walks, and any walk from u to w contains a path from u to w (delete the segment between any two occurrences of a repeated vertex, repeatedly; this strictly shortens the walk each time so it terminates, by well-ordering, Chapter 7).

By the partition theorem of Chapter 8, the classes partition V , and by construction each class is a maximal connected set. ■

That proposition is why `find_components` via BFS or union-find is correct: the components are genuinely a partition, so every vertex belongs to exactly one, and you can label them independently.

10.3. Trees

Definition 10.7 (tree).

A **tree** is a connected acyclic undirected graph. A **forest** is an acyclic graph (each component a tree).

Theorem 10.8 (characterisations of a tree).

For a graph G with n vertices, the following are equivalent:

- (i) G is a tree.
- (ii) G is connected and has exactly $n - 1$ edges.
- (iii) G is acyclic and has exactly $n - 1$ edges.
- (iv) There is exactly one path between every pair of vertices.
- (v) G is connected, and removing any edge disconnects it.
- (vi) G is acyclic, and adding any edge creates a cycle.

Proof. We prove (i) $\Rightarrow$ (iv) $\Rightarrow$ (v), and (i) $\Rightarrow$ (ii), which is the substance; the rest follow by similar arguments.

(i) $\Rightarrow$ (iv). Connectedness gives at least one path. If there were two distinct paths from u to v , walk along the first until it first diverges from the second, then follow it to the first vertex where the two meet again (they must, since both end at v). The two segments between those points form a cycle — contradiction.

(iv) $\Rightarrow$ (v). Connectedness is immediate. Remove edge $\{u, v\}$. If u and v were still connected there would be a path from u to v avoiding the edge, which together with the edge gives two distinct paths, contradicting uniqueness.

(i) $\Rightarrow$ (ii). Induction on n . For $n = 1$ a tree has 0 edges. Suppose the claim for all trees with fewer than n vertices, and let T have $n \geq 2$ vertices. T has a **leaf** (a vertex of degree 1): follow a longest path in T ; its endpoint cannot have another neighbour, since that neighbour is either on the path (creating a cycle) or off it (making the path longer). Delete that leaf and its edge. The result is still connected and acyclic — a tree on $n - 1$ vertices — so by hypothesis it has $n - 2$ edges. Adding the leaf back gives $n - 1$. ■

The idea: a tree is the minimum connectivity.

$n - 1$ edges is exactly the amount of connection needed and no more. Below it you are disconnected; above it you have a cycle, meaning redundancy.

This is why: a spanning tree is the cheapest way to connect a network; why a minimum spanning tree has $n - 1$ edges; why an n -node distributed system needs $n - 1$ links to be connected and any extra link creates an alternate route; and why a rooted tree with n nodes has $n - 1$ parent pointers, one per non-root node.

10.4. Bipartite graphs

Definition 10.9 (bipartite).

G is **bipartite** if V splits into two disjoint sets V_1, V_2 such that every edge joins a vertex of V_1 to one of V_2 . Equivalently, G is 2-colourable.

Theorem 10.10 (König's characterisation).

G is bipartite if and only if it contains no cycle of odd length.

Proof. ($\Rightarrow$) Walking along a cycle alternates sides at every step, so returning to the start requires an even number of steps.

($\Leftarrow$) Assume G is connected (otherwise argue per component). Pick any vertex r and define

$$V_1 = \{v : d(r, v) \text{ even}\}, \quad V_2 = \{v : d(r, v) \text{ odd}\},$$

where d is the shortest-path distance. Suppose some edge $\{u, v\}$ had both endpoints in the same part, so $d(r, u)$ and $d(r, v)$ have the same parity. Then the shortest path $r \rightsquigarrow u$, the edge uv , and the reverse of the shortest path $v \rightsquigarrow r$ form a closed walk of odd length, which must contain an odd cycle. Contradiction. ■

That proof is literally the algorithm: run BFS, colour by the parity of the level, and check every edge. If you find an edge inside a level, you have found an odd cycle. Bipartiteness testing is a BFS with one extra check, and the theorem is why.

10.5. Eulerian and Hamiltonian

Two questions that look similar and are wildly different in difficulty — a distinction worth internalising, because it recurs everywhere in complexity.

Definition 10.11.

An **Eulerian circuit** uses every **edge** exactly once and returns to the start. A **Hamiltonian cycle** visits every **vertex** exactly once and returns to the start.

Theorem 10.12 (Euler, 1736).

A connected graph has an Eulerian circuit if and only if every vertex has even degree.

Proof. ($\Rightarrow$) Each time the circuit visits a vertex it enters by one edge and leaves by another, pairing up the edges at that vertex. So every degree is even.

($\Leftarrow$) Suppose all degrees are even. Start anywhere and walk, never reusing an edge. You can never get stuck at a vertex other than the start: arriving at $v \neq \text{start}$ uses up one edge, leaving an odd number of unused edges at v , so at least one remains to leave by. So the walk terminates only by returning to the start, giving a closed circuit C .

If C uses every edge, done. Otherwise, since the graph is connected, some vertex w on C has unused edges. The unused edges also form a graph with all degrees even (we removed a circuit, which removes an even number at each vertex), so by the same argument there is a circuit C' through w in the leftovers. Splice C' into C at w . Repeat; the number of unused edges strictly decreases, so this terminates. ■

That proof is Hierholzer's algorithm, and it runs in linear time.

Hamiltonicity, meanwhile, has **no** such characterisation. Deciding whether a Hamiltonian cycle exists is NP-complete. Nobody knows a condition that is both necessary and sufficient and checkable quickly.

The idea: why edges are easy and vertices are hard.

The Euler proof works because the constraint is **local**: each vertex only cares about its own degree, and local constraints can be verified and repaired independently. The splicing step is possible precisely because fixing one place does not break another.

Hamiltonicity is a **global** constraint — whether you can visit everything once depends on the whole structure at once, and a local fix anywhere can destroy the solution elsewhere. That gap between local and global verifiability is, informally, the gap between P and NP.

10.6. Planarity and colouring

Definition 10.13 (planar).

A graph is **planar** if it can be drawn in the plane with no edges crossing.

Theorem 10.14 (Euler's formula).

For a connected planar graph drawn in the plane with V vertices, E edges, and F faces (including the unbounded outer face),

$$V - E + F = 2.$$

Proof. Induction on E . If the graph is a tree, then $E = V - 1$ and $F = 1$ (no cycles, so no bounded faces), giving $V - (V - 1) + 1 = 2$.

Otherwise the graph has a cycle; delete one edge of it. This keeps the graph connected (the cycle provided an alternate route) and merges the two faces adjacent to that edge into one. So E and F each drop by 1 and the alternating sum is unchanged. Repeat until a tree remains. ■

Corollary 10.15.

A simple planar graph with $V \geq 3$ has $E \leq 3V - 6$.

Proof. Each face is bounded by at least 3 edges, and each edge borders at most 2 faces, so $3F \leq 2E$, giving $F \leq 2E/3$. Substituting into Euler's formula: $2 = V - E + F \leq V - E + 2E/3 = V - E/3$, hence $E \leq 3V - 6$. ■

So K_5 (five vertices, all 10 edges) is not planar: $10 > 3 \cdot 5 - 6 = 9$. This linear bound is why planar graphs are sparse, and it is why algorithms on planar graphs (road networks, circuit layouts, meshes) can afford to be $O(V)$ rather than $O(V^2)$.

The famous **four colour theorem** — every planar graph is 4-colourable — is true, was proved in 1976 with substantial computer assistance, and has never been given a proof a human can check unaided. The weaker **six colour** version is a two-line consequence of the corollary above (Exercise).

10.7. Directed graphs and DAGs

Definition 10.16 (DAG).

A **directed acyclic graph** is a digraph with no directed cycles.

Theorem 10.17 (topological ordering).

A finite digraph has a topological ordering — a linear order of the vertices in which every edge points forwards — if and only if it is a DAG.

Proof. ($\Rightarrow$) If there were a directed cycle, every vertex on it would have to precede the next, and the last would have to precede the first — impossible in a linear order.

($\Leftarrow$) Suppose G is a DAG. It has a vertex of in-degree 0: otherwise every vertex has a predecessor, so following predecessors backwards forever from any starting vertex must revisit a vertex (there are finitely many), producing a cycle. Output that vertex, delete it, and repeat on the rest, which is still a DAG. ■

Again the proof is the algorithm (Kahn's algorithm), and the failure mode is informative: if at some point no vertex has in-degree 0 and vertices remain, there is a cycle. That is precisely how a build tool reports circular dependencies.

Definition 10.18 (strongly connected).

A digraph is **strongly connected** if there is a directed path from every vertex to every other. The **strongly connected components** are the equivalence classes of the relation “ u reaches v and v reaches u ”.

Contracting each strongly connected component to a single node always yields a DAG — the **condensation**. This is the standard structural decomposition of a digraph: cycles live inside components, and the components themselves have a clean layered order.

10.8. Graphs as matrices

The bridge to Part IV, and worth planting now.

Definition 10.19 (adjacency matrix).

For a graph on vertices $1, \dots, n$, the adjacency matrix A is the $n \times n$ matrix with $A_{ij} = 1$ if there is an edge from i to j , and 0 otherwise.

Theorem 10.20 (powers of the adjacency matrix count walks).

$(A^k)_{ij}$ is the number of walks of length exactly k from i to j .

Proof. Induction on k . For $k = 1$ this is the definition. Suppose it holds for k . A walk of length $k + 1$ from i to j consists of a walk of length k from i to some vertex m , followed by an edge $m \rightarrow j$. Summing over the intermediate vertex,

$$(\text{number of such walks}) = \sum_m (A^k)_{im} A_{mj} = (A^{k+1})_{ij},$$

which is exactly the definition of matrix multiplication. ■

The idea: matrix multiplication is path composition.

That theorem explains what matrix multiplication **is**. The formula $\sum_m A_{im} B_{mj}$ looks arbitrary until you read it as: to get from i to j in two stages, sum over all possible middle points. Composition of relations, and of functions, and of transformations, all have that shape.

Replace $(+, \times)$ with $(\min, +)$ and the same product computes shortest paths — that is the Floyd–Warshall algorithm, which is matrix multiplication over a different semiring. Replace them with (or, and) and it computes reachability, which is Warshall’s transitive closure.

One algorithm; three semirings.

The **Laplacian** $L = D - A$ (where D is the diagonal matrix of degrees) is where graph theory and linear algebra fuse. Its eigenvalues (Chapter 23) encode connectivity: the number of zero eigenvalues equals the number of connected components, and the second-smallest eigenvalue — the **algebraic connectivity** — measures how hard the graph is to cut in two. Spectral clustering, graph partitioning, and a good deal of modern network analysis are applications of that one fact.

10.9. Exercises

10.1 A graph has 7 vertices with degrees 3, 3, 3, 3, 3, 3, 3. Show no such graph exists.

10.2 Prove that every tree with at least two vertices has at least two leaves.

- 10.3** Prove that a graph in which every vertex has degree at least 2 contains a cycle. (Consider a longest path.)
- 10.4** Determine whether the following graph is bipartite, and if so give the two parts: vertices $\{1, \dots, 6\}$, edges 12, 23, 34, 45, 56, 61. What if you add the edge 14?
- 10.5** The classic Königsberg bridge problem: four landmasses joined by seven bridges, with degrees 3, 3, 3, 5. Explain in one sentence, using Euler's theorem, why no walk crosses every bridge exactly once and returns to the start. Then state the condition under which such a walk exists if you are **not** required to return to the start.
- 10.6** Use $E \leq 3V - 6$ to prove that every simple planar graph has a vertex of degree at most 5. Then use that, with induction, to prove every planar graph is 6-colourable.
- 10.7** Let A be the adjacency matrix of an undirected graph. What does $(A^2)_{ii}$ count? What does $(A^3)_{ii}$ count, and what does $\text{tr}(A^3)/6$ therefore compute?
- 10.8** Prove that in any DAG, the relation “ u can reach v ” is a partial order. Which property of partial orders would fail if the graph had a cycle?

CHAPTER 11.

Recurrences and Asymptotics

11.1. Two halves of one subject

You want to know how long a program takes. That question splits into two mathematical questions.

First, **set up**: express the cost of size- n input in terms of the cost of smaller inputs. That is a **recurrence**.

Second, **solve and simplify**: turn the recurrence into a growth rate, and throw away the detail that does not matter. That is **asymptotics**.

This chapter does both, properly. Most treatments give you the Master Theorem as a lookup table; here we will prove it, because the proof tells you what to do when the table does not apply.

11.2. Asymptotic notation

Definition 11.1 (big-O, big-Omega, big-Theta).

For functions $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$:

$$f(n) = O(g(n))$$

means there exist constants $C > 0$ and n_0 such that $f(n) \leq Cg(n)$ for all $n \geq n_0$.

$$f(n) = \Omega(g(n))$$

means $f(n) \geq cg(n)$ for some $c > 0$ and all large n . (Equivalently, $g(n) = O(f(n))$.)

$$f(n) = \Theta(g(n))$$

means both: $cg(n) \leq f(n) \leq Cg(n)$ for large n .

Trap: the equals sign is a lie.

$f(n) = O(g(n))$ is not an equation. $O(g(n))$ denotes a **set** of functions, and the honest notation is $f \in O(g)$. The traditional abuse means:

- it is not symmetric. $n = O(n^2)$ is true; $n^2 = O(n)$ is false. You cannot read it backwards.
- it is not transitive in the way an equation would be. From $n = O(n^2)$ and $n^2 = O(n^2)$ you may not conclude $n = n^2$.
- $O(n) + O(n) = O(n)$ is a true statement in which the same symbol denotes different things on the two sides.

Read = as “is” throughout, in the sense of set membership, and no harm comes of it.

The idea: what each one is for.

O is an **upper bound**: it will take no longer than this. Use it for guarantees.

Ω is a **lower bound**: it will take at least this long. Use it for impossibility results — “no comparison sort beats $\Omega(n \log n)$ ”.

Θ is **both**: the growth rate is exactly this. Use it when you actually know.

Most people write O when they mean Θ . Saying quicksort is $O(n^2)$ is true and useless; saying it is $\Theta(n \log n)$ on average and $\Theta(n^2)$ in the worst case is informative. Be precise about which you mean and you will think more clearly.

There are also the strict versions: $f = o(g)$ means $f(n)/g(n) \rightarrow 0$ (“strictly smaller order”), and $f = \omega(g)$ the reverse. And $f \sim g$ means $f(n)/g(n) \rightarrow 1$ — asymptotic equality, which keeps the constant. Stirling’s formula $n! \sim \sqrt{2\pi n}(n/e)^n$ is a $\sim$ statement, and it would be much weaker as a Θ statement.

11.2.1. Working with it

Proposition 11.2 (the rules you will use).

- (a) $O(f) + O(g) = O(\max(f, g))$.
- (b) $O(f) \cdot O(g) = O(fg)$.
- (c) $c \cdot O(f) = O(f)$ for constant $c > 0$.
- (d) A polynomial of degree d is $\Theta(n^d)$.
- (e) $\log_a n = \Theta(\log_b n)$ for any bases (Chapter 4).

Proof. We do (a) and (d); the rest are similar.

(a) Suppose $f_1 \leq C_1 f$ and $g_1 \leq C_2 g$ for large n . Then $f_1 + g_1 \leq C_1 f + C_2 g \leq (C_1 + C_2) \max(f, g)$.

(d) Let $p(n) = a_d n^d + \dots + a_0$ with $a_d > 0$. For the upper bound, $p(n) \leq (|a_d| + \dots + |a_0|)n^d$ for $n \geq 1$, since each $n^k \leq n^d$. For the lower bound, note $p(n)/n^d \rightarrow a_d$, so for large enough n it exceeds $a_d/2$, giving $p(n) \geq (a_d/2)n^d$. ■

The growth hierarchy from Chapter 4 in this notation:

$$1 \ll \log \log n \ll \log n \ll n^\epsilon \ll n \ll n \log n \ll n^2 \ll 2^n \ll n! \ll n^n$$

where $\ll$ now means $o(\cdot)$.

Trap: asymptotics is a statement about the limit, not about your data.

O hides constants and only describes behaviour for large n . Both caveats bite in practice.

Matrix multiplication has an $O(n^{2.37})$ algorithm. Nobody uses it: the constant is astronomical and the crossover with the $O(n^3)$ method is beyond any matrix that will ever be multiplied.

Insertion sort is $\Theta(n^2)$ and merge sort is $\Theta(n \log n)$, and every serious sort implementation switches to insertion sort below about $n = 16$, because the constants dominate there.

Asymptotics answers “how does the cost scale?”, which is a genuinely important question and is not the same question as “how long does it take?”

11.3. Solving recurrences

11.3.1. Method 1: unrolling

Expand the recurrence a few times, spot the pattern, sum the series, verify by induction.

Example 11.3 (unrolling a linear recurrence).

$$T(n) = T(n-1) + n, T(0) = 0.$$

$$\begin{aligned}
T(n) &= T(n-1) + n \\
&= T(n-2) + (n-1) + n \\
&= T(n-3) + (n-2) + (n-1) + n \\
&= \dots = T(0) + 1 + 2 + \dots + n = \frac{n(n+1)}{2} = \Theta(n^2).
\end{aligned}$$

The pattern after k steps is $T(n) = T(n-k) + \sum_{i=n-k+1}^n i$; setting $k = n$ terminates it. This is the cost of insertion sort and of the naive quicksort worst case.

Example 11.4 (unrolling a divide-and-conquer recurrence).

$T(n) = 2T(n/2) + n$, $T(1) = 1$. Take $n = 2^k$.

$$\begin{aligned}
T(n) &= 2T\left(\frac{n}{2}\right) + n \\
&= 2\left[2T\left(\frac{n}{4}\right) + \frac{n}{2}\right] + n = 4T\left(\frac{n}{4}\right) + 2n \\
&= 8T\left(\frac{n}{8}\right) + 3n = \dots = 2^k T(n/2^k) + kn.
\end{aligned}$$

With $k = \lg n$: $T(n) = nT(1) + n \lg n = \Theta(n \lg n)$.

Read the structure: at every level of the recursion the total work is n , and there are $\lg n$ levels. That picture — **work per level, times number of levels** — is the whole content of the Master Theorem.

11.3.2. Method 2: the recursion tree

Draw the recursion as a tree. Each node is a subproblem, labelled with the work done **at that node** excluding recursive calls. The total is the sum over all nodes, and the useful way to sum is level by level.

For $T(n) = aT(n/b) + f(n)$:

- level 0: one node, work $f(n)$;
- level 1: a nodes, work $af(n/b)$;
- level i : a^i nodes, work $a^i f(n/b^i)$;
- the tree has depth $\log_b n$, and $a^{\log_b n} = n^{\log_b a}$ leaves.

Total:

$$T(n) = \sum_{i=0}^{\log_b n - 1} a^i f\left(\frac{n}{b^i}\right) + \Theta(n^{\log_b a}).$$

The behaviour depends entirely on whether that geometric-ish sum is dominated by its first term, its last term, or spread evenly. That trichotomy is the Master Theorem.

11.3.3. Method 3: the Master Theorem

Theorem 11.5 (Master Theorem).

Let $T(n) = aT(n/b) + f(n)$ with $a \geq 1$, $b > 1$. Write $c = \log_b a$ (so n^c is the number of leaves).

- (i) If $f(n) = O(n^{c-\varepsilon})$ for some $\varepsilon > 0$, then $T(n) = \Theta(n^c)$. (**Leaves dominate.**)

- (ii) If $f(n) = \Theta(n^c)$, then $T(n) = \Theta(n^c \log n)$. (**Every level costs the same.**)
 (iii) If $f(n) = \Omega(n^{c+\varepsilon})$ for some $\varepsilon > 0$, and $af(n/b) \leq kf(n)$ for some $k < 1$ and large n (a regularity condition), then $T(n) = \Theta(f(n))$. (**The root dominates.**)

Proof. From the recursion tree, $T(n) = \Theta(n^c) + \sum_{i=0}^{\log_b n - 1} a^i f(n/b^i)$. Call the sum S .

Case (ii). $f(n) = \Theta(n^c)$ gives $a^i f(n/b^i) = \Theta(a^i (n/b^i)^c) = \Theta(n^c a^i / b^{ic})$. But $b^c = b^{\log_b a} = a$, so $a^i / b^{ic} = 1$: every level contributes $\Theta(n^c)$. There are $\log_b n$ levels, so $S = \Theta(n^c \log n)$, which dominates the $\Theta(n^c)$ leaf term.

Case (i). $f(n) = O(n^{c-\varepsilon})$ gives $a^i f(n/b^i) = O(n^{c-\varepsilon} a^i / b^{i(c-\varepsilon)}) = O(n^{c-\varepsilon} b^{i\varepsilon})$, using $a^i / b^{ic} = 1$ again. Summing the geometric series in $b^\varepsilon > 1$:

$$S = O\left(n^{c-\varepsilon} \frac{b^{\varepsilon \log_b n} - 1}{b^\varepsilon - 1}\right) = O(n^{c-\varepsilon} \cdot n^\varepsilon) = O(n^c).$$

So S does not exceed the leaf term, and $T(n) = \Theta(n^c)$.

Case (iii). The regularity condition says each level is at most k times the previous, with $k < 1$. So $S \leq f(n) \sum_{i \geq 0} k^i = f(n)/(1-k) = O(f(n))$, and $S \geq f(n)$ trivially from the $i = 0$ term. Hence $S = \Theta(f(n))$, and one checks this dominates n^c under the case hypothesis. ■

Example 11.6 (the standard applications).

Algorithm	a	b	$f(n)$	Result
Binary search	1	2	$\Theta(1)$	$c = 0$, case (ii): $\Theta(\log n)$
Merge sort	2	2	$\Theta(n)$	$c = 1$, case (ii): $\Theta(n \log n)$
Naive matrix mult.	8	2	$\Theta(n^2)$	$c = 3$, case (i): $\Theta(n^3)$
Strassen	7	2	$\Theta(n^2)$	$c = \lg 7$, case (i): $\Theta(n^{2.807})$
Karatsuba	3	2	$\Theta(n)$	$c = \lg 3$, case (i): $\Theta(n^{1.585})$

Strassen's algorithm is the Master Theorem applied as a **design** tool: the naive method has $a = 8$, and reducing a to 7 at the cost of extra additions (which stay in f) lowers the exponent, because the exponent is $\log_2 a$ and nothing else.

Trap: the Master Theorem does not always apply.

There is a gap between the cases: if $f(n)$ is larger than n^c but not **polynomially** larger, none of the three cases fires. The standard example is $T(n) = 2T(n/2) + n \log n$, where $c = 1$ and $f(n) = n \log n$ — bigger than n by only a logarithmic factor.

Here the recursion tree still works: every level costs $\Theta(n \log n)$ near the top and there are $\log n$ levels, giving $\Theta(n \log^2 n)$. When the theorem fails, go back to the tree.

11.3.4. Method 4: substitution

Guess the answer, prove it by induction. Unglamorous and always available.

Example 11.7.

Show $T(n) = 2T(\lfloor n/2 \rfloor) + n$ is $O(n \lg n)$.

Guess: $T(n) \leq Cn \lg n$ for $n \geq 2$ and some constant C .

Step: assume it for all smaller arguments. Then

$$\begin{aligned} T(n) &\leq 2C \left\lfloor \frac{n}{2} \right\rfloor \lg \left\lfloor \frac{n}{2} \right\rfloor + n \\ &\leq 2C \left(\frac{n}{2} \right) \lg \left(\frac{n}{2} \right) + n \\ &= Cn(\lg n - 1) + n = Cn \lg n - Cn + n. \end{aligned}$$

This is at most $Cn \lg n$ provided $-Cn + n \leq 0$, i.e. $C \geq 1$. Choose C large enough to also cover the base case, and the induction goes through.

Trap: the classic substitution error.

Do not “prove” $T(n) = O(n)$ for $T(n) = 2T(n/2) + n$ by writing $T(n) \leq 2C(n/2) + n = Cn + n = O(n)$.

The final step is wrong: you needed to establish $T(n) \leq Cn$, and you established $T(n) \leq Cn + n$, which is bigger. The inductive hypothesis must be reproduced **exactly**, with the same constant, not merely up to big-O. Sloppiness here proves false statements, and this particular false statement is a rite of passage.

11.4. Linear recurrences with constant coefficients

A different family, and one with a complete closed-form theory.

Definition 11.8.

A **linear homogeneous recurrence with constant coefficients** has the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}.$$

Theorem 11.9 (solution by characteristic roots).

Form the **characteristic polynomial**

$$x^k - c_1 x^{k-1} - c_2 x^{k-2} - \cdots - c_k.$$

If it has k distinct roots $r_1, \dots, r_k$, then every solution has the form

$$a_n = A_1 r_1^n + A_2 r_2^n + \cdots + A_k r_k^n,$$

with the constants A_i determined by the initial conditions. A root of multiplicity m contributes

$$(A_0 + A_1 n + \cdots + A_{m-1} n^{m-1}) r^n.$$

Proof. **Why the form works:** substitute $a_n = r^n$ into the recurrence. Dividing through by r^{n-k} turns it into exactly the characteristic equation. So r^n solves the recurrence precisely when r is a characteristic root.

Why any combination works: the recurrence is linear, so a linear combination of solutions is a solution (check by substituting).

Why every solution has this form: a solution is determined by its first k values, so the solution space has dimension k (Chapter 21). The k functions r_i^n are linearly independent when the r_i are distinct, so they span it. ■

Example 11.10 (Fibonacci in closed form).

$$F_n = F_{n-1} + F_{n-2}, \quad F_0 = 0, \quad F_1 = 1.$$

Characteristic polynomial: $x^2 - x - 1$, with roots

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618, \quad \psi = \frac{1 - \sqrt{5}}{2} \approx -0.618.$$

So $F_n = A\varphi^n + B\psi^n$. From $F_0 = 0$: $A + B = 0$. From $F_1 = 1$: $A\varphi + B\psi = 1$, so $A(\varphi - \psi) = 1$ and $\varphi - \psi = \sqrt{5}$. Hence $A = 1/\sqrt{5}$, $B = -1/\sqrt{5}$, giving **Binet's formula**:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Since $|\psi| < 1$, the second term vanishes fast: F_n is the nearest integer to $\varphi^n/\sqrt{5}$ for every $n \geq 0$. So Fibonacci grows exponentially with base φ — which is why naive recursive `fib` is $\Theta(\varphi^n)$, not $\Theta(2^n)$ as people often say.

The idea: characteristic roots are eigenvalues.

Write the recurrence as a matrix acting on the state vector:

$$\begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_{n-1} \\ F_{n-2} \end{pmatrix}.$$

Then F_n is read off from M^n applied to the initial state. Chapter 23 will show that computing M^n amounts to diagonalising M , and its eigenvalues are the roots of $\det(M - \lambda I) = \lambda^2 - \lambda - 1$ — the characteristic polynomial, appearing under its proper name.

So “solving a linear recurrence” and “diagonalising a matrix” are the same operation. It is also the reason you can compute F_n in $O(\log n)$ time by fast matrix exponentiation.

11.5. Amortised analysis, briefly

Definition 11.11 (amortised cost).

The **amortised** cost of an operation is the total cost of a worst-case sequence of m operations, divided by m . It is a statement about sequences, not about single calls.

Three standard techniques: **aggregate** (count the total directly), **accounting** (overcharge cheap operations to bank credit for expensive ones), and **potential** (define a function of the data structure state whose drop pays for expensive operations).

Example 11.12 (dynamic array doubling, three ways).

Aggregate. Chapter 2 already did it: n insertions cost $n + (1 + 2 + 4 + \dots) < n + 2n = 3n$, so $O(1)$ amortised.

Accounting. Charge 3 units per insertion. One pays for the insert itself; one is banked to pay for copying this element at the next resize; one is banked to pay for copying an element from the first

half, which had already spent its own credit at the previous resize. The bank never goes negative, so 3 units per operation suffices.

Potential. Let $\Phi = 2 \cdot (\text{size}) - (\text{capacity})$. A non-resizing insert raises Φ by 2, so its amortised cost is $1 + 2 = 3$. A resizing insert copies s elements at real cost $s + 1$, while Φ falls from $2s - s = s$ to $2(s + 1) - 2s = 2$, a drop of $s - 2$; amortised cost $(s + 1) - (s - 2) = 3$. Uniformly 3.

Amortised $O(1)$ is genuinely weaker than worst-case $O(1)$: individual operations still take $\Theta(n)$, which matters for latency and is why real-time systems use incremental resizing instead.

11.6. Exercises

- 11.1** Prove directly from the definition that $3n^2 + 100n + 7 = \Theta(n^2)$, by exhibiting explicit constants c , C , and n_0 .
- 11.2** Order these by growth rate, and justify each comparison: $n \log n$, $2^{\log n}$, $n^{1.5}$, $n(\log n)^2$, 2^n , $n!$, $\log(n!)$, $\sqrt{n}$.
- 11.3** Solve $T(n) = T(n - 1) + 1/n$, $T(1) = 1$, by unrolling. (Your answer will involve the harmonic numbers; state the asymptotic growth.)
- 11.4** Apply the Master Theorem where possible: (a) $T(n) = 4T(n/2) + n$; (b) $T(n) = 4T(n/2) + n^2$; (c) $T(n) = 4T(n/2) + n^3$; (d) $T(n) = 2T(n/2) + n/\log n$. For any case where it does not apply, say why and solve it with a recursion tree.
- 11.5** Prove by substitution that $T(n) = T(n/3) + T(2n/3) + n$ is $O(n \log n)$. (Note the two subproblems have different sizes, so the Master Theorem does not apply.)
- 11.6** Solve $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1$, $a_1 = 4$, using characteristic roots.
- 11.7** Solve $a_n = 4a_{n-1} - 4a_{n-2}$ with $a_0 = 1$, $a_1 = 6$. Note the repeated root and use the correct form.
- 11.8** A stack supports **push**, **pop**, and **multipop(k)** which pops $m \in (k, \text{size})$ items. Show by the accounting method that all three have amortised cost $O(1)$, even though a single **multipop** can cost $\Theta(n)$.

CHAPTER 12.

Number Theory and Modular Arithmetic

12.1. Why this is a programmer's subject

Number theory was the standard example of beautiful, useless mathematics for two thousand years. Then it turned out to be the foundation of cryptography, hashing, error correction, and random number generation, and now it runs underneath every secure connection you make.

It is also, quietly, the arithmetic your CPU actually does. Chapter 1 warned that a 64-bit integer lives in $\mathbb{Z}/2^{64}\mathbb{Z}$, not $\mathbb{Z}$. This chapter is the theory of $\mathbb{Z}/n\mathbb{Z}$.

12.2. Divisibility

Definition 12.1 (divides).

For integers a, b with $a \neq 0$, we say $a \mid b$ (“ a divides b ”) if there is an integer k with $b = ak$.

Note this says nothing about remainders — it is exact division. Immediate consequences, all one-liners: if $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for any integers x, y ; divisibility is transitive; and $a \mid b$ with $b \neq 0$ forces $|a| \leq |b|$.

Theorem 12.2 (division algorithm).

For integers a and $n > 0$ there exist **unique** integers q and r with

$$a = qn + r, \quad 0 \leq r < n.$$

Proof. Existence. Consider the set $S = \{a - qn : q \in \mathbb{Z}, a - qn \geq 0\}$. It is nonempty (take q very negative). By well-ordering (Chapter 7) it has a least element $r = a - qn \geq 0$. If $r \geq n$ then $r - n = a - (q + 1)n$ is a smaller nonnegative member of S , contradicting minimality. So $r < n$.

Uniqueness. Suppose $a = q_1n + r_1 = q_2n + r_2$ with both remainders in $[0, n)$. Subtracting, $n(q_1 - q_2) = r_2 - r_1$. The right side has absolute value less than n ; the left is a multiple of n . The only multiple of n with absolute value less than n is 0. So $r_1 = r_2$ and then $q_1 = q_2$. ■

Trap: your language's % is probably not this \$r\$.

The theorem guarantees $0 \leq r < n$. In C, C++, Java, Rust, and Go, `-7 % 3` is `-1`, not `2` — these languages **truncate** the quotient towards zero, so the remainder takes the sign of the dividend.

Python and Haskell use floored division, so `-7 % 3` is `2`, matching the mathematics.

This is a real bug source in hashing and cyclic indexing: `arr[i % n]` will index negatively if `i` can be negative. The portable fix is `((i % n) + n) % n`. Know which convention your language uses; there is no agreement.

12.3. Greatest common divisors

Definition 12.3 (gcd).

$\gcd(a, b)$ is the largest integer dividing both a and b , with $\gcd(0, 0)$ left undefined. a and b are **coprime** if $\gcd(a, b) = 1$.

Theorem 12.4 (Euclid's algorithm is correct).

For $b > 0$,

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Proof. Write $a = qb + r$. We show the two pairs have **exactly the same** set of common divisors, which makes their greatest elements equal.

If $d \mid a$ and $d \mid b$, then $d \mid (a - qb) = r$, so d divides b and r .

Conversely if $d \mid b$ and $d \mid r$, then $d \mid (qb + r) = a$, so d divides a and b . ■

Euclid's algorithm

```
function gcd(a, b):
    while b != 0:
        (a, b) := (b, a mod b)
    return a
```

Termination is well-ordering: b strictly decreases and stays nonnegative.

Proposition 12.5 (Euclid is fast).

Euclid's algorithm on (a, b) with $a > b$ terminates in $O(\log b)$ iterations.

Proof. Claim: after two iterations the smaller argument has at least halved. Let $a = qb + r$. If $r \leq b/2$ we are done in one step. If $r > b/2$, then $q = 1$ (since $a - b = r < b$ would need $q = 1$; more carefully, $r > b/2$ and $r < b$), and the next step computes $b \bmod r = b - r < b/2$.

So the second argument halves every two steps, giving at most $2 \lg b$ iterations. ■

The worst case is consecutive Fibonacci numbers — which is Lamé's theorem, and it is why φ shows up in the analysis of an algorithm about divisors.

Theorem 12.6 (Bézout's identity).

For any integers a, b not both zero, there exist integers x, y with

$$ax + by = \gcd(a, b).$$

Moreover $\gcd(a, b)$ is the **smallest** positive integer expressible in that form.

Proof. Let $S = \{ax + by : x, y \in \mathbb{Z}\} \cap \mathbb{Z}_{>0}$, which is nonempty (it contains $|a|$ or $|b|$). By well-ordering let $d = ax_0 + by_0$ be its least element.

d divides a . Divide: $a = qd + r$ with $0 \leq r < d$. Then

$$r = a - qd = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0),$$

which is of the form $ax + by$. If $r > 0$ it would be a member of S smaller than d — impossible. So $r = 0$ and $d \mid a$. By symmetry $d \mid b$.

d is the greatest such. Any common divisor c of a and b divides $ax_0 + by_0 = d$, hence $c \leq d$. ■

The **extended** Euclidean algorithm computes x and y alongside the gcd, by tracking the coefficients through each step. It is the workhorse: it gives modular inverses, solves linear Diophantine equations, and implements RSA key generation.

Extended Euclid

```
function extgcd(a, b):
    if b == 0: return (a, 1, 0)          # gcd = a = a*1 + b*0
    (g, x, y) := extgcd(b, a mod b)
    return (g, y, x - (a div b) * y)
```

Proof. Correctness by strong induction on b . Base $b = 0$: $\text{gcd} = a$ and $a \cdot 1 + 0 \cdot 0 = a$. Otherwise, by hypothesis the recursive call returns $g = \text{gcd}(b, a \bmod b) = \text{gcd}(a, b)$ with $bx + (a \bmod b)y = g$. Substituting $a \bmod b = a - \lfloor a/b \rfloor b$:

$$g = bx + \left(a - \left\lfloor \frac{a}{b} \right\rfloor b\right)y = ay + b\left(x - \left\lfloor \frac{a}{b} \right\rfloor y\right),$$

which is exactly what the code returns. ■

12.4. Primes

Definition 12.7 (prime).

An integer $p \geq 2$ is **prime** if its only positive divisors are 1 and p . Otherwise it is **composite**.

1 is deliberately excluded, for a reason that becomes visible in a moment.

Lemma 12.8 (Euclid's lemma).

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. Suppose $p \mid ab$ and $p \nmid a$. Since p is prime and does not divide a , $\text{gcd}(p, a) = 1$. By Bézout there are x, y with $px + ay = 1$. Multiply through by b :

$$b = bpx + aby.$$

p divides the first term obviously, and divides the second because $p \mid ab$. So $p \mid b$. ■

Note that this fails for composite numbers: $6 \mid (4 \cdot 9)$ but 6 divides neither.

Theorem 12.9 (fundamental theorem of arithmetic).

Every integer $n \geq 2$ factors into primes, and the factorisation is unique up to the order of the factors.

Proof. **Existence** is the strong induction of Chapter 7.

Uniqueness. Suppose some integer has two distinct factorisations; by well-ordering take the smallest such n , say

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

p_1 divides the right-hand product, so by Euclid's lemma (applied repeatedly) p_1 divides some q_j . Both are prime, so $p_1 = q_j$. Cancel it from both sides. The result is a smaller integer with two distinct factorisations — contradicting minimality, unless the two factorisations were the same all along. ■

The idea: why 1 is not prime.

If 1 were prime, $12 = 2 \cdot 2 \cdot 3 = 1 \cdot 2 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 2 \cdot 3$ and uniqueness would be destroyed.

This is the same “choose the definition that makes the theorem clean” instinct as $0! = 1$ and $a^0 = 1$. The definitions are downstream of the theorems, not the other way round.

12.5. Modular arithmetic

Definition 12.10 (congruence).

For $n \geq 1$, we write

$$a \equiv b \pmod{n}$$

to mean $n \mid (a - b)$.

By Chapter 8's vocabulary this is an equivalence relation — reflexive, symmetric, transitive — so it partitions $\mathbb{Z}$ into n classes, the **residues** $[0], [1], \dots, [n-1]$. The set of classes is written $\mathbb{Z}/n\mathbb{Z}$, and it is exactly the integer type of width n .

Proposition 12.11 (congruence respects arithmetic).

If $a \equiv b$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d, \quad a - c \equiv b - d, \quad ac \equiv bd \pmod{n}.$$

Proof. Write $a = b + kn$ and $c = d + \ell n$. Then $a + c = (b + d) + (k + \ell)n$, and

$$ac = (b + kn)(d + \ell n) = bd + n(bl + dk + k\ell n),$$

so both differ from the claimed value by a multiple of n . ■

This is what makes modular arithmetic usable: you may reduce mod n at **any point** in a chain of additions and multiplications without changing the answer. That is why you can compute a hash incrementally, and why you can keep intermediate values small in modular exponentiation.

Trap: division is the exception.

Cancellation fails. $2 \cdot 3 \equiv 2 \cdot 8 \pmod{10}$, since both sides are 6 and $16 \equiv 6$. But $3 \not\equiv 8 \pmod{10}$. You may not cancel the 2.

The reason is Chapter 2's zero divisors: $2 \cdot 5 \equiv 0 \pmod{10}$ with neither factor zero. $\mathbb{Z}/10\mathbb{Z}$ is a ring but not a field, so not everything is invertible.

Theorem 12.12 (modular inverses).

a has a multiplicative inverse mod n — an x with $ax \equiv 1$ — if and only if $\gcd(a, n) = 1$. When it exists it is unique mod n .

Proof. ($\Leftarrow$) If $\gcd(a, n) = 1$, Bézout gives $ax + ny = 1$, so $ax \equiv 1 \pmod{n}$. The extended Euclidean algorithm computes x .

($\Rightarrow$) If $ax \equiv 1$ then $ax - 1 = kn$, so $ax - kn = 1$, and any common divisor of a and n divides 1.

Uniqueness: if $ax \equiv ax' \equiv 1$ then $x \equiv x(ax') \equiv (xa)x' \equiv x'$. ■

Corollary 12.13.

$\mathbb{Z}/p\mathbb{Z}$ is a field exactly when p is prime.

Proof. If p is prime, every $a \in \{1, \dots, p-1\}$ has $\gcd(a, p) = 1$, so all nonzero elements are invertible. If n is composite, $n = ab$ with $1 < a, b < n$, and then a is a zero divisor, hence not invertible. ■

This field is written $\mathbb{F}_p$, and it is where error-correcting codes, secret sharing, and elliptic-curve cryptography live.

12.6. Fermat, Euler, and fast exponentiation

Theorem 12.14 (Fermat's little theorem).

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ numbers $a, 2a, 3a, \dots, (p-1)a$ reduced mod p .

None is 0, since p divides neither a nor any of $1, \dots, p-1$ (Euclid's lemma). They are pairwise distinct: if $ia \equiv ja$ then multiplying by a^{-1} (which exists) gives $i \equiv j$.

So they are $p-1$ distinct nonzero residues — that is, a permutation of $1, 2, \dots, p-1$. Multiply them all together:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

$(p-1)!$ is coprime to p , so it is invertible and cancels, leaving $a^{p-1} \equiv 1$. ■

Definition 12.15 (Euler's totient).

$\varphi(n)$ is the number of integers in $\{1, \dots, n\}$ coprime to n . For p prime, $\varphi(p) = p-1$ and $\varphi(p^k) = p^k - p^{k-1}$. φ is **multiplicative**: $\varphi(mn) = \varphi(m)\varphi(n)$ when $\gcd(m, n) = 1$.

Theorem 12.16 (Euler's theorem).

If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

The proof is the same permutation argument, run over the $\varphi(n)$ invertible residues instead of all of them. Fermat is the case $n = p$.

Fast modular exponentiation (square and multiply)

```
function powmod(base, exp, m):
    result := 1
    base   := base mod m
    while exp > 0:
```

```

    if exp is odd:
        result := (result * base) mod m
    base := (base * base) mod m
    exp := exp div 2
return result

```

Proof. The loop invariant is: $\text{result} \times \text{base}^{\text{exp}} \equiv b^e \pmod{m}$ for the original b, e .

Initialisation: result is 1, base is b , exp is e . Holds.

Maintenance: if exp is even, $\text{base}^{\text{exp}} = (\text{base}^2)^{\text{exp}/2}$, and the body replaces base by base^2 and exp by $\text{exp}/2$ — the product is unchanged. If exp is odd, one factor of base is peeled off into result first, then the even case applies.

Termination: exp reaches 0, so the invariant reads $\text{result} \equiv b^e$. ■

The running time is $O(\log e)$ multiplications, since exp halves each iteration. Computing b^e naively would take e multiplications and produce astronomically large intermediate values; reducing mod m at every step keeps everything bounded. Both facts are essential and both are one-liners once you have the theory.

Theorem 12.17 (Chinese remainder theorem).

Let $n_1, \dots, n_k$ be pairwise coprime and $N = n_1 \cdots n_k$. Then the system

$$x \equiv a_1 \pmod{n_1}, \quad \dots, \quad x \equiv a_k \pmod{n_k}$$

has a unique solution mod N .

Proof. **Existence, constructively.** For each i let $N_i = N/n_i$. Then $\gcd(N_i, n_i) = 1$ (every prime factor of N_i divides some n_j with $j \neq i$, and those are coprime to n_i), so N_i has an inverse M_i mod n_i . Set

$$x = \sum_{i=1}^k a_i N_i M_i.$$

Modulo n_j , every term with $i \neq j$ vanishes (since $n_j \mid N_i$), leaving $x \equiv a_j N_j M_j \equiv a_j \pmod{n_j}$.

Uniqueness. If x and x' both work, then $n_i \mid (x - x')$ for every i ; since the n_i are pairwise coprime, $N \mid (x - x')$. ■

CRT is why RSA decryption can be done about four times faster by working mod p and mod q separately, and it is the basis of residue number systems for fast big-integer arithmetic.

12.7. A worked application: RSA in outline

Enough theory has accumulated to see the whole thing.

1. Pick large primes p, q ; set $N = pq$ and $\varphi(N) = (p-1)(q-1)$.
2. Pick e coprime to $\varphi(N)$ — commonly 65537.
3. Compute $d \equiv e^{-1} \pmod{\varphi(N)}$ by extended Euclid.
4. Public key (N, e) ; private key d .
5. Encrypt: $c = m^e \bmod N$. Decrypt: $m = c^d \bmod N$.

Proof. Why decryption inverts encryption. By construction $ed = 1 + k\varphi(N)$ for some integer k . So, for m coprime to N ,

$$c^d \equiv m^{ed} = m^{1+k\varphi(N)} = m \cdot (m^{\varphi(N)})^k \equiv m \cdot 1^k = m \pmod{N}$$

by Euler's theorem. (The case where m shares a factor with N is handled by CRT and also works out.) ■

The security rests on the belief that recovering d requires knowing $\varphi(N)$, which requires factoring N — and that factoring large integers is hard. Note that this is a **belief**, not a theorem: nobody has proved factoring is hard, and Shor's algorithm factors in polynomial time on a quantum computer.

12.8. Primes in practice

Density. The prime number theorem says the number of primes below x is approximately $x/\ln x$. So a random 2048-bit number is prime with probability about $1/\ln(2^{2048}) \approx 1/1420$ — and about twice that if you skip evens. Generating an RSA prime is: pick at random, test, repeat a few hundred times.

Testing. Deterministic primality testing is polynomial (the AKS algorithm) but slow. In practice, Miller–Rabin: a randomised test that uses Fermat's little theorem as a **necessary** condition. If $a^{n-1} \not\equiv 1 \pmod{n}$ for some a , then n is definitely composite. The converse is not quite true — the **Carmichael numbers** fool the naive Fermat test for every a — and Miller–Rabin patches the hole by also checking for nontrivial square roots of 1.

Hash tables. Using a prime modulus spreads keys better when the keys have structure. If your table size is 2^k , then `h % size` keeps only the low k bits, so any pattern in the low bits of your hash becomes a pattern in your bucket distribution. A prime modulus mixes all the bits into the result. The modern alternative is to keep a power-of-two size, which allows a cheap mask instead of a division, and fix the hash function instead — but then the hash function had better actually mix.

12.9. Exercises

- 12.1** Compute $\gcd(1071, 462)$ by hand with Euclid's algorithm, showing each step, and then run the extended version to find x, y with $1071x + 462y = \gcd$.
- 12.2** Find the multiplicative inverse of 17 modulo 43, and verify it.
- 12.3** Solve the system $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$, using the CRT construction. Give the unique solution mod 105.
- 12.4** Compute $7^{222} \bmod 11$ two ways: by Fermat's little theorem, and by square-and-multiply. Confirm they agree.
- 12.5** Prove that if $\gcd(a, n) = d > 1$ then the congruence $ax \equiv 1 \pmod{n}$ has no solution. (This is half of the modular-inverse theorem; write it out yourself.)
- 12.6** Prove that $n^2 \equiv 0$ or $1 \pmod{4}$ for every integer n , and use this to show that no integer of the form $4k + 3$ is a sum of two squares.
- 12.7** Show that $\varphi(n)$ is even for every $n > 2$. Hint: pair each a coprime to n with $n - a$.
- 12.8** Your language computes `-17 % 5`. Give the value under truncated division and under floored division, and write a one-line expression that returns the mathematically correct residue in $[0, 5)$ under either convention.

PART III

Calculus

What happens to a quantity when you nudge it. Limits, derivatives, integrals, infinite series, and then the same ideas in many variables — which is where the gradient comes from, and therefore where all of machine learning comes from.

CHAPTER 13.

Limits and Continuity

13.1. The problem calculus was invented to solve

Two questions, both about two thousand years old, both apparently unanswerable.

What is the slope of a curve at a point? Slope is rise over run, but at a single point there is no run. You get $0/0$.

What is the area under a curve? Area is base times height, but the height keeps changing. You get an infinite sum of infinitely thin things.

Both questions have the same shape: a quantity that is undefined at the target, but which seems to be **approaching** something as you get close. The concept that makes “approaching something” precise is the limit, and it is the single foundational idea of all of calculus.

The idea: what a limit is, before any symbols.

$\lim_{x \rightarrow a} f(x) = L$ says:

You can make $f(x)$ as close to L as anyone demands, provided you take x close enough to a .

Not “ $f(a) = L$ ”. Not “ f eventually equals L ”. It is a promise about a negotiation: you name a tolerance, and I can name a neighbourhood of a inside which I meet it. Every time. No matter how small your tolerance.

Note especially: **the value of f at a is irrelevant**. The limit is about the neighbourhood of a with a itself deleted. That is not a technicality — it is the entire point, because the interesting cases are exactly the ones where $f(a)$ is undefined.

13.2. The definition

Definition 13.1 (limit).

$$\lim_{x \rightarrow a} f(x) = L$$

means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all x ,

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Read it with the vocabulary of Chapter 7. It is $\forall \varepsilon \exists \delta \forall x : (...)$, and the order matters: δ is allowed to depend on ε (and on a), which is exactly the negotiation described above.

Read $|x - a|$ and $|f(x) - L|$ as **distances**, per Chapter 1. Then the statement is: for any target distance ε around L , there is a distance δ around a that guarantees it. The strict inequality $0 < |x - a|$ is what excludes $x = a$ itself.

Example 13.2 (an epsilon-delta proof, in full).

Claim: $\lim_{x \rightarrow 3} (2x + 1) = 7$.

Scratch work (this is not the proof; it is how you find it). We want $|(2x + 1) - 7| < \varepsilon$, i.e. $|2x - 6| < \varepsilon$, i.e. $2|x - 3| < \varepsilon$, i.e. $|x - 3| < \varepsilon/2$. So $\delta = \varepsilon/2$ should work.

Proof. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon/2$. Suppose $0 < |x - 3| < \delta$. Then

$$|(2x + 1) - 7| = 2|x - 3| < 2\delta = \varepsilon.$$

Since ε was arbitrary, the limit is 7.

The structure is fixed and worth memorising: **let ε be given; choose $\delta = \dots$; suppose $|x - a| < \delta$; derive $|f(x) - L| < \varepsilon$** . All the creativity is in the scratch work, which you do backwards and then discard.

Example 13.3 (the limit that motivates everything).

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}.$$

At $x = 2$ this is $0/0$ — genuinely undefined, the function has no value there. But for every $x \neq 2$,

$$\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2,$$

and the cancellation is legal precisely because $x \neq 2$, which is exactly the region the limit looks at. So the limit is 4.

This is the shape of every derivative: a quotient that is $0/0$ at the point, but which simplifies to something well-behaved everywhere else.

13.3. Limit laws

Theorem 13.4 (algebra of limits).

Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then

$$\lim(f + g) = L + M, \quad \lim(fg) = LM,$$

$$\lim(cf) = cL, \quad \lim\left(\frac{f}{g}\right) = \frac{L}{M} \text{ if } M \neq 0.$$

Proof. We prove the sum; it shows the technique.

Let $\varepsilon > 0$. Since $f \rightarrow L$, there is δ_1 with $|f(x) - L| < \varepsilon/2$ whenever $0 < |x - a| < \delta_1$. Similarly there is δ_2 for g with tolerance $\varepsilon/2$.

Let $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - a| < \delta$, both hold, and by the triangle inequality (Chapter 1)

$$|(f(x) + g(x)) - (L + M)| \leq |f(x) - L| + |g(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

■

The idea: the epsilon-over-2 trick.

Splitting the budget — give each source of error half the tolerance — is the standard move in every analysis proof. You will see $\varepsilon/3$ when there are three error terms, and $\varepsilon/2^n$ when there are infinitely many.

It is exactly error-budget reasoning: total error must stay under ε , so allocate a share to each contributor.

13.4. One-sided limits, and when limits fail

Definition 13.5 (one-sided limits).

$\lim_{x \rightarrow a^+} f(x)$ requires the condition only for $a < x < a + \delta$; $\lim_{x \rightarrow a^-} f(x)$ only for $a - \delta < x < a$.

Proposition 13.6.

$\lim_{x \rightarrow a} f(x) = L$ if and only if both one-sided limits exist and equal L .

A limit can fail to exist in three distinguishable ways:

The two sides disagree. $\operatorname{sgn}(x)$ at 0: the left limit is -1 , the right is $+1$. This is a **jump discontinuity**.

It blows up. $1/x^2$ at 0. We write $\lim = +\infty$, which is shorthand for “exceeds any bound”, not a claim that the limit exists.

It oscillates. $\sin(1/x)$ at 0 oscillates infinitely fast and never settles. There is no L and no infinity.

Theorem 13.7 (squeeze theorem).

If $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a) and $\lim_{x \rightarrow a} g = \lim_{x \rightarrow a} h = L$, then $\lim_{x \rightarrow a} f = L$.

Proof. Let $\varepsilon > 0$. Choose δ small enough that both $|g(x) - L| < \varepsilon$ and $|h(x) - L| < \varepsilon$ hold. Then

$$L - \varepsilon < g(x) \leq f(x) \leq h(x) < L + \varepsilon,$$

so $|f(x) - L| < \varepsilon$. ■

Example 13.8 (the squeeze in action).

$\lim_{x \rightarrow 0} x^2 \sin(1/x)$. The sine oscillates wildly and has no limit, so the product law does not apply.

But $-1 \leq \sin(1/x) \leq 1$, so

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2,$$

and both outer functions tend to 0. Hence the limit is 0.

This is also the standard example of a function that is differentiable everywhere but whose derivative is not continuous, once you get to Chapter 14.

Theorem 13.9 (the fundamental trigonometric limit).

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$

Proof. Take $0 < \theta < \pi/2$ and compare three areas in the unit circle: the triangle with vertices $(0, 0), (1, 0), (\cos \theta, \sin \theta)$; the circular sector of angle θ ; and the triangle with vertices $(0, 0), (1, 0), (1, \tan \theta)$. Nesting gives

$$\frac{1}{2} \sin \theta \leq \frac{1}{2} \theta \leq \frac{1}{2} \tan \theta.$$

Divide through by $(1/2) \sin \theta > 0$:

$$1 \leq \frac{\theta}{\sin \theta} \leq \frac{1}{\cos \theta}.$$

Take reciprocals (reversing the inequalities):

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1.$$

As $\theta \rightarrow 0^+$, $\cos \theta \rightarrow 1$, so the squeeze theorem gives the limit from the right. The function is even, so the left limit agrees. ■

This is where the radian convention pays for itself (Chapter 5): the middle term is the **arc length** of the sector, and it is θ only in radians. Every derivative of a trigonometric function traces back to this limit.

13.5. Limits at infinity

Definition 13.10.

$\lim_{x \rightarrow \infty} f(x) = L$ means: for every $\varepsilon > 0$ there is M such that $x > M$ implies $|f(x) - L| < \varepsilon$.

Same negotiation, with “close to a ” replaced by “large”. These limits are where asymptotic analysis lives: $f(n) = o(g(n))$ from Chapter 11 is exactly $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$.

For rational functions, divide top and bottom by the highest power in the denominator:

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 5x}{2x^2 - 1} = \lim_{x \rightarrow \infty} \frac{3 + 5/x}{2 - 1/x^2} = \frac{3}{2}.$$

13.6. Continuity

Definition 13.11 (continuous).

f is **continuous at a** if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

This bundles three requirements: $f(a)$ is defined, the limit exists, and they agree. f is **continuous on a set** if it is continuous at each point of it.

The idea: continuity is 'no surprises'.

The informal “you can draw it without lifting the pen” is a decent picture and a bad definition (it fails for functions on disconnected domains, and it smuggles in the intermediate value theorem).

The engineering reading is better: **continuity means a small change in input produces a small change in output**. A discontinuous function is one where an arbitrarily tiny nudge to the

input can swing the output by a fixed amount — which is exactly the situation in which floating-point error, sensor noise, or a rounding decision becomes catastrophic rather than negligible.

Chapter 33's notion of a **well-conditioned** problem is a quantitative version of this.

By the limit laws, sums, products, quotients (where the denominator is nonzero), and compositions of continuous functions are continuous. Polynomials are continuous everywhere; rational functions wherever defined; $\sin$, $\cos$, $\exp$ everywhere; $\ln$ on $(0, \infty)$.

13.6.1. The three big theorems

These are the payoff for the completeness axiom of Chapter 1, and all three are false over $\mathbb{Q}$.

Theorem 13.12 (intermediate value theorem).

If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then there is $c \in [a, b]$ with $f(c) = y$.

Proof. Assume $f(a) < y < f(b)$ (the other case is symmetric). Let

$$S = \{x \in [a, b] : f(x) < y\}.$$

S is nonempty ($a \in S$) and bounded above by b , so by completeness it has a supremum $c = \sup S$.

Not $f(c) > y$: by continuity there would be a neighbourhood of c where $f > y$, so points slightly below c are not in S , making a smaller number an upper bound — contradicting leastness.

Not $f(c) < y$: then $c < b$, and by continuity there is a neighbourhood of c where $f < y$, so some point above c is in S — contradicting that c is an upper bound.

By trichotomy, $f(c) = y$. ■

Example 13.13 (the IVT is the bisection algorithm).

To solve $f(x) = 0$ with f continuous, $f(a) < 0 < f(b)$: evaluate at the midpoint, keep whichever half still brackets a sign change, repeat.

The IVT is the **correctness proof**: at every stage the bracket contains a root, and the bracket width halves each iteration, so after k steps you have located a root to within $(b - a)/2^k$. Guaranteed convergence, no assumptions beyond continuity.

And this is why it fails over $\mathbb{Q}$: $f(x) = x^2 - 2$ changes sign on $[1, 2]$ but has no rational root. Bisection would run forever, narrowing on nothing. Completeness is what puts something at the limit.

Theorem 13.14 (extreme value theorem).

A continuous function on a **closed, bounded** interval $[a, b]$ attains a maximum and a minimum.

Both hypotheses are essential. On the open interval $(0, 1)$, $f(x) = x$ has no maximum. On the unbounded $[0, \infty)$, $f(x) = x$ has no maximum. On $[0, 1]$ with $f(x) = 1/x$ for $x > 0$ and $f(0) = 0$, discontinuity kills it. Remove any one hypothesis and there is a counterexample.

The EVT is why “find the maximum of f on $[a, b]$ ” is a well-posed problem at all — Chapter 15's optimisation procedure would be searching for something that might not exist.

Theorem 13.15 (Bolzano–Weierstrass).

Every bounded sequence of reals has a convergent subsequence.

Proof. Let (a_n) lie in $[A, B]$. Bisect the interval; at least one half contains infinitely many terms of the sequence — pick it. Bisect again, and so on. This produces nested intervals of width $(B - A)/2^k$, each containing infinitely many terms. Choose a_{n_1} from the first, a_{n_2} from the second with $n_2 > n_1$, and so on.

The interval endpoints converge to a common point L by completeness, and $|a_{n_k} - L| \leq (B - A)/2^k \rightarrow 0$. ■

Theorem 13.16 (monotone convergence).

A bounded, monotone sequence converges.

Proof. Say (a_n) is increasing and bounded above. Let $L = \sup\{a_n\}$, which exists by completeness. Given $\varepsilon > 0$, $L - \varepsilon$ is not an upper bound, so some $a_N > L - \varepsilon$. Since the sequence is increasing, for all $n \geq N$ we have $L - \varepsilon < a_N \leq a_n \leq L$, hence $|a_n - L| < \varepsilon$. ■

This is the theorem you use constantly without noticing: it is why an increasing, bounded loop variable settles; why Newton’s method converges once it is monotone; and why the definition of e in Chapter 4 makes sense at all ($(1 + 1/n)^n$ is increasing and bounded above).

13.7. Uniform continuity, briefly

Definition 13.17.

f is **uniformly continuous** on S if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in S : |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Compare with ordinary continuity: there, δ was allowed to depend on the point a . Here one δ must work everywhere. That is Chapter 7’s warning about quantifier order, made concrete.

$f(x) = 1/x$ on $(0, 1)$ is continuous but not uniformly so: near zero the function is so steep that no single δ suffices. $f(x) = x^2$ on all of $\mathbb{R}$ has the same problem at infinity.

Theorem 13.18.

A continuous function on a closed bounded interval is uniformly continuous.

This matters computationally: uniform continuity is what lets you pick a single step size for a numerical method and have the error bound hold across the whole interval. Without it you would need an adaptive mesh — which, when uniform continuity genuinely fails, is exactly what adaptive quadrature does (Chapter 33).

13.8. Exercises

13.1 Prove from the definition that $\lim_{x \rightarrow 1} (3x - 2) = 1$, exhibiting your δ explicitly in terms of ε .

13.2 Prove from the definition that $\lim_{x \rightarrow 2} x^2 = 4$. (You will need to bound $|x + 2|$; do so by first restricting $\delta \leq 1$.)

- 13.3** Evaluate, showing your reasoning: (a) $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$; (b) $\lim_{x \rightarrow 0} (\sqrt{1+x} - 1)/x$ (multiply by the conjugate); (c) $\lim_{x \rightarrow 0} \sin(5x)/x$; (d) $\lim_{x \rightarrow \infty} (4x^3 - x)/(2x^3 + 7)$.
- 13.4** Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x \cos(1/x^2)$.
- 13.5** Give an example of functions f and g where neither $\lim_{x \rightarrow 0} f$ nor $\lim_{x \rightarrow 0} g$ exists, but $\lim_{x \rightarrow 0} (f + g)$ does. Explain why this does not contradict the sum law.
- 13.6** Determine the value of c that makes

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ c & x = 1 \end{cases}$$

continuous at $x = 1$.

- 13.7** Use the intermediate value theorem to prove that $x^3 - x - 1 = 0$ has a root in $[1, 2]$, and state how many bisection steps are needed to locate it to within 10^{-6} .
- 13.8** Prove that every polynomial of odd degree has at least one real root. (Use the IVT plus the behaviour of the leading term as $x \rightarrow \pm\infty$.)

CHAPTER 14.

Derivatives

14.1. The definition, and why it looks like that

Definition 14.1 (derivative).

The **derivative** of f at a is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

provided the limit exists, in which case f is **differentiable** at a .

The quotient inside is the slope of the **secant** line through the points $(a, f(a))$ and $(a+h, f(a+h))$: rise over run. As $h \rightarrow 0$ the second point slides towards the first and the secant line pivots towards the **tangent**. The derivative is the slope of that limiting line.

Note it is exactly the 0/0 situation of Chapter 13: numerator and denominator both go to zero, and everything depends on their **rate**.

The idea: four readings of the same object.

You need all four, because different problems present it differently.

Geometric: the slope of the tangent line.

Physical: the instantaneous rate of change. If $s(t)$ is position, $s'(t)$ is velocity.

Approximation: near a , $f(a+h) \approx f(a) + f'(a)h$. The derivative is **the best linear approximation** to f at a . This is the reading that generalises to many variables and is the one that matters for machine learning.

Sensitivity: if you perturb the input by a small amount, the output moves by $f'(a)$ times as much. The derivative is a gain, an amplification factor. This is the reading that explains exploding and vanishing gradients.

The linear-approximation reading deserves its own statement, because it is what “differentiable” really means.

Proposition 14.2 (differentiability is local linearity).

f is differentiable at a with derivative m if and only if

$$f(a+h) = f(a) + mh + r(h) \quad \text{where} \quad \lim_{h \rightarrow 0} \frac{r(h)}{h} = 0.$$

Proof. Define $r(h) = f(a+h) - f(a) - mh$. Then

$$\frac{r(h)}{h} = \frac{f(a+h) - f(a)}{h} - m,$$

and this tends to 0 exactly when the difference quotient tends to m . ■

So f is differentiable at a precisely when it can be approximated by a straight line with an error that vanishes **faster than linearly**. That is the form which generalises: in Chapter 18, “derivative” in many variables will mean exactly the same thing with mh replaced by a matrix times a vector.

14.2. Notation

Several notations coexist, each good for something.

Notation	Name	Best for
$f'(x), f''(x)$	Lagrange	compact, when the variable is obvious
$\frac{dy}{dx}$	Leibniz	chain rule and substitution; the “fractions” cancel suggestively
$\dot{x}$	Newton	derivatives with respect to time, in physics
$Df, \partial_x f$	operator	when you want to emphasise that differentiation is a function of functions

Leibniz notation is the most useful and the most philosophically dubious. $(dy)/(dx)$ is **not** a fraction — dy and dx are not numbers. But it behaves like one often enough that the notation is worth its danger, and Chapter 18 will explain when it stops working.

14.3. Computing derivatives from the definition

Do this a few times and then never again.

Example 14.3 (the power rule for small n).

$f(x) = x^2$:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \rightarrow 2x.$$

The cancellation of h is the whole trick, and it is legal because the limit never evaluates at $h = 0$.

Theorem 14.4 (power rule).

For any real n ,

$$\frac{d}{dx} x^n = nx^{n-1}.$$

Proof. For positive integer n , use the difference-of-powers identity from Chapter 2 with $a = x + h$, $b = x$:

$$\frac{(x+h)^n - x^n}{h} = \frac{h \cdot \sum_{k=0}^{n-1} (x+h)^{n-1-k} x^k}{h} = \sum_{k=0}^{n-1} (x+h)^{n-1-k} x^k.$$

As $h \rightarrow 0$ each of the n terms tends to x^{n-1} , giving nx^{n-1} .

The general real case follows from the chain rule and the derivative of $\exp$, by writing $x^n = e^{n \ln x}$ for $x > 0$. ■

Theorem 14.5 (derivative of the exponential).

$$\frac{d}{dx}e^x = e^x.$$

Proof.

$$\frac{e^{x+h} - e^x}{h} = e^x \frac{e^h - 1}{h}.$$

By the definition of e in Chapter 4, the second factor tends to 1. ■

Theorem 14.6 (derivative of sine).

$$\frac{d}{dx} \sin x = \cos x.$$

Proof. Using the angle addition formula of Chapter 5,

$$\begin{aligned} \frac{\sin(x+h) - \sin x}{h} &= \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}. \end{aligned}$$

The second bracket tends to 1 by the fundamental trigonometric limit of Chapter 13. For the first, multiply by the conjugate:

$$\frac{\cos h - 1}{h} = \frac{\cos^2 h - 1}{h(\cos h + 1)} = \frac{-\sin^2 h}{h(\cos h + 1)} = -\frac{\sin h}{h} \cdot \frac{\sin h}{\cos h + 1} \rightarrow -1 \cdot \frac{0}{2} = 0.$$

So the whole expression tends to $\cos x$. ■

Everything else is built by rules, which is the point of having rules.

14.4. The rules

Theorem 14.7 (linearity).

$$(f + g)' = f' + g' \text{ and } (cf)' = cf'.$$

Proof. Immediate from the limit laws of Chapter 13, since the difference quotient of a sum is the sum of the difference quotients. ■

Theorem 14.8 (product rule).

$$(fg)' = f'g + fg'.$$

Proof. The trick is to add and subtract a cross term:

$$\begin{aligned} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} &= \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \frac{f(x+h) - f(x)}{h} \cdot g(x+h) + f(x) \cdot \frac{g(x+h) - g(x)}{h}. \end{aligned}$$

As $h \rightarrow 0$ the first quotient tends to $f'(x)$, and $g(x+h) \rightarrow g(x)$ because differentiability implies continuity. The second tends to $g'(x)$. ■

The idea: why the product rule is not $f'g'$.

Think of a rectangle with sides f and g ; its area is the product. Grow each side a little: f by Δf and g by Δg . The new area exceeds the old by three pieces — a strip of size $\Delta f \cdot g$, a strip of size $f \cdot \Delta g$, and a tiny corner of size $\Delta f \cdot \Delta g$.

Divide by h and take the limit: the two strips give $f'g + fg'$, and the corner is **second order** — it involves the product of two small things, so it vanishes faster than h and contributes nothing.

“Second-order terms vanish” is the recurring theme of all of calculus.

Theorem 14.9 (quotient rule).

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g \neq 0.$$

Proof. Write $f = (f/g) \cdot g$ and differentiate with the product rule:

$$f' = \left(\frac{f}{g}\right)' g + \left(\frac{f}{g}\right) g'.$$

Solve for $(f/g)'$:

$$\left(\frac{f}{g}\right)' = \frac{f' - (f/g)g'}{g} = \frac{f'g - fg'}{g^2}.$$

Theorem 14.10 (chain rule).

If g is differentiable at x and f is differentiable at $g(x)$, then

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz notation, with $u = g(x)$:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Proof. Use the local-linearity form. Near x ,

$$g(x+h) = g(x) + g'(x)h + r_1(h), \quad r_1 \frac{h}{h} \rightarrow 0,$$

and writing $k = g'(x)h + r_1(h)$ (which tends to 0 with h), near $g(x)$,

$$f(g(x) + k) = f(g(x)) + f'(g(x))k + r_2(k), \quad r_2 \frac{k}{k} \rightarrow 0.$$

Substituting,

$$f(g(x+h)) = f(g(x)) + f'(g(x))[g'(x)h + r_1(h)] + r_2(k).$$

Divide by h . The main term is $f'(g(x))g'(x)$. The term $f'(g(x))r_1(h)/h \rightarrow 0$. And $r_2(k)/h = (r_2(k)/k)(k/h)$ where $k/h \rightarrow g'(x)$ is bounded and $r_2(k)/k \rightarrow 0$. So the error terms vanish. ■

The idea: the chain rule is gain composition, and it is all of backprop.

Read the derivative as **sensitivity**, the fourth reading above. If nudging x moves u by a factor of 3, and nudging u moves y by a factor of 5, then nudging x moves y by a factor of 15. Gains multiply along a chain.

Now stack a hundred of them. The gain of the whole chain is the product of a hundred numbers. If they average slightly below 1, the product goes exponentially to zero — **vanishing gradients**. If slightly above, exponentially to infinity — **exploding gradients**.

That is not an analogy. A deep network is a composition of a hundred functions, backpropagation is the chain rule applied to it, and the numerical pathology of deep learning is the numerical behaviour of a long product. Chapter 36 does this properly.

Theorem 14.11 (inverse function rule).

If f is differentiable and invertible near a , with $f'(a) \neq 0$, then

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}.$$

Proof. Differentiate $f(f^{-1}(y)) = y$ with respect to y using the chain rule:

$$f'(f^{-1}(y)) \cdot (f^{-1})'(y) = 1.$$

Solve. ■

Example 14.12 (the derivative of the logarithm).

$\ln$ is the inverse of $\exp$. By the rule, with $y = e^x$,

$$(\ln)'(y) = \frac{1}{(\exp)'(x)} = \frac{1}{e^x} = \frac{1}{y}.$$

So

$$\frac{d}{dx} \ln x = \frac{1}{x}.$$

Notice what happened: the derivative of a transcendental function turned out to be a simple rational function. This is the reason $\ln$ appears whenever you integrate $1/x$ (Chapter 16) and the reason harmonic sums are logarithmic.

14.5. The standard table

$$\begin{array}{ll}
 \frac{d}{dx} x^n = nx^{n-1} & \frac{d}{dx} e^x = e^x \\
 \frac{d}{dx} \ln|x| = \frac{1}{x} & \frac{d}{dx} a^x = a^x \ln a \\
 \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \cos x = -\sin x \\
 \frac{d}{dx} \tan x = \sec^2 x & \frac{d}{dx} \arctan x = \frac{1}{1+x^2} \\
 \frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \sigma(x) = \sigma(x)(1-\sigma(x))
 \end{array}$$

That last one is worth deriving, since it is the reason sigmoid was computationally attractive.

Example 14.13 (the sigmoid derivative).

$\sigma(x) = (1 + e^{-x})^{-1}$. By the chain rule,

$$\sigma'(x) = -(1 + e^{-x})^{-2} \cdot (-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2}.$$

Now rewrite:

$$\frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \cdot \frac{e^{-x}}{1 + e^{-x}} = \sigma(x) \cdot (1 - \sigma(x)),$$

using $\frac{e^{-x}}{1 + e^{-x}} = \frac{1 + e^{-x} - 1}{1 + e^{-x}} = 1 - \sigma(x)$.

So the derivative is expressible in terms of the **output value alone** — no need to keep the input around. In a network, the forward pass already computed $\sigma(x)$; the backward pass gets the derivative for one multiplication. That efficiency is why sigmoid and tanh dominated for so long, and $\tanh' = 1 - \tanh^2$ has the same property.

14.6. Differentiability versus continuity

Theorem 14.14.

Differentiable at $a \Rightarrow$ continuous at a .

Proof.

$$\lim_{h \rightarrow 0} [f(a+h) - f(a)] = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot h = f'(a) \cdot 0 = 0.$$

■

The converse is false, and the counterexamples matter.

Example 14.15 (three ways to be continuous but not differentiable).

A corner. $f(x) = |x|$ at 0. The left difference quotient is -1 , the right is $+1$; they disagree, so no limit. This is exactly ReLU's kink, and it is why frameworks adopt the convention $\text{ReLU}'(0) = 0$ — a choice, not a derivative.

A vertical tangent. $f(x) = x^{1/3}$ at 0. The difference quotient is $h^{-2/3} \rightarrow \infty$. The tangent line exists but is vertical, so its slope is not a real number.

A cusp. $f(x) = x^{2/3}$ at 0: the two sides go to $+\infty$ and $-\infty$.

And in the extreme: the **Weierstrass function** is continuous everywhere and differentiable nowhere. In a precise sense, such functions are the typical case — smooth functions are the rare exception, which is a good thing to know before you assume anything about a function you did not construct.

14.7. Higher derivatives

f'' is the derivative of f' , and so on. $f^{(n)}$ denotes the n -th.

Physical: if s is position, s' is velocity, s'' is acceleration.

Geometric: f'' measures **concavity**. Where $f'' > 0$ the curve bends upwards (convex, holds water); where $f'' < 0$ it bends downwards. A sign change is an **inflection point**.

Optimisation: Chapter 34 is essentially the study of f'' . First derivatives find candidate optima; second derivatives tell you which kind you found and how fast an algorithm will converge. In many variables f'' becomes the Hessian matrix, and its eigenvalues describe the local shape of the loss surface.

14.8. Implicit and logarithmic differentiation

Sometimes y is defined by an equation rather than a formula.

Example 14.16 (implicit differentiation).

The circle $x^2 + y^2 = 25$. Differentiate both sides with respect to x , treating y as a function of x and using the chain rule on y^2 :

$$2x + 2y \frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$ the slope is $-3/4$, which is perpendicular to the radius — as it must be.

No need to solve for y first, which is fortunate, since for most implicit equations you cannot.

Example 14.17 (logarithmic differentiation).

To differentiate $y = x^x$ (where the power rule does not apply, since the exponent is not constant, and the exponential rule does not apply, since the base is not constant): take logs first.

$$\ln y = x \ln x.$$

Differentiate both sides:

$$\left(\frac{1}{y}\right) \frac{dy}{dx} = \ln x + 1, \quad \text{so} \quad \frac{dy}{dx} = x^x (\ln x + 1).$$

The same technique turns any big product into a sum before differentiating, which is exactly why log-likelihood is the thing that gets differentiated in statistics rather than likelihood itself (Chapter 31).

14.9. Numerical differentiation, and its trap

You will sometimes want f' without a formula. The obvious approach is to take the definition and stop before the limit:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}.$$

Trap: there is an optimal h and it is not tiny.

Two errors fight each other.

Truncation error. Taylor's theorem (Chapter 17) gives $(f(x+h) - f(x))/h = f'(x) + (h/2)f''(\xi)$, so this error is $O(h)$: it **shrinks** as h shrinks.

Roundoff error. $f(x+h)$ and $f(x)$ are nearly equal, so subtracting them cancels leading digits. If each is computed with relative error $\varepsilon_{\text{mach}} \approx 10^{-16}$, the difference carries absolute error about $\varepsilon_{\text{mach}}|f(x)|$, and dividing by h inflates it to $\varepsilon_{\text{mach}}|f(x)|/h$: this error **grows** as h shrinks.

Total error is roughly $Ch + \varepsilon_{\text{mach}}/h$, minimised (by Chapter 15's technique) at $h \approx \sqrt{\varepsilon_{\text{mach}}} \approx 10^{-8}$, giving about 8 correct digits — half your precision, gone.

The **central difference** $(f(x+h) - f(x-h))/(2h)$ has truncation error $O(h^2)$, giving optimal $h \approx \varepsilon_{\text{mach}}^{1/3} \approx 10^{-5}$ and about 10 correct digits. Better, still not great.

This is why automatic differentiation exists. AD applies the chain rule symbolically to the computation graph and computes derivatives to **full** machine precision with no step size at all. It is not numerical differentiation and it is not symbolic differentiation; it is the chain rule executed alongside the program. Chapter 36 builds one.

14.10. Exercises

- 14.1** Compute from the definition (not from the rules): the derivative of $f(x) = 1/x$ at a general $x \neq 0$.
- 14.2** Differentiate: (a) x^3e^x ; (b) $(2x+1)/(x^2-3)$; (c) $\sin(x^2+1)$; (d) $e^{\cos x}$; (e) $\ln(\ln x)$; (f) $\sqrt{1+\tan x}$.
- 14.3** Prove the quotient rule directly from the definition (not via the product rule), by adding and subtracting a suitable term as in the product-rule proof.
- 14.4** Use implicit differentiation to find $(dy)/(dx)$ for $x^3 + y^3 = 6xy$ (the folium of Descartes), and find the points where the tangent is horizontal.
- 14.5** Use logarithmic differentiation to find the derivative of

$$y = \frac{(x^2+1)^3 \sqrt{x-1}}{(x+4)^5}.$$

- 14.6** Show that $\tanh'(x) = 1 - \tanh^2(x)$, and explain the computational significance of the derivative depending only on the output.
- 14.7** Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Show that $f'(0)$ exists (compute it from the definition), and show that f' is not continuous at 0. This is a function that is differentiable everywhere but not continuously so.

- 14.8** For $f(x) = e^x$ at $x = 1$, estimate the total error of the forward difference at $h = 10^{-3}$, 10^{-8} , and 10^{-14} , using the error model $Ch + \varepsilon_{\text{mach}}/h$ with $C \approx e/2$ and $\varepsilon_{\text{mach}} = 2.2 \times 10^{-16}$. Which h wins?

CHAPTER 15.

What Derivatives Are For

15.1. The theorem everything rests on

Almost every use of the derivative depends on one theorem, and it is worth seeing why.

Theorem 15.1 (Fermat's theorem on interior extrema).

If f has a local maximum or minimum at an interior point c of its domain, and $f'(c)$ exists, then $f'(c) = 0$.

Proof. Suppose c is a local maximum, so $f(c+h) \leq f(c)$ for all small h .

For $h > 0$: $(f(c+h) - f(c))/h \leq 0$, so the right-hand limit $f'(c) \leq 0$.

For $h < 0$: the numerator is still ≤ 0 but the denominator is negative, so the quotient is ≥ 0 , giving $f'(c) \geq 0$.

Both hold, so $f'(c) = 0$. The minimum case reverses the inequalities. ■

Trap: the converse is false, and both hypotheses matter.

$f'(c) = 0$ does **not** imply an extremum: $f(x) = x^3$ has $f'(0) = 0$ and no extremum at 0. Points with $f' = 0$ are **candidates**, called **critical points**, and must be tested.

“Interior” matters: on $[0, 1]$, $f(x) = x$ has its maximum at $x = 1$, where the derivative is 1, not 0. Endpoints must be checked separately.

“ $f'(c)$ exists” matters: $|x|$ has a minimum at 0 where there is no derivative at all.

So the full recipe for optimising f on $[a, b]$ is: evaluate f at every point where $f' = 0$, at every point where f' fails to exist, and at the two endpoints. Compare. The Extreme Value Theorem (Chapter 13) is what guarantees the answer is in that finite list.

Theorem 15.2 (Rolle's theorem).

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof. By the Extreme Value Theorem f attains a maximum and a minimum on $[a, b]$. If both occur at endpoints, then since $f(a) = f(b)$ the max and min are equal, so f is constant and $f' = 0$ everywhere inside. Otherwise one of them occurs at an interior point c , and Fermat's theorem gives $f'(c) = 0$. ■

Theorem 15.3 (mean value theorem).

If f is continuous on $[a, b]$ and differentiable on (a, b) , then there is $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Let $L(x)$ be the straight line through $(a, f(a))$ and $(b, f(b))$, and set $g = f - L$. Then $g(a) = g(b) = 0$, so Rolle gives c with $g'(c) = 0$, i.e. $f'(c) = L'(c) = (f(b) - f(a))/(b - a)$. ■

The idea: the MVT is the bridge from local to global.

In words: **somewhere in the interval, the instantaneous rate equals the average rate.** If you averaged 60 mph over an hour, at some instant your speedometer read exactly 60.

Its importance is structural. The derivative is a purely **local** object — it knows only about an infinitesimal neighbourhood. The MVT is the tool that converts local information into global conclusions, and essentially every such conversion goes through it:

- $f' = 0$ everywhere $\Rightarrow f$ is constant.
- $f' > 0$ everywhere $\Rightarrow f$ is increasing.
- $|f'| \leq M$ everywhere $\Rightarrow |f(x) - f(y)| \leq M|x - y|$ (Lipschitz).

Each of these is proved by applying the MVT on an arbitrary subinterval, and none of them can be proved without it.

Corollary 15.4.

If $f' > 0$ on an interval, f is strictly increasing there.

Proof. Take $x < y$ in the interval. By the MVT there is c between them with $f(y) - f(x) = f'(c)(y - x) > 0$. ■

The Lipschitz corollary above is worth naming, since it is the standard hypothesis in numerical analysis and in optimisation:

Definition 15.5 (Lipschitz).

f is **L -Lipschitz** on S if $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in S$. By the MVT, $|f'| \leq L$ implies L -Lipschitz.

A Lipschitz function cannot change faster than a fixed rate. Gradient descent converges (Chapter 34) precisely under the assumption that the **gradient** is Lipschitz, and the learning rate you are allowed to use is $1/L$.

15.2. Optimisation

Finding the extrema of f on $[a, b]$

1. Compute f' .
2. Solve $f'(x) = 0$ for critical points in (a, b) .
3. Find points in (a, b) where f' does not exist.
4. Evaluate f at all points from 2, 3, and at a and b .
5. The largest value is the max; the smallest is the min.

To classify an interior critical point without evaluating everything:

Theorem 15.6 (second derivative test).

Let $f'(c) = 0$. If $f''(c) > 0$ then c is a local minimum; if $f''(c) < 0$, a local maximum. If $f''(c) = 0$ the test is inconclusive.

Proof. Suppose $f''(c) > 0$. Then f' is increasing near c (previous corollary applied to f'). Since $f'(c) = 0$, f' is negative just left of c and positive just right. So f decreases into c and increases out of it: a local minimum. ■

$f(x) = x^4$ and $f(x) = x^3$ both have $f''(0) = 0$; the first has a minimum and the second does not. Hence “inconclusive”.

Example 15.7 (a real optimisation).

A rectangular box with a square base and no lid must hold 32 cubic units. Minimise the material used.

Let the base be $x \times x$ and the height h . The constraint is $x^2h = 32$, so $h = 32/x^2$. The surface area is

$$A = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}, \quad x > 0.$$

Differentiate: $A' = 2x - 128/x^2$. Setting $A' = 0$ gives $2x^3 = 128$, so $x = 4$, and then $h = 2$.

Check: $A'' = 2 + 256/x^3 > 0$ for $x > 0$, so this is a minimum, and since it is the only critical point on $(0, \infty)$ it is the global one. The minimum area is $16 + 32 = 48$.

Note the shape of the work: **express the objective in one variable using the constraint, then differentiate**. Chapter 34 shows how to avoid the substitution using Lagrange multipliers, which is essential when the constraint cannot be solved.

15.3. L'Hôpital's rule

Theorem 15.8 (L'Hôpital's rule).

Suppose $f(a) = g(a) = 0$ (or both tend to $\pm\infty$), f and g are differentiable near a , and $g' \neq 0$ near a . If $\lim_{x \rightarrow a} f'(x)/g'(x)$ exists, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Proof. Sketch, for the $0/0$ case with f, g continuous at a . Apply the **Cauchy mean value theorem** — a two-function version of the MVT which says there is c between a and x with

$$(f(x) - f(a))g'(c) = (g(x) - g(a))f'(c).$$

Since $f(a) = g(a) = 0$ this becomes $f(x)g'(c) = g(x)f'(c)$, so

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}.$$

As $x \rightarrow a$, c is squeezed to a too, and the right side tends to the assumed limit. ■

The idea: why differentiating both parts is legitimate.

Near a , $f(x) \approx f'(a)(x - a)$ and $g(x) \approx g'(a)(x - a)$, since both vanish at a . The ratio is then approximately $f'(a)/g'(a)$ — the $(x - a)$ factors cancel.

So L'Hôpital is not a trick; it is the statement that the ratio of two vanishing quantities is the ratio of the **rates** at which they vanish.

Trap: the ways people misuse it.

Check the indeterminate form first. $\lim_{x \rightarrow 0} (x+1)/(x+2)$ is $1/2$. Applying L'Hôpital blindly gives $1/1 = 1$, which is wrong. The rule requires $0/0$ or ∞/∞ .

It can loop. $\lim_{x \rightarrow \infty} (e^x)/(e^x)$ differentiates to itself forever.

It can fail to help even when valid. Some limits get worse with each application. Series expansion (Chapter 17) is often better.

Other indeterminate forms need conversion first. $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , ∞^0 must be rewritten as a quotient — typically by taking logs or by combining into a single fraction — before the rule applies.

Example 15.9 (proving the growth hierarchy of Chapter 4).

Logarithms lose to every power. For $k > 0$,

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^k} = \lim_{x \rightarrow \infty} \frac{1/x}{kx^{k-1}} = \lim_{x \rightarrow \infty} \frac{1}{kx^k} = 0.$$

Every power loses to every exponential. For $b > 1$ and positive integer k , apply L'Hôpital k times:

$$\lim_{x \rightarrow \infty} \frac{x^k}{b^x} = \lim_{x \rightarrow \infty} \frac{kx^{k-1}}{b^x \ln b} = \dots = \lim_{x \rightarrow \infty} \frac{k!}{b^x (\ln b)^k} = 0.$$

Those two computations justify the entire asymptotic hierarchy of Chapter 11.

15.4. Linear approximation and Newton's method

Definition 15.10 (linearisation).

The **linearisation** of f at a is

$$L(x) = f(a) + f'(a)(x - a).$$

For x near a , $f(x) \approx L(x)$, with error $o(|x - a|)$.

This is the first-order Taylor polynomial and, per Chapter 14, is what differentiability **means**. Two large consequences.

15.4.1. Error propagation

If x is known with uncertainty Δx , then $f(x)$ has uncertainty approximately $|f'(x)|\Delta x$. The derivative is the amplification factor for error.

In relative terms, dividing by $f(x)$:

$$\frac{\text{relative error in } f}{\text{relative error in } x} \approx \left| \frac{xf'(x)}{f(x)} \right|.$$

That quantity is the **condition number** of f at x , and Chapter 33 is largely about it. A function with a large condition number destroys precision no matter how carefully you implement it.

15.4.2. Newton's method

To solve $f(x) = 0$: replace f by its linearisation at the current guess and solve **that** exactly.

$$0 = f(x_n) + f'(x_n)(x - x_n) \Rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Theorem 15.11 (quadratic convergence).

If f is twice continuously differentiable, $f(r) = 0$, $f'(r) \neq 0$, and x_0 is close enough to r , then the errors $e_n = x_n - r$ satisfy

$$|e_{n+1}| \leq C|e_n|^2$$

for a constant C near $|f''(r)|/(2|f'(r)|)$.

Proof. Taylor-expand f about x_n and evaluate at r (Chapter 17 supplies the remainder form):

$$0 = f(r) = f(x_n) + f'(x_n)(r - x_n) + \frac{f''(\xi)}{2}(r - x_n)^2$$

for some ξ between. Divide by $f'(x_n)$ and rearrange:

$$r - \left[x_n - \frac{f(x_n)}{f'(x_n)} \right] = -\frac{f''(\xi)}{2f'(x_n)}(r - x_n)^2.$$

The bracket is x_{n+1} , so $|e_{n+1}| = |f''(\xi)/(2f'(x_n))| \cdot e_n^2$. ■

The idea: what quadratic convergence buys you.

The number of correct digits **doubles** each iteration. From 3 correct digits you get 6, then 12, then 24. In practice Newton's method reaches full double precision in four or five iterations from a decent start.

Compare bisection, which gains one **bit** per iteration — about 50 iterations for full precision. The difference is the difference between linear and quadratic convergence, and it is why Newton is the default wherever a derivative is available.

Trap: Newton's failure modes are all real.

Bad start. Convergence is only guaranteed locally. Far away, Newton can diverge, oscillate between two points, or jump to a different root.

$f'(x_n) \approx 0$. The step is enormous and you are thrown off.

Multiple roots. If $f'(r) = 0$ too (a repeated root), convergence degrades to linear.

Production root-finders (Brent's method, for one) hybridise: use Newton or secant steps when they look productive, fall back to bisection when they do not. That gives Newton's speed with bisection's guarantee, and it is the right engineering answer.

Newton's method generalises directly to optimisation: to minimise f , solve $f'(x) = 0$ by Newton, giving

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}.$$

In many variables the derivative becomes the gradient and the second derivative becomes the Hessian, and that is Chapter 34.

15.5. Curve sketching

A checklist, useful mostly as a way of organising what the derivatives tell you.

1. **Domain** and any points of discontinuity.
2. **Intercepts:** solve $f(x) = 0$ and evaluate $f(0)$.
3. **Symmetry:** even, odd, periodic (Chapter 3).
4. **Asymptotes:** vertical where the function blows up; horizontal from $\lim_{x \rightarrow \pm\infty} f$; slant if deg of numerator exceeds denominator by exactly one.
5. **First derivative:** critical points, intervals of increase and decrease.
6. **Second derivative:** inflection points, intervals of concavity.
7. Assemble.

The two derivative rows are the informative ones: f' gives the **direction** of travel, f'' gives the **bend**. Everything else is bookkeeping.

15.6. Related rates

If several quantities are linked by an equation and all vary with time, differentiating the equation with respect to t links their rates.

Example 15.12.

A ladder 10 units long leans against a wall. Its base slides away at 1 unit per second. How fast is the top descending when the base is 6 units from the wall?

Let x be the base distance and y the height. Then $x^2 + y^2 = 100$ at all times. Differentiate with respect to t :

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0.$$

At the instant in question $x = 6$, so $y = 8$, and $(dx)/(dt) = 1$. Substituting: $12 + 16(dy)/(dt) = 0$, so $(dy)/(dt) = -3/4$. Descending at 0.75 units per second.

The pattern: write the **constraint** that holds at all times, then differentiate it. Do not substitute the instantaneous values until after differentiating — substituting first turns variables into constants and destroys their derivatives. This is the single most common error in the topic.

15.7. Exercises

- 15.1 Find the absolute maximum and minimum of $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 4]$, following the full recipe including endpoints.
- 15.2 Verify that $f(x) = x^3 - x$ satisfies the hypotheses of Rolle's theorem on $[-1, 1]$ and find every c the theorem promises.
- 15.3 Use the mean value theorem to prove that $|\sin a - \sin b| \leq |a - b|$ for all reals a, b .
- 15.4 Evaluate with L'Hôpital where valid, and say explicitly which indeterminate form each is: (a) $\lim_{x \rightarrow 0} (e^x - 1 - x)/x^2$; (b) $\lim_{x \rightarrow 0^+} x \ln x$; (c) $\lim_{x \rightarrow \infty} (1 + 1/x)^x$; (d) $\lim_{x \rightarrow 0} (\tan x - x)/x^3$.
- 15.5 A closed cylindrical can must hold a fixed volume V . Find the ratio of height to radius that minimises the surface area, and check with the second derivative test.
- 15.6 Carry out three iterations of Newton's method on $f(x) = x^2 - 2$ starting from $x_0 = 1$. Record the error at each step and verify that the number of correct digits roughly doubles.

- 15.7** Show that Newton's method applied to $f(x) = x^{1/3}$ from any $x_0 \neq 0$ diverges, by computing the iteration explicitly. Which hypothesis of the convergence theorem fails?
- 15.8** Sand falls into a conical pile whose height always equals its base radius, at a rate of 10 cubic units per second. How fast is the height increasing when the pile is 5 units high? (The volume of a cone is $V = \pi r^2 h/3$.)

CHAPTER 16.

Integration

16.1. Two completely different ideas that turn out to be one

Integration comes in two flavours, and confusing them is a common source of misery. They are genuinely different questions, and the fact that they have the same answer is a **theorem** — arguably the most important theorem in mathematics.

The definite integral is a limit of sums. It answers: what is the total accumulated amount? Area under a curve, total distance from a velocity, total charge from a density.

The antiderivative is the reverse of differentiation. It answers: which function has this derivative?

There is no obvious reason these should be related. The Fundamental Theorem of Calculus says they are, and that is what makes areas computable.

16.2. The definite integral

Definition 16.1 (Riemann sum).

Partition $[a, b]$ into n subintervals with points $a = x_0 < x_1 < \dots < x_n = b$. Choose a sample point x_i^* in each subinterval $[x_{i-1}, x_i]$, and write $\Delta x_i = x_i - x_{i-1}$. The **Riemann sum** is

$$\sum_{i=1}^n f(x_i^*) \Delta x_i.$$

Each term is (height) $\times$ (width) — the area of a thin rectangle. The sum approximates the area under the curve; the approximation improves as the rectangles get thinner.

Definition 16.2 (definite integral).

$$\int_a^b f(x) \, dx = \lim_{\max \Delta x_i \rightarrow 0} \sum_{i=1}^n f(x_i^*) \Delta x_i,$$

provided the limit exists and does not depend on the partitions or sample points chosen. Then f is **Riemann integrable** on $[a, b]$.

The idea: reading the notation.

The symbol $\int$ is an elongated S, for **sum**. The dx is what is left of Δx after the limit. So $\int f(x) \, dx$ is literally “sum of (value $\times$ width)”, with the widths going to zero.

The dx is not decoration and not a multiplication. It tells you **which variable is being integrated over** — essential once there are several — and it carries the units. If f is a velocity in m/s and x is a time in s, the integral is in metres. Dimensional analysis of an integral works exactly as you would hope, and it is a good check on whether you have set one up correctly.

Theorem 16.3 (integrability).

Every continuous function on $[a, b]$ is Riemann integrable. So is every bounded function with only finitely many discontinuities.

Not everything is integrable. The **Dirichlet function** — 1 on rationals, 0 on irrationals — has upper Riemann sums always 1 and lower sums always 0, so the limit does not exist. Repairing this is what the Lebesgue integral does, and it is the reason measure theory exists. For everything in this book, Riemann is enough.

16.2.1. Properties

$$\begin{aligned}\int_a^b (f + g) &= \int_a^b f + \int_a^b g \\ \int_a^b cf &= c \int_a^b f \\ \int_a^b f &= \int_a^c f + \int_c^b f \\ \int_a^b f &= - \int_b^a f, \quad \int_a^a f = 0\end{aligned}$$

The last line is a convention chosen, as usual, to make the additivity rule hold without caring about the order of a , b , c .

Also: if $f \leq g$ on $[a, b]$ then $\int f \leq \int g$; and

$$\left| \int_a^b f \right| \leq \int_a^b |f|,$$

which is the integral form of the triangle inequality and is used constantly to bound things.

16.3. The Fundamental Theorem

Theorem 16.4 (Fundamental Theorem of Calculus, Part I).

Let f be continuous on $[a, b]$ and define

$$F(x) = \int_a^x f(t) \, dt.$$

Then F is differentiable on (a, b) and $F'(x) = f(x)$.

Proof. Compute the difference quotient. By the additivity property,

$$F(x+h) - F(x) = \int_x^{x+h} f(t) \, dt.$$

Since f is continuous on $[x, x+h]$, it attains a minimum m_h and a maximum M_h there (Extreme Value Theorem, Chapter 13). The integral of a function over an interval of width h is between the width times the minimum and the width times the maximum:

$$m_h h \leq \int_x^{x+h} f(t) \, dt \leq M_h h.$$

Dividing by $h > 0$:

$$m_h \leq \frac{F(x+h) - F(x)}{h} \leq M_h.$$

As $h \rightarrow 0$, both m_h and M_h tend to $f(x)$, by continuity of f . The squeeze theorem gives $F'(x) = f(x)$. ■

Theorem 16.5 (Fundamental Theorem of Calculus, Part II).

If f is continuous on $[a, b]$ and G is **any** antiderivative of f (that is, $G' = f$), then

$$\int_a^b f(x) \, dx = G(b) - G(a).$$

Proof. Let $F(x) = \int_a^x f$ as in Part I, so $F' = f = G'$. Then $(F - G)' = 0$ on (a, b) , so by the mean value theorem corollary of Chapter 15, $F - G$ is constant: $F(x) = G(x) + C$.

Now $F(a) = 0$, so $C = -G(a)$. Therefore

$$\int_a^b f = F(b) = G(b) - G(a).$$

The idea: why these two things are the same, in one picture.

Think of $F(x)$ as the accumulated area up to x . Push x forward by a tiny h . The area gains a sliver of width h and height about $f(x)$, so it gains about $f(x)h$. Therefore the **rate** at which accumulated area grows is $f(x)$.

That is Part I in a sentence: **the rate of accumulation is the thing being accumulated.**

Part II is then the telescoping observation. Chop $[a, b]$ into pieces. The total change in G is the sum of the changes over the pieces — almost everything cancels — and each small change in G is about G' times the width, which is f times the width, which is a Riemann sum. So the total change in G equals the integral of f .

The Fundamental Theorem is a telescoping sum in the limit. Chapter 2's telescoping identity was the discrete rehearsal.

Definition 16.6 (indefinite integral).

$$\int f(x) \, dx = F(x) + C$$

denotes the family of all antiderivatives of f . The $+C$ is not decoration: by the MVT corollary, two antiderivatives on an interval differ by a constant, and **every** constant occurs.

Trap: the $+C$ has a real edge case.

“Antiderivatives differ by a constant” is true **on an interval**. On a disconnected domain it is false: you get an independent constant on each component.

$\int (dx)/x = \ln|x| + C$ is the standard answer, but the domain is $\mathbb{R} \setminus \{0\}$, and the honest answer has different constants on the two sides. This bites in practice when a symbolic integrator hands you

an antiderivative and you evaluate it across a singularity — getting a confident, finite, completely wrong number for a divergent integral.

16.4. Techniques

16.4.1. The basic table

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int e^x dx &= e^x + C \\ \int \sin x dx &= -\cos x + C \\ \int \cos x dx &= \sin x + C \\ \int \frac{1}{1+x^2} dx &= \arctan x + C \\ \int \frac{1}{\sqrt{1-x^2}} dx &= \arcsin x + C\end{aligned}$$

Note the exception at $n = -1$: the power rule would divide by zero. That gap is exactly where the logarithm lives, and it is why $\ln$ is unavoidable.

16.4.2. Substitution: the chain rule backwards

Theorem 16.7 (substitution).

$$\int f(g(x))g'(x) dx = \int f(u) du, \quad u = g(x).$$

For a definite integral, change the limits too:

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Proof. Let $F' = f$. By the chain rule, $(F \circ g)'(x) = f(g(x))g'(x)$. So $F \circ g$ is an antiderivative of the left integrand, and by Part II the definite integral is $F(g(b)) - F(g(a))$, which is the right-hand side. ■

Example 16.8.

$$\int 2xe^{x^2} dx.$$

Put $u = x^2$, so $du = 2x dx$ — and the $2x dx$ is sitting right there. The integral becomes $\int e^u du = e^u + C = e^{x^2} + C$.

Substitution works exactly when the integrand contains an inner function **and** its derivative as a factor. Training yourself to spot that pattern is most of the skill.

16.4.3. Integration by parts: the product rule backwards

Theorem 16.9 (integration by parts).

$$\int u \, dv = uv - \int v \, du,$$

or explicitly

$$\int f g' = fg - \int f' g.$$

Proof. Integrate the product rule $(fg)' = f'g + fg'$ over $[a, b]$ and apply Part II to the left side:

$$f(b)g(b) - f(a)g(a) = \int_a^b f'g + \int_a^b fg'.$$

Rearrange. ■

The idea: what parts is for, and how to choose u and v .

Integration by parts trades one integral for another. It is useful when the new integral is easier — which happens when differentiating one factor simplifies it.

The standard mnemonic for choosing u (the factor to differentiate) is **LIATE**: Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — earlier in the list makes a better u . The reason is that logs and inverse trig get **much** simpler when differentiated, while exponentials do not change at all, so they are better left as dv .

Example 16.10.

$$\int x e^x \, dx.$$

Take $u = x$ (algebraic, differentiates to 1) and $dv = e^x \, dx$. Then $du = dx$ and $v = e^x$, so

$$\int x e^x \, dx = x e^x - \int e^x \, dx = x e^x - e^x + C.$$

Example 16.11 (the trick where the integral comes back).

$$I = \int e^x \sin x \, dx.$$

Parts with $u = \sin x$, $dv = e^x \, dx$:

$$I = e^x \sin x - \int e^x \cos x \, dx.$$

Parts again on the new integral, with $u = \cos x$:

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I.$$

Substituting back:

$$I = e^x \sin x - e^x \cos x - I \Rightarrow 2I = e^{x(\sin x - \cos x)},$$

so $I = e^x(\sin x - \cos x)/2 + C$.

Solving for the unknown integral algebraically is a legitimate and underused move.

16.4.4. Partial fractions

Chapter 2 decomposed a rational function into simple pieces. Now the reason: each piece integrates to a logarithm or an arctangent.

Example 16.12.

$$\int \frac{3x+1}{x^2-x-2} \, dx.$$

Chapter 2 gave the decomposition

$$\frac{7/3}{x-2} + \frac{2/3}{x+1},$$

so the integral is

$$\frac{7}{3} \ln|x-2| + \frac{2}{3} \ln|x+1| + C.$$

Any rational function can be integrated this way in principle: factor the denominator (possible over $\mathbb{R}$ into linear and irreducible quadratic factors, by Chapter 6's fundamental theorem of algebra), decompose, and integrate each piece. Linear factors give logs, irreducible quadratics give arctangents.

16.4.5. Trigonometric substitution

When the integrand contains $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$, substitute $x = a \sin \theta$, $x = a \tan \theta$, or $x = a \sec \theta$ respectively. Each choice makes the Pythagorean identity collapse the square root.

Example 16.13.

$$\int \sqrt{1-x^2} \, dx.$$

Put $x = \sin \theta$, $dx = \cos \theta \, d\theta$. Then $\sqrt{1 - \sin^2 \theta} = \cos \theta$, and

$$\int \cos^2 \theta \, d\theta = \int \frac{1 + \cos 2\theta}{2} \, d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C,$$

using the half-angle formula of Chapter 5. Converting back with $\theta = \arcsin x$ and $\sin 2\theta = 2x\sqrt{1-x^2}$:

$$\int \sqrt{1-x^2} \, dx = \frac{\arcsin x}{2} + \frac{x\sqrt{1-x^2}}{2} + C.$$

Evaluating from -1 to 1 gives $\pi/2$ — the area of a semicircle of radius 1, as it must be.

16.5. Improper integrals

Definition 16.14.

If the interval is infinite or the integrand blows up, define the integral as a limit:

$$\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f, \quad \int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{\sqrt{x}} dx.$$

If the limit exists and is finite, the integral **converges**; otherwise it **diverges**.

Example 16.15 (the p-test, which you will use for series too).

$$\int_1^\infty \frac{1}{x^p} dx = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \quad (\text{converges}) \\ \infty & \text{if } p \leq 1 \quad (\text{diverges}) \end{cases}.$$

In particular $\int_1^\infty (dx)/x = \lim_{b \rightarrow \infty} \ln b = \infty$ diverges, while $\int_1^\infty (dx)/x^2 = 1$ converges. The boundary is exactly at $p = 1$, and $1/x$ is on the wrong side of it.

Note the mirror image near zero: $\int_0^1 x^{-p} dx$ converges iff $p < 1$. The same exponent that saves you at infinity kills you at the origin.

16.6. Applications

Area between curves. $\int_a^b [f(x) - g(x)] dx$ where $f \geq g$.

Volume by slicing. If $A(x)$ is the cross-sectional area at position x , the volume is $\int_a^b A(x) dx$. For a solid of revolution about the x -axis, $A(x) = \pi f(x)^2$, giving the **disc method**.

Arc length. A tiny piece of curve has length $\sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (y')^2} dx$, so

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx.$$

Average value.

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx.$$

This is the continuous version of the arithmetic mean, and the **mean value theorem for integrals** says a continuous f actually attains its average somewhere.

Probability. If $p(x)$ is a probability density (Chapter 28), then $\int_a^b p = \mathbb{P}(a \leq X \leq b)$, $\int_{-\infty}^\infty p = 1$, and $\mathbb{E}[X] = \int x p(x) dx$. Every formula in continuous probability is an integral, which is why Part V waits until now.

16.7. Numerical integration

Most integrals have no closed form. $\int e^{-x^2} dx$ has no elementary antiderivative — and that is a theorem, not a failure of ingenuity. So you compute numerically.

Rule	Formula on $[a, b]$	Error
Midpoint	$(b-a)f\left(\frac{a+b}{2}\right)$	$O(h^2)$ per interval

Rule	Formula on $[a, b]$	Error
Trapezoid	$(b - a) \frac{f(a) + f(b)}{2}$	$O(h^2)$
Simpson	$\frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$	$O(h^4)$

The idea: why Simpson's rule is so much better.

Trapezoid fits a straight line through two points. Simpson fits a **parabola** through three. Since a parabola matches the function to one more order, the error drops by two powers of h rather than one — Simpson is exact for all cubics, not just quadratics, by a symmetry cancellation.

For a composite rule with n subintervals, halving h improves trapezoid by a factor of 4 and Simpson by a factor of 16. That gap compounds, and it is why Simpson is the default for smooth integrands.

For **non-smooth** integrands all of this collapses: the error bounds involve f'' or $f^{(4)}$, and if those are large or nonexistent the rules are no better than their worst behaviour. That is when you reach for adaptive quadrature, which subdivides only where the estimated error is large.

Chapter 33 develops this properly, including Gaussian quadrature, which achieves $O(h^{2n})$ accuracy by choosing the sample points cleverly instead of evenly.

16.8. Exercises

- 16.1** Evaluate directly from the definition, as a limit of Riemann sums with equal subintervals and right endpoints: $\int_0^1 x^2 dx$. (You will need the sum-of-squares formula from Chapter 2.)
- 16.2** Compute: (a) $\int x\sqrt{x^2+1} dx$; (b) $\int (\ln x)/x dx$; (c) $\int_0^{\pi/2} \sin^3 x \cos x dx$; (d) $\int (dx)/(x^2+9)$.
- 16.3** Use integration by parts to compute $\int x^2 e^{-x} dx$. (You will need to apply it twice.)
- 16.4** Compute $\int \ln x dx$. Hint: take $u = \ln x$ and $dv = dx$ — parts works even when there appears to be only one factor.
- 16.5** Decompose and integrate

$$\int \frac{x+5}{x^2+x-6} dx.$$

- 16.6** Determine whether each converges, and evaluate if so: (a) $\int_1^\infty (dx)/x^{3/2}$; (b) $\int_0^1 (dx)/x$; (c) $\int_0^\infty e^{-x} dx$.
- 16.7** Find the area between $y = x^2$ and $y = x + 2$, and then the volume obtained by rotating the region between $y = \sqrt{x}$ and the x -axis, for $0 \leq x \leq 4$, about the x -axis.
- 16.8** Estimate $\int_0^1 e^{-x^2} dx$ using the trapezoid rule and Simpson's rule, each with 4 subintervals. The true value is about 0.746824. Compare the two errors and check they are roughly in the ratio the error orders predict.

CHAPTER 17.

Sequences, Series, and Taylor Expansion

17.1. Why infinite sums

Every non-trivial function your computer evaluates — $\sin$, $\exp$, $\ln$, $\sqrt{\cdot}$ — is computed as a finite chunk of an infinite sum. There is no other way; a CPU has addition and multiplication and nothing else.

So this chapter is not abstraction for its own sake. It is the theory of how a transcendental function becomes a polynomial you can actually evaluate, and of how much error you incur when you stop.

17.2. Sequences

Definition 17.1 (convergence of a sequence).

(a_n) **converges** to L , written $\lim_{n \rightarrow \infty} a_n = L$, if for every $\varepsilon > 0$ there is N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. Otherwise it **diverges**.

Same negotiation as Chapter 13, with “close to a ” replaced by “far enough along”. The tools transfer directly: limit laws, squeeze theorem, and the two theorems that do the real work:

- **Monotone convergence** (Chapter 13): bounded and monotone implies convergent.
- **Bolzano–Weierstrass**: bounded implies a convergent subsequence.

Example 17.2 (the sequences to know).

$$\begin{aligned}\lim n^{1/n} &= 1, & \lim \left(1 + \frac{x}{n}\right)^n &= e^x, \\ \lim r^n &= \begin{cases} 0 & |r| < 1 \\ 1 & r = 1 \\ \text{diverges} & \text{otherwise} \end{cases}, \\ \lim \frac{n^k}{b^n} &= 0 \quad (b > 1), & \lim \frac{\ln n}{n^\varepsilon} &= 0.\end{aligned}$$

The last two are Chapter 15’s L’Hôpital computations, restated for sequences.

17.3. Series

Definition 17.3 (series and partial sums).

Given (a_n) , the **series** $\sum_{n=1}^{\infty} a_n$ is defined as the limit of its **partial sums**

$$s_N = \sum_{n=1}^N a_n.$$

If (s_N) converges to S , the series converges to S ; otherwise it diverges.

Trap: a series is not a sum.

An infinite series is **defined** as a limit. It is not “addition, done forever” — that phrase means nothing. Everything surprising about series comes from forgetting this.

In particular, the familiar laws of addition do not automatically apply. Commutativity fails: a **conditionally** convergent series can be rearranged to converge to any value you like (the Riemann rearrangement theorem, below). Associativity fails for divergent series: grouping $1 - 1 + 1 - 1 + \dots$ as $(1 - 1) + (1 - 1) + \dots$ gives 0 and as $1 - (1 - 1) - (1 - 1) - \dots$ gives 1, which proves only that the series diverges.

Theorem 17.4 (divergence test).

If $\sum a_n$ converges then $a_n \rightarrow 0$. Contrapositively: if $a_n \not\rightarrow 0$, the series diverges.

Proof. $a_n = s_n - s_{n-1}$, and both partial sums tend to the same S , so the difference tends to 0. ■

Trap: the converse is false and this is the classic error.

$a_n \rightarrow 0$ does **not** imply convergence. The harmonic series $\sum 1/n$ has terms tending to zero and diverges.

The divergence test can only ever prove divergence. It never proves convergence.

17.3.1. The two series you must know

Theorem 17.5 (geometric series).

For $|r| < 1$,

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

For $|r| \geq 1$ it diverges.

Proof. From Chapter 2, $s_N = (1 - r^{N+1})/(1 - r)$. If $|r| < 1$ then $r^{N+1} \rightarrow 0$, so $s_N \rightarrow 1/(1 - r)$. If $|r| \geq 1$ then r^{N+1} does not tend to 0, so the terms fail the divergence test. ■

Theorem 17.6 (the harmonic series diverges).

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Proof. Group the terms in blocks of doubling length:

$$1 + \underbrace{\frac{1}{2}}_{\geq 1/2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq 2 \cdot 1/4 = 1/2} + \underbrace{\frac{1}{5} + \dots + \frac{1}{8}}_{\geq 4 \cdot 1/8 = 1/2} + \dots$$

Each block sums to at least $1/2$, and there are infinitely many blocks, so the partial sums exceed any bound. ■

The idea: the harmonic series shows up in your code.

$H_n = \sum_{k=1}^n 1/k \approx \ln n + \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The approximation comes from comparing the sum to $\int_1^n (dx)/x$.

This appears constantly in algorithm analysis:

- The expected number of comparisons in randomised quicksort is $\approx 2n \ln n$, and the $\ln$ comes from a harmonic sum over pivot positions.
- The **coupon collector** problem — how many random draws to see all n coupons — has expectation $nH_n \approx n \ln n$.
- The expected depth of a random binary search tree is $\approx 2 \ln n$.

Every one of those is the harmonic series in disguise, and its slow logarithmic divergence is why these bounds are $n \log n$ and not n or n^2 .

17.4. Convergence tests

Theorem 17.7 (p-series).

$\sum_{n=1}^{\infty} 1/n^p$ converges if and only if $p > 1$.

Proof. Compare with $\int_1^{\infty} x^{-p} dx$ from Chapter 16. Precisely: since $1/x^p$ is decreasing,

$$\int_1^{N+1} \frac{dx}{x^p} \leq \sum_{n=1}^N \frac{1}{n^p} \leq 1 + \int_1^N \frac{dx}{x^p},$$

by comparing each term to the area of a unit-width rectangle. Both bounds are finite iff the integral converges, which by Chapter 16 happens iff $p > 1$. ■

That argument is the **integral test** in general: for positive decreasing f , $\sum f(n)$ and $\int_1^{\infty} f$ converge together.

Theorem 17.8 (comparison tests).

Suppose $0 \leq a_n \leq b_n$ for all large n .

- If $\sum b_n$ converges, so does $\sum a_n$.
- If $\sum a_n$ diverges, so does $\sum b_n$.

Limit comparison: if $a_n, b_n > 0$ and $\lim a_n/b_n = c$ with $0 < c < \infty$, then the two series converge or diverge together.

Theorem 17.9 (ratio test).

Let $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$. If $L < 1$ the series converges absolutely; if $L > 1$ it diverges; if $L = 1$ the test says nothing.

Proof. Suppose $L < 1$; pick r with $L < r < 1$. For large n , $|a_{n+1}| \leq r|a_n|$, so from some index N onwards $|a_{N+k}| \leq r^k|a_N|$. The tail is dominated by a convergent geometric series, so it converges by comparison.

If $L > 1$, terms eventually grow, so $a_n \nrightarrow 0$ and the divergence test applies. ■

The ratio test is the workhorse for anything containing factorials or exponentials, because those simplify beautifully in a ratio. It fails (returning $L = 1$) exactly on p -series, where you need the integral test instead.

Definition 17.10 (absolute versus conditional convergence).

$\sum a_n$ converges **absolutely** if $\sum |a_n|$ converges, and **conditionally** if it converges but $\sum |a_n|$ does not.

Theorem 17.11 (absolute convergence implies convergence).

If $\sum |a_n|$ converges then $\sum a_n$ converges.

Theorem 17.12 (alternating series test).

If $b_n > 0$, b_n is decreasing, and $b_n \rightarrow 0$, then $\sum (-1)^n b_n$ converges. Moreover the error from truncating after N terms is at most b_{N+1} — the first omitted term.

The alternating harmonic series $\sum (-1)^{n+1}/n = \ln 2$ is conditionally convergent: it converges, but the absolute series diverges.

Trap: conditional convergence is genuinely unstable.

Riemann rearrangement theorem: the terms of a conditionally convergent series can be permuted to sum to **any** real number, or to $\pm\infty$.

The mechanism: the positive terms alone diverge to $+\infty$ and the negative terms alone to $-\infty$. So to hit a target T , take positive terms until you overshoot, then negative terms until you undershoot, and repeat. Since the terms tend to zero, the oscillation shrinks and you converge to T .

This is not a curiosity. It means the **order of summation matters** for a conditionally convergent series — so any parallel or reordered summation of such a series is meaningless. Absolute convergence is exactly the condition under which reordering is safe. This is the mathematical statement underneath “do not reduce a numerically delicate sum in arbitrary order”.

17.5. Power series

Definition 17.13 (power series).

$$\sum_{n=0}^{\infty} c_n (x - a)^n.$$

There is a **radius of convergence** $R \in [0, \infty]$ such that the series converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$. Behaviour at $|x - a| = R$ must be checked separately.

R is usually found by the ratio test. Within the radius, a power series can be differentiated and integrated term by term, and the result has the same radius — which is what makes them so pleasant to work with.

17.6. Taylor series

Here is the payoff.

Definition 17.14 (Taylor series).

If f is infinitely differentiable at a , its **Taylor series** about a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

With $a = 0$ it is also called the **Maclaurin series**.

The idea: where the coefficients come from.

Suppose $f(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots$ and ask what the c_n must be.

Set $x = 0$: everything but c_0 vanishes, so $c_0 = f(0)$.

Differentiate once, then set $x = 0$: $f'(0) = c_1$.

Differentiate twice: the x^2 term becomes $2c_2$, so $c_2 = f''(0)/2$.

Differentiate n times: the x^n term becomes $n!c_n$ and everything else is either gone or still has an x . Hence $c_n = f^{(n)}(0)/n!$.

So the Taylor coefficients are **forced**. The $n!$ is not decoration — it is exactly the factor that differentiating x^n down to a constant produces. The polynomial is the unique one matching f 's value, slope, curvature, and every higher derivative at the point.

Theorem 17.15 (Taylor's theorem with Lagrange remainder).

If f is $(n+1)$ -times differentiable on an interval containing a and x , then

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}_{P_n(x)} + R_n(x),$$

where for some ξ between a and x ,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}.$$

Proof. Sketch. Fix x and define $g(t) = f(x) - P_n^{[t]}(x) - M(x-t)^{n+1}$, where $P_n^{[t]}$ is the Taylor polynomial expanded about t and M is chosen so that $g(a) = 0$. Note $g(x) = 0$ as well. By Rolle's theorem (Chapter 15) there is ξ between with $g'(\xi) = 0$. Differentiating, almost everything telescopes and one is left with

$$0 = -\frac{f^{(n+1)}(\xi)}{n!} (x-\xi)^n + M(n+1)(x-\xi)^n,$$

giving $M = f^{(n+1)}(\xi)/(n+1)!$. Substituting M into $g(a) = 0$ yields the stated formula. ■

The idea: the remainder is the whole point.

The Taylor **polynomial** is what you compute. The **remainder** is what tells you whether the answer is any good.

Read the remainder formula: the error is controlled by (a) the size of the next derivative and (b) $(x-a)^{n+1}/(n+1)!$. The second factor collapses very fast — factorial in the denominator beats

everything — so as long as the derivatives do not blow up, adding terms helps enormously and staying near a helps enormously.

This is why `sin(x)` in a standard library first reduces x modulo 2π into a small range: the error term is $x^{n+1}/(n+1)!$, so keeping $|x|$ small is worth many terms of the series.

17.6.1. The standard expansions

$$\begin{aligned}
 e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots & (R = \infty) \\
 \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots & (R = \infty) \\
 \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots & (R = \infty) \\
 \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n & (R = 1) \\
 \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots & (R = 1) \\
 (1+x)^\alpha &= \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n & (R = 1)
 \end{aligned}$$

The last is the **binomial series**, valid for any real α with $\binom{\alpha}{n} = \alpha(\alpha-1)\cdots(\alpha-n+1)/n!$. For a nonnegative integer α it terminates and reduces to Chapter 9's binomial theorem.

Example 17.16 (proving Euler's formula).

Chapter 6 promised this. Substitute $x = i\theta$ into the exponential series and split by parity, using $i^2 = -1$:

$$\begin{aligned}
 e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\
 &= 1 - \underbrace{\frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots}_{\cos \theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right)}_{\sin \theta} \\
 &= \cos \theta + i \sin \theta.
 \end{aligned}$$

The even powers of i are real and alternate in sign; the odd powers are imaginary and alternate. Splitting the single exponential series by parity produces exactly the cosine and sine series.

The rearrangement is legitimate because the series converges absolutely everywhere — which is precisely where the Riemann rearrangement warning above bites, and precisely why absolute convergence was worth defining.

Trap: a Taylor series can converge to the wrong function.

Let

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

Every derivative of f at 0 is zero (the exponential decays faster than any polynomial grows). So its Taylor series about 0 is $0 + 0x + 0x^2 + \dots = 0$, which converges everywhere — to the zero function, which is not f .

A function whose Taylor series converges to it on a neighbourhood is called **analytic**. Infinitely differentiable is strictly weaker than analytic, and this is the standard example. Over the **complex** numbers the distinction vanishes: complex differentiability once implies analyticity, which is one of the reasons complex analysis is a nicer subject than real analysis.

17.7. Series in practice

Function evaluation. $\exp(x)$ in a library: reduce $x = k \ln 2 + r$ with $|r|$ small, compute e^r by a short polynomial (often a minimax polynomial, not literally Taylor — Taylor is optimal **at a point**, and you want uniform accuracy over an interval), then multiply by 2^k by adjusting the exponent field directly.

Small-angle approximations. $\sin x \approx x$, $\cos x \approx 1 - x^2/2$, $e^x \approx 1 + x$, $\ln(1+x) \approx x$, $(1+x)^\alpha \approx 1 + \alpha x$. These are first- or second-order truncations, and physics and graphics run on them.

Numerical stability. $\ln(1+x)$ for tiny x : computing `1.0 + x` first rounds away x 's low bits, then the log of something near 1 loses more. That is why `log1p(x)` exists — it evaluates the series $x - x^2/2 + \dots$ directly. Likewise `expm1(x)` for $e^x - 1$.

Generating functions. Encode a sequence (a_n) as the coefficients of a power series $A(x) = \sum a_n x^n$. Then recurrences on the sequence become **algebraic equations** on A . For Fibonacci one gets $A(x) = x/(1 - x - x^2)$, and partial fractions (Chapter 2) plus the geometric series recover Binet's formula from Chapter 11. It is a whole technique, and this is the bridge to it.

17.8. Exercises

- 17.1** Determine convergence or divergence, naming the test used: (a) $\sum n/(n^2 + 1)$; (b) $\sum 1/(n^2 + 1)$; (c) $\sum n!/n^n$; (d) $\sum (-1)^n/\sqrt{n}$; (e) $\sum (2^n)/(n!)$.
- 17.2** Find the sum of $\sum_{n=0}^{\infty} 3/4^n$ and of $\sum_{n=2}^{\infty} (2/3)^n$.
- 17.3** Find the radius and interval of convergence of $\sum_{n=1}^{\infty} (x-2)^n/(n3^n)$, checking both endpoints.
- 17.4** Compute the Maclaurin series of $f(x) = 1/(1+x^2)$ two ways: by substituting into the geometric series, and by differentiating repeatedly. Then integrate it term by term to derive the series for $\arctan x$, and set $x = 1$ to obtain a (very slowly converging) series for $\pi/4$.
- 17.5** Use Taylor's theorem to bound the error in approximating $e^{0.5}$ by the first four terms of the exponential series. Compare with the true error.
- 17.6** Use the series for $\sin$ to evaluate $\lim_{x \rightarrow 0} (\sin x - x)/x^3$ without L'Hôpital.
- 17.7** Show that the series $\sum (-1)^{n+1}/n$ converges, and estimate how many terms are needed to compute $\ln 2$ to within 10^{-3} . Then explain why nobody computes $\ln 2$ this way.
- 17.8** Prove that $\sum 1/(n(\ln n)^p)$ converges iff $p > 1$, using the integral test with the substitution $u = \ln x$.

CHAPTER 18.

Calculus in Many Variables

18.1. The jump

Every real problem has more than one input. A loss function has a million parameters. A physical field depends on three coordinates and time. A rendering equation depends on position and direction.

The good news: nothing conceptually new happens. The derivative is still **the best linear approximation**, and the integral is still **a limit of sums**. What changes is that “linear approximation” now needs a vector or a matrix to write down, and the bookkeeping grows.

This chapter is where calculus and linear algebra meet. It is also where the gradient comes from, and therefore where machine learning comes from.

Trap: read Chapters 19-20 first if you need to.

From here on we use vectors and matrices freely. If they are unfamiliar, read the first two chapters of Part IV now and come back. Nothing is lost by reading Part IV before finishing Part III — the dependency genuinely runs that way.

18.2. Functions of several variables

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ takes a vector and returns a number: a **scalar field**. A loss function is one of these.

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ returns a vector: a **vector field** or a **map**. A neural network layer is one of these.

Visualisation for $n = 2$: the graph is a surface in $\mathbb{R}^3$. More useful in practice is the **contour plot** — the curves $f(x, y) = c$ for various c , exactly like a topographic map. Contours are how you should picture a loss landscape, and they make the geometry of gradient descent obvious.

18.3. Partial derivatives

Definition 18.1 (partial derivative).

$$\frac{\partial f}{\partial x_i}(a) = \lim_{h \rightarrow 0} \frac{f(a + he_i) - f(a)}{h},$$

where e_i is the i -th standard basis vector. That is: differentiate with respect to x_i , holding all other variables fixed.

Computationally this is trivially easy — treat the other variables as constants and use every rule from Chapter 14.

Example 18.2.

$$f(x, y) = x^2y + \sin(xy).$$

$$\frac{\partial f}{\partial x} = 2xy + y \cos(xy), \quad \frac{\partial f}{\partial y} = x^2 + x \cos(xy).$$

Trap: partial derivatives alone do not mean much.

Here is a function whose partials both exist at the origin but which is not even continuous there:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & \text{otherwise} \end{cases}.$$

Along either axis f is identically 0, so both partials at the origin are 0. But along the line $y = x$, $f = 1/2$ everywhere except at the origin — so the limit as you approach the origin depends on the direction, and f is discontinuous.

In one variable, differentiable implied continuous. In many variables, **having partial derivatives does not**. Partial derivatives only probe n special directions; they can miss what happens in between. The correct notion of differentiability is stronger, and it is next.

18.4. The gradient and total differentiability

Definition 18.3 (gradient).

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **gradient** is the vector of partials

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)^T.$$

Definition 18.4 (differentiable, properly).

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **differentiable** at a if there is a vector g with

$$f(a + h) = f(a) + g^T h + r(h), \quad \text{where} \quad \frac{r(h)}{\|h\|} \rightarrow 0 \text{ as } h \rightarrow 0.$$

When this holds, $g = \nabla f(a)$.

Compare Chapter 14's local-linearity proposition. It is the **same definition**, with the scalar mh replaced by the dot product $g^T h$. That is the whole generalisation.

Theorem 18.5.

If all partial derivatives of f exist and are continuous near a , then f is differentiable at a .

So the pathological example above is ruled out by continuity of the partials, and for every function you will meet in practice, “compute the partials and stack them” is legitimate.

Definition 18.6 (directional derivative).

For a unit vector u ,

$$D_u f(a) = \lim_{h \rightarrow 0} \frac{f(a + hu) - f(a)}{h} = \nabla f(a)^T u.$$

Theorem 18.7 (the gradient points uphill, steepest).

Among all unit vectors u , $D_u f(a)$ is largest when u points in the direction of $\nabla f(a)$, and its value is then $\|\nabla f(a)\|$.

Proof. $D_u f = \nabla f^T u = \|\nabla f\| \|u\| \cos \theta = \|\nabla f\| \cos \theta$ where θ is the angle between u and ∇f (Chapter 19). This is maximised at $\cos \theta = 1$, i.e. $\theta = 0$. ■

The idea: everything about the gradient, in one place.

- **Direction:** ∇f points in the direction of steepest **increase**.
- **Magnitude:** $\|\nabla f\|$ is the rate of increase in that direction.
- **Descent:** $-\nabla f$ is steepest decrease. That is the entire idea of gradient descent: $x_{k+1} = x_k - \eta \nabla f(x_k)$.
- **Orthogonality:** ∇f is perpendicular to the level set (contour) through the point. Proof: moving along a contour keeps f constant, so the directional derivative along it is 0, so $\nabla f^T u = 0$.
- **Zero at optima:** at a local max or min, $\nabla f = 0$ — Fermat's theorem (Chapter 15) applied in each coordinate.

The contour picture makes gradient descent intuitive: you are standing on a hillside in fog, you can feel which way is downhill, you take a step. If the contours are long thin ellipses (an **ill-conditioned** problem), the gradient points mostly **across** the valley rather than along it, and you zigzag. Chapter 34 quantifies exactly how badly, in terms of the condition number.

18.5. The Jacobian

Definition 18.8 (Jacobian).

For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with components $f_1, \dots, f_m$, the **Jacobian** is the $m \times n$ matrix

$$J = \frac{\partial f}{\partial x} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}, \quad J_{ij} = \frac{\partial f_i}{\partial x_j}.$$

The defining property is exactly the same as before:

$$f(a + h) \approx f(a) + Jh.$$

The Jacobian is the derivative. Row i is the gradient of the i -th output; column j says how every output responds to input j . For $m = 1$ the Jacobian is the gradient written as a row.

Theorem 18.9 (multivariate chain rule).

If $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}^p$, then

$$J_{f \circ g}(x) = J_f(g(x)) \cdot J_g(x),$$

a $p \times m$ matrix times an $m \times n$ matrix.

The idea: the chain rule is matrix multiplication, and that is backprop.

In one variable, gains multiplied. Here, **Jacobians multiply** — and matrix multiplication is the correct generalisation because each output can depend on each input through every intermediate.

Now consider a network: $x \rightarrow h_1 \rightarrow h_2 \rightarrow \dots \rightarrow h_L \rightarrow L$ where the final loss L is a scalar. The chain rule gives

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial h_L} \frac{\partial h_L}{\partial h_{L-1}} \cdots \frac{\partial h_1}{\partial x}.$$

You can multiply that chain of matrices left-to-right or right-to-left, and it matters enormously.

Right to left (forward mode): you multiply matrices together, cost dominated by matrix–matrix products.

Left to right (reverse mode): the leftmost factor is a $1 \times n$ row vector (since L is a scalar), so every product is **vector times matrix** — far cheaper. You get the gradient with respect to **all** parameters at roughly the cost of one forward pass.

That asymmetry — one output, many inputs — is why reverse mode wins for machine learning, and reverse-mode automatic differentiation is exactly what backpropagation is. It is a statement about the associativity of matrix multiplication and nothing more. Chapter 36 works it out in full.

The special case worth writing separately: if $z = f(x, y)$ with $x = x(t)$, $y = y(t)$, then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Sum over every path by which t influences z . That is how you should read every chain rule: **sum over paths, multiply along each path** — which is exactly the matrix product of Chapter 10’s walk-counting theorem, again.

18.6. Second derivatives: the Hessian

Definition 18.10 (Hessian).

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **Hessian** is the $n \times n$ matrix of second partials

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Theorem 18.11 (Clairaut / Schwarz).

If the second partials are continuous, then

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x},$$

so the Hessian is **symmetric**.

Symmetry is not a technicality: Chapter 23 shows symmetric matrices have real eigenvalues and orthogonal eigenvectors, and that is what makes the second derivative test below work.

Theorem 18.12 (second derivative test in n dimensions).

Let $\nabla f(a) = 0$. Then:

- $H(a)$ positive definite (all eigenvalues > 0) $\Rightarrow$ local minimum;
- $H(a)$ negative definite (all eigenvalues < 0) $\Rightarrow$ local maximum;
- $H(a)$ indefinite (eigenvalues of both signs) $\Rightarrow$ **saddle point**;
- $H(a)$ singular $\Rightarrow$ inconclusive.

The idea: saddle points are the real obstacle in high dimensions.

In one variable a critical point is a max or a min (or an inflection). In n dimensions there are n eigenvalue signs to get right, so there are many more ways to be neither.

Heuristically, if the signs were random, a critical point would be a local minimum with probability about 2^{-n} . For n in the millions, local minima are vanishingly rare and **saddle points overwhelmingly dominate**.

This overturns the folk story that neural network training is hard because it gets stuck in bad local minima. The actual difficulty is saddle points and long flat regions, where the gradient is nearly zero but you are not at a minimum. Momentum-based optimisers exist largely to escape them.

The multivariate Taylor expansion, to second order:

$$f(a+h) \approx f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H(a) h.$$

The quadratic form $h^T H h$ is the local shape of the surface, and its eigenvalues are the curvatures along the principal axes. When the ratio of largest to smallest eigenvalue is huge, the bowl is a long thin valley — the ill-conditioning of the intuition box above.

18.7. Optimisation with constraints: Lagrange multipliers

Theorem 18.13 (Lagrange multipliers).

To find the extrema of $f(x)$ subject to $g(x) = 0$, solve

$$\nabla f = \lambda \nabla g, \quad g(x) = 0$$

for x and the scalar λ .

The idea: why the gradients must be parallel.

Picture the contours of f and the constraint curve $g = 0$.

If the constraint curve **crosses** a contour of f , you can slide along the curve to a higher contour — so you are not at an extremum. At an extremum the constraint curve must be **tangent** to a contour of f .

Tangency of the curves means their normals are parallel. The normal to a contour of f is ∇f ; the normal to $g = 0$ is ∇g . So $\nabla f = \lambda \nabla g$.

The multiplier λ is not junk either: it is the **shadow price**, the rate at which the optimal value changes as you relax the constraint. In an economics problem it is the marginal value of the resource; in a machine learning problem it is the trade-off weight in the corresponding penalty term. Ridge regression and its constrained form are related exactly by this λ .

Example 18.14.

Maximise $f(x, y) = xy$ subject to $x + y = 10$.

$\nabla f = (y, x)$, $g = x + y - 10$, $\nabla g = (1, 1)$. The conditions are $y = \lambda$, $x = \lambda$, $x + y = 10$, giving $x = y = 5$ and the maximum 25.

Sanity check with AM–GM from Chapter 2: $\sqrt{xy} \leq (x + y)/2 = 5$, so $xy \leq 25$ with equality iff $x = y$. The two methods agree, as they must.

For several constraints, one multiplier each: $\nabla f = \sum_i \lambda_i \nabla g_i$. With inequality constraints the conditions become the Karush–Kuhn–Tucker conditions, which are the foundation of constrained convex optimisation (Chapter 34).

18.8. Multiple integrals

Definition 18.15 (double integral).

$$\iint_R f(x, y) \, dA = \lim \sum f(x_i^*, y_j^*) \Delta A_{ij},$$

a limit of Riemann sums over small rectangles tiling the region R .

Theorem 18.16 (Fubini's theorem).

If f is continuous on a rectangle $[a, b] \times [c, d]$, then

$$\iint f \, dA = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.$$

So a double integral is computed as two nested single integrals, in either order. Choosing the convenient order can turn an impossible integral into an easy one, and for non-rectangular regions the inner limits become functions of the outer variable.

Theorem 18.17 (change of variables).

Under a substitution $x = T(u)$ with Jacobian matrix J_T ,

$$\iint_R f(x) \, dx = \iint_{T^{-1}(R)} f(T(u)) |\det J_T| \, du.$$

The idea: the determinant is a volume scale factor.

In one variable, substitution introduced a factor $(dx)/(du)$. Here that factor becomes $|\det J_T|$, and the reason is Chapter 22's interpretation of the determinant: **$|\det A|$ is the factor by which the linear map A scales volume.**

Locally, T looks like the linear map J_T . So a tiny box of volume du maps to a tiny parallelepiped of volume $|\det J_T| du$. The determinant is the exchange rate between the two coordinate systems.

Example 18.18 (polar coordinates, and the Gaussian integral).

With $x = r \cos \theta$, $y = r \sin \theta$,

$$J = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det J = r(\cos^2 + \sin^2) = r,$$

so $dA = r \, dr \, d\theta$. The extra r is not a convention to memorise; it is the determinant, and it says that a patch far from the origin subtends more area for the same $d\theta$.

Now the famous application. Let $I = \int_{-\infty}^{\infty} e^{-x^2} dx$, which has no elementary antiderivative. Square it and convert to polar:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA \\ &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \cdot \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi. \end{aligned}$$

So $I = \sqrt{\pi}$.

That r from the Jacobian is exactly what makes the inner integral elementary — re^{-r^2} integrates by substitution while e^{-r^2} does not. This computation is why the normal distribution has a $\sqrt{2\pi}$ in its denominator (Chapter 28), and it is one of the most satisfying pieces of mathematics in this book.

18.9. Vector calculus, briefly

For completeness, since the notation appears in physics, graphics, and PDE literature.

Definition 18.19.

For a vector field $F = (F_1, F_2, F_3)$ and scalar field f :

$$\text{grad } f = \nabla f \quad (\text{vector: direction of steepest ascent})$$

$$\text{div } F = \nabla \cdot F = \sum_i \frac{\partial F_i}{\partial x_i} \quad (\text{scalar: net outflow per volume})$$

$$\text{curl } F = \nabla \times F \quad (\text{vector: local rotation})$$

$$\text{Laplacian} = \nabla^2 f = \text{div}(\text{grad } f) = \sum_i \frac{\partial^2 f}{\partial x_i^2} \quad (\text{scalar: how far } f \text{ is from its local average})$$

The Laplacian reading is the useful one: $\nabla^2 f$ at a point is proportional to (average of f on a small sphere around the point) minus (f at the point). It is a **smoothing detector**, which is why it governs heat diffusion, why $\nabla^2 f = 0$ (Laplace's equation) means “equal to your own average”, and why the discrete Laplacian is an edge-detection kernel in image processing — and why the graph Laplacian of Chapter 10 has the same name.

The three great theorems — Green's, Stokes', and the divergence theorem — are all the Fundamental Theorem of Calculus in higher dimensions: **the integral of a derivative over a region equals something evaluated on the boundary**. Chapter 16's $\int_a^b f' = f(b) - f(a)$ is the one-dimensional case, where the “boundary” of $[a, b]$ is the two points a and b .

18.10. Exercises

- 18.1** For $f(x, y) = x^3y - 4xy^2 + y$, compute both partial derivatives, the gradient at $(1, 2)$, and the Hessian.
- 18.2** Find the directional derivative of $f(x, y) = x^2 + 3xy$ at $(1, 2)$ in the direction of the vector $(3, 4)$. (Remember to normalise.)
- 18.3** Verify Clairaut's theorem for $f(x, y) = x^2e^{xy}$ by computing both mixed partials.

- 18.4** Find and classify all critical points of $f(x, y) = x^3 - 3x + y^2$, using the Hessian.
- 18.5** Use the chain rule to find $(dz)/(dt)$ where $z = x^2y$, $x = \cos t$, $y = \sin t$. Then verify by substituting first and differentiating directly.
- 18.6** Use Lagrange multipliers to find the point on the plane $2x + 3y + z = 12$ closest to the origin. (Minimise the squared distance, not the distance — and say why that is legitimate.)
- 18.7** Evaluate $\int_0^1 \int_0^2 (x + y^2) dx dy$, then swap the order and confirm you get the same answer.
- 18.8** Use polar coordinates to evaluate $\iint_R (x^2 + y^2) dA$ where R is the unit disc.
- 18.9** A network computes $L = f(W_2 \sigma(W_1 x))$ with $x \in \mathbb{R}^n$, $W_1 \in \mathbb{R}^{h \times n}$, $W_2 \in \mathbb{R}^{1 \times h}$. Write the chain rule for $\partial L / \partial W_1$ as a product of Jacobians, and state the dimensions of each factor. Then say which multiplication order is cheaper and by how much.

PART IV

Linear Algebra

The most useful mathematics ever invented for a programmer. Vectors, matrices as functions, solving systems, determinants, eigenvalues, orthogonality and least squares, the singular value decomposition, and matrix calculus.

CHAPTER 19.

Vectors and Geometry

19.1. What a vector is

Three answers, all correct, all needed.

The physicist's: a quantity with magnitude and direction — an arrow.

The programmer's: an ordered list of numbers, `[3.0, -1.5, 2.0]`.

The mathematician's: an element of a **vector space** — any set where you can add elements and scale them by numbers, subject to some axioms.

The third looks like abstraction for its own sake, and it is the one that pays off. Its point is that a great many things behave like arrows without being arrows: polynomials, functions, matrices, images, random variables, solutions of differential equations. Every theorem proved for abstract vector spaces applies to all of them at once.

Definition 19.1 (vector space).

A **vector space** over $\mathbb{R}$ is a set V with an addition $V \times V \rightarrow V$ and a scalar multiplication $\mathbb{R} \times V \rightarrow V$, such that for all $u, v, w \in V$ and $a, b \in \mathbb{R}$:

- (V1) $u + v = v + u$
- (V2) $(u + v) + w = u + (v + w)$
- (V3) there is a zero vector 0 with $v + 0 = v$
- (V4) each v has an inverse $-v$ with $v + (-v) = 0$
- (V5) $a(u + v) = au + av$
- (V6) $(a + b)v = av + bv$
- (V7) $(ab)v = a(bv)$
- (V8) $1v = v$

Compare Chapter 8: (V1)–(V4) say V is an abelian group under addition; the rest say the scalars act compatibly. If you find yourself reaching for a property not on that list, it does not hold in general.

Example 19.2 (vector spaces you already use).

- $\mathbb{R}^n$, tuples of n reals. The prototype.
- $\mathbb{R}^{m \times n}$, all $m \times n$ matrices.
- $\mathcal{P}_n$, polynomials of degree at most n . Adding polynomials and scaling them behaves exactly like adding arrows.
- $C[0, 1]$, continuous functions on $[0, 1]$. Infinite-dimensional.
- Solutions of a homogeneous linear ODE or recurrence — which is why Chapter 11's “linear combination of characteristic roots” was legitimate.
- Images: an RGB image is a vector in $\mathbb{R}^{3hw}$, and averaging two images is vector addition.

Trap: not everything is a vector space.

Vectors in $\mathbb{R}^3$ under the **cross product** are not a vector space under that operation — it is not associative.

Probability distributions over n outcomes are **not** a vector space: they are not closed under scaling (doubling breaks the sum-to-one constraint) or under general addition. They form a **simplex**, a convex set, which is why optimising over distributions requires convex optimisation rather than plain linear algebra.

Unsigned image pixel values in $[0, 255]$ are not a vector space either. Clipping is exactly the failure of closure.

19.2. $\mathbb{R}^n$ and its geometry

Write vectors as columns:

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n.$$

Addition and scaling are componentwise. Geometrically, addition is tip-to-tail; scaling stretches (and flips, if negative).

Definition 19.3 (dot product).

$$u \cdot v = u^T v = \sum_{i=1}^n u_i v_i.$$

Definition 19.4 (norm).

$$\|v\| = \sqrt{v \cdot v} = \sqrt{\sum_i v_i^2}.$$

This is the **Euclidean** or ℓ_2 norm — the length of the arrow, by Pythagoras.

Theorem 19.5 (the geometric meaning of the dot product).

$$u \cdot v = \|u\| \|v\| \cos \theta,$$

where θ is the angle between u and v .

Proof. Apply the law of cosines (Chapter 5) to the triangle with sides u , v , and $v - u$:

$$\|v - u\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\| \cos \theta.$$

Expand the left side algebraically:

$$\|v - u\|^2 = (v - u) \cdot (v - u) = \|v\|^2 - 2u \cdot v + \|u\|^2.$$

Comparing the two gives the result. ■

The idea: the dot product measures agreement.

Read $u \cdot v = \|u\| \|v\| \cos \theta$ as: **how much of u points along v , times how big v is.**

- $u \cdot v > 0$: the vectors broadly agree, angle under 90° .

- $u \cdot v = 0$: perpendicular. This is the definition of **orthogonal**, and it is the single most useful condition in linear algebra.
- $u \cdot v < 0$: they oppose.

This is the reason cosine similarity is **the** similarity measure for embeddings. Two word vectors are similar when they point the same way; their magnitudes usually encode frequency rather than meaning, so you divide them out:

$$\text{cos-sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|} \in [-1, 1].$$

And when your embeddings are pre-normalised to unit length — as they usually are — cosine similarity **is** the dot product, which is why similarity search is a matrix multiplication.

Theorem 19.6 (Cauchy–Schwarz inequality).

$$|u \cdot v| \leq \|u\| \|v\|,$$

with equality iff u and v are parallel.

Proof. If $v = 0$ both sides are zero. Otherwise consider, for real t ,

$$p(t) = \|u - tv\|^2 = \|u\|^2 - 2t(u \cdot v) + t^2\|v\|^2.$$

This is a quadratic in t that is never negative (it is a squared norm). A quadratic $at^2 + bt + c$ with $a > 0$ is nonnegative for all t exactly when its discriminant is ≤ 0 (Chapter 2):

$$4(u \cdot v)^2 - 4\|v\|^2\|u\|^2 \leq 0.$$

Rearranged, that is the claim. Equality holds iff the quadratic has a real root, i.e. $u = tv$ for some t . ■

Note this proof does not use the geometric formula, so it holds in **any** space with an inner product — including infinite-dimensional function spaces. It is also what guarantees $\cos \theta \in [-1, 1]$, so that the angle is well-defined in the first place.

Corollary 19.7 (triangle inequality in $\mathbb{R}^n$).

$$\|u + v\| \leq \|u\| + \|v\|.$$

Proof.

$$\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2,$$

using Cauchy–Schwarz in the middle. Take square roots. ■

19.3. Other norms

$$\|v\|_p = \left(\sum_i |v_i|^p \right)^{1/p}.$$

- $p = 2$: Euclidean. Rotation-invariant, differentiable away from zero, the default.
- $p = 1$: $\|v\|_1 = \sum |v_i|$, “Manhattan” or taxicab distance.
- $p = \infty$: $\|v\|_\infty = \max_i |v_i|$, the limiting case.

The idea: why ℓ_1 produces sparsity.

Picture the unit balls in $\mathbb{R}^2$: the ℓ_2 ball is a circle, the ℓ_1 ball is a diamond with corners on the axes, the ℓ_∞ ball is a square.

Now think about a regularised regression, which minimises error subject to the coefficient vector lying in a small ball. Geometrically you inflate the error contours until they first touch the ball.

A circle has no corners, so the touch point is generically somewhere arbitrary — no coordinate is zero. A diamond has corners **on the axes**, and a generic contour is much more likely to touch a corner. A corner is a point where some coordinates are exactly zero.

That is the entire geometric explanation of why lasso (ℓ_1) gives sparse models and ridge (ℓ_2) does not. It is one picture and it is worth drawing.

19.4. Linear combinations, span, independence

Definition 19.8 (linear combination and span).

A **linear combination** of $v_1, \dots, v_k$ is $\sum_i c_i v_i$ for scalars c_i . Their **span** is the set of all such combinations:

$$\text{span}\{v_1, \dots, v_k\} = \left\{ \sum_i c_i v_i : c_i \in \mathbb{R} \right\}.$$

The span is always a subspace: it contains 0 and is closed under addition and scaling. In $\mathbb{R}^3$, the span of one nonzero vector is a line, of two independent vectors a plane, of three a copy of all of $\mathbb{R}^3$.

Definition 19.9 (linear independence).

$v_1, \dots, v_k$ are **linearly independent** if

$$c_1 v_1 + \dots + c_k v_k = 0 \Rightarrow c_1 = \dots = c_k = 0.$$

Otherwise they are **dependent**.

The idea: independence means no redundancy.

Dependent means one of the vectors can be written in terms of the others — it adds no new directions. Independent means every vector contributes something the rest cannot produce.

The definition is phrased via “the only way to get zero is trivially” because that formulation is symmetric in the vectors and easy to check: it is a homogeneous linear system, and Chapter 21 will solve it by elimination.

Programmer’s reading: a dependent set is a schema with a derived column — storing it is redundant, and it will make your normal equations singular.

Definition 19.10 (basis and dimension).

A **basis** of a vector space V is a linearly independent set that spans V . Every basis of V has the same number of elements, called the **dimension** $\dim V$.

Theorem 19.11 (unique representation).

If $B = \{v_1, \dots, v_n\}$ is a basis of V , every $v \in V$ has **exactly one** expression $v = \sum c_i v_i$.

Proof. Existence is spanning. For uniqueness, suppose $\sum c_i v_i = \sum d_i v_i$. Subtracting, $\sum (c_i - d_i) v_i = 0$, and independence forces every $c_i - d_i = 0$. ■

The idea: a basis is a coordinate system.

The unique coefficients $(c_1, \dots, c_n)$ are the **coordinates** of v in the basis B . Choosing a basis is choosing how to name points with numbers.

The standard basis $e_1, \dots, e_n$ of $\mathbb{R}^n$ (ones in one slot, zeros elsewhere) is one choice among infinitely many, and often not the best one. Much of applied linear algebra is **finding the right basis**:

- Eigenvectors (Chapter 23): the basis in which a linear map is diagonal.
- Principal components (Chapter 25): the basis in which your data has uncorrelated coordinates, ordered by variance.
- Fourier basis: the basis in which convolution becomes multiplication.
- Wavelets: the basis in which natural images are sparse — which is what JPEG-2000 exploits.

Same objects, different names for them, radically different difficulty.

19.5. Subspaces

Definition 19.12 (subspace).

A subset $W \subseteq V$ is a **subspace** if it contains 0 and is closed under addition and scalar multiplication.

Checking those three conditions is the standard exercise, and the first one does most of the work: a set not containing the origin is never a subspace. The plane $x + y + z = 1$ is not a subspace; the plane $x + y + z = 0$ is. That distinction — **affine** versus **linear** — is the same one that distinguishes $y = mx + b$ from $y = mx$, and it is why neural networks carry a separate bias term.

19.6. Projection

Definition 19.13 (projection onto a vector).

The projection of u onto the direction of $v \neq 0$ is

$$\text{proj}_v(u) = \left(\frac{u \cdot v}{v \cdot v} \right) v.$$

Proof. We want the multiple tv closest to u ; equivalently, the one for which the error $u - tv$ is orthogonal to v . Setting $(u - tv) \cdot v = 0$ gives $u \cdot v = t(v \cdot v)$, so $t = (u \cdot v)/(v \cdot v)$. ■

The idea: orthogonality of the residual is the whole story.

The condition that made the projection optimal was: **the error is perpendicular to the thing you projected onto.**

That single principle, repeated, gives:

- least squares regression (Chapter 24): the residual is orthogonal to the column space of the design matrix;
- Gram–Schmidt: subtract off projections until what remains is orthogonal;
- Fourier coefficients: each is a projection onto a basis sinusoid;
- conditional expectation in probability (Chapter 29): $\mathbb{E}[Y|X]$ is the orthogonal projection of Y onto the space of functions of X , and the “error is orthogonal” statement is the law of total expectation.

If you take one thing from Part IV, take this.

19.7. The cross product (three dimensions only)

Definition 19.14.

For $u, v \in \mathbb{R}^3$,

$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}.$$

Properties: $u \times v$ is orthogonal to both u and v ; $\|u \times v\| = \|u\| \|v\| \sin \theta$, which is the area of the parallelogram they span; the direction follows the right-hand rule; and $u \times v = -(v \times u)$, so it is anticommutative and definitely not associative.

Uses: surface normals in graphics, torque and angular momentum in physics, testing which side of a line a point falls on (the sign of a 2×2 determinant, which is the z -component of a cross product) — the primitive underneath every convex hull and polygon-triangulation algorithm.

The cross product exists only in $\mathbb{R}^3$ (and, in a degenerate way, $\mathbb{R}^7$). The general-dimension replacement is the **wedge product** of exterior algebra, and the determinant of Chapter 22 is its shadow.

19.8. Exercises

- 19.1** Let $u = (1, -2, 3)$ and $v = (4, 0, -1)$. Compute $u \cdot v$, $\|u\|$, $\|v\|$, the angle between them, $\text{proj}_v(u)$, and $u \times v$. Verify that $u \times v$ is orthogonal to both.
- 19.2** Determine whether each is a subspace of $\mathbb{R}^3$, justifying: (a) $\{(x, y, z) : x + 2y = 0\}$; (b) $\{(x, y, z) : x \geq 0\}$; (c) $\{(x, y, z) : xy = 0\}$; (d) $\{(x, y, z) : x = y = z\}$.
- 19.3** Determine whether $(1, 2, 3), (2, 1, 0), (4, 5, 6)$ are linearly independent. If not, exhibit an explicit dependence relation.
- 19.4** Prove that any set of vectors containing the zero vector is linearly dependent.
- 19.5** Prove the **parallelogram law**: $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$. Then explain what it says geometrically about a parallelogram’s diagonals.
- 19.6** Show that $\|v\|_\infty \leq \|v\|_2 \leq \|v\|_1$ for every $v \in \mathbb{R}^n$, and find vectors achieving equality in each.
- 19.7** Show that $\{1, x, x^2\}$ is a basis for the space of polynomials of degree at most 2, and find the coordinates of $p(x) = 3 - x + 2x^2$ in that basis. Then find its coordinates in the basis $\{1, 1 + x, (1 + x)^2\}$.
- 19.8** Prove that if $u \cdot v = 0$ then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$, and identify the classical theorem this is.

CHAPTER 20.

Matrices Are Functions

20.1. The single idea of this chapter

A matrix is not a table of numbers. A matrix is a **function** — specifically, a linear function — written down in a particular way. Everything strange about matrices becomes obvious once you believe that, and stays strange forever if you do not.

Definition 20.1 (linear map).

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if for all u, v and scalars c :

$$T(u + v) = T(u) + T(v), \quad T(cu) = cT(u).$$

Equivalently, $T(au + bv) = aT(u) + bT(v)$: **linear maps preserve linear combinations**. Note $T(0) = 0$ always, so $x \mapsto x + 1$ is not linear (it is **affine**).

Theorem 20.2 (every linear map is a matrix).

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and let $e_1, \dots, e_n$ be the standard basis of $\mathbb{R}^n$. Define A to be the $m \times n$ matrix whose j -th column is $T(e_j)$. Then $T(x) = Ax$ for every x .

Proof. Write $x = x_1e_1 + \dots + x_ne_n$. By linearity,

$$T(x) = x_1T(e_1) + x_2T(e_2) + \dots + x_nT(e_n).$$

The right-hand side is precisely the linear combination of the columns of A with coefficients x_i , which is the definition of Ax . ■

The idea: this theorem is why matrices exist.

A general function $\mathbb{R}^n \rightarrow \mathbb{R}^m$ needs infinite information to specify. A **linear** one is pinned down by what it does to n basis vectors — because linearity forces everything else.

So a matrix is a finite encoding of an infinite object, and the encoding is exactly “where each basis vector goes”. Read that way:

- The columns of A are the images of the basis vectors. To see what a matrix does, look at its columns as destinations.
- Ax is a **linear combination of the columns of A** , weighted by the entries of x . This is the single most useful way to read matrix–vector multiplication, more useful than the row-by-row dot product recipe.
- The set of possible outputs is the span of the columns — the **column space**.

Example 20.3 (reading a matrix off its geometry).

What matrix rotates $\mathbb{R}^2$ by angle θ ?

Do not derive; just ask where the basis vectors go. $e_1 = (1, 0)$ lands on $(\cos \theta, \sin \theta)$, straight off the unit circle. $e_2 = (0, 1)$ is a quarter turn ahead, so it lands on $(-\sin \theta, \cos \theta)$. Those are the columns:

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Similarly, scaling by s_x, s_y is $\begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}$; reflection in the x -axis is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$; the shear that slides x by ky is $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ — because $e_1 \mapsto e_1$ and $e_2 \mapsto (k, 1)$.

You can now write down any linear transformation you can picture, with no algebra.

20.2. Matrix multiplication is composition

Definition 20.4 (matrix product).

For $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$, the product AB is the $m \times n$ matrix with

$$(AB)_{ij} = \sum_{k=1}^p A_{ik} B_{kj}.$$

Theorem 20.5.

AB is the matrix of the composition: $(AB)x = A(Bx)$ for all x .

Proof. Compute the i -th entry of $A(Bx)$:

$$(A(Bx))_i = \sum_k A_{ik} (Bx)_k = \sum_k A_{ik} \sum_j B_{kj} x_j = \sum_j \left(\sum_k A_{ik} B_{kj} \right) x_j = ((AB)x)_i.$$

■

The idea: why the formula looks like that.

The formula $\sum_k A_{ik} B_{kj}$ is not arbitrary and it is not a convention. It is: **to get from input j to output i , sum over every intermediate slot k ; along each route, multiply the two gains.**

Sum over paths, multiply along paths. That is the same reading as:

- Chapter 10's theorem that $(A^k)_{ij}$ counts walks of length k ;
- Chapter 18's multivariate chain rule, where Jacobians multiply;
- composition of relations, and of probability transition matrices.

All four are the same computation. Once you see composition, the definition is forced.

Consequences that now need no memorisation:

Dimensions must match. $(m \times p)(p \times n) = (m \times n)$. You cannot compose $g \circ f$ unless f 's outputs are g 's inputs.

$AB \neq BA$ in general. Rotating then scaling differs from scaling then rotating. Function composition is not commutative, so matrix multiplication is not either. Do not expect it to be, ever.

$A(BC) = (AB)C$. Composition is associative (Chapter 3), so this is free. It is also the most economically important identity in numerical computing: the two groupings can differ by orders of magnitude in cost. Multiplying $(1000 \times 1) \cdot (1 \times 1000) \cdot (1000 \times 1)$ left-to-right costs a million

operations; right-to-left costs two thousand. This is why reverse-mode autodiff wins (Chapter 18) and why matrix-chain ordering is a classic dynamic programming problem.

$AB = 0$ does not imply $A = 0$ or $B = 0$. Two nonzero maps can compose to nothing if the first sends everything into the second's null space.

20.3. The basic operations

Definition 20.6.

- **Transpose:** $(A^T)_{ij} = A_{ji}$. Flip across the main diagonal.
- **Identity:** I_n , ones on the diagonal, zeros elsewhere; $AI = IA = A$.
- **Inverse:** A^{-1} with $AA^{-1} = A^{-1}A = I$, when it exists.
- **Trace:** $\text{tr}(A) = \sum_i A_{ii}$, defined for square A .

Proposition 20.7 (transpose rules).

$(A + B)^T = A^T + B^T$, $(cA)^T = cA^T$, $(A^T)^T = A$, and crucially

$$(AB)^T = B^T A^T.$$

Proof. $((AB)^T)_{ij} = (AB)_{ji} = \sum_k A_{jk} B_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. ■

The order reversal is the same phenomenon as $(AB)^{-1} = B^{-1}A^{-1}$: undoing a composition means undoing the last step first. Socks then shoes; shoes off then socks off.

Proposition 20.8 (trace is cyclic).

$\text{tr}(AB) = \text{tr}(BA)$, and more generally $\text{tr}(ABC) = \text{tr}(CAB)$.

Proof. $\text{tr}(AB) = \sum_i \sum_k A_{ik} B_{ki} = \sum_k \sum_i B_{ki} A_{ik} = \text{tr}(BA)$, just by swapping the order of two finite sums. ■

Cyclicity looks minor and is used constantly in matrix calculus (Chapter 26) to move factors around inside a trace until a derivative is recognisable.

20.4. Special matrices worth recognising

Type	Definition and significance
Diagonal	Nonzero only on the diagonal. Acts by independent scaling of each coordinate. Trivial to invert, exponentiate, and multiply.
Symmetric	$A^T = A$. Real eigenvalues, orthogonal eigenvectors (Ch. 23). Covariance matrices and Hessians are symmetric.
Orthogonal	$Q^T Q = I$, so $Q^{-1} = Q^T$. Preserves lengths and angles: a rotation or reflection. Numerically ideal — never amplifies error.
Triangular	Zeros above (lower) or below (upper) the diagonal. Solvable by back-substitution in $O(n^2)$. The output of LU and QR.

Type	Definition and significance
Positive definite	Symmetric with $x^T A x > 0$ for all $x \neq 0$. Defines a genuine “energy”. Cholesky-factorisable; the Hessian at a strict minimum.
Sparse	Mostly zeros. Not a mathematical class but the dominant practical one — graphs, meshes, and discretised PDEs give sparse matrices, and exploiting that is the difference between feasible and impossible.
Permutation	A single 1 in each row and column. Reorders coordinates. Orthogonal, and its inverse is its transpose.

Proposition 20.9 (orthogonal matrices preserve geometry).

If $Q^T Q = I$ then $\|Qx\| = \|x\|$ and $(Qx) \cdot (Qy) = x \cdot y$.

Proof.

$$(Qx) \cdot (Qy) = (Qx)^T (Qy) = x^T Q^T Q y = x^T I y = x \cdot y.$$

Setting $y = x$ and taking square roots gives the norm statement. ■

This is why numerical algorithms prefer orthogonal transformations wherever possible: they cannot magnify an error, since they do not magnify anything. Gaussian elimination can lose precision badly; QR factorisation, built from orthogonal steps, cannot. Chapter 33 makes that precise.

20.5. Block matrices

A matrix can be partitioned into submatrices and multiplied blockwise, as long as the block dimensions are compatible:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} AX + BY \\ CX + DY \end{pmatrix}.$$

This is not just notation. It is the basis of cache-blocked matrix multiplication (operate on tiles that fit in cache), of Strassen’s algorithm (Chapter 11), and of how a transformer’s multi-head attention is implemented as one big matrix multiply rather than many small ones.

20.6. The four fundamental subspaces

Definition 20.10.

For $A \in \mathbb{R}^{m \times n}$:

$$\text{Col}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m \quad (\text{column space, or range})$$

$$\text{Null}(A) = \{x : Ax = 0\} \subseteq \mathbb{R}^n \quad (\text{null space, or kernel})$$

$$\text{Row}(A) = \text{Col}(A^T) \subseteq \mathbb{R}^n$$

$$\text{Null}(A^T) \subseteq \mathbb{R}^m \quad (\text{left null space})$$

Each is a subspace (check the three conditions). The two that matter most:

The column space is what outputs are achievable. $Ax = b$ has a solution iff $b \in \text{Col}(A)$.

The null space is what inputs get destroyed. If $x \in \text{Null}(A)$, then $A(x_0 + x) = Ax_0$, so solutions are never unique when the null space is bigger than $\{0\}$.

Theorem 20.11 (solutions to a linear system).

If $Ax_0 = b$, then the full solution set is

$$\{x_0 + z : z \in \text{Null}(A)\}.$$

Proof. If $Az = 0$ then $A(x_0 + z) = b$. Conversely if $Ax = b$ then $A(x - x_0) = 0$, so $x - x_0 \in \text{Null}(A)$. ■

So a linear system has either no solution ($b \notin \text{Col}(A)$), exactly one ($b \in \text{Col}(A)$ and $\text{Null}(A) = \{0\}$), or infinitely many. Never exactly two. That trichotomy is a complete answer to “what can happen”, and Chapter 21 turns it into an algorithm.

The idea: injective, surjective, bijective — again.

Chapter 3’s three words, now for matrices:

- A is **injective** iff $\text{Null}(A) = \{0\}$ iff the columns are independent.
- A is **surjective** iff $\text{Col}(A) = \mathbb{R}^m$ iff the columns span $\mathbb{R}^m$.
- A is **bijective** (invertible) iff both. This requires $m = n$.

Chapter 3 said linear algebra is what happens to function theory when the functions are linear. This is that promise being kept.

20.7. Inverses

Definition 20.12.

A square A is **invertible** (or **nonsingular**) if there is A^{-1} with $AA^{-1} = A^{-1}A = I$.

Theorem 20.13 (the invertible matrix theorem, first instalment).

For square $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

- (i) A is invertible.
- (ii) $Ax = b$ has a unique solution for every b .
- (iii) $Ax = 0$ has only the solution $x = 0$.
- (iv) The columns of A are linearly independent.
- (v) The columns of A span $\mathbb{R}^n$.
- (vi) $\text{rank}(A) = n$.
- (vii) $\det A \neq 0$ (Chapter 22).
- (viii) 0 is not an eigenvalue of A (Chapter 23).

The equivalences are proved as the machinery arrives; the list is here so you have it in one place. Learn it as a single fact with eight faces — in practice you check whichever face is cheapest and conclude all the others.

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

valid when $ad - bc \neq 0$. That quantity is the determinant.

Trap: never actually compute an inverse.

To solve $Ax = b$, do **not** compute A^{-1} and multiply. Factor A (LU or QR) and solve by substitution.

Three reasons. It is about three times more arithmetic. It is numerically worse — forming the inverse introduces error that solving directly avoids. And it destroys sparsity: the inverse of a sparse matrix is generally dense, so a matrix that fit in memory produces an inverse that does not.

When mathematical writing says $x = A^{-1}b$, read it as “solve the system”, not as an instruction to invert. Numerical libraries make this distinction in their API for exactly this reason: `solve(A, b)`, not `inv(A) @ b`.

20.8. Matrices as data versus matrices as maps

Both readings are legitimate and you should know which one you are using.

As a map: a rotation, a projection, a network layer. Multiplication is composition, and the natural questions are about eigenvalues, invertibility, and geometry.

As data: a table of m observations by n features, an adjacency matrix, a grayscale image. Multiplication does something less obvious, and the natural questions are about low-rank approximation and factorisation.

The bridge between the two is the SVD (Chapter 25), which says that **every** matrix, read either way, is a rotation, then a scaling, then a rotation.

20.9. Exercises

20.1 Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Compute AB and BA and confirm they differ. Describe in words what B does geometrically.

20.2 Write the 3×3 matrix that reflects $\mathbb{R}^3$ in the xy -plane, and the one that rotates $\mathbb{R}^3$ by 90° about the z -axis. Find their product both ways and describe the two resulting transformations.

20.3 Prove that if A and B are invertible then AB is invertible with $(AB)^{-1} = B^{-1}A^{-1}$. Then show by example that $A + B$ need not be invertible even when both are.

20.4 Show that $\text{tr}(A^T A) = \sum_{i,j} A_{ij}^2$, and explain why this means $\text{tr}(A^T A) = 0$ forces $A = 0$. (This quantity is the squared **Frobenius norm** of A .)

20.5 Find the null space and column space of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

State the dimension of each and check they sum to the number of columns.

20.6 Prove that if Q_1 and Q_2 are orthogonal then so is $Q_1 Q_2$. (Together with the inverse property this makes the orthogonal matrices a group, Chapter 8.)

20.7 You must compute ABC where A is 100×5 , B is 5×100 , and C is 100×1 . Count the scalar multiplications for both association orders and state the ratio.

- 20.8** Prove that the set of 2×2 matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ is closed under addition and multiplication, and show that this set behaves exactly like the complex numbers of Chapter 6. What matrix corresponds to i , and what is its square?

CHAPTER 21.

Solving Linear Systems

21.1. The problem

Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, find every x with $Ax = b$.

Chapter 20 already told us the **shape** of the answer: no solutions, one, or infinitely many, and in the last case the solution set is a shifted copy of the null space. This chapter gives the algorithm that finds it, and the theory that falls out of the algorithm.

That ordering is deliberate. Gaussian elimination is not merely a computational recipe; running it is how you **discover** rank, independence, dimension, and the rank–nullity theorem. The theory is the algorithm’s exhaust.

21.2. Gaussian elimination

Definition 21.1 (elementary row operations).

- (R1) Swap two rows.
- (R2) Multiply a row by a nonzero scalar.
- (R3) Add a multiple of one row to another.

Theorem 21.2.

Elementary row operations do not change the solution set of $Ax = b$.

Proof. Each operation is reversible — swap back, divide back, subtract back — so the new system implies the old and vice versa. Two systems that imply each other have the same solution set. (This is Chapter 2’s distinction between equivalent and one-way transformations, applied to systems.) ■

Definition 21.3 (row echelon form).

A matrix is in **row echelon form** (REF) if: all zero rows are at the bottom; the first nonzero entry of each row (the **pivot**) is strictly to the right of the pivot above it.

It is in **reduced row echelon form** (RREF) if additionally every pivot is 1 and is the only nonzero entry in its column.

Gaussian elimination with partial pivoting

```
for each column j = 1..n:
    find row i >= current with the largest |A[i][j]|      # pivoting
    if that entry is 0: continue      # no pivot in this column
    swap row i with the current row
    for each row k below the current:
        factor := A[k][j] / A[current][j]
        row_k := row_k - factor * row_current
    advance to the next row
```

This produces REF in $O(n^3)$ operations for a square system — roughly $2n^3/3$ multiply-adds. Continuing upwards to clear above the pivots gives RREF (Gauss–Jordan), which costs about 50% more and is usually not worth it.

Trap: pivoting is not optional.

In exact arithmetic you may pick any nonzero pivot. In floating point you must pick the **largest available**, and the reason is Chapter 1’s warning about cancellation.

Consider

$$\begin{pmatrix} 10^{-20} & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Eliminating with the tiny pivot multiplies the second row by 10^{20} , which swamps its original entries: the value $1 - 10^{20}$ rounds to exactly -10^{20} , and the information in that row is destroyed. You get $y = 1$, $x = 0$ — badly wrong.

Swapping the rows first makes the pivot 1, the multiplier tiny, and the answer accurate to full precision. The matrix was perfectly well-conditioned; only the pivot choice was bad. This is why every library routine pivots by default, and why writing your own solver without pivoting is a mistake you make exactly once.

21.3. Reading the answer off the echelon form

Definition 21.4 (pivot and free variables).

Variables whose columns contain a pivot are **pivot variables**; the rest are **free variables**.

Solving from RREF

1. If a row reads $[0 \ 0 \ \dots \ 0 \mid c]$ with $c \neq 0$, the system is inconsistent: no solutions. Stop.
2. Otherwise, set each free variable to an arbitrary parameter.
3. Solve for each pivot variable in terms of the parameters by back-substitution.

Example 21.5 (a full worked system).

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 0 & 8 \\ -1 & -2 & 2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}.$$

Augment and eliminate. Row 2 minus 2(Row 1), Row 3 plus Row 1:

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 2 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array} \right)$$

Row 3 minus $1/2$ (Row 2):

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 2 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right)$$

Pivots in columns 1, 3, 4; column 2 is free. Back-substituting: $x_4 = 1$; then $2x_3 + 2 = 2$ so $x_3 = 0$; then $x_1 + 2x_2 - 0 + 3 = 2$ so $x_1 = -1 - 2x_2$.

With $x_2 = t$ free, the solution set is

$$x = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Exactly the structure Chapter 20 predicted: one particular solution plus the null space, which here is one-dimensional and spanned by $(-2, 1, 0, 0)$.

21.4. Rank

Definition 21.6 (rank).

$\text{rank}(A)$ is the number of pivots in any echelon form of A .

Theorem 21.7 (rank is well defined and symmetric).

The number of pivots does not depend on the elimination path, and

$$\text{rank}(A) = \dim \text{Col}(A) = \dim \text{Row}(A) = \text{rank}(A^T).$$

The idea: row rank equals column rank is not obvious.

The columns live in $\mathbb{R}^m$ and the rows live in $\mathbb{R}^n$ — different spaces, possibly wildly different sizes. That the maximum number of independent columns equals the maximum number of independent rows is a genuinely surprising theorem.

One clean proof: if $\text{rank}(A) = r$ by columns, then A can be written as $A = CR$ where C is $m \times r$ (a basis of the column space) and R is $r \times n$ (the coefficients). But then every **row** of A is a combination of the r rows of R , so the row rank is at most r too. Applying the same argument to A^T gives the reverse inequality.

The factorisation $A = CR$ is worth noticing on its own: **a rank- r matrix factors through $\mathbb{R}^r$** . An $m \times n$ matrix of rank r needs only $r(m+n)$ numbers instead of mn . That is the compression at the heart of low-rank approximation, of recommender systems, and of LoRA fine-tuning — where you freeze a large weight matrix and learn only a rank- r update.

Theorem 21.8 (rank-nullity).

For $A \in \mathbb{R}^{m \times n}$,

$$\text{rank}(A) + \dim \text{Null}(A) = n.$$

Proof. Run elimination to RREF. Each of the n columns is either a pivot column or a free column, so

$$(\text{number of pivot columns}) + (\text{number of free columns}) = n.$$

The pivot count is $\text{rank}(A)$ by definition. And each free variable contributes exactly one basis vector to the null space — set that variable to 1, the other free variables to 0, and solve — so the free count is $\dim \text{Null}(A)$. These null-space vectors are independent (each has a 1 in a slot where the others have 0) and they span it (any solution is determined by its free values). ■

The idea: rank–nullity is a conservation law.

You start with n dimensions of input. The map A either **keeps** a direction (contributing to the rank, i.e. to the output) or **crushes** it (contributing to the null space). Nothing else can happen. So kept + crushed = started with.

Reading it as information: a linear map cannot create dimensions, and whatever it does not preserve, it destroys. A projection from $\mathbb{R}^3$ to a plane has rank 2 and nullity 1 — the direction perpendicular to the plane is exactly what is lost, and it is lost irrecoverably.

Corollary 21.9.

A square matrix is injective iff surjective iff invertible.

Proof. For $m = n$: injective means nullity 0, which by rank–nullity means rank n , which means the column space is all of $\mathbb{R}^n$, which is surjectivity. And Chapter 20 recorded that both together give invertibility. ■

This is a purely finite-dimensional phenomenon and it is worth knowing that it fails in infinite dimensions: the shift operator on infinite sequences, $(a_1, a_2, \dots) \mapsto (0, a_1, a_2, \dots)$, is injective and not surjective.

21.5. LU decomposition

Elimination can be recorded as a factorisation, and that is what you actually want in practice.

Theorem 21.10 (LU with pivoting).

For any square A there is a permutation matrix P , a unit lower triangular L , and an upper triangular U with

$$PA = LU.$$

The idea: U is the echelon form; L stores the multipliers used to get there; P records the row swaps.

The idea: why factor rather than just solve.

Elimination costs $O(n^3)$. Solving a triangular system by substitution costs $O(n^2)$.

If you need to solve $Ax = b$ for many different b with the **same** A — which happens constantly: time-stepping a simulation, Newton iterations, multiple right-hand sides — then factoring once for $O(n^3)$ and reusing it for $O(n^2)$ per solve is an enormous saving.

$$Ax = b \Rightarrow \underbrace{LU}_{y}x = Pb : \text{ solve } Ly = Pb \text{ (forward), then } Ux = y \text{ (backward).}$$

This is exactly why `lu_factor` and `lu_solve` are separate calls in every numerical library, and why `solve(A, b)` in a loop over b is a performance bug.

Related factorisations, each specialised:

- **Cholesky**, $A = LL^T$, for symmetric positive definite A . Half the work of LU, needs no pivoting, and **fails** precisely when A is not positive definite — which makes it the standard numerical test for positive definiteness.
- **QR**, $A = QR$ with Q orthogonal (Chapter 24). More expensive but numerically superior, and the right tool for least squares.
- **SVD** (Chapter 25). Most expensive, most informative, works for any shape.

21.6. Homogeneous systems and the null space

$Ax = 0$ always has the solution $x = 0$. The question is whether it has others.

Proposition 21.11.

$Ax = 0$ has a nonzero solution iff $\text{rank}(A) < n$; in particular, always when $m < n$ (more unknowns than equations).

Proof. Nonzero solutions exist iff the nullity is positive, which by rank–nullity means $\text{rank}(A) < n$. And $\text{rank}(A) \leq \min(m, n)$, so $m < n$ forces it. ■

“More unknowns than equations implies infinitely many solutions” is the formal version of a fact you already use when counting degrees of freedom.

21.7. Conditioning

Solvability is binary; **usefulness** is not. A system can be technically solvable and practically worthless.

Definition 21.12 (condition number).

For invertible A ,

$$\kappa(A) = \|A\| \cdot \|A^{-1}\| \geq 1,$$

where $\|\cdot\|$ is a matrix norm (Chapter 25 identifies the natural choice as the ratio of largest to smallest singular value).

Theorem 21.13 (error amplification).

If $Ax = b$ and $A\hat{x} = b + \delta b$, then

$$\frac{\|\hat{x} - x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$

So $\kappa(A)$ is the worst-case factor by which relative input error is magnified in the output.

The idea: what an ill-conditioned system looks like.

$$\begin{pmatrix} 1 & 1 \\ 1 & 1.0001 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 2.0001 \end{pmatrix}.$$

The exact solution is $x = y = 1$. Perturb the right-hand side by 0.0001 in the second entry and the solution jumps to $x = 0$, $y = 2$. A relative input change of about 5×10^{-5} produced a relative output change of order 1: an amplification of roughly 10^4 , which is κ .

Geometrically, the two equations are two nearly parallel lines. Their intersection point is well defined but slides a long way when either line is nudged. The matrix is not singular — it is **nearly** singular, and near is bad enough.

The rule of thumb: with $\kappa(A) \approx 10^k$, you lose about k decimal digits of accuracy. Double precision gives you 16, so $\kappa = 10^{12}$ leaves 4 — and $\kappa = 10^{16}$ leaves nothing. This is not a defect of your algorithm; no algorithm can do better, because the **problem** is sensitive. Chapter 33 separates these two kinds of blame properly.

21.8. Iterative methods, briefly

Direct elimination is $O(n^3)$ and destroys sparsity through **fill-in**: zeros become nonzeros as the algorithm proceeds. For the enormous sparse systems that come out of discretised PDEs, meshes, and graphs, that is fatal.

Iterative methods only ever **multiply** by A — which is cheap and preserves sparsity — and converge to the solution.

- **Jacobi and Gauss–Seidel**: split A into easy and hard parts and iterate. Simple, slow, converge under diagonal dominance.
- **Conjugate gradient**: for symmetric positive definite A . In exact arithmetic it terminates in n steps, and in practice it gets a good answer in far fewer — roughly $\sqrt{\kappa}$ iterations for a given accuracy, which is why **preconditioning** (transforming to reduce κ) matters so much.
- **GMRES**: the general non-symmetric workhorse.

The theme: for large sparse problems, an approximate answer obtained by repeated matrix–vector products beats an exact answer you cannot afford to compute.

21.9. Exercises

21.1 Solve by Gaussian elimination, giving the complete solution set:

$$x + 2y + z = 4$$

$$2x + y - z = 1$$

$$3x + 3y = 5$$

21.2 Find the rank and a basis for the null space of

$$A = \begin{pmatrix} 1 & 3 & 1 & 4 \\ 2 & 6 & 3 & 9 \\ 1 & 3 & 2 & 5 \end{pmatrix}.$$

Verify rank–nullity explicitly.

21.3 For which values of k does the system $x + ky = 1$, $kx + 4y = 2$ have (a) a unique solution, (b) no solution, (c) infinitely many?

21.4 Show that if A is 3×5 , then $Ax = 0$ must have a nonzero solution. What is the smallest possible dimension of the null space?

21.5 Compute the LU factorisation (without pivoting) of

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix},$$

and then use it to solve $Ax = (4, 10, 24)^T$ by forward and back substitution.

21.6 Carry out elimination on

$$\begin{pmatrix} 10^{-4} & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

first without pivoting and then with, working to 3 significant figures throughout. Compare both answers to the exact solution.

21.7 Prove that $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$. Hint: the column space of AB is contained in the column space of A ; for the other bound use transposes.

21.8 Let A be $m \times n$ with $m > n$ and $\text{rank}(A) = n$. Show that $Ax = b$ has at most one solution, and explain what it means when it has none. (This is the overdetermined case that motivates least squares in Chapter 24.)

CHAPTER 22.

Determinants

22.1. What the determinant measures

The determinant is usually introduced as a formula, which is the worst possible way in: the formula for a 3×3 is ugly and the formula for an $n \times n$ has $n!$ terms. Nobody could guess what it means. Start with the meaning instead.

The idea: the determinant is a volume scale factor, with a sign.

A matrix A is a linear map. Feed it the unit cube — the box spanned by $e_1, \dots, e_n$, of volume 1. It comes out as a parallelepiped.

$$|\det A| = \text{the volume of that parallelepiped.}$$

In other words, A multiplies every volume by $|\det A|$. That is the definition worth carrying.

The **sign** records orientation. $\det A > 0$ means the map preserves handedness; $\det A < 0$ means it flips it, like a mirror. In 2D, positive means counterclockwise stays counterclockwise.

And then the single most important fact is immediate: $\det A = 0$ means the cube is squashed **flat** — into something of lower dimension, with zero volume. Information has been destroyed, the map is not invertible, and the null space is nontrivial. Everything about singularity follows from “flattened”.

22.2. The defining properties

Rather than a formula, characterise the determinant by what it must do. This is how mathematicians actually think about it, and it makes proofs short.

Definition 22.1 (determinant, axiomatically).

$\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is the unique function satisfying:

- (D1) **Multilinear in the columns.** Holding the other columns fixed, $\det$ is linear in each column:
 $\det(\dots, au + bv, \dots) = a \det(\dots, u, \dots) + b \det(\dots, v, \dots)$.
- (D2) **Alternating.** If two columns are equal, $\det = 0$.
- (D3) **Normalised.** $\det I = 1$.

From these three, everything follows.

Proposition 22.2 (immediate consequences).

- (a) Swapping two columns negates the determinant.
- (b) Adding a multiple of one column to another leaves it unchanged.
- (c) Scaling one column by c scales the determinant by c .
- (d) If any column is zero, $\det = 0$.
- (e) If the columns are linearly dependent, $\det = 0$.

Proof. (a) Consider $\det(\dots, u + v, \dots, u + v, \dots)$, which is 0 by (D2). Expanding by multilinearity gives four terms; the two with a repeated column vanish, leaving

$$\det(\dots, u, \dots, v, \dots) + \det(\dots, v, \dots, u, \dots) = 0.$$

(b) $\det(\dots, u + cv, \dots, v, \dots) = \det(\dots, u, \dots, v, \dots) + c \det(\dots, v, \dots, v, \dots)$, and the second term is 0 by (D2).

(c) Immediate from (D1). (d) Take $c = 0$ in (c).

(e) If dependent, one column is a combination of the others; subtract that combination using (b), producing a zero column, then apply (d). ■

Property (b) is the practically important one: **elimination does not change the determinant**. So the way to compute a determinant is to run Gaussian elimination and multiply the pivots — $O(n^3)$, not $O(n!)$.

Theorem 22.3 (determinant of a triangular matrix).

If A is triangular, $\det A = \prod_i A_{ii}$.

Theorem 22.4 (computing determinants in practice).

If $PA = LU$ is the LU factorisation with k row swaps, then

$$\det A = (-1)^k \prod_i U_{ii}.$$

Proof. $\det L = 1$ (L is unit lower triangular), $\det U$ is the product of its diagonal by the previous theorem, $\det P = (-1)^k$ by property (a) applied k times, and $\det$ is multiplicative by the theorem below. ■

22.3. The formula, for completeness

Definition 22.5 (Leibniz formula).

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i, \sigma(i)},$$

where the sum runs over all $n!$ permutations of $\{1, \dots, n\}$ and $\text{sgn}(\sigma)$ is +1 for an even permutation, −1 for odd.

For $n = 2$ this is $ad - bc$. For $n = 3$ it is the six-term rule of Sarrus. For $n = 20$ it has 2.4×10^{18} terms and is entirely useless as an algorithm.

Cofactor expansion along a row or column,

$$\det A = \sum_j (-1)^{i+j} A_{ij} M_{ij},$$

where M_{ij} is the determinant of A with row i and column j deleted, is the recursive version. It costs $O(n!)$ too. It is useful for hand computation on small or sparse matrices — expand along a row with many zeros — and for proving things, and for nothing else.

22.4. The multiplicative property

Theorem 22.6 (determinant is multiplicative).

$$\det(AB) = \det(A) \det(B).$$

Proof. Read it as volume scaling. B scales volumes by $\det B$, then A scales by $\det A$; composing scales by the product. Signs compose the same way: two flips make no flip.

(A rigorous proof fixes B and checks that $A \mapsto \det(AB)/\det B$ satisfies the three axioms as a function of the columns of A ; uniqueness then forces it to equal $\det A$. The volume argument is the reason it is true.) ■

Corollary 22.7.

- (a) $\det(A^{-1}) = 1/\det(A)$.
- (b) $\det(A^T) = \det(A)$.
- (c) $\det(cA) = c^n \det(A)$ for $n \times n$ matrices.
- (d) Similar matrices have equal determinants: $\det(P^{-1}AP) = \det(A)$.

Proof. (a) $\det(A) \det(A^{-1}) = \det(I) = 1$.

(c) Scaling all n columns by c multiplies by c each time — this catches people out, since $\det(2A) = 2^n \det A$, not $2 \det A$.

(d) $\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \det(A)$ by (a) and commutativity of scalar multiplication. ■

Item (d) matters: similar matrices are **the same linear map written in different bases** (Chapter 23), and the determinant does not care which basis you chose. Volume scaling is a property of the map, not of its coordinates.

22.5. The big theorem

Theorem 22.8 (determinant and invertibility).

A is invertible $\Leftrightarrow \det A \neq 0$.

Proof. ($\Rightarrow$) If A^{-1} exists then $\det(A) \det(A^{-1}) = 1$, so $\det A \neq 0$.

($\Leftarrow$) Contrapositive. If A is not invertible, its columns are linearly dependent (Chapter 20), so $\det A = 0$ by property (e). ■

This adds a line to the invertible matrix theorem, and it is the entry people reach for first — for small matrices.

Trap: do not test for singularity with the determinant.

Numerically, $\det A \neq 0$ is nearly useless as a test.

The determinant scales like c^n : a perfectly well-conditioned 100×100 matrix with entries around 0.1 has determinant around 10^{-100} , which underflows to zero. Meanwhile $10^{-6}I$ in 100 dimensions has determinant 10^{-600} and is trivially invertible.

Conversely, a matrix can have a healthy-looking determinant and be catastrophically ill-conditioned.

The determinant measures **volume**, which is a product of all n singular values. Conditioning depends on the **ratio** of the largest to the smallest. Those are different questions. To test invertibility numerically, look at the smallest singular value (Chapter 25) or the condition number — **rcond**, not **det**.

22.6. The determinant in calculus and probability

Three places you will meet it outside linear algebra.

Change of variables (Chapter 18). $dx = |\det J| du$. The Jacobian determinant is the local volume scale factor of the coordinate change, which is exactly the interpretation above applied to the linearisation.

Probability density transformation. If $Y = g(X)$ with g invertible and differentiable, then

$$p_Y(y) = p_X(g^{-1}(y)) |\det J_{g^{-1}}(y)|.$$

Densities are “probability per unit volume”, so changing coordinates requires dividing by the volume change. This formula is the entire basis of **normalising flows** in machine learning, where the model is a chain of invertible maps and the training objective requires the log-determinant of each Jacobian — which is why such models are designed to have triangular or otherwise cheap Jacobians.

The multivariate normal. Its density carries a factor $1/\sqrt{\det \Sigma}$, which is the normalising constant making the total probability 1. $\sqrt{\det \Sigma}$ is the volume of the ellipsoid of one-standard-deviation spread (Chapter 29).

22.7. Cramer’s rule, and why not to use it

Theorem 22.9 (Cramer’s rule).

If A is invertible, the solution of $Ax = b$ has

$$x_i = \frac{\det A_i}{\det A},$$

where A_i is A with its i -th column replaced by b .

Proof. Let X_i be the identity matrix with its i -th column replaced by x . Then $AX_i = A_i$ — check column by column: the i -th column of AX_i is $Ax = b$, and every other column of AX_i is the corresponding column of A . Taking determinants, $\det(A) \det(X_i) = \det(A_i)$. But X_i is triangular-ish with x_i on the diagonal in position i and ones elsewhere, so $\det X_i = x_i$. ■

Beautiful, and never used for computation: it requires $n + 1$ determinants, so $O(n^4)$ against elimination’s $O(n^3)$, and it is numerically poor. Its value is theoretical — it shows the solution depends **smoothly** (indeed rationally) on the entries of A and b , which matters when you want to differentiate through a linear solve.

22.8. Exercises

22.1 Compute $\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ and $\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. For the second, explain the answer in terms of the columns.

22.2 Use row reduction (not cofactor expansion) to compute

$$\det \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 3 \end{pmatrix},$$

keeping track of any sign changes from row swaps.

22.3 Let A be 4×4 with $\det A = 5$. Find $\det(2A)$, $\det(A^T)$, $\det(A^{-1})$, $\det(A^3)$, and $\det(A^T A)$.

22.4 Prove that if A is orthogonal then $\det A = \pm 1$. Which sign corresponds to a rotation and which to a reflection?

22.5 Show that the area of the triangle with vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) is

$$\frac{1}{2} \left| \det \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} \right|.$$

Then explain how the **sign** of that determinant is used to decide the orientation of three points — the primitive behind convex hull algorithms.

22.6 Prove that if A is $n \times n$ with $A^2 = I$, then $\det A = \pm 1$. Give an example of each sign.

22.7 Compute the Jacobian determinant of the spherical coordinate map $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, $z = \rho \cos \varphi$, and confirm it equals $\rho^2 \sin \varphi$.

22.8 A matrix has entries drawn independently and uniformly from $[0, 1]$ and is 200×200 . Estimate the order of magnitude of its determinant, and use that to explain in one paragraph why `det(A) == 0` is a bad singularity test.

CHAPTER 23.

Eigenvalues and Eigenvectors

23.1. The question

A matrix moves vectors around: it rotates them, stretches them, shears them. Most vectors change direction.

Some do not. For those, the matrix acts as pure scaling — multiplication by a number. Those directions are the **eigenvectors**, and the numbers are the **eigenvalues**, and finding them is the closest thing linear algebra has to a central problem.

Definition 23.1 (eigenvalue and eigenvector).

For square A , a nonzero vector v is an **eigenvector** with **eigenvalue** λ if

$$Av = \lambda v.$$

The set of all eigenvectors for a given λ , together with 0 , is the **eigenspace** — a subspace.

We require $v \neq 0$ because $A \cdot 0 = \lambda \cdot 0$ for every λ , which would make the definition vacuous. Eigenvalues may be zero; eigenvectors may not.

The idea: why anyone cares.

Along an eigenvector, a matrix is **just a number**. All the complexity of the linear map collapses to scalar multiplication.

If you can find a whole basis of eigenvectors, then in **that** basis the matrix is diagonal — the coordinates do not interact at all, and every question about the matrix becomes n independent scalar questions.

That is why they matter. Eigen-decomposition is the search for the coordinate system in which a problem falls apart into independent pieces. Applications:

- A^k becomes λ^k : matrix powers, Markov chains, linear recurrences (Chapter 11).
- Stability: the system runs away iff some $|\lambda| > 1$ (discrete) or $\text{Re}(\lambda) > 0$ (continuous).
- Principal components: the eigenvectors of the covariance matrix are the directions of greatest variance.
- PageRank: the ranking vector is the dominant eigenvector of the link matrix.
- Vibration and resonance: eigenvalues are the natural frequencies.

23.2. Finding them

$Av = \lambda v$ rearranges to $(A - \lambda I)v = 0$, which asks for a nonzero null-space vector of $A - \lambda I$. By Chapter 22 that happens exactly when the matrix is singular:

Definition 23.2 (characteristic polynomial).

$$p_A(\lambda) = \det(A - \lambda I).$$

Its roots are exactly the eigenvalues of A . It is a degree- n polynomial in λ .

Eigen-decomposition by hand (small matrices only)

1. Form $\det(A - \lambda I)$ and expand: a degree- n polynomial.
2. Find its roots. These are the eigenvalues.
3. For each eigenvalue, solve $(A - \lambda I)v = 0$ by elimination.
The null space basis vectors are the eigenvectors.

Example 23.3 (a worked 2×2).

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10,$$

with roots $\lambda = 5$ and $\lambda = 2$.

For $\lambda = 5$: solve $\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}v = 0$, giving $v_1 = v_2$, so $v = (1, 1)^T$ (any nonzero multiple).

For $\lambda = 2$: solve $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}v = 0$, giving $2v_1 = -v_2$, so $v = (1, -2)^T$.

Sanity checks: the eigenvalues sum to $7 = \text{tr}(A)$ and multiply to $10 = \det A$. Always do this; it catches most arithmetic errors.

Theorem 23.4 (trace and determinant from eigenvalues).

With eigenvalues $\lambda_1, \dots, \lambda_n$ counted with multiplicity,

$$\text{tr}(A) = \sum_i \lambda_i, \quad \det(A) = \prod_i \lambda_i.$$

Proof. $p_A(\lambda) = \det(A - \lambda I) = \prod_i (\lambda_i - \lambda)$ by the factor theorem (Chapter 2), since the λ_i are exactly its roots and the leading coefficient works out.

Setting $\lambda = 0$ gives $\det A = \prod_i \lambda_i$.

Comparing coefficients of λ^{n-1} on both sides gives $\text{tr}(A) = \sum \lambda_i$, by Vieta's relations (Chapter 2) applied to the characteristic polynomial. This is the promise made in Chapter 2, now delivered. ■

Trap: the characteristic polynomial is a terrible algorithm.

For anything larger than 3×3 , do not compute eigenvalues by finding roots of p_A .

Polynomial root-finding is severely ill-conditioned: tiny changes in the coefficients can move the roots enormously (**Wilkinson's polynomial** is the standard horror story). And going from A to its characteristic polynomial throws away structure the algorithm could have used.

Real algorithms — the QR algorithm, and Krylov methods like Lanczos and Arnoldi for large sparse problems — work on the matrix directly with orthogonal transformations, which cannot amplify error (Chapter 20). The characteristic polynomial is a definition, not a method.

Amusingly, the relationship runs the other way in practice: to find roots of a polynomial, numerical libraries build its **companion matrix** and compute eigenvalues.

23.3. Diagonalisation

Definition 23.5 (diagonalisable).

A is **diagonalisable** if there is an invertible P and diagonal D with

$$A = PDP^{-1}.$$

Theorem 23.6.

A is diagonalisable iff it has n linearly independent eigenvectors. In that case the columns of P are those eigenvectors and the diagonal entries of D are the corresponding eigenvalues.

Proof. Suppose $v_1, \dots, v_n$ are independent eigenvectors with eigenvalues λ_i . Let $P = [v_1 \mid \dots \mid v_n]$. Then

$$AP = [Av_1 \mid \dots \mid Av_n] = [\lambda_1 v_1 \mid \dots \mid \lambda_n v_n] = PD.$$

P is invertible since its columns are independent, so $A = PDP^{-1}$.

Conversely, if $A = PDP^{-1}$ then $AP = PD$, and reading that column by column says each column of P is an eigenvector. P invertible makes them independent. ■

The idea: diagonalisation is a change of coordinates.

Read $A = PDP^{-1}$ right to left, as a pipeline applied to a vector x :

1. $P^{-1}x$: rewrite x in the eigenvector basis. “What are your coordinates in the natural coordinate system for this problem?”
2. D : scale each coordinate independently. Trivial.
3. P : translate back to the original coordinates.

So A is a diagonal matrix; you were just looking at it in the wrong basis. Similar matrices A and $P^{-1}AP$ are the same map described by two different observers.

Once you see this, the computational payoff is obvious:

$$A^k = (PDP^{-1})^k = PD^kP^{-1},$$

because all the interior $P^{-1}P$ pairs cancel. And D^k is just each diagonal entry raised to the k . An $O(n^3k)$ computation becomes $O(n^3)$ plus n scalar powers.

Example 23.7 (Fibonacci, closed form, again).

Chapter 11 solved $F_n = F_{n-1} + F_{n-2}$ with characteristic roots. Here is the same thing as diagonalisation.

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$\text{So } \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$\det(M - \lambda I) = \lambda^2 - \lambda - 1$, whose roots are φ and ψ — the **same polynomial** as Chapter 11’s characteristic equation, and now you can see why: a linear recurrence is a matrix power, and its characteristic equation is the matrix’s characteristic polynomial.

Diagonalising and extracting the second component gives Binet's formula. And since M^n can be computed by repeated squaring, F_n is available in $O(\log n)$ big-integer multiplications.

Theorem 23.8 (distinct eigenvalues give independent eigenvectors).

Eigenvectors corresponding to distinct eigenvalues are linearly independent. In particular, a matrix with n distinct eigenvalues is diagonalisable.

Proof. Induction on the number of eigenvectors. Suppose $c_1 v_1 + \dots + c_k v_k = 0$ with distinct λ_i , and that any $k-1$ of them are independent.

Apply A : $\sum c_i \lambda_i v_i = 0$. Also multiply the original by λ_k : $\sum c_i \lambda_k v_i = 0$. Subtract:

$$\sum_{i=1}^{k-1} c_i (\lambda_i - \lambda_k) v_i = 0.$$

By the inductive hypothesis these $k-1$ vectors are independent, so every $c_i (\lambda_i - \lambda_k) = 0$; since the eigenvalues are distinct, $c_i = 0$ for $i < k$. Then $c_k v_k = 0$ forces $c_k = 0$ too. ■

23.3.1. When diagonalisation fails

Example 23.9 (a defective matrix).

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

$\det(A - \lambda I) = (1 - \lambda)^2$, so $\lambda = 1$ with **algebraic multiplicity** 2. But solving $(A - I)v = 0$ gives $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v = 0$, i.e. $v_2 = 0$: the eigenspace is only one-dimensional. **Geometric multiplicity** 1.

There is no basis of eigenvectors, so A is not diagonalisable. Such a matrix is called **defective**.

Geometrically, A is a shear. A shear fixes one direction and tilts everything else; it has no second invariant direction to find.

The general repair is the **Jordan normal form**: every matrix is similar to a block-diagonal matrix with eigenvalues on the diagonal and some 1s on the superdiagonal. It is theoretically complete and numerically useless — the Jordan structure is destroyed by any perturbation, since an arbitrarily small change makes the eigenvalues distinct. In practice, use the SVD (Chapter 25), which always exists and is stable.

23.4. The symmetric case, which is the good case

Theorem 23.10 (spectral theorem).

Let A be real and symmetric ($A^T = A$). Then:

- (i) All eigenvalues of A are real.
- (ii) Eigenvectors for distinct eigenvalues are **orthogonal**.
- (iii) A is always diagonalisable, by an **orthogonal** matrix:

$$A = Q \Lambda Q^T, \quad Q^T Q = I.$$

Proof. (i) Let $Av = \lambda v$ with v possibly complex, and let $\bar{v}^T$ denote the conjugate transpose. Then

$$\bar{v}^T Av = \lambda \bar{v}^T v = \lambda \|v\|^2.$$

Taking the conjugate transpose of the left side, and using $A^T = A$ with A real:

$$\overline{\bar{v}^T Av} = v^T A \bar{v} = \bar{\lambda} \|v\|^2.$$

But the left side is a 1×1 real-symmetric quantity equal to its own conjugate, so $\lambda \|v\|^2 = \bar{\lambda} \|v\|^2$. Since $\|v\|^2 > 0$, $\lambda = \bar{\lambda}$: real.

(ii) Let $Au = \lambda u$, $Av = \mu v$ with $\lambda \neq \mu$. Then

$$\lambda(u \cdot v) = (Au) \cdot v = u^T A^T v = u^T Av = u \cdot (Av) = \mu(u \cdot v),$$

using symmetry in the middle. So $(\lambda - \mu)(u \cdot v) = 0$, and since $\lambda \neq \mu$, $u \cdot v = 0$.

(iii) Requires an induction on dimension (restrict A to the orthogonal complement of one eigenvector and repeat); the content is that even repeated eigenvalues get a full orthogonal eigenspace in the symmetric case. ■

The idea: why symmetric matrices are the ones you actually meet.

The spectral theorem is one of the most useful theorems in applied mathematics, and it applies to almost everything you care about:

- **Covariance matrices** $\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T]$ are symmetric by construction. Their eigenvectors are the principal components.
- **Hessians** are symmetric by Clairaut (Chapter 18). Their eigenvalues are the curvatures.
- **Graph Laplacians** $L = D - A$ are symmetric for undirected graphs.
- **Gram matrices** $A^T A$ are always symmetric, whatever A is.
- Kernel matrices, adjacency matrices, distance matrices, inertia tensors.

In each case you get: real eigenvalues (no complex bookkeeping), an **orthonormal** eigenbasis (so $P^{-1} = P^T$ — inversion is free and numerically perfect), and the guarantee that diagonalisation never fails.

Definition 23.11 (positive definite).

A symmetric A is **positive definite** if $x^T Ax > 0$ for all $x \neq 0$, and **positive semidefinite** if $x^T Ax \geq 0$.

Theorem 23.12.

A symmetric matrix is positive definite iff all its eigenvalues are positive.

Proof. Write $x = Qy$ in the eigenbasis. Then

$$x^T Ax = y^T Q^T Q \Lambda Q^T Q y = y^T \Lambda y = \sum_i \lambda_i y_i^2.$$

This is positive for all $y \neq 0$ iff every $\lambda_i > 0$: sufficiency is clear, and if some $\lambda_j \leq 0$ take $y = e_j$. ■

That computation is worth keeping. It says a quadratic form, in the eigenbasis, is just a weighted sum of squares — so the geometry of $x^T Ax = 1$ is an ellipsoid with axes along the eigenvectors and semi-axis lengths $1/\sqrt{\lambda_i}$. Chapter 34 uses exactly this picture for optimisation.

23.5. Powers, stability, and the dominant eigenvalue

Definition 23.13 (spectral radius).

$$\rho(A) = \max_i |\lambda_i|.$$

Theorem 23.14.

$A^k \rightarrow 0$ as $k \rightarrow \infty$ if and only if $\rho(A) < 1$.

Proof. For diagonalisable A , $A^k = P\Lambda^k P^{-1}$ and $\Lambda^k = \text{diag}(\lambda_i^k)$, which tends to 0 iff every $|\lambda_i| < 1$. (The defective case needs a Jordan-form argument but the conclusion is the same.) ■

Example 23.15 (Markov chains and PageRank).

A **stochastic matrix** has nonnegative entries and columns summing to 1: it transports probability. Then:

- $\lambda = 1$ is always an eigenvalue (the all-ones vector is a left eigenvector, since columns sum to 1).
- All eigenvalues satisfy $|\lambda| \leq 1$ (probability cannot grow).
- The eigenvector for $\lambda = 1$ is the **stationary distribution** π with $P\pi = \pi$.

Perron–Frobenius guarantees that if the chain is irreducible and aperiodic, $\lambda = 1$ is simple and every other eigenvalue is strictly smaller in modulus. Then from any start, $P^k x \rightarrow \pi$, and the **rate** of convergence is governed by $|\lambda_2|$: the **spectral gap** $1 - |\lambda_2|$ is the mixing rate.

PageRank is exactly this: a random surfer's transition matrix, damped to guarantee irreducibility, and the ranking is the dominant eigenvector. It is computed by **power iteration**, which is the crudest possible eigenvalue algorithm and here the right one.

Power iteration

$x := \text{random nonzero vector}$

repeat:

$x := A x$

$x := x / \|x\|$ # renormalise to avoid overflow

converges to the dominant eigenvector; the Rayleigh quotient

$x^T A x / (x^T x)$ converges to the dominant eigenvalue.

Proof. Expand $x_0 = \sum c_i v_i$ in the eigenbasis. Then

$$A^k x_0 = \sum_i c_i \lambda_i^k v_i = \lambda_1^k \left(c_1 v_1 + \sum_{i \geq 2} c_i \left(\frac{\lambda_i}{\lambda_1} \right)^k v_i \right).$$

If $|\lambda_1| > |\lambda_i|$ for all $i \geq 2$, every ratio tends to zero and the direction converges to v_1 . The error decays like $(|\lambda_2|/|\lambda_1|)^k$ — so a large spectral gap means fast convergence, and a small one means slow.

The renormalisation exists only to stop λ_1^k from overflowing; it does not affect the direction. ■

23.6. Exercises

23.1 Find the eigenvalues and eigenvectors of

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and verify that the eigenvectors are orthogonal, as the spectral theorem promises.

23.2 Find the eigenvalues of

$$\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}.$$

They are complex — interpret the matrix geometrically and explain what the eigenvalues say about it. (Compare Chapter 6.)

23.3 Show that

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

is not diagonalisable, by computing the algebraic and geometric multiplicities of its eigenvalue.

23.4 Prove that A and A^T have the same eigenvalues. (Consider the characteristic polynomial and use $\det(M^T) = \det(M)$.) Do they have the same eigenvectors?

23.5 Let A be invertible with eigenvalue λ and eigenvector v . Show that A^{-1} has eigenvalue $1/\lambda$ with the same eigenvector, and that A^2 has eigenvalue λ^2 .

23.6 Diagonalise

$$A = \begin{pmatrix} 5 & -2 \\ -2 & 5 \end{pmatrix}$$

and use the result to compute A^{10} without multiplying matrices ten times.

23.7 Determine whether

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

are positive definite, by computing eigenvalues. Then check your answers by testing $x^T A x$ on a well-chosen x .

23.8 A Markov chain has transition matrix

$$P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}$$

(columns sum to 1). Find its stationary distribution as the eigenvector for $\lambda = 1$, find λ_2 , and state how many steps it takes for the deviation from stationarity to shrink by a factor of 100.

CHAPTER 24.

Orthogonality, Projection, and Least Squares

24.1. Why orthogonality is the good case

Chapter 19 introduced the dot product and noticed that $u \cdot v = 0$ — being perpendicular — is a special condition. This chapter is about why it is **the** special condition.

The idea: orthogonal means independent in the strongest sense.

Linear independence says no vector is a combination of the others. Orthogonality says something much stronger: **the vectors do not interact at all.**

Concretely, with an orthonormal basis:

- Finding coordinates requires no solving. It is one dot product per coordinate, in $O(n)$, instead of an $O(n^3)$ elimination.
- Lengths decompose by Pythagoras: $\|\sum c_i q_i\|^2 = \sum c_i^2$.
- Changing basis is done by a matrix whose inverse is its transpose. No inversion, no numerical loss.
- Errors in one coordinate do not leak into the others.

Every good numerical algorithm is built from orthogonal transformations, and this is why.

Definition 24.1 (orthogonal and orthonormal sets).

A set $\{v_1, \dots, v_k\}$ is **orthogonal** if $v_i \cdot v_j = 0$ for $i \neq j$, and **orthonormal** if additionally $\|v_i\| = 1$ for all i .

Proposition 24.2.

An orthogonal set of nonzero vectors is linearly independent.

Proof. Suppose $\sum c_i v_i = 0$. Take the dot product with v_j :

$$0 = v_j \cdot \sum_i c_i v_i = \sum_i c_i (v_j \cdot v_i) = c_j \|v_j\|^2,$$

since all cross terms vanish. As $v_j \neq 0$, $c_j = 0$. This holds for every j . ■

Theorem 24.3 (coordinates in an orthonormal basis are free).

If $\{q_1, \dots, q_n\}$ is an orthonormal basis of V and $v \in V$, then

$$v = \sum_{i=1}^n (v \cdot q_i) q_i.$$

Proof. Write $v = \sum c_i q_i$ (possible, since it is a basis). Dot with q_j : all terms vanish except $i = j$, leaving $v \cdot q_j = c_j \|q_j\|^2 = c_j$. ■

That formula is a Fourier series in miniature. The coefficients of a function in the basis of sines and cosines are computed by exactly this recipe, with the integral $\int fg$ playing the role of the dot product.

24.2. Orthogonal complements

Definition 24.4.

For a subspace $W \subseteq \mathbb{R}^n$,

$$W^\perp = \{v \in \mathbb{R}^n : v \cdot w = 0 \text{ for all } w \in W\}.$$

Theorem 24.5 (orthogonal decomposition).

For any subspace W of $\mathbb{R}^n$, every $v \in \mathbb{R}^n$ decomposes **uniquely** as

$$v = w + w^\perp, \quad w \in W, w^\perp \in W^\perp,$$

and $\dim W + \dim W^\perp = n$.

Theorem 24.6 (the fundamental theorem of linear algebra).

For any $A \in \mathbb{R}^{m \times n}$:

$$\text{Null}(A) = \text{Row}(A)^\perp \quad \text{in } \mathbb{R}^n, \quad \text{Null}(A^T) = \text{Col}(A)^\perp \quad \text{in } \mathbb{R}^m.$$

Proof. $Ax = 0$ says every row of A dots to zero with x — which is exactly the statement $x \in \text{Row}(A)^\perp$. Applying this to A^T gives the second. ■

The idea: the four subspaces, pictured.

Chapter 20 named four subspaces; now they lock together.

$\mathbb{R}^n$ (the input space) splits into two orthogonal pieces: the row space (dimension r) and the null space (dimension $n - r$).

$\mathbb{R}^m$ (the output space) splits into the column space (dimension r) and the left null space (dimension $m - r$).

And A maps the row space **bijectionally** onto the column space — both have dimension r — while crushing the null space to zero.

So every matrix, however complicated it looks, does one thing: it throws away a subspace, and acts invertibly on what is left. Rank–nullity is the dimension count of that picture; the SVD (Chapter 25) will give explicit orthonormal bases for all four pieces at once.

24.3. Projection onto a subspace

Theorem 24.7 (projection formula).

Let $W = \text{Col}(A)$ with the columns of A linearly independent. The orthogonal projection of b onto W is

$$\hat{b} = A(A^T A)^{-1} A^T b,$$

and $P = A(A^T A)^{-1} A^T$ is the **projection matrix**.

Proof. $\hat{b} \in W$ means $\hat{b} = Ax$ for some x . The defining property of orthogonal projection is that the residual $b - Ax$ is orthogonal to W , i.e. orthogonal to every column of A :

$$A^T(b - Ax) = 0 \Rightarrow A^T Ax = A^T b.$$

Independence of the columns makes $A^T A$ invertible (see below), so $x = (A^T A)^{-1} A^T b$ and $\hat{b} = Ax$. ■

Lemma 24.8.

If the columns of A are independent, $A^T A$ is invertible.

Proof. Suppose $A^T Ax = 0$. Then $x^T A^T Ax = 0$, i.e. $\|Ax\|^2 = 0$, so $Ax = 0$. Independence of the columns gives $x = 0$. So $A^T A$ has trivial null space and, being square, is invertible. ■

Proposition 24.9 (properties of a projection matrix).

$P^2 = P$ (idempotent) and $P^T = P$ (symmetric). Its eigenvalues are 0 and 1: it acts as the identity on W and as zero on $W^\perp$.

Proof. Symmetry: $P^T = (A(A^T A)^{-1} A^T)^T = A((A^T A)^{-1})^T A^T = P$, since $A^T A$ is symmetric and so is its inverse.

Idempotence: $P^2 = A(A^T A)^{-1} \underbrace{A^T A(A^T A)^{-1}}_I A^T = P$. Which is what projection should do: projecting twice is projecting once. ■

24.4. Least squares

Now the payoff, and the most-used application of linear algebra there is.

Definition 24.10 (the least squares problem).

Given $A \in \mathbb{R}^{m \times n}$ with $m > n$ (more equations than unknowns) and $b \in \mathbb{R}^m$, the system $Ax = b$ generally has **no** solution. Instead find

$$\hat{x} = \arg \min_x \|Ax - b\|^2.$$

Theorem 24.11 (normal equations).

$\hat{x}$ minimises $\|Ax - b\|$ iff

$$A^T A \hat{x} = A^T b.$$

If A has independent columns this has the unique solution $\hat{x} = (A^T A)^{-1} A^T b$.

Proof. Geometric proof. Ax ranges over $\text{Col}(A)$ as x varies. So we are looking for the point of $\text{Col}(A)$ closest to b — which is exactly the orthogonal projection. The residual must be orthogonal to the column space: $A^T(b - A\hat{x}) = 0$, which rearranges to the normal equations.

Calculus proof. Let $f(x) = \|Ax - b\|^2 = x^T A^T Ax - 2b^T Ax + b^T b$. By Chapter 26's rules, $\nabla f = 2A^T Ax - 2A^T b$. Setting it to zero gives the same equations, and the Hessian $2A^T A$ is positive semidefinite, so the critical point is a minimum.

The two proofs agreeing is not a coincidence: “the gradient vanishes” and “the residual is orthogonal” are the same statement. ■

The idea: why *least squares* and not least absolute error.

Minimising $\sum |r_i|$ is a perfectly reasonable thing to want, and it is more robust to outliers. Why is squared error the default?

It is the only choice that makes the problem linear. The derivative of a square is linear, so setting the gradient to zero gives a **linear system** — solvable exactly, in closed form, in $O(mn^2)$. The absolute value has no derivative at zero and gives a linear program instead: solvable, but iteratively and more slowly.

It is geometry. Squared error is squared Euclidean distance, so the solution is an orthogonal projection, and the entire Pythagorean apparatus applies.

It is maximum likelihood under Gaussian noise. If $b = Ax + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, then maximising the likelihood is exactly minimising $\|Ax - b\|^2$ (Chapter 31). So “least squares” encodes an assumption — that errors are Gaussian — and when that assumption is wrong (heavy tails, outliers), least squares is genuinely the wrong estimator and you should know it.

Example 24.12 (fitting a line).

Fit $y = mx + c$ to points (x_i, y_i) , $i = 1, \dots, N$. Set

$$A = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} m \\ c \end{pmatrix}, \quad b = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}.$$

Then

$$A^T A = \begin{pmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{pmatrix}, \quad A^T b = \begin{pmatrix} \sum x_i y_i \\ \sum y_i \end{pmatrix},$$

and solving the 2×2 system gives the familiar formulas

$$m = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}, \quad c = \bar{y} - m\bar{x}.$$

Those formulas are usually presented as something to memorise. They are just Cramer’s rule on the normal equations. And the same setup with more columns fits a polynomial, a multiple regression, or any linear-in-parameters model — the framework does not change at all.

Trap: do not form $A^T A$.

The normal equations are the right **theory** and the wrong **algorithm**.

Forming $A^T A$ squares the condition number: $\kappa(A^T A) = \kappa(A)^2$. If A has condition number 10^8 — unremarkable for a polynomial fit — then $A^T A$ has 10^{16} , and in double precision you have no digits left.

The correct method is QR factorisation. Write $A = QR$ with Q orthogonal and R upper triangular. Then

$$\|Ax - b\| = \|QRx - b\| = \|Rx - Q^T b\|$$

(orthogonal matrices preserve norms), and the minimiser solves the triangular system $Rx = Q^T b$ by back-substitution. Condition number untouched. This is what `numpy.linalg.lstsq` and every serious library actually do.

For rank-deficient or nearly rank-deficient A , use the SVD (Chapter 25), which handles the degenerate case gracefully by truncating tiny singular values.

24.5. Gram–Schmidt and QR

Theorem 24.13 (Gram–Schmidt).

Given independent $v_1, \dots, v_n$, define

$$\begin{aligned} u_1 &= v_1, & q_1 &= u_1 / \|u_1\| \\ u_k &= v_k - \sum_{j < k} (v_k \cdot q_j) q_j, & q_k &= u_k / \|u_k\|. \end{aligned}$$

Then $\{q_1, \dots, q_n\}$ is orthonormal and spans the same space, with $\text{span}\{q_1, \dots, q_k\} = \text{span}\{v_1, \dots, v_k\}$ for every k .

Proof. By induction. Suppose $q_1, \dots, q_{k-1}$ are orthonormal with the right spans. Then for $i < k$,

$$u_k \cdot q_i = v_k \cdot q_i - \sum_{j < k} (v_k \cdot q_j) (q_j \cdot q_i) = v_k \cdot q_i - v_k \cdot q_i = 0,$$

using orthonormality to kill all but the $j = i$ term. So u_k is orthogonal to all previous q 's. It is nonzero because v_k is not in the span of the earlier vectors (independence), and normalising gives q_k . ■

The idea in one line: **subtract off everything that is already accounted for, keep what is new**. The sum is exactly the projection onto the span of what came before.

Theorem 24.14 (QR factorisation).

Every $A \in \mathbb{R}^{m \times n}$ with independent columns factors as $A = QR$ with $Q \in \mathbb{R}^{m \times n}$ having orthonormal columns and $R \in \mathbb{R}^{n \times n}$ upper triangular with positive diagonal.

Proof. Run Gram–Schmidt on the columns of A to get the columns of Q . Since each v_k is a combination of $q_1, \dots, q_k$ only — never of later ones — the coefficient matrix R with $R_{jk} = v_k \cdot q_j$ is upper triangular, and $A = QR$ by construction. ■

Trap: classical Gram–Schmidt is numerically unstable.

Written exactly as stated, Gram–Schmidt loses orthogonality badly in floating point: the computed q_k drift away from perpendicular as k grows, and for ill-conditioned input they can end up almost parallel.

Modified Gram–Schmidt — subtract each projection immediately and update the running vector, rather than computing all the projections against the original v_k — is mathematically identical and substantially more stable.

Better still are **Householder reflections**, which build Q as a product of reflections rather than by orthogonalising. Householder QR is backward-stable and is what libraries implement. Gram-Schmidt is how you **understand** QR; Householder is how you **compute** it.

24.6. Function spaces and Fourier, briefly

Everything above needed only an **inner product** — a way to multiply two vectors into a scalar with the right properties. So it all works in any space that has one, including spaces of functions.

Definition 24.15.

On $C[-\pi, \pi]$, define

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x) \, dx.$$

Proposition 24.16.

The functions $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ are **orthogonal** under this inner product.

Proof. Each integral $\int_{-\pi}^{\pi} \cos(mx) \cos(nx) \, dx$ with $m \neq n$ is evaluated by the product-to-sum identity (Chapter 5), giving integrals of $\cos((m \pm n)x)$ over a whole number of periods, which vanish. Similarly for the other combinations. ■

So the Fourier coefficients

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) \, dx$$

are exactly the coordinate formula of this chapter, in the basis of sinusoids. Truncating the Fourier series to N terms is **orthogonal projection onto the span of the first N basis functions**, which is why the truncation is the best possible N -term approximation in the mean-square sense. That is the whole conceptual content of Fourier analysis. The rest is computation, and the FFT (Chapter 6) is how you do it fast.

24.7. Exercises

- 24.1** Apply Gram-Schmidt to $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$ and verify the result is orthonormal.
- 24.2** Find the orthogonal projection of $b = (1, 2, 3)$ onto the plane spanned by $(1, 0, 1)$ and $(0, 1, 1)$, and verify that the residual is orthogonal to both spanning vectors.
- 24.3** Fit a least-squares line to the points $(1, 1), (2, 3), (3, 4), (4, 6)$ by setting up and solving the normal equations by hand.
- 24.4** Prove that if P is a projection matrix then $I - P$ is also a projection matrix, and identify which subspace it projects onto.
- 24.5** Let A be $m \times n$ with independent columns. Prove that $\text{Null}(A^T A) = \text{Null}(A)$, and deduce that $\text{rank}(A^T A) = \text{rank}(A)$ for **any** A , independent columns or not.

24.6 Show that for an orthogonal matrix Q , $\kappa(Q) = 1$ in the 2-norm. Then explain in one sentence why this is the reason QR-based least squares beats the normal equations.

24.7 Find the QR factorisation of

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

using Gram–Schmidt, and use it to solve the least-squares problem $Ax \approx (1, 2, 3)^T$.

24.8 Verify by direct integration that $\int_{-\pi}^{\pi} \sin(2x) \cos(3x) \, dx = 0$, and state the general principle it illustrates.

CHAPTER 25.

The Singular Value Decomposition

25.1. The best theorem in linear algebra

Eigen-decomposition is powerful and fragile. It needs a square matrix, it can fail to exist (defective matrices, Chapter 23), it can produce complex eigenvalues for a real matrix, and the eigenvector basis is generally not orthogonal so the change of basis is numerically dangerous.

The SVD has none of those problems. It exists for **every** matrix, of any shape, always, with real nonnegative singular values and **orthonormal** bases on both sides. If you learn one factorisation, learn this one.

Theorem 25.1 (singular value decomposition).

Every $A \in \mathbb{R}^{m \times n}$ factors as

$$A = U\Sigma V^T$$

where $U \in \mathbb{R}^{m \times m}$ is orthogonal, $V \in \mathbb{R}^{n \times n}$ is orthogonal, and $\Sigma \in \mathbb{R}^{m \times n}$ is “diagonal” with nonnegative entries

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots$$

down its main diagonal, where $r = \text{rank}(A)$.

The σ_i are the **singular values**; the columns of U are the **left singular vectors** and those of V the **right singular vectors**.

The idea: every matrix is a rotation, a scaling, and a rotation.

Read $A = U\Sigma V^T$ right to left as a pipeline:

1. V^T : an orthogonal map — a rotation (possibly with a reflection). No stretching, no distortion.
2. Σ : scale each coordinate axis independently by σ_i . Some axes may be scaled by zero, which is how dimensions get destroyed.
3. U : another rotation, into the output space.

That is **every linear map that has ever existed**. Rotate, stretch along axes, rotate. Nothing else can happen.

The geometric statement: **A maps the unit sphere to an ellipsoid**. The right singular vectors v_i are the directions on the sphere that map to the ellipsoid’s axes; the left singular vectors u_i are those axes’ directions; and the σ_i are the semi-axis lengths.

Once you can picture that, the SVD stops being a formula and becomes obvious.

Theorem 25.2 (existence, via the symmetric case).

The SVD exists for every real matrix.

Proof. $A^T A$ is $n \times n$, symmetric, and positive semidefinite: $x^T A^T A x = \|Ax\|^2 \geq 0$. By the spectral theorem (Chapter 23) it has an orthonormal eigenbasis $v_1, \dots, v_n$ with real eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$.

Define $\sigma_i = \sqrt{\lambda_i}$, and for the r indices with $\sigma_i > 0$ set

$$u_i = \frac{Av_i}{\sigma_i}.$$

These u_i are orthonormal:

$$u_i \cdot u_j = \frac{v_i^T A^T A v_j}{\sigma_i \sigma_j} = \frac{\lambda_j (v_i \cdot v_j)}{\sigma_i \sigma_j},$$

which is 0 for $i \neq j$ and $\lambda_i/\sigma_i^2 = 1$ for $i = j$. Extend $u_1, \dots, u_r$ to an orthonormal basis of $\mathbb{R}^m$.

Finally $Av_i = \sigma_i u_i$ for $i \leq r$, and $Av_i = 0$ for $i > r$ (since $\|Av_i\|^2 = \lambda_i = 0$). Stacking these column relations is exactly $AV = U\Sigma$, i.e. $A = U\Sigma V^T$. ■

Proposition 25.3 (relations to eigen-decompositions).

$$A^T A = V\Sigma^T \Sigma V^T, \quad AA^T = U\Sigma \Sigma^T U^T.$$

So the right singular vectors are eigenvectors of $A^T A$, the left ones are eigenvectors of AA^T , and the nonzero eigenvalues of both are σ_i^2 .

Trap: computing the SVD from $A^T A$ is a bad idea.

The proof above is a construction, and like the characteristic polynomial in Chapter 23 it is a proof rather than an algorithm.

Forming $A^T A$ squares the condition number, so singular values smaller than $\sqrt{\varepsilon_{\text{mach}}} \sigma_1 \approx 10^{-8} \sigma_1$ are lost entirely. Real algorithms (Golub–Kahan bidiagonalisation followed by an implicit QR sweep) work on A directly and resolve singular values down to $\varepsilon_{\text{mach}} \sigma_1$.

Use the library. But know that the library is doing something cleverer than the textbook proof.

25.2. What the SVD tells you

Theorem 25.4 (the SVD reveals everything).

With $A = U\Sigma V^T$ and r nonzero singular values:

- (a) $\text{rank}(A) = r$.
- (b) $u_1, \dots, u_r$ is an orthonormal basis of $\text{Col}(A)$.
- (c) $u_{r+1}, \dots, u_m$ is an orthonormal basis of $\text{Null}(A^T)$.
- (d) $v_1, \dots, v_r$ is an orthonormal basis of $\text{Row}(A)$.
- (e) $v_{r+1}, \dots, v_n$ is an orthonormal basis of $\text{Null}(A)$.
- (f) $\|A\|_2 = \sigma_1$ (the largest factor by which A can stretch a vector).
- (g) $\|A\|_F = \sqrt{\sum_i \sigma_i^2}$ (the Frobenius norm).
- (h) For square invertible A , $\kappa_2(A) = \sigma_1/\sigma_n$.

So the four fundamental subspaces of Chapter 24, which needed elimination and careful book-keeping, all fall out of one factorisation with **orthonormal** bases attached.

The idea: the condition number, finally explained.

$\kappa = \sigma_1/\sigma_n$ is the ratio of the longest to the shortest axis of the ellipsoid A maps the unit sphere to.

If that ratio is 1, the sphere maps to a sphere — A is a scaled orthogonal matrix, perfectly conditioned, no direction is favoured.

If the ratio is 10^{12} , the sphere maps to a needle. Going forwards, everything gets squashed into the long direction; going backwards, the tiny direction has to be blown up by 10^{12} , and so does any error in it.

That is the entire content of ill-conditioning, and it is a **geometric** statement about the matrix, not a defect of any algorithm.

25.3. Low-rank approximation

Here is the result that makes the SVD indispensable in practice.

Definition 25.5 (truncated SVD).

Writing the SVD as a sum of rank-one pieces,

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T,$$

the **rank- k truncation** is

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T.$$

Theorem 25.6 (Eckart–Young–Mirsky).

A_k is the **best** rank- k approximation to A , in both the spectral and Frobenius norms:

$$\min_{\text{rank}(B) \leq k} \|A - B\|_2 = \|A - A_k\|_2 = \sigma_{k+1},$$

$$\min_{\text{rank}(B) \leq k} \|A - B\|_F = \|A - A_k\|_F = \sqrt{\sum_{i>k} \sigma_i^2}.$$

The idea: why this theorem is remarkable.

There are infinitely many rank- k matrices. The theorem says the best one is found by a **completely mechanical** procedure — compute the SVD, keep the top k terms, discard the rest — with no search and no optimisation.

Optimisation problems almost never have closed-form solutions. This one does, and the residual error is exactly σ_{k+1} , so you know in advance how good a rank- k approximation can possibly be.

Practically: plot the singular values. If they decay fast, your matrix is **approximately** low rank and can be compressed with little loss. If they decay slowly, it cannot, and no method will do better — Eckart–Young says so.

Example 25.7 (what this is used for).

Image compression. A grayscale image is a matrix. Keeping the top k singular values stores $k(m+n+1)$ numbers instead of mn . Natural images have rapidly decaying spectra, so $k=50$ on a 512×512 image is a $10 \times$ compression and often visually indistinguishable. (Actual codecs

use DCT, which is cheaper and does not require per-image factorisation — but the principle is identical.)

Principal component analysis. Centre your data matrix X (subtract the column means). Then $X^T X / (n - 1)$ is the covariance matrix, and the right singular vectors of X are its eigenvectors — the **principal components**. Projecting onto the first k gives the best k -dimensional linear summary of the data, by exactly the Eckart–Young criterion. And $\sigma_i^2 / \sum_j \sigma_j^2$ is the fraction of variance explained by component i .

Latent semantic analysis / recommender systems. A user–item matrix is huge, sparse, and approximately low rank because preferences are driven by a small number of latent factors. Truncating the SVD both compresses and **fills in** the missing entries, which is the recommendation.

Noise removal. Signal tends to live in the large singular values, noise is spread across all of them. Truncating keeps the signal and discards a proportional share of the noise.

LoRA. Fine-tuning a large model by learning $W + BA$ with B, A thin — a low-rank update. The bet is that the weight **change** needed for a new task is approximately low rank, so a few million parameters suffice where billions would otherwise be needed. Whether that bet holds is an empirical question about the spectrum of the update.

25.4. The pseudoinverse

Definition 25.8 (Moore–Penrose pseudoinverse).

For $A = U\Sigma V^T$, define

$$A^+ = V\Sigma^+ U^T,$$

where Σ^+ is formed by transposing Σ and replacing each nonzero σ_i with $1/\sigma_i$ (leaving the zeros alone).

Theorem 25.9.

$\hat{x} = A^+ b$ is the least-squares solution of $Ax = b$ of **minimum norm**. When A is invertible, $A^+ = A^{-1}$; when A has independent columns, $A^+ = (A^T A)^{-1} A^T$.

The idea: the pseudoinverse does the only sensible thing.

A general $Ax = b$ has two possible pathologies: no exact solution (inconsistent) and too many (a nontrivial null space). A^+ handles both.

On the directions A preserves ($\sigma_i > 0$), it inverts: divide by σ_i .

On the directions A destroyed ($\sigma_i = 0$), there is no information to invert, so it returns zero rather than inventing something. That choice is what makes the answer minimum-norm.

Note it does **not** try to invert tiny singular values into enormous ones — or rather, it does, which is why in practice you use a **truncated** pseudoinverse, zeroing every σ_i below a tolerance. That is exactly what `numpy.linalg.pinv(A, rcond=...)` does, and the `rcond` parameter is the line between “small but real” and “numerical noise”. Choosing it is a judgement call, and it is the same judgement as choosing k in a truncated SVD.

The truncated pseudoinverse is also a form of **regularisation**: it stabilises an ill-posed problem by refusing to amplify the directions in which the data carries no information. Ridge regression does the same thing more smoothly, by replacing $1/\sigma_i$ with $\sigma_i/(\sigma_i^2 + \lambda)$ — which is close to $1/\sigma_i$ for large σ_i and close to zero for small ones.

25.5. Matrix norms

Definition 25.10.

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sigma_1 \quad (\text{spectral / operator norm})$$

$$\|A\|_F = \sqrt{\sum_{ij} A_{ij}^2} = \sqrt{\sum_i \sigma_i^2} \quad (\text{Frobenius})$$

$$\|A\|_* = \sum_i \sigma_i \quad (\text{nuclear / trace norm})$$

The spectral norm is the worst-case amplification factor — the right norm for error analysis. The Frobenius norm treats the matrix as a long vector — the right norm when you want something differentiable and cheap. The nuclear norm is the convex relaxation of **rank**, in the same way that the ℓ_1 norm is the convex relaxation of “number of nonzeros” (Chapter 19); minimising it is how matrix completion problems are made tractable.

25.6. Exercises

25.1 Compute the SVD of

$$A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$$

by inspection. What are U , Σ , V ? (Note that singular values must be nonnegative — where does the sign go?)

25.2 Find the singular values of

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

by computing the eigenvalues of $A^T A$. Then compute $\kappa_2(A)$.

25.3 Let A be $m \times n$ with $m > n$ and rank n . Show that $A^+ = (A^T A)^{-1} A^T$ by substituting the SVD into both sides.

25.4 Prove that $\|A\|_F^2 = \text{tr}(A^T A)$, and then that this equals $\sum_i \sigma_i^2$. (Use the cyclic property of the trace and the orthogonality of U and V .)

25.5 A 1000×800 matrix has singular values that decay like $\sigma_i = 100/i$. How many singular values must you keep to capture 90% of the Frobenius norm squared? Roughly how much storage does that truncation save?

25.6 Show that for any A , the matrices $A^T A$ and $A A^T$ have the same nonzero eigenvalues. (Hint: if $A^T A v = \lambda v$ with $\lambda \neq 0$, consider $A v$.)

25.7 Explain, using the SVD, why $\kappa(A^T A) = \kappa(A)^2$. Then state in one sentence the practical consequence for solving least-squares problems.

- 25.8** A data matrix X (rows = samples, columns = features) has been centred. Show that the first principal component — the unit vector w maximising the variance of Xw — is the first right singular vector of X . (Maximise $\|Xw\|^2$ subject to $\|w\| = 1$, using the SVD.)

CHAPTER 26.

Matrix Calculus

26.1. The notation ML papers are written in

Chapter 18 established that the derivative of a multivariable function is a gradient or a Jacobian. In practice you rarely compute those entry by entry; you manipulate whole matrices, using a small set of identities.

This chapter is that toolkit. It is short, it is mostly a table, and having it is the difference between reading a paper and staring at one.

Trap: layout conventions differ, and they will confuse you.

There are two conventions for arranging the derivative of a vector with respect to a vector.

Numerator layout (Jacobian layout): $\partial y / \partial x$ has shape (size of y) $\times$ (size of x). The derivative of a scalar with respect to a column vector is a **row** vector.

Denominator layout (gradient layout): the transpose of that. The gradient of a scalar is a **column** vector.

Neither is standard. Papers switch between them, sometimes within a section.

This book uses denominator layout, so that ∇f is a column vector and shapes line up naturally in optimisation. When you read anything else, the reliable defence is not to memorise the convention but to **check the dimensions** — there is generally only one arrangement that type-checks.

26.2. The building blocks

Throughout, $x \in \mathbb{R}^n$ is a column vector, $a \in \mathbb{R}^n$ a constant vector, $A \in \mathbb{R}^{n \times n}$ a constant matrix.

Theorem 26.1 (the essential gradients).

$$\nabla_x (a^T x) = a$$

$$\nabla_x (x^T A x) = (A + A^T)x, \quad \text{and } = 2Ax \text{ if } A \text{ is symmetric}$$

$$\nabla_x \|x\|^2 = 2x$$

$$\nabla_x \|Ax - b\|^2 = 2A^T(Ax - b)$$

Proof. **First.** $a^T x = \sum_i a_i x_i$, so $\partial / \partial x_j = a_j$. Stacking gives a .

Second. $x^T A x = \sum_i \sum_j A_{ij} x_i x_j$. Differentiate with respect to x_k ; the terms involving x_k are those with $i = k$ (giving $\sum_j A_{kj} x_j$), those with $j = k$ (giving $\sum_i A_{ik} x_i$), and the diagonal term $A_{kk} x_k^2$ (whose derivative $2A_{kk} x_k$ is exactly what the first two contribute at $i = j = k$, so no double count). So

$$\frac{\partial}{\partial x_k} x^T A x = \sum_j A_{kj} x_j + \sum_i A_{ik} x_i = (Ax)_k + (A^T x)_k.$$

Third. Special case of the second with $A = I$.

Fourth. Expand $\|Ax - b\|^2 = x^T A^T Ax - 2b^T Ax + b^T b$. The first term uses the second identity with the symmetric matrix $A^T A$, giving $2A^T Ax$; the second uses the first identity with $a = A^T b$, giving $-2A^T b$. Adding: $2A^T(Ax - b)$. ■

The idea: the analogy that actually works.

Compare to single-variable calculus:

$$\begin{aligned} a^T x &\sim ax && \Rightarrow \text{derivative } a \\ x^T Ax &\sim ax^2 && \Rightarrow \text{derivative } 2ax \\ \|Ax - b\|^2 &\sim (ax - b)^2 && \Rightarrow \text{derivative } 2a(ax - b) \end{aligned}$$

The pattern is the same; the only new content is **where the transposes go**, and the transposes are forced by dimensional consistency. If a formula type-checks and reduces to the scalar case when $n = 1$, it is almost certainly right.

Theorem 26.2 (traces and determinants).

$$\begin{aligned} \frac{\partial}{\partial A} \text{tr}(AB) &= B^T \\ \frac{\partial}{\partial A} \text{tr}(A^T B) &= B \\ \frac{\partial}{\partial A} \text{tr}(A^T A) &= 2A \\ \frac{\partial}{\partial A} \log \det A &= (A^{-1})^T = (A^T)^{-1} \\ \frac{\partial}{\partial A} \det A &= \det(A)(A^{-1})^T \end{aligned}$$

Proof. First. $\text{tr}(AB) = \sum_i \sum_j A_{ij} B_{ji}$, so the derivative with respect to A_{ij} is $B_{ji} = (B^T)_{ij}$.

Last two. By cofactor expansion (Chapter 22) along row i , $\det A = \sum_j A_{ij} C_{ij}$ where the cofactors C_{ij} do not involve row i . Hence $\partial \det A / \partial A_{ij} = C_{ij}$. The adjugate formula $A^{-1} = \text{adj}(A) / \det A$ with $\text{adj}(A)_{ji} = C_{ij}$ then gives $\partial \det A / \partial A = \det(A)(A^{-1})^T$. Dividing by $\det A$ gives the log version. ■

The log det derivative is not exotic: it appears every time you fit a Gaussian (the log-likelihood contains $-\log \det \Sigma$), and in every normalising-flow model.

26.3. The chain rule in matrix form

The rule from Chapter 18, restated for practical use. If $y = f(u)$ and $u = g(x)$, then in denominator layout

$$\frac{\partial y}{\partial x} = \left(\frac{\partial u}{\partial x} \right)^T \frac{\partial y}{\partial u} \dots$$

— and this is exactly where the layout confusion bites hardest. The reliable method is different, and it is worth adopting permanently.

The idea: differentials: the technique that never gets the shapes wrong.

Instead of manipulating derivative arrays, manipulate **differentials** and read the derivative off at the end.

The rules are the ones you already know, applied to matrices:

$$\begin{aligned}d(A + B) &= dA + dB \\d(AB) &= (dA)B + A(dB) \quad (\text{order matters!}) \\d(A^T) &= (dA)^T \\d(A^{-1}) &= -A^{-1}(dA)A^{-1} \\d \operatorname{tr}(A) &= \operatorname{tr}(dA) \\d \log \det A &= \operatorname{tr}(A^{-1} dA)\end{aligned}$$

Then use the **identification rule**: for a scalar f ,

$$df = \operatorname{tr}(G^T dA) \Rightarrow \frac{\partial f}{\partial A} = G.$$

Manipulate with the trace's cyclic property (Chapter 20) until the differential is isolated on the right, and whatever multiplies it is the derivative.

Once you work this way, the layout question disappears: you never write a Jacobian, you just push differentials around and read off the answer at the end.

Example 26.3 (least squares, by differentials).

$$f(x) = \|Ax - b\|^2 = (Ax - b)^T (Ax - b).$$

$$\begin{aligned}df &= (A dx)^T (Ax - b) + (Ax - b)^T (A dx) \\&= 2(Ax - b)^T A dx \quad (\text{both terms are equal scalars}) \\&= \operatorname{tr}\left((2A^T(Ax - b))^T dx\right).\end{aligned}$$

By the identification rule, $\nabla f = 2A^T(Ax - b)$. Setting it to zero gives the normal equations of Chapter 24 in three lines.

Example 26.4 (the derivative of the inverse).

Differentiate $AA^{-1} = I$. The right side is constant, so

$$(dA)A^{-1} + A d(A^{-1}) = 0,$$

$$\text{hence } d(A^{-1}) = -A^{-1}(dA)A^{-1}.$$

Compare the scalar $d(1/a) = -(1/a^2) da$. Same shape; the matrix version just cannot commute the factors, so the perturbation sits in the middle.

26.4. The three derivatives you need for neural networks

Everything in Chapter 36 rests on these.

Proposition 26.5 (a linear layer).

Let $y = Wx + b$ with $W \in \mathbb{R}^{m \times n}$. If L is a scalar loss and $g = \partial L / \partial y \in \mathbb{R}^m$ is known, then

$$\frac{\partial L}{\partial x} = W^T g, \quad \frac{\partial L}{\partial W} = g x^T, \quad \frac{\partial L}{\partial b} = g.$$

Proof. $dy = (dW)x + W dx + db$, and $dL = g^T dy = g^T (dW)x + g^T W dx + g^T db$.

Reading off the dx term: $\partial L / \partial x = W^T g$.

For dW : $g^T (dW)x = \text{tr}(g^T (dW)x) = \text{tr}(x g^T dW)$ by cyclicity, which is $\text{tr}((g x^T)^T dW)$. So $\partial L / \partial W = g x^T$. ■

The idea: read the shapes and you can reconstruct these.

You do not need to memorise these. Reconstruct them from shapes.

$\partial L / \partial W$ must be $m \times n$, same as W . You have $g \in \mathbb{R}^m$ and $x \in \mathbb{R}^n$ available. The only way to combine them into an $m \times n$ matrix is $g x^T$. Done.

$\partial L / \partial x$ must be $n \times 1$. You have W ($m \times n$) and g ($m \times 1$). The only product that works is $W^T g$. Done.

This shape-matching heuristic is remarkably reliable in matrix calculus, because there are usually only one or two dimensionally valid arrangements and the correct one is among them. It is not a proof, but it will get you through a paper.

Proposition 26.6 (an elementwise nonlinearity).

Let $y = \varphi(x)$ applied elementwise. Then the Jacobian is diagonal, and

$$\frac{\partial L}{\partial x} = \varphi'(x) \circ \frac{\partial L}{\partial y},$$

where $\circ$ is the elementwise (Hadamard) product.

Proof. y_i depends only on x_i , so $\partial y_i / \partial x_j = 0$ for $i \neq j$ and $\varphi'(x_i)$ for $i = j$. Multiplying by a diagonal matrix is elementwise scaling. ■

That is why activation functions are cheap in the backward pass: the Jacobian is diagonal, so you never form it — you just multiply elementwise.

Proposition 26.7 (softmax and cross-entropy).

Let $p = \text{softmax}(z)$ and $L = -\sum_i y_i \log p_i$ for a one-hot target y . Then

$$\frac{\partial L}{\partial z} = p - y.$$

Proof. First the Jacobian of softmax. With $p_i = e^{z_i} / S$, $S = \sum_k e^{z_k}$:

$$\frac{\partial p_i}{\partial z_j} = \begin{cases} p_i(1 - p_i) & i = j \\ -p_i p_j & i \neq j \end{cases} = p_i(\delta_{ij} - p_j).$$

Now the loss. $L = -\sum_i y_i \log p_i$, so

$$\frac{\partial L}{\partial z_j} = -\sum_i \frac{y_i}{p_i} \frac{\partial p_i}{\partial z_j} = -\sum_i \frac{y_i}{p_i} p_i(\delta_{ij} - p_j) = -\sum_i y_i(\delta_{ij} - p_j).$$

Expanding: $-y_j + p_j \sum_i y_i = p_j - y_j$, using $\sum_i y_i = 1$. ■

The idea: why $p - y$ is such a nice gradient.

The gradient of the loss with respect to the logits is **predicted minus actual**. Nothing could be simpler, and there is no division by anything that might be near zero.

Note what did **not** happen: the $1/p_i$ from the log's derivative was cancelled exactly by the p_i from the softmax's derivative. That cancellation is why softmax and cross-entropy are always implemented as a **single fused operation**. Computing them separately means computing a logarithm, then a division, and losing precision to a cancellation that should never have been performed.

This is also why the same $p - y$ appears in logistic regression and in linear regression with squared loss. It is not a coincidence: all three are generalised linear models whose loss is the negative log-likelihood of an exponential-family distribution, and for that whole family the gradient is always “prediction minus observation” (Chapter 31).

26.5. A reference table

Expression	Derivative
$a^T x$	a
$x^T a$	a
$x^T x$	$2x$
$x^T A x$	$(A + A^T)x$; $2Ax$ if symmetric
$a^T A b$ w.r.t. A	ab^T
$\ Ax - b\ ^2$	$2A^T(Ax - b)$
$\text{tr}(AB)$ w.r.t. A	B^T
$\text{tr}(A^T B)$ w.r.t. A	B
$\log \det A$ w.r.t. A	$(A^{-1})^T$
A^{-1} (differential)	$-A^{-1}(\mathrm{d}A)A^{-1}$
$Wx + b$ w.r.t. W	gx^T (with upstream g)
$\varphi(x)$ elementwise	$\varphi'(x) \circ g$
softmax + cross-entropy	$p - y$

26.6. Exercises

26.1 Compute $\nabla_x (x^T A x)$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $x = (x_1, x_2)^T$ by expanding the quadratic form fully and differentiating entry by entry. Check your answer against $(A + A^T)x$.

26.2 Use differentials to derive $\nabla_x \|x\|$ (not the square) for $x \neq 0$. What goes wrong at $x = 0$, and what is the standard fix in optimisation code?

- 26.3** Show that $\nabla_x(x^T Ay) = Ay$ and $\nabla_y(x^T Ay) = A^T x$.
- 26.4** Derive the gradient of ridge regression, $L(x) = \|Ax - b\|^2 + \lambda\|x\|^2$, and solve $\nabla L = 0$ for x . Explain what the λI term does to the conditioning of the system.
- 26.5** Use the differential rules to show that $d \log \det A = \text{tr}(A^{-1} dA)$, starting from $d \det A = \det(A) \text{tr}(A^{-1} dA)$.
- 26.6** For a two-layer network $y = W_2 \varphi(W_1 x)$ with scalar loss L , write out $\partial L / \partial W_1$ and $\partial L / \partial W_2$ in terms of $g = \partial L / \partial y$, using the three propositions of this chapter. State the shape of each factor.
- 26.7** Verify the softmax Jacobian formula $\partial p_i / \partial z_j = p_i(\delta_{ij} - p_j)$ directly for the case $n = 2$, by writing out p_1 and p_2 explicitly and differentiating.
- 26.8** Show that the softmax Jacobian is symmetric and positive semidefinite, and that it is singular. What vector is in its null space, and what does that say about the parameterisation?

PART V

Probability and Statistics

Reasoning under uncertainty, from the axioms up: sample spaces, random variables, distributions, expectation, the limit theorems that make statistics possible, estimation and inference, and information theory.

CHAPTER 27.

Probability

27.1. What probability is a theory of

Probability is the mathematics of **not knowing**. That is a broader remit than gambling, and the two standard interpretations of what a probability **means** disagree about how broad.

Frequentist: a probability is a long-run frequency. “This coin has $\mathbb{P}(\text{heads}) = 0.5$ ” means that in many tosses, about half come up heads. Clean, but it says nothing about one-off events.

Bayesian: a probability is a degree of belief. “There is a 30% chance this service fails today” is meaningful even though today happens once. Beliefs update as evidence arrives (Bayes’ theorem, below).

The **mathematics** is identical under both readings — the axioms are the same — and the disagreement is about interpretation and about which statistical procedures follow. Chapter 31 returns to it. Until then, nothing depends on which camp you sit in.

27.2. The axioms

Definition 27.1 (probability space).

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$:

- Ω , the **sample space**: the set of all possible outcomes.
- $\mathcal{F}$, a collection of subsets of Ω called **events**, closed under complement and countable union.
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$, satisfying Kolmogorov’s axioms:
 - (P1) $\mathbb{P}(A) \geq 0$ for every event A ;
 - (P2) $\mathbb{P}(\Omega) = 1$;
 - (P3) if $A_1, A_2, \dots$ are pairwise **disjoint**, then $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$.

That is the whole foundation. Three axioms.

The idea: why events are sets, and why that is useful.

An **outcome** is a single result: this die shows 4. An **event** is a **set** of outcomes: “the die shows an even number” is $\{2, 4, 6\}$.

Making events sets means the logic of Chapter 7 and the set operations of Chapter 8 transfer directly:

English	Set operation
A or B	$A \cup B$
A and B	$A \cap B$
not A	$\overline{A}$
A but not B	$A \setminus B$
A and B cannot both happen	$A \cap B = \emptyset$

So probability inherits an entire algebra for free, and De Morgan's laws are as useful here as they were there.

Proposition 27.2 (immediate consequences).

- (a) $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$.
- (b) $\mathbb{P}(\emptyset) = 0$.
- (c) If $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- (d) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof. (a) A and $\bar{A}$ are disjoint with union Ω , so by (P3) and (P2), $\mathbb{P}(A) + \mathbb{P}(\bar{A}) = 1$.

(c) Write $B = A \cup (B \setminus A)$, a disjoint union, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$ by (P1).

(d) Write $A \cup B = A \cup (B \setminus A)$, disjoint, and $B = (A \cap B) \cup (B \setminus A)$, also disjoint. Then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

■

Item (d) is Chapter 8's inclusion–exclusion, now for probabilities. The general n -set version holds verbatim.

Definition 27.3 (uniform probability on a finite space).

If Ω is finite and all outcomes are equally likely,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This is the bridge to Chapter 9: under the equally-likely assumption, every probability question is a counting question. Most introductory probability is combinatorics wearing a different hat.

Trap: 'equally likely' is an assumption, not a default.

It has to be justified, and it is often wrong.

Two dice: the sums 2 through 12 are **not** equally likely, because the underlying equally-likely objects are the 36 ordered pairs, and more of them sum to 7 than to 2.

Similarly, “the next request either hits cache or misses, so it is 50/50” is nonsense. Symmetry is what justifies uniformity, and you must be able to point to the symmetry.

27.3. Conditional probability

Definition 27.4 (conditional probability).

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The idea: conditioning is zooming in.

Learning that B happened means the sample space is no longer Ω ; it is B . So you renormalise: the probability of A is now the share of B that also lies in A .

The division by $\mathbb{P}(B)$ is exactly that renormalisation — it makes the probabilities within B sum to 1 again. Conditioning is restricting your universe and rescaling.

It is also, incidentally, a perfectly good probability measure in its own right: $\mathbb{P}(\cdot | B)$ satisfies all three Kolmogorov axioms. So everything you know about probability applies inside a conditional world, and you can condition on more things without any new theory.

Rearranging gives the **multiplication rule**

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A),$$

and by induction the **chain rule**

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \mathbb{P}(A_2 | A_1) \mathbb{P}(A_3 | A_1 \cap A_2) \dots$$

which is the identity underneath every autoregressive language model: the probability of a sequence is the product of the conditional probabilities of each token given its predecessors.

Definition 27.5 (independence).

A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B),$$

equivalently (when $\mathbb{P}(B) > 0$) if $\mathbb{P}(A | B) = \mathbb{P}(A)$: knowing B tells you nothing about A .

Trap: independent is not the same as disjoint.

They are almost opposites, and the words get confused constantly.

Disjoint means $A \cap B = \emptyset$: they cannot both happen. So if A happens, B definitely did not — which is a very strong dependence. Two disjoint events with positive probability are **never** independent.

Independent means they do not inform each other. Independent events with positive probability **must** overlap, since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) > 0$.

Also worth knowing: **pairwise** independence does not imply **mutual** independence. Toss two fair coins and let A = first is heads, B = second is heads, C = the two agree. Any two of these are independent; but $\mathbb{P}(A \cap B \cap C) = 1/4 \neq 1/8$. Knowing any two determines the third.

27.4. Total probability and Bayes

Theorem 27.6 (law of total probability).

If $B_1, \dots, B_n$ partition Ω (disjoint, union is everything, each with positive probability), then for any A :

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

Proof. The sets $A \cap B_i$ are disjoint and their union is A , so by (P3) $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i)$. Apply the multiplication rule to each term. ■

This is **case analysis for probabilities** — split on which B_i happened, work out the probability in each case, and average with the case probabilities as weights.

Theorem 27.7 (Bayes' theorem).

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B) \mathbb{P}(B)}{\mathbb{P}(A)},$$

and expanding the denominator by total probability over a partition $\{B_i\}$:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}.$$

Proof. Both $\mathbb{P}(A \mid B)\mathbb{P}(B)$ and $\mathbb{P}(B \mid A)\mathbb{P}(A)$ equal $\mathbb{P}(A \cap B)$. Equate and divide. ■

The proof is one line. The importance is entirely in what it lets you do.

The idea: Bayes reverses the direction of a conditional.

You usually know $\mathbb{P}(\text{evidence} \mid \text{hypothesis})$ — that is what a test, a model, or a physical process gives you. You want $\mathbb{P}(\text{hypothesis} \mid \text{evidence})$ — that is what you actually need to decide anything.

Bayes converts one into the other, at the cost of requiring a **prior** $\mathbb{P}(\text{hypothesis})$.

The vocabulary:

$$\underbrace{\mathbb{P}(H \mid E)}_{\text{posterior}} = \frac{\overbrace{\mathbb{P}(E \mid H)}^{\text{likelihood}} \cdot \overbrace{\mathbb{P}(H)}^{\text{prior}}}{\underbrace{\mathbb{P}(E)}_{\text{evidence}}}.$$

And the reason the prior cannot be dropped is the single most important practical lesson in probability, which is the next example.

Example 27.8 (the medical test, and why base rates dominate).

A disease affects 1 in 10,000. A test has 99% sensitivity ($\mathbb{P}(\text{positive} \mid \text{disease}) = 0.99$) and 99% specificity ($\mathbb{P}(\text{negative} \mid \text{no disease}) = 0.99$).

You test positive. What is the probability you have the disease?

The intuitive answer is 99%. The correct answer:

$$\begin{aligned} \mathbb{P}(D \mid +) &= \frac{\mathbb{P}(+ \mid D) \mathbb{P}(D)}{\mathbb{P}(+ \mid D) \mathbb{P}(D) + \mathbb{P}(+ \mid \overline{D}) \mathbb{P}(\overline{D})} \\ &= \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + 0.01 \times 0.9999} \\ &= \frac{0.000099}{0.000099 + 0.009999} \approx 0.0098. \end{aligned}$$

About 1%. Not 99%.

The reason: out of a million people, 100 have the disease and 99 test positive; 999,900 are healthy and 9,999 of them **also** test positive. False positives outnumber true positives 100 to 1, because there are so many more healthy people. The test is good; the base rate is brutal.

This is **base rate neglect**, and it is not a curiosity. It is why:

- a rare-event alert with a 1% false positive rate is mostly false alarms, and your on-call engineers will learn to ignore it;
- a fraud detector with excellent accuracy still floods review queues;
- a “99.9% accurate” screening test for anything rare is nearly useless alone.

The fix in every case is the same: compute the posterior, not the accuracy.

Example 27.9 (Monty Hall).

Three doors, a car behind one. You pick door 1. The host — who **knows** where the car is and always opens a different, empty door — opens door 3. Switch to door 2, or stay?

Let C_i be “car behind door i ”, each with prior $1/3$, and let H_3 be “host opens door 3”.

$\mathbb{P}(H_3 \mid C_1) = 1/2$ (the host may open 2 or 3 freely). $\mathbb{P}(H_3 \mid C_2) = 1$ (the host must avoid your door and the car). $\mathbb{P}(H_3 \mid C_3) = 0$.

By Bayes,

$$\mathbb{P}(C_1 \mid H_3) = \frac{(1/2)(1/3)}{(1/2)(1/3) + 1 \cdot (1/3) + 0} = \frac{1/6}{1/2} = \frac{1}{3},$$

and $\mathbb{P}(C_2 \mid H_3) = 2/3$. **Switch.**

Where the intuition fails: the host’s action is not random. He is constrained by knowledge, so his choice carries information. If a **bystander** opened door 3 at random and happened to reveal a goat, then $\mathbb{P}(H_3 \mid C_1) = \mathbb{P}(H_3 \mid C_2) = 1/2$ and the posterior really is $1/2$ each — switching gains nothing.

Same observed outcome, different probability, because the **process** that generated the observation differs. Getting that right is most of applied probability.

27.5. Simulation, and how to think about randomness in code

Two practical notes that belong here rather than in a footnote.

Pseudorandom generators are deterministic. `rand()` is a deterministic function of hidden state. That is a feature: seeding makes experiments reproducible. It is also a hazard: linear congruential generators have detectable structure, and using one for anything security-related is a vulnerability. Use a cryptographic generator when an adversary is involved and a well-tested one (PCG, xoshiro, Mersenne Twister) otherwise.

Sampling from a distribution. If $U \sim \text{Uniform}(0, 1)$ and F is a cumulative distribution function (Chapter 28), then $F^{-1}(U)$ has distribution F . This is **inverse transform sampling**, and it is why a single uniform generator suffices for everything — when F^{-1} is available. When it is not (the normal distribution, for one), you use rejection sampling or a special-purpose method like Box–Muller.

Example 27.10 (estimating π by Monte Carlo).

Sample points uniformly in $[0, 1]^2$ and count the fraction landing inside the quarter disc $x^2 + y^2 \leq 1$. That fraction estimates the area, $\pi/4$.

With n samples the standard error is $O(1/\sqrt{n})$ — Chapter 30 proves this. So $100 \times$ more samples buys $10 \times$ more accuracy: four digits costs about 10^8 samples. Monte Carlo is a **terrible** way to compute π .

Its virtue is that the $O(1/\sqrt{n})$ rate is **independent of dimension**. Deterministic quadrature (Chapter 16) in d dimensions needs n^d grid points for comparable accuracy — the curse of dimensionality. So in high dimensions Monte Carlo wins by an enormous margin, which is why it dominates in finance, rendering, and Bayesian inference, and why nobody uses Simpson's rule on a 50-dimensional integral.

27.6. Exercises

- 27.1** Two fair dice are rolled. Find the probability that (a) the sum is 7, (b) the sum is 7 given that the first die shows 3, (c) at least one die shows 6, (d) the sum is 7 and at least one die shows 6. Are the events in (a) and (c) independent?
- 27.2** Prove that if A and B are independent, then so are A and $\bar{B}$.
- 27.3** A bag has 5 red and 3 blue balls. Two are drawn without replacement. Find the probability that (a) both are red, (b) the second is red, (c) the first is red given the second is red. Comment on why (b) has the answer it does.
- 27.4** A spam filter flags 95% of spam and misflags 2% of legitimate mail. If 40% of incoming mail is spam, what is the probability that a flagged message is actually spam? Redo the calculation for a 1% spam rate and comment on the difference.
- 27.5** Three events A , B , C are pairwise independent but not mutually independent. Construct an explicit example on a sample space of size 4 and verify both claims.
- 27.6** In a room of n people, use the birthday calculation of Chapter 9 to find the smallest n for which the probability of a shared birthday exceeds $1/2$. Then find the n at which it exceeds 0.99.
- 27.7** You have two coins: one fair, one double-headed. You pick one at random and flip it three times, getting three heads. What is the probability you picked the double-headed coin?
- 27.8** Prove the law of total probability implies Bayes' theorem in the expanded form given, and explain in one sentence what the denominator represents.

CHAPTER 28.

Random Variables and Distributions

28.1. From events to numbers

Chapter 27 assigned probabilities to sets. That is the right foundation and a clumsy way to work: you usually care about a **number** attached to an outcome — the sum of the dice, the latency of the request, the loss of the model.

Definition 28.1 (random variable).

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$.

Trap: it is neither random nor a variable.

A random variable is a **deterministic function**. The randomness lives in which $\omega \in \Omega$ occurs; X just reports a number about it.

Programmer's reading: ω is the seed, and X is a pure function of it. Two random variables on the same Ω are two pure functions of the same seed — which is exactly why they can be dependent. This picture makes dependence obvious and it is worth keeping.

Notation: $\mathbb{P}(X \leq 3)$ is shorthand for $\mathbb{P}(\{\omega \in \Omega : X(\omega) \leq 3\})$. The set is suppressed, which is convenient and occasionally confusing.

Definition 28.2 (discrete and continuous).

X is **discrete** if it takes countably many values. Its **probability mass function** is $p(x) = \mathbb{P}(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$.

X is **continuous** if there is a **probability density function** f with

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) \, dx,$$

where $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f = 1$.

Trap: a density is not a probability.

For continuous X , $\mathbb{P}(X = c) = 0$ for **every** single value c — the integral over a point is zero. That does not mean the value is impossible; it means single values have no probability mass.

So $f(x)$ is not $\mathbb{P}(X = x)$. It is probability **per unit length**, and it can exceed 1: a uniform distribution on $[0, 0.1]$ has density 10 everywhere on that interval. Only $\int f$ over a set is a probability.

Consequence: for continuous variables, $\mathbb{P}(X \leq a)$ and $\mathbb{P}(X < a)$ are equal, and endpoint conventions never matter. For discrete ones they do.

Definition 28.3 (cumulative distribution function).

$$F(x) = \mathbb{P}(X \leq x).$$

F is non-decreasing, right-continuous, with $F(-\infty) = 0$ and $F(\infty) = 1$. For continuous X , $F' = f$ by the Fundamental Theorem of Calculus.

The CDF exists for every random variable, discrete or continuous, which makes it the right object for general statements. It is also what you invert for inverse-transform sampling (Chapter 27) and what you compare in a Kolmogorov–Smirnov test.

28.2. Expectation

Definition 28.4 (expectation).

$$\mathbb{E}[X] = \begin{cases} \sum_x x p(x) & \text{discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{continuous} \end{cases}$$

when the sum or integral converges absolutely.

The idea: expectation is a weighted average, and also a centre of mass.

Each value, weighted by how likely it is. If you ran the experiment infinitely often and averaged, you would get $\mathbb{E}[X]$ — that is the law of large numbers (Chapter 30), and it is a theorem, not the definition.

The mechanical picture is useful: put a point mass $p(x)$ at position x on a rod. $\mathbb{E}[X]$ is where the rod balances. It follows immediately that the expectation need not be an attainable value — the expected number of pips on a die is 3.5 — and that it is dragged hard by distant mass, which is why the mean is not robust to outliers.

Theorem 28.5 (linearity of expectation).

For **any** random variables X, Y on the same space and constants a, b ,

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y].$$

No independence is required.

Proof. In the discrete case, summing over the underlying outcomes:

$$\mathbb{E}[aX + bY] = \sum_{\omega} (aX(\omega) + bY(\omega)) \mathbb{P}(\omega) = a \sum_{\omega} X(\omega) \mathbb{P}(\omega) + b \sum_{\omega} Y(\omega) \mathbb{P}(\omega).$$

The continuous case is the same computation with an integral, and linearity of the integral (Chapter 16) doing the work. ■

The idea: linearity without independence is the most useful fact in probability.

This is the workhorse. It lets you compute expectations of horribly dependent things by chopping them into simple pieces.

The standard technique is **indicator variables**: let $I_A = 1$ if event A happens and 0 otherwise. Then $\mathbb{E}[I_A] = \mathbb{P}(A)$, and any count is a sum of indicators.

Example. Expected number of fixed points of a random permutation of n items. Let I_i indicate that item i stays put. The I_i are strongly dependent — but linearity does not care:

$$\mathbb{E} \left[\sum_i I_i \right] = \sum_i \mathbb{P}(I_i = 1) = n \cdot \frac{1}{n} = 1.$$

Exactly one, for every n . Compare Chapter 9's derangement count, which needed inclusion-exclusion; the **expected** value needs one line.

Example. Coupon collector: to collect all n coupons, the time to get the k -th new one after having $k - 1$ is geometric with success probability $(n - k + 1)/n$, hence expectation $n/(n - k + 1)$. Summing,

$$\mathbb{E}[T] = \sum_{k=1}^n \frac{n}{n - k + 1} = nH_n \approx n \ln n,$$

the harmonic sum of Chapter 17.

Theorem 28.6 (law of the unconscious statistician).

For any function g ,

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad \int g(x)f(x) \, dx.$$

The name is a joke about people using it without noticing it needs proof. Note what it says: you do **not** need the distribution of $g(X)$ to compute its expectation. And note what it does **not** say: in general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.

Theorem 28.7 (Jensen's inequality).

If g is convex, $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$. If concave, the inequality reverses.

The idea: Jensen, and why averages of ratios lie.

A convex function bends upwards, so averaging inputs and then applying it undershoots; applying it and then averaging catches the curvature.

Consequences worth having:

- $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ — which rearranges to $\text{Var}(X) \geq 0$, so variance being nonnegative **is** Jensen.
- $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$ for positive X . So the average of speeds is not the speed of the average, and average latency does not give you average throughput.
- $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$, which is why the log-likelihood bound in variational inference (the ELBO) goes the way it does.
- The AM–GM inequality of Chapter 2 is Jensen applied to log.

28.3. Variance

Definition 28.8 (variance and standard deviation).

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}[X], \quad \sigma = \sqrt{\text{Var}(X)}.$$

Proposition 28.9 (computational formula).

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. Expand and use linearity:

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

■

Trap: that formula is numerically dangerous.

$\mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is a difference of two nearly equal large numbers when the mean is large relative to the spread — catastrophic cancellation (Chapter 1). Computing the variance of values near 10^9 with a spread of 1 can return a negative number.

Use **Welford's algorithm**, which maintains the mean and the sum of squared deviations incrementally and never subtracts large quantities:

```
n := 0; mean := 0; M2 := 0
for each x:
    n      := n + 1
    d      := x - mean
    mean   := mean + d / n
    M2     := M2 + d * (x - mean)    # note: new mean
variance := M2 / (n - 1)
```

Same answer in exact arithmetic, vastly better in floating point. This is the standard example of an algebraically valid rearrangement being numerically unacceptable.

Proposition 28.10 (properties).

$\text{Var}(aX + b) = a^2 \text{Var}(X)$, and if X, Y are **independent**, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Note the asymmetry with expectation: linearity of expectation needs nothing; additivity of variance needs independence. Shifting does not change spread; scaling changes it quadratically, which is why standard deviation (in the same units as X) is often the more interpretable quantity.

28.4. The distributions worth knowing

28.4.1. Discrete

Definition 28.11 (Bernoulli).

$X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$. $\mathbb{E}[X] = p$, $\text{Var}(X) = p(1 - p)$.

The variance is maximised at $p = 1/2$ — maximum uncertainty — and zero at $p = 0$ or 1 . A single yes/no trial. Every classification label is one of these.

Definition 28.12 (binomial).

$X \sim \text{Binomial}(n, p)$ counts successes in n independent Bernoulli trials.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

Proof. The pmf: choose which k trials succeed ($\binom{n}{k}$ ways, Chapter 9), each such pattern having probability $p^k(1-p)^{n-k}$ by independence.

The mean and variance: $X = \sum_i X_i$ with each X_i Bernoulli. Linearity gives $\mathbb{E}[X] = np$; independence gives $\text{Var}(X) = np(1-p)$. ■

Definition 28.13 (geometric).

X counts trials until the first success. $\mathbb{P}(X = k) = (1-p)^{k-1}p$, $\mathbb{E}[X] = 1/p$, $\text{Var}(X) = (1-p)/p^2$.

The geometric distribution is **memoryless**: $\mathbb{P}(X > s+t \mid X > s) = \mathbb{P}(X > t)$. Having waited does not bring success closer. It is the only discrete distribution with that property, and it is why “this hash probe has failed five times, surely the next will work” is wrong.

Definition 28.14 (Poisson).

$X \sim \text{Poisson}(\lambda)$ counts events in a fixed interval when they occur independently at constant average rate λ :

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \mathbb{E}[X] = \text{Var}(X) = \lambda.$$

Proof. As a limit of the binomial. Take n trials with $p = \lambda/n$ and let $n \rightarrow \infty$:

$$\binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\rightarrow 1} \cdot \frac{\lambda^k}{k!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\rightarrow e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\rightarrow 1},$$

using $\lim (1 + x/n)^n = e^x$ from Chapter 17. ■

That derivation says what Poisson **is**: many opportunities, each unlikely, constant rate. Requests arriving at a server, packets dropped, mutations, typos per page, cosmic-ray bit flips. The variance equalling the mean is a strong signature — real count data with variance much larger than its mean is **overdispersed**, and Poisson is the wrong model for it.

28.4.2. Continuous

Definition 28.15 (uniform).

$X \sim \text{Uniform}(a, b)$ with $f(x) = 1/(b-a)$ on $[a, b]$. $\mathbb{E}[X] = (a+b)/2$, $\text{Var}(X) = (b-a)^2/12$.

Definition 28.16 (exponential).

$X \sim \text{Exp}(\lambda)$ with $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. $\mathbb{E}[X] = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$.

The exponential is the continuous memoryless distribution — the waiting time between Poisson events. It is the standard model for time-to-failure with constant hazard rate, and for inter-arrival times in queueing theory. Its memorylessness is why “the process has been running an hour, it must be due to finish” is wrong under this model, and why the exponential is often a **bad** model for things that genuinely age.

Definition 28.17 (normal (Gaussian)).

$X \sim \mathcal{N}(\mu, \sigma^2)$ with

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}.$$

$\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$.

Proposition 28.18 (the density integrates to 1).

The normalising constant $1/(\sigma\sqrt{2\pi})$ is exactly right.

Proof. Substituting $z = (x - \mu)/\sigma$ reduces the claim to $\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}$, and substituting $z = u\sqrt{2}$ reduces that to $\int e^{-u^2} du = \sqrt{\pi}$ — the Gaussian integral computed in Chapter 18 by converting to polar coordinates. That $\sqrt{2\pi}$ has been waiting for this since then. ■

The idea: why the normal distribution is everywhere.

Three separate reasons, and it is worth knowing they are separate.

The central limit theorem (Chapter 30). Sums of many independent contributions are approximately normal, regardless of what the contributions look like. Since a great many measured quantities are sums of small independent effects, a great many are approximately normal.

Maximum entropy. Among all distributions with a given mean and variance, the normal has the largest entropy (Chapter 32). It is the least presumptuous choice given only those two facts.

Mathematical convenience. Sums of independent normals are normal. Linear transforms of normals are normal. Marginals and conditionals of a multivariate normal are normal. The maximum-likelihood estimate under normal noise is least squares. Nothing else is this well behaved.

The rule of thumb: about 68% of the mass within 1σ , 95% within 2σ , 99.7% within 3σ .

Trap: assuming normality is a real assumption.

Normal distributions have very thin tails — the density decays like e^{-x^2} , so a 5σ event has probability about 3×10^{-7} .

Many real distributions do not. Financial returns, file sizes, network traffic, city populations, and word frequencies have **heavy tails**, often power-law. Under a power law, extreme events are orders of magnitude more likely than a normal model predicts, and the sample mean converges slowly or — if the tail index is small enough — the mean may not exist at all.

Every “six sigma event” that occurs twice a decade is a normality assumption being wrong, not a miracle. Check your tails: plot the empirical distribution on log-log axes (Chapter 4) and see whether it straightens.

28.5. Transformations

Theorem 28.19 (change of variable for densities).

If $Y = g(X)$ with g monotone and differentiable, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Proof. For increasing g ,

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiate with respect to y using the chain rule. For decreasing g the inequality flips and the derivative is negative, which the absolute value absorbs. ■

The Jacobian factor is Chapter 22’s volume interpretation: densities are probability per unit length, so stretching the axis must dilute the density. In several variables the factor becomes $|\det J|$, and that is the foundation of normalising flows.

Example 28.20 (standardisation).

If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$. Every normal computation reduces to the standard normal by this shift and scale, which is why tables and library functions only ever provide the standard one.

28.6. Exercises

- 28.1** A fair die is rolled. Let X be the result. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var}(X)$.
- 28.2** Let $X \sim \text{Binomial}(10, 0.3)$. Compute $\mathbb{P}(X = 3)$, $\mathbb{E}[X]$, and $\text{Var}(X)$. Then compute $\mathbb{P}(X \geq 1)$ using complementary counting.
- 28.3** Prove that the geometric distribution is memoryless: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$. Start by showing $\mathbb{P}(X > k) = (1 - p)^k$.
- 28.4** Requests arrive at a mean rate of 3 per second, Poisson distributed. Find the probability of (a) exactly 5 in a given second, (b) none in a given second, (c) more than 1 in a given half-second.
- 28.5** Let $X \sim \text{Uniform}(0, 1)$ and $Y = -\ln(X)/\lambda$. Use the change of variable formula to show $Y \sim \text{Exp}(\lambda)$. (This is how exponential variates are generated in practice.)
- 28.6** Use indicator variables and linearity of expectation to find the expected number of “records” in a random permutation of n distinct numbers — a record being an element larger than everything before it. (You will get a harmonic number.)
- 28.7** Show that $\text{Var}(X) = 0$ if and only if X is constant with probability 1.
- 28.8** A hash table with m buckets receives n keys uniformly at random. Use indicators to find the expected number of empty buckets, and evaluate it for $n = m$.
- 28.9** Use Jensen’s inequality to prove that for positive X that is not constant, $\mathbb{E}[1/X] > 1/\mathbb{E}[X]$ strictly. Then give a concrete two-point example showing how large the gap can be.

CHAPTER 29.

Several Random Variables

29.1. Joint distributions

One variable at a time is rarely enough. The interesting questions are about how quantities move together.

Definition 29.1 (joint distribution).

For discrete X, Y : the **joint pmf** $p(x, y) = \mathbb{P}(X = x, Y = y)$, with $\sum_x \sum_y p(x, y) = 1$.

For continuous: the **joint pdf** $f(x, y)$ with

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y) \, dx \, dy.$$

Definition 29.2 (marginal and conditional).

The **marginals** are obtained by summing (or integrating) out the other variable:

$$p_X(x) = \sum_y p(x, y), \quad f_X(x) = \int f(x, y) \, dy.$$

The **conditional** is

$$p_{Y|X}(y | x) = \frac{p(x, y)}{p_X(x)}, \quad f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}.$$

The idea: marginalising is a `GROUP BY`.

Think of the joint distribution as a two-dimensional table of probabilities.

The **marginal** of X is the row sums — collapse the table along Y . (The name is literal: the totals were written in the margins.) In database terms, `SELECT x, SUM(p) FROM joint GROUP BY x`.

The **conditional** is one row, renormalised to sum to 1: `WHERE x = ...`, then divide by the row total.

Everything in multivariate probability is one of those two operations, plus Bayes to swap which variable you are conditioning on.

Definition 29.3 (independence of random variables).

X and Y are **independent** if

$$p(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y$$

(or the same with densities). Equivalently, the joint **factors**.

Theorem 29.4.

If X and Y are independent then $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$.

Proof.

$$\mathbb{E}[XY] = \sum_x \sum_y xy p(x, y) = \sum_x \sum_y xy p_X(x) p_Y(y) = \left(\sum_x x p_X(x) \right) \left(\sum_y y p_Y(y) \right).$$

The converse is false, which is the subject of the next section.

29.2. Covariance and correlation

Definition 29.5 (covariance).

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Definition 29.6 (correlation).

$$\rho = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

Proof. That $|\rho| \leq 1$ is Cauchy–Schwarz (Chapter 19) applied to the inner product $\langle U, V \rangle = \mathbb{E}[UV]$ on centred random variables. Equality holds iff $Y - \mu_Y$ is a constant multiple of $X - \mu_X$ — that is, iff the relationship is exactly linear. ■

The idea: covariance is a dot product.

Treat centred random variables as vectors, with $\mathbb{E}[UV]$ as the inner product. Then:

- $\text{Cov}(X, Y)$ is the dot product.
- $\text{Var}(X) = \text{Cov}(X, X)$ is the squared norm.
- σ_X is the length.
- ρ is $\cos \theta$ between them.
- **Uncorrelated means orthogonal.**

Everything from Chapter 24 now transfers. Projecting Y onto X gives the least-squares regression line, with slope $\text{Cov}(X, Y) / \text{Var}(X)$ — literally the projection formula. The residual is orthogonal to X , which is why regression residuals are uncorrelated with the predictor. And ρ^2 is the fraction of variance explained, which is $\cos^2 \theta$.

Correlation is geometry.

Trap: uncorrelated does not mean independent.

Correlation only detects **linear** relationship.

Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Then Y is a deterministic function of X — as dependent as it is possible to be. But

$$\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0 - 0 = 0,$$

since odd moments of a symmetric distribution vanish. Zero correlation, total dependence.

Independence implies uncorrelated. The converse holds only in special cases — most importantly for **jointly normal** variables, where uncorrelated does imply independent. That exception is why the distinction is so easy to forget: in the Gaussian world it does not bite.

Practical consequence: a correlation matrix of zeros does not license you to treat your features as independent. Mutual information (Chapter 32) detects dependence of any kind, and is the right tool when you actually need independence.

Proposition 29.7 (variance of a sum, in general).

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y),$$

and more generally

$$\text{Var}\left(\sum_i a_i X_i\right) = \sum_i \sum_j a_i a_j \text{Cov}(X_i, X_j).$$

Proof. Expand $\mathbb{E}[(X - \mu_X) + (Y - \mu_Y)]^2$ and use linearity of expectation. ■

Example 29.8 (why diversification works, and when it stops).

Average n measurements each with variance σ^2 and pairwise correlation ρ . Then

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} + \frac{(n-1)}{n} \rho \sigma^2 \rightarrow \sigma^2 \rho \quad \text{as } n \rightarrow \infty.$$

If $\rho = 0$, the variance goes to zero like $1/n$: averaging independent noise works, and this is why ensembles help and why more data reduces error.

If $\rho > 0$, the variance floors out at $\rho \sigma^2$ **no matter how many measurements you take**. Correlated errors cannot be averaged away.

This is why ensembling near-identical models buys almost nothing; why replicating a service across one datacentre does not protect against a datacentre failure; and why a portfolio of correlated assets is not diversified. The interesting quantity is never n alone, it is ρ .

29.3. The covariance matrix

Definition 29.9 (covariance matrix).

For a random vector $X = (X_1, \dots, X_n)^T$ with mean μ ,

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T], \quad \Sigma_{ij} = \text{Cov}(X_i, X_j).$$

Proposition 29.10.

Σ is symmetric and positive semidefinite.

Proof. Symmetry is immediate from $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$. For semidefinite, take any $a \in \mathbb{R}^n$:

$$a^T \Sigma a = a^T \mathbb{E}[(X - \mu)(X - \mu)^T] a = \mathbb{E}[(a^T (X - \mu))^2] = \text{Var}(a^T X) \geq 0.$$

That computation is worth keeping: $a^T \Sigma a$ is the variance of the projection of X onto direction a . So the covariance matrix is a machine that reports the spread of your data in any direction you ask about.

Theorem 29.11 (linear transformation).

If $Y = AX + b$ then

$$\mathbb{E}[Y] = A\mu + b, \quad \Sigma_Y = A\Sigma A^T.$$

Proof. The mean is linearity. For the covariance,

$$\Sigma_Y = \mathbb{E}[(A(X - \mu))(A(X - \mu))^T] = A \mathbb{E}[(X - \mu)(X - \mu)^T] A^T = A\Sigma A^T.$$

The $A\Sigma A^T$ form is the multivariate version of $\text{Var}(aX) = a^2 \text{Var}(X)$ — the “square” becomes a sandwich because covariance is a bilinear form. ■

The idea: principal component analysis, in three lines.

Σ is symmetric, so by the spectral theorem (Chapter 23), $\Sigma = Q\Lambda Q^T$ with orthonormal eigenvectors and nonnegative eigenvalues.

Since $a^T \Sigma a$ is the variance in direction a , and since (from Chapter 23) that quadratic form is maximised over unit a by the top eigenvector with value λ_1 :

The eigenvectors of the covariance matrix are the directions of greatest variance, and the eigenvalues are the variances in those directions.

Those directions are the principal components. Rotating your data into that basis, $Y = Q^T X$, gives $\Sigma_Y = Q^T Q \Lambda Q^T Q = \Lambda$ — diagonal. The new coordinates are **uncorrelated**.

So PCA is: find the basis in which your data has no correlations, ordered by how much they vary. It is diagonalisation, applied to the covariance matrix, and it is the same computation as the SVD of the centred data matrix (Chapter 25).

Definition 29.12 (multivariate normal).

$X \sim \mathcal{N}(\mu, \Sigma)$ with density

$$f(x) = \frac{1}{(2\pi)^{n/2} (\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1} (x - \mu)\right).$$

The exponent contains $(x - \mu)^T \Sigma^{-1} (x - \mu)$, the **Mahalanobis distance** squared. It is the natural distance for correlated data: it measures how many standard deviations away a point is, **accounting for the correlations**. Level sets are ellipsoids with axes along the eigenvectors of Σ and semi-axis lengths proportional to $\sqrt{\lambda_i}$ — exactly the geometry of Chapter 23. The $\sqrt{\det \Sigma}$ in the normalising constant is the volume of that ellipsoid (Chapter 22).

Key properties, all worth knowing:

- Every marginal of a multivariate normal is normal.
- Every conditional is normal, with a mean that is a **linear** function of the conditioning variables.
- Any linear transform $AX + b$ is normal, with parameters given by the theorem above.
- For jointly normal variables, uncorrelated **does** imply independent (since Σ diagonal makes the density factor).

That last one is why Gaussians are so pleasant and so misleading as a mental model for probability in general.

29.4. Conditional expectation

Definition 29.13.

$$\mathbb{E}[Y \mid X = x] = \sum_y y p_{Y|X}(y|x) \quad \text{or} \quad \int y f_{Y|X}(y|x) dy.$$

Note $\mathbb{E}[Y|X]$, without a specific value, is a **random variable**: it is a function of X .

Theorem 29.14 (law of total expectation (tower property)).

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

Proof.

$$\mathbb{E}[\mathbb{E}[Y|X]] = \sum_x \mathbb{E}[Y|X = x] p_X(x) = \sum_x \left(\sum_y y \frac{p(x,y)}{p_X(x)} \right) p_X(x) = \sum_y y \sum_x p(x,y) = \sum_y y p_Y(y).$$

■

The idea: condition on the thing you wish you knew.

The tower property is the probabilistic version of case analysis, and it is the standard technique for computing expectations that look intractable: condition on something that makes the problem easy, then average over it.

Example. Expected number of coin flips to get two heads in a row. Condition on the first flip. Let E be the answer.

- Tails (prob $1/2$): one flip wasted, start over. Cost $1 + E$.
- Heads then tails (prob $1/4$): two flips wasted, start over. Cost $2 + E$.
- Heads then heads (prob $1/4$): done in 2.

So $E = (1/2)(1 + E) + (1/4)(2 + E) + (1/4)(2)$, giving $E = 6$.

Recursive expectation computations like this are the probabilistic analogue of recurrence relations (Chapter 11), and they are solved the same way.

Theorem 29.15 (law of total variance).

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}(\mathbb{E}[Y|X]).$$

Read: total variance = (average within-group variance) + (variance of the group means). This decomposition is the foundation of analysis of variance, of the bias–variance decomposition in machine learning, and of variance reduction techniques in Monte Carlo — if you can find an X that explains a lot of the variation, conditioning on it shrinks the first term.

Theorem 29.16 (conditional expectation is a projection).

Among all functions g , the one minimising $\mathbb{E}[(Y - g(X))^2]$ is $g(X) = \mathbb{E}[Y|X]$.

Proof. Write $Y - g(X) = (Y - \mathbb{E}[Y|X]) + (\mathbb{E}[Y|X] - g(X))$ and expand the square. The cross term is

$$2 \mathbb{E}[(Y - \mathbb{E}[Y|X])(\mathbb{E}[Y|X] - g(X))].$$

Condition on X : the second factor is then constant and the first has conditional expectation zero, so the whole thing vanishes by the tower property. Hence

$$\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y|X])^2] + \mathbb{E}[(\mathbb{E}[Y|X] - g(X))^2],$$

and the second term is minimised (at zero) by $g = \mathbb{E}[Y|X]$. ■

The idea: this is why regression is what it is.

That proof is the Pythagorean argument of Chapter 24, verbatim, in the space of random variables with inner product $\mathbb{E}[UV]$.

$\mathbb{E}[Y|X]$ is the **orthogonal projection** of Y onto the space of functions of X . The residual is orthogonal to everything you could have computed from X — which is exactly the statement that you have extracted all the predictive information X carries.

So: the best possible predictor of Y from X , under squared loss, is the conditional mean. That is what a regression model is trying to approximate, and it is why squared loss and conditional expectation always appear together. Train with a different loss and you converge to a different functional — absolute loss gives the conditional **median**, which is a genuinely different (and more robust) thing to predict.

29.5. Sums of independent random variables

Theorem 29.17 (convolution).

If X and Y are independent with densities f_X, f_Y , then $Z = X + Y$ has density

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

Proof. $\mathbb{P}(Z \leq z) = \iint_{x+y \leq z} f_X(x) f_Y(y) dx dy$. Substituting $y = t - x$ and differentiating with respect to z gives the stated integral. ■

Convolution appears in image processing, signal processing, and neural networks under the same name and for the same reason: it is what happens when two independent things combine additively.

Definition 29.18 (moment generating function).

$$M_X(t) = \mathbb{E}[e^{tX}].$$

Proposition 29.19.

$M_{X+Y} = M_X \cdot M_Y$ for independent X, Y ; and $\mathbb{E}[X^n] = M_X^{(n)}(0)$.

Proof. The first is $\mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}]$ by independence.

The second: expand $e^{tX} = \sum_n (tX)^n/n!$ (Chapter 17), so $M_X(t) = \sum_n \mathbb{E}[X^n] t^n/n!$, and the n -th derivative at 0 picks out $\mathbb{E}[X^n]$. ■

MGFs turn the awkward operation of convolution into multiplication — the same trick as logarithms, and the same trick as the Fourier transform. They are the standard tool for proving

that sums of familiar distributions stay familiar, and they are the machinery behind the central limit theorem in Chapter 30.

Example 29.20.

Sums of independent normals are normal. The MGF of $\mathcal{N}(\mu, \sigma^2)$ is $\exp(\mu t + \sigma^2 t^2/2)$. Multiplying two such gives $\exp((\mu_1 + \mu_2)t + (\sigma_1^2 + \sigma_2^2)t^2/2)$, which is the MGF of $\mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Since the MGF determines the distribution, done.

Compare doing this by convolution: a two-dimensional Gaussian integral, completing the square. The MGF turns it into multiplying two exponentials.

29.6. Exercises

- 29.1** The joint pmf of (X, Y) is: $p(0, 0) = 0.1$, $p(0, 1) = 0.2$, $p(1, 0) = 0.3$, $p(1, 1) = 0.4$. Find both marginals, $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\text{Cov}(X, Y)$, and determine whether X and Y are independent.
- 29.2** Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Verify directly that $\text{Cov}(X, Y) = 0$, and explain in one sentence why this does not contradict their obvious dependence.
- 29.3** Prove that $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$, and deduce that correlation is unchanged by shifting and by positive scaling.
- 29.4** Two assets each have return variance $\sigma^2 = 0.04$ and correlation $\rho = 0.6$. Compute the variance of an equally weighted portfolio, and compare to the $\rho = 0$ case. Then find the limiting variance of an equally-weighted portfolio of n such assets as $n \rightarrow \infty$.
- 29.5** A random vector has covariance matrix

$$\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

Find the variance of $X_1 + X_2$, the variance of $X_1 - X_2$, and the direction of maximum variance (the top eigenvector).

- 29.6** Use the tower property to find the expected number of coin flips needed to see the pattern HTH. (Set up conditional expectations on the progress made so far — there are four states.)
- 29.7** A number N of items is Poisson with mean λ , and each item independently has weight with mean μ and variance τ^2 . Use the laws of total expectation and total variance to find the mean and variance of the total weight.
- 29.8** Compute the MGF of the exponential distribution with rate λ , state for which t it converges, and use it to derive the mean and variance.

CHAPTER 30.

The Limit Theorems

30.1. Why these two theorems matter more than the rest

Everything in statistics rests on two results.

The **law of large numbers** says sample averages converge to the true mean. It is what licenses estimating anything from data at all — without it, a measurement tells you nothing about the underlying quantity.

The **central limit theorem** says the **error** in that estimate is approximately normal, with a size you can calculate. It is what licenses error bars, confidence intervals, hypothesis tests, and A/B testing.

Together: you can measure things, and you can say how wrong you probably are.

30.2. Inequalities first

The limit theorems need tools for bounding tail probabilities. These inequalities are also useful in their own right — they appear all over the analysis of randomised algorithms.

Theorem 30.1 (Markov's inequality).

If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Split the expectation according to whether $X \geq a$:

$$\mathbb{E}[X] = \mathbb{E}[X \cdot I_{X \geq a}] + \mathbb{E}[X \cdot I_{X < a}] \geq \mathbb{E}[X \cdot I_{X \geq a}] \geq a \mathbb{E}[I_{X \geq a}] = a \mathbb{P}(X \geq a),$$

using $X \geq 0$ for the first inequality and $X \geq a$ on the event for the second. ■

The idea: Markov, in words.

A nonnegative quantity cannot often be much larger than its average.

At most 1/10 of a population can earn ten times the mean income, for the simple reason that they would otherwise account for more than all the income.

It is a weak bound — it uses only the mean and nonnegativity — and its virtue is that it needs almost nothing. Every sharper bound below is Markov applied to a cleverly chosen function of X .

Theorem 30.2 (Chebyshev's inequality).

For any X with finite variance,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Proof. Apply Markov to the nonnegative variable $(X - \mu)^2$ with threshold $k^2\sigma^2$:

$$\mathbb{P}((X - \mu)^2 \geq k^2 \sigma^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2 \sigma^2} = \frac{\sigma^2}{k^2 \sigma^2} = \frac{1}{k^2}.$$

The event $(X - \mu)^2 \geq k^2 \sigma^2$ is the same as $|X - \mu| \geq k\sigma$. ■

So at most 25% of any distribution lies beyond 2σ , at most 11% beyond 3σ — for **any** distribution with finite variance, however weird. For a normal distribution the true figures are 4.6% and 0.3%; Chebyshev is loose, but it is universal, and that is the trade.

Theorem 30.3 (Chernoff bound, for sums of independent bounded variables).

Let $X = \sum_{i=1}^n X_i$ with the X_i independent and in $[0, 1]$, and $\mu = \mathbb{E}[X]$. Then for $\delta > 0$,

$$\mathbb{P}(X \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2 + \delta}\right), \quad \mathbb{P}(X \leq (1 - \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2}\right).$$

The idea: why Chernoff is exponentially better.

Chebyshev gives a bound decaying like $1/k^2$. Chernoff gives one decaying like e^{-k^2} . The difference is enormous and it comes from using **more information**: Chebyshev uses only the variance, Chernoff uses the full moment generating function and independence.

Mechanically: apply Markov to e^{tX} rather than to $(X - \mu)^2$, use $\mathbb{E}[e^{tX}] = \prod \mathbb{E}[e^{tX_i}]$ by independence (Chapter 29), and then optimise over t . Every “concentration inequality” in the literature is that recipe with different bookkeeping.

This is the workhorse for randomised algorithms. It is why a randomised quicksort is $O(n \log n)$ **with high probability**, not merely in expectation; why load balancing n jobs across n machines gives a maximum load of $O(\log n / \log \log n)$; and why a Bloom filter’s false positive rate is tightly concentrated rather than merely correct on average.

30.3. The law of large numbers

Definition 30.4 (sample mean).

For i.i.d. $X_1, \dots, X_n$ (independent and identically distributed) with mean μ ,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Proposition 30.5.

$\mathbb{E}[\bar{X}_n] = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$.

Proof. Linearity gives the mean. For the variance, independence gives $\text{Var}(\sum X_i) = n\sigma^2$, and scaling by $1/n$ divides the variance by n^2 . ■

That σ^2/n is the whole story of statistics in one expression: the spread of your estimate shrinks like $1/n$, so its **standard deviation** shrinks like $1/\sqrt{n}$.

Theorem 30.6 (weak law of large numbers).

For i.i.d. X_i with finite mean μ and variance, for every $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof. Apply Chebyshev to $\bar{X}_n$, whose standard deviation is $\sigma/\sqrt{n}$:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

■

Three lines, and it is the entire justification for empirical measurement.

The **strong** law says more: $\bar{X}_n \rightarrow \mu$ with probability 1, meaning the sequence of averages converges for almost every infinite sequence of draws. The proof is harder and the practical difference is small.

Trap: the gambler's fallacy is a misreading of the LLN.

The LLN says the **average** converges. It does **not** say deviations get corrected.

After 100 tosses with 60 heads, the excess is +10. The LLN does not predict a compensating run of tails. It predicts that the excess becomes **negligible relative to n** : after a million tosses, an excess of 10 — or even 1000 — is a rounding error in the average.

Deviations are not cancelled, they are **diluted**. Indeed the expected absolute deviation from $n/2$ **grows**, like $\sqrt{n}$; it just grows slower than n .

This also disposes of the “law of averages” in system design: a service that has been unusually reliable this month is not “due” for an outage.

30.4. The central limit theorem

Theorem 30.7 (central limit theorem).

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and finite variance $\sigma^2 > 0$. Then

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow \mathcal{N}(0, 1)$$

in distribution: $\mathbb{P}(Z_n \leq z) \rightarrow \Phi(z)$ for every z , where Φ is the standard normal CDF.

Proof. Sketch via moment generating functions (Chapter 29). Let $Y_i = (X_i - \mu)/\sigma$, so $\mathbb{E}[Y_i] = 0$ and $\text{Var}(Y_i) = 1$, and $Z_n = (1/\sqrt{n}) \sum Y_i$.

Expand the MGF of Y as a power series (Chapter 17):

$$M_Y(t) = 1 + t \mathbb{E}[Y] + \frac{t^2}{2} \mathbb{E}[Y^2] + O(t^3) = 1 + \frac{t^2}{2} + O(t^3).$$

By independence,

$$M_{Z_n}(t) = (M_Y(t/\sqrt{n}))^n = \left(1 + \frac{t^2}{2n} + O(n^{-3/2})\right)^n.$$

Taking logs and using $\ln(1 + u) \approx u$ for small u :

$$\ln M_{Z_n}(t) = n \ln \left(1 + \frac{t^2}{2n} + \dots \right) \rightarrow \frac{t^2}{2}.$$

So $M_{Z_n}(t) \rightarrow e^{t^2/2}$, which is the MGF of $\mathcal{N}(0,1)$. A continuity theorem for MGFs converts convergence of MGFs into convergence in distribution. ■

The idea: what the CLT is really saying, and what it is not.

Look at where the normal came from in that proof. Only the first two moments survived: everything past $t^2/2$ vanished in the limit. **The distribution of the individual X_i is completely forgotten** — all that remains is the mean and the variance.

That is why the normal is universal. Whatever the underlying distribution — uniform, exponential, some grotesque bimodal thing — averaging washes out every feature except the first two moments, and what is left is Gaussian.

Two things the CLT does **not** say:

It is not about the data. The CLT says the **sample mean** is approximately normal. Your data can be as non-normal as it likes; that is a separate question.

It needs finite variance. Averages of Cauchy-distributed variables are **not** asymptotically normal — the average of n Cauchy variables has exactly the same distribution as one of them, so more data buys nothing at all. Heavy-tailed data with infinite variance genuinely breaks this theorem, and the correct limits are the **stable distributions**.

Example 30.8 (how fast is 'asymptotically'?).

The usual rule of thumb is $n \geq 30$, which is folklore rather than a theorem. The truth depends on skewness:

- Symmetric, light-tailed (uniform): $n = 5$ already looks normal.
- Moderately skewed (exponential): $n \approx 30$ is reasonable.
- Heavily skewed (a Bernoulli with $p = 0.01$): you need $np \geq 10$ or so, i.e. $n \geq 1000$, before the normal approximation to the count is any good.

The **Berry–Esseen theorem** makes this quantitative: the error in the CLT approximation is at most $C\rho/(\sigma^3\sqrt{n})$ where ρ is the third absolute moment. Note the $1/\sqrt{n}$: convergence is **slow**, and skewness makes it slower.

For rare events, use the Poisson approximation (Chapter 28) instead — it is the right limit for that regime, and the normal is the wrong one.

30.5. Standard error and confidence intervals

Definition 30.9 (standard error).

The **standard error** of the sample mean is

$$\text{SE} = \frac{\sigma}{\sqrt{n}},$$

estimated by $s/\sqrt{n}$ when σ is unknown, with s the sample standard deviation.

The idea: the tyranny of the square root.

Standard error shrinks like $1/\sqrt{n}$, not $1/n$. To halve your uncertainty you need **four times** the data. To gain one decimal digit you need a hundred times.

This single fact governs the economics of measurement:

- A/B tests with small effects need enormous samples. Detecting a 1% lift with reasonable power typically needs tens of thousands of users per arm, and there is no clever way around it.
- Benchmarking: 10 runs give you about 30% better precision than 6. Getting to $10\times$ better precision needs 100 times the runs. Reducing the **variance** of each run — pinning cores, disabling turbo, controlling the environment — is almost always cheaper than more runs.
- Monte Carlo integration converges at $1/\sqrt{n}$ regardless of dimension (Chapter 27). Terrible in 1D, unbeatable in 50D.

Variance reduction beats sample size. Always look there first.

Theorem 30.10 (confidence interval for a mean).

By the CLT, an approximate 95% confidence interval for μ is

$$\bar{X} \pm 1.96 \cdot \frac{s}{\sqrt{n}},$$

since $\mathbb{P}(|Z| \leq 1.96) \approx 0.95$ for standard normal Z .

Trap: what a confidence interval does not mean.

A 95% confidence interval does **not** mean “there is a 95% probability that μ lies in this interval”. Under the frequentist reading, μ is a fixed unknown constant — it is either in the interval or it is not, with no probability involved.

What it means: **the procedure** produces an interval containing μ 95% of the time, across hypothetical repetitions of the experiment. The randomness is in the interval, not in μ .

The statement people actually want — “given the data, there is a 95% probability that μ is in here” — is a **credible interval**, and it is a Bayesian object requiring a prior (Chapter 31). The two often coincide numerically, which is why the misinterpretation persists and usually does no damage. But they are different claims, and they can diverge sharply when the prior matters.

30.6. Convergence, and the different kinds of it

Worth naming, since the literature distinguishes them and the distinctions matter.

Definition 30.11.

- **In probability:** $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ for every ε . (Weak LLN.)
- **Almost surely:** $\mathbb{P}(\lim X_n = X) = 1$. (Strong LLN.) Stronger.
- **In distribution:** $F_n(x) \rightarrow F(x)$ at every continuity point. (CLT.) Weakest of the three — it says nothing about the variables being close, only their distributions.
- **In L^2 :** $\mathbb{E}[(X_n - X)^2] \rightarrow 0$. Implies convergence in probability.

Almost sure $\Rightarrow$ in probability $\Rightarrow$ in distribution, and none of the reverse implications hold. When a paper says an estimator “is consistent” it means convergence in probability to the true value; when

it says “asymptotically normal” it means a CLT-style convergence in distribution of the rescaled error.

30.7. The bootstrap

A modern technique that makes the CLT unnecessary in many practical cases, and which is worth knowing because it is so easy to implement.

Bootstrap confidence interval

```
given a sample  $x[1..n]$  and a statistic  $T$ :
  repeat  $B$  times ( $B = 10000$ , say):
    draw  $n$  items from  $x$  WITH replacement
    compute  $T$  on that resample
  the 2.5th and 97.5th percentiles of the  $B$  values
  form a 95% confidence interval for  $T$ 
```

The idea: why resampling your own data works.

You want to know how much T would vary if you drew a new sample from the population. You cannot — you only have one sample.

The bootstrap’s move: **use the empirical distribution as a stand-in for the population.** Resampling with replacement from your data simulates drawing fresh samples from that stand-in, and the variability you observe estimates the variability you care about.

It works because the empirical distribution converges to the true one (a uniform version of the LLN), so the simulated variability converges to the real variability.

Its virtues: it needs no formula for the standard error, so it works for medians, ratios, correlations, and arbitrary statistics with no analytic theory. Its costs: it needs the sample to be representative, it struggles for statistics that depend on the extreme tails (the maximum, for instance), and it is B times the compute — which, for a modern machine, is usually nothing.

30.8. Exercises

- 30.1** A random variable has mean 10 and standard deviation 2. Use Chebyshev to bound $\mathbb{P}(X \geq 20)$, then compare with the exact value if X were normal. Comment on the gap.
- 30.2** Use Markov’s inequality to show that for a nonnegative X with $\mathbb{E}[X] = 1$, $\mathbb{P}(X \geq 100) \leq 0.01$. Then construct a distribution achieving equality, showing the bound is tight.
- 30.3** You measure a latency 100 times and get $\bar{x} = 250$ ms with $s = 40$ ms. Give a 95% confidence interval for the mean latency. How many measurements would you need to halve the interval width?
- 30.4** A fair coin is tossed 10,000 times. Use the CLT to estimate the probability of getting between 4900 and 5100 heads.
- 30.5** Prove that if $X_1, \dots, X_n$ are i.i.d. with variance σ^2 , then $\text{Var}(\bar{X}) = \sigma^2/n$. Then redo the computation when the X_i have pairwise correlation ρ , and find the limit as $n \rightarrow \infty$.
- 30.6** Explain why the CLT does not apply to the sample mean of Cauchy-distributed variables, and state what actually happens to $\bar{X}_n$ in that case.

- 30.7** You want to detect a 2% relative improvement in a conversion rate currently at 5%. Using the normal approximation, estimate the sample size per arm needed for the standard error of the difference to be small enough that a 2% relative lift is two standard errors.
- 30.8** You have 200 measurements and want a confidence interval for their **median**, not their mean. Explain why the standard formula does not apply, and describe how you would use the bootstrap instead.

CHAPTER 31.

Estimation and Inference

31.1. The inverse problem

Chapters 27–30 went forwards: given a model, what does the data look like?

Statistics goes backwards: given the data, what is the model? That is a harder question — it has no unique answer, only better and worse ones — and the whole subject is about what “better” should mean.

Definition 31.1 (statistical model and estimator).

A **model** is a family of distributions $p(x; \theta)$ indexed by a parameter θ (possibly a vector).

An **estimator** $\hat{\theta}(X_1, \dots, X_n)$ is any function of the data used to guess θ . Being a function of random data, an estimator is itself a random variable.

Definition 31.2 (how to judge an estimator).

- **Bias:** $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. Zero bias means right **on average**.
- **Variance:** $\text{Var}(\hat{\theta})$. How much it jumps around between samples.
- **Mean squared error:** $\text{MSE} = \mathbb{E}[(\hat{\theta} - \theta)^2]$.
- **Consistency:** $\hat{\theta} \rightarrow \theta$ in probability as $n \rightarrow \infty$.

Theorem 31.3 (bias–variance decomposition).

$$\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}).$$

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)$ and square. The cross term is $2 \mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]] \cdot (\mathbb{E}[\hat{\theta}] - \theta) = 0$, since the first factor has expectation zero. ■

The idea: unbiased is not the goal.

Beginners treat unbiasedness as a requirement. It is not; it is one term of two, and trading a little bias for a lot of variance reduction is usually a win.

This is exactly the bias–variance trade-off in machine learning. A high-capacity model has low bias (it can represent the truth) and high variance (it fits the noise). Regularisation deliberately introduces bias — ridge regression shrinks coefficients towards zero, which is **wrong on average** — and reduces variance enough that total error falls.

The James–Stein estimator is the famous formal example: for estimating three or more means simultaneously, a **biased** shrinkage estimator has strictly lower MSE than the obvious unbiased one, for every value of the parameters. Unbiasedness is not free and it is not sacred.

31.2. Maximum likelihood

Definition 31.4 (likelihood).

Given observed data $x_1, \dots, x_n$, the **likelihood** is

$$\mathcal{L}(\theta) = \prod_{i=1}^n p(x_i; \theta),$$

and the **log-likelihood** is $\ell(\theta) = \sum_i \log p(x_i; \theta)$.

The **maximum likelihood estimator** is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \mathcal{L}(\theta) = \arg \max_{\theta} \ell(\theta).$$

The idea: read the likelihood in the right direction.

$\mathcal{L}(\theta)$ is the probability of the data you actually saw, as a function of the parameter. The data is fixed; θ varies. That is the reverse of how you read $p(x; \theta)$, and getting the direction right is most of understanding likelihood.

MLE picks the parameter that makes the observed data least surprising.

Two reasons to take logs, both from Chapter 4. Numerically, a product of thousands of small probabilities underflows and a sum of logs does not. Analytically, sums differentiate far more pleasantly than products. And the maximiser is unchanged, because log is strictly increasing — a “safe transformation” in the sense of Chapter 2.

Example 31.5 (MLE for a Bernoulli).

n coin flips, k heads. Then

$$\ell(p) = k \log p + (n - k) \log(1 - p).$$

Differentiate and set to zero:

$$\ell'(p) = \frac{k}{p} - \frac{n - k}{1 - p} = 0 \Rightarrow k(1 - p) = (n - k)p \Rightarrow \hat{p} = \frac{k}{n}.$$

The sample proportion. Reassuring, and it did not have to be: the **method** produced the obvious answer rather than the answer being assumed.

Check it is a maximum: $\ell''(p) = -k/p^2 - (n - k)/(1 - p)^2 < 0$, so ℓ is concave and the critical point is the global maximum.

Example 31.6 (MLE for a normal).

With $x_1, \dots, x_n$ from $\mathcal{N}(\mu, \sigma^2)$,

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2.$$

$\partial \ell / \partial \mu = 0$ gives $\sum (x_i - \mu) = 0$, so $\hat{\mu} = \bar{x}$.

$\partial \ell / \partial \sigma^2 = 0$ gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \hat{\mu})^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2.$$

Note the $1/n$. This estimator is **biased**: its expectation is $((n - 1)/n)\sigma^2$, because $\bar{x}$ was itself estimated from the same data and sits slightly closer to the points than the true μ does. Dividing

by $n - 1$ instead gives the unbiased **sample variance** — and that is where the mysterious $n - 1$ comes from. One degree of freedom was consumed estimating the mean.

Theorem 31.7 (least squares is maximum likelihood under Gaussian noise).

If $y_i = f(x_i; \theta) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independent, then

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \sum_i (y_i - f(x_i; \theta))^2.$$

Proof.

$$\ell(\theta) = \sum_i \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y_i - f(x_i; \theta))^2}{2\sigma^2} \right].$$

The first term does not involve θ ; maximising the rest is minimising the sum of squares. ■

The idea: every loss function is a distributional assumption.

This theorem is the reason to care about likelihood even if you never do statistics. It says that **choosing a loss function is choosing a noise model**, whether or not you realise it:

Loss	Implied noise model
Squared error	Gaussian
Absolute error	Laplace (heavier tails, robust)
Cross-entropy	Categorical / Bernoulli
Poisson deviance	Poisson counts
Huber	Gaussian centre, Laplace tails

So “my model is sensitive to outliers” is not a mysterious pathology — it is the Gaussian assumption in your squared loss telling you it did not expect outliers, and correctly concluding that an extreme point must be very informative. Switching to absolute or Huber loss is switching to a noise model with heavier tails, which discounts extremes.

Similarly, adding ℓ_2 regularisation is placing a Gaussian **prior** on the parameters, and ℓ_1 regularisation is a Laplace prior. Chapter 34’s regularisers all have this reading.

Theorem 31.8 (properties of the MLE).

Under regularity conditions, as $n \rightarrow \infty$ the MLE is:

- **consistent:** $\hat{\theta} \rightarrow \theta_0$ in probability;
- **asymptotically normal:** $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow \mathcal{N}(0, I(\theta_0)^{-1})$;
- **asymptotically efficient:** no consistent estimator has smaller asymptotic variance.

Definition 31.9 (Fisher information).

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p(X; \theta) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log p(X; \theta) \right].$$

Theorem 31.10 (Cramér–Rao lower bound).

For any unbiased estimator,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}.$$

The idea: information is curvature.

Fisher information is (minus) the expected **curvature** of the log-likelihood at the truth.

A sharply peaked log-likelihood means the data strongly discriminates between nearby parameter values: high curvature, high information, low achievable variance. A flat log-likelihood means many parameter values explain the data about equally well: low information, and no estimator can do better than badly.

The Cramér–Rao bound is the statistical analogue of a complexity lower bound. It says: this is the best any unbiased estimator can do, so stop looking. And the MLE attains it asymptotically, which is why the MLE is the default.

In practice, the inverse Hessian of the negative log-likelihood at the optimum estimates I^{-1} , which is how standard errors are computed for fitted models — including for a logistic regression, where the reported coefficient standard errors are exactly this.

31.3. Bayesian inference

Definition 31.11 (the Bayesian setup).

Treat θ as a random variable with a **prior** $p(\theta)$. After observing data D , Bayes’ theorem (Chapter 27) gives the **posterior**

$$p(\theta \mid D) = \frac{p(D|\theta)p(\theta)}{p(D)} \propto \mathcal{L}(\theta)p(\theta).$$

Definition 31.12 (point estimates from a posterior).

- **MAP** (maximum a posteriori): $\arg \max_{\theta} p(\theta \mid D)$ — the mode.
- **Posterior mean**: $\mathbb{E}[\theta \mid D]$ — minimises posterior squared error (Chapter 29).
- **Credible interval**: a region containing, say, 95% of the posterior mass. This **does** mean “95% probability the parameter is in here”, unlike a confidence interval.

Proposition 31.13 (MAP is regularised MLE).

$$\arg \max_{\theta} [\log \mathcal{L}(\theta) + \log p(\theta)].$$

A Gaussian prior $\mathcal{N}(0, \tau^2 I)$ contributes $-\|\theta\|^2/(2\tau^2)$ — exactly ℓ_2 regularisation. A Laplace prior contributes $-\|\theta\|_1/b$ — exactly ℓ_1 .

So ridge regression is MAP estimation with a Gaussian prior, and lasso is MAP with a Laplace prior. The regularisation strength λ is the inverse prior variance: strong regularisation means a confident prior that the coefficients are small.

Example 31.14 (Beta-Bernoulli, worked).

Estimate a coin's bias p from k heads in n flips, with prior $p \sim \text{Beta}(\alpha, \beta)$, whose density is proportional to $p^{\alpha-1}(1-p)^{\beta-1}$.

$$p(p \mid D) \propto \underbrace{p^k(1-p)^{n-k}}_{\text{likelihood}} \cdot \underbrace{p^{\alpha-1}(1-p)^{\beta-1}}_{\text{prior}} = p^{k+\alpha-1}(1-p)^{n-k+\beta-1},$$

which is $\text{Beta}(\alpha + k, \beta + n - k)$. The posterior is in the same family as the prior — the Beta is **conjugate** to the Bernoulli — so updating is just adding counts.

Posterior mean:

$$\frac{\alpha + k}{\alpha + \beta + n}.$$

Read the prior parameters as **pseudo-counts**: α imagined heads and β imagined tails, seen before any data. With $\alpha = \beta = 1$ (a uniform prior) the estimate is $(k + 1)/(n + 2)$ — **Laplace's rule of succession**, and a far better default than k/n when n is small.

With $n = 0$ the MLE k/n is undefined; with $k = 0$ it says the event is impossible, which is a strong claim from no evidence. The Bayesian estimate degrades gracefully in both cases, which is exactly why additive smoothing is standard practice in naive Bayes classifiers and in language modelling.

The idea: frequentist versus Bayesian, practically.

The philosophical argument is old and mostly unproductive. The practical differences:

Priors. Bayesian methods require one. That is a cost (it is a choice you must defend) and a benefit (you can encode real knowledge, and you get sensible answers from small samples).

What you get out. A frequentist gives you a point estimate and a procedure with long-run guarantees. A Bayesian gives you a full distribution over parameters, which composes: uncertainty propagates naturally through downstream calculations.

Compute. Frequentist methods are usually cheap and closed-form. Bayesian posteriors usually are not, requiring MCMC or variational approximation. This was decisive before about 1990 and matters much less now.

Convergence. With enough data they agree: the likelihood swamps any reasonable prior, and the posterior concentrates around the MLE. The disagreement is about the small-data regime — which is, unfortunately, the regime most real decisions are made in.

31.4. Hypothesis testing

Definition 31.15 (the setup).

A **null hypothesis** H_0 (the boring explanation) and an **alternative** H_1 . Choose a test statistic T , compute its distribution **assuming H_0 is true**, and see whether the observed value is extreme.

The **p-value** is

$$p = \mathbb{P}(T \text{ at least as extreme as observed} \mid H_0).$$

Reject H_0 if $p < \alpha$, a threshold fixed in advance.

Definition 31.16 (the two errors).

- **Type I** (false positive): rejecting a true H_0 . Probability α .
- **Type II** (false negative): failing to reject a false H_0 . Probability β ; the **power** is $1 - \beta$.

Trap: what a p-value is not.

A p-value is $\mathbb{P}(\text{data} \mid H_0)$. It is **not** $\mathbb{P}(H_0 \mid \text{data})$. Those differ by exactly the Bayes factor of Chapter 27, and confusing them is the base-rate fallacy in another costume.

Specifically, a p-value is not:

- the probability the null hypothesis is true;
- the probability the result was due to chance;
- a measure of effect size — with enough data, a trivially small effect gives a tiny p-value;
- evidence **for** H_0 when it is large; failing to reject is not accepting.

And **p-hacking** is a real mechanism, not a moralism. Test 20 independent hypotheses at $\alpha = 0.05$ and the probability of at least one false positive is $1 - 0.95^{20} \approx 64\%$. Multiple comparisons must be corrected for — Bonferroni (α/m , conservative) or Benjamini–Hochberg (controls the false discovery rate, more powerful) — and the number of tests includes the ones you tried and did not report.

Example 31.17 (the standard tests, and when each applies).

Test	Use
z -test	Mean, known variance, large n
t -test	Mean, unknown variance — uses the t distribution, which has heavier tails than the normal to account for estimating σ
Paired t -test	Before/after on the same units; removes between-unit variance
Chi-squared	Goodness of fit; independence in a contingency table
Mann–Whitney	Two groups, no distributional assumption (rank-based)
Kolmogorov–Smirnov	Does a sample come from a given distribution
ANOVA	Comparing more than two group means at once

Example 31.18 (an A/B test, done properly).

Control: 1000 users, 50 conversions ($\hat{p}_A = 0.05$). Treatment: 1000 users, 65 conversions ($\hat{p}_B = 0.065$).

Under H_0 (no difference), pool the rates: $\hat{p} = 115/2000 = 0.0575$. The standard error of the difference is

$$\text{SE} = \sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)} = \sqrt{0.0575 \times 0.9425 \times 0.002} \approx 0.0104.$$

The test statistic is

$$z = \frac{0.065 - 0.05}{0.0104} \approx 1.44,$$

giving a two-sided p-value of about 0.15. Not significant at $\alpha = 0.05$.

The right conclusion is **not** “there is no effect”. It is “this experiment could not distinguish a 1.5-point lift from noise”. A power calculation done **before** running it would have said so: detecting a lift of this size with 80% power needs roughly 3500 users per arm.

Running the test first and computing power afterwards — or worse, peeking at the results daily and stopping when $p < 0.05$ — inflates the false positive rate substantially. Sequential testing requires its own machinery (alpha spending, or always-valid confidence sequences).

31.5. Exercises

- 31.1** Derive the MLE for the rate λ of a Poisson distribution given observations $x_1, \dots, x_n$. Verify it is a maximum.
- 31.2** Derive the MLE for the parameter λ of an exponential distribution, and check whether it is unbiased.
- 31.3** Show that the sample variance with divisor $n - 1$ is unbiased, i.e. $\mathbb{E}[(1/(n - 1)) \sum (X_i - \bar{X})^2] = \sigma^2$. (Expand and use $\mathbb{E}[\bar{X}^2] = \mu^2 + \sigma^2/n$.)
- 31.4** A coin is flipped 10 times, giving 7 heads. Give the MLE for p , and the posterior mean under a $\text{Beta}(1, 1)$ prior and under a $\text{Beta}(10, 10)$ prior. Explain the difference.
- 31.5** Show that MAP estimation with a Gaussian prior $\mathcal{N}(0, \tau^2 I)$ and Gaussian likelihood gives exactly ridge regression, and identify λ in terms of τ^2 and the noise variance σ^2 .
- 31.6** You run 40 independent A/B tests, all with no true effect, at $\alpha = 0.05$. What is the expected number of “significant” results, and the probability of at least one? What Bonferroni threshold would keep the family-wise error rate at 5%?
- 31.7** Compute the Fisher information for a Bernoulli parameter p , and use the Cramér–Rao bound to state the minimum possible variance of an unbiased estimator from n samples. Compare with the actual variance of $\hat{p} = k/n$.
- 31.8** Design an A/B test to detect a relative lift of 10% on a baseline conversion rate of 2%, with 80% power at $\alpha = 0.05$. Estimate the required sample size per arm using the normal approximation.

CHAPTER 32.

Information Theory

32.1. Measuring surprise

Shannon's question in 1948: how much information is in a message?

The answer turns out to be a single formula, derivable from a handful of requirements, and it explains simultaneously why compression has a hard limit, why decision trees split the way they do, and why the loss function in every classifier is what it is.

The idea: information is surprise.

Start from what “information” ought to mean.

Learning that something **certain** happened tells you nothing. Learning that something **unlikely** happened tells you a lot. So information should be a decreasing function of probability.

And information from **independent** sources should add: learning two unrelated facts should give you the sum. But independent probabilities **multiply**.

So we need a function turning multiplication into addition. Chapter 4 has exactly one: the logarithm.

$$I(x) = -\log p(x) = \log \frac{1}{p(x)}.$$

The minus sign makes it positive (probabilities are at most 1, so their logs are at most 0). With base 2 the unit is the **bit**; with base e , the **nat**.

Sanity check: an event of probability $1/2$ carries 1 bit. An event of probability $1/1024$ carries 10 bits. A certain event carries 0. Those are the right answers.

32.2. Entropy

Definition 32.1 (entropy).

$$H(X) = -\sum_x p(x) \log p(x) = \mathbb{E}[-\log p(X)].$$

The **average** surprise. Convention: $0 \log 0 = 0$, justified since $\lim_{p \rightarrow 0} p \log p = 0$.

Example 32.2 (a fair coin, a biased coin, a die).

Fair coin: $H = -2 \times (1/2) \log_2(1/2) = 1$ bit.

Biased coin, $p = 0.9$: $H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.47$ bits. Less uncertain, so less information per flip.

Certain coin, $p = 1$: $H = 0$. No information at all.

Fair die: $H = \log_2 6 \approx 2.58$ bits.

Theorem 32.3 (entropy is maximised by the uniform distribution).

For a distribution on n outcomes, $H(X) \leq \log n$, with equality iff the distribution is uniform.

Proof. By Jensen's inequality (Chapter 28) applied to the concave function $\log$:

$$H(X) = \sum_x p(x) \log \frac{1}{p(x)} \leq \log \left(\sum_x p(x) \cdot \frac{1}{p(x)} \right) = \log n.$$

Equality in Jensen for a strictly concave function requires the argument to be constant, i.e. all $p(x)$ equal. ■

The idea: entropy is the cost of a description.

The operational meaning, and the reason entropy is not just a formula.

Shannon's source coding theorem: the average number of bits needed to encode symbols from a source with entropy H is at least H per symbol, and codes exist achieving arbitrarily close to it.

So entropy is **the compression limit**. You cannot do better, and you can get as close as you like.

This also explains Huffman coding: assign short codes to frequent symbols and long codes to rare ones, with length approximately $-\log_2 p(x)$. The average length is then approximately H , which is optimal. Arithmetic coding does better still by removing the integer-length constraint.

And it explains why random data does not compress: a uniform distribution over 2^n possibilities has entropy exactly n bits, so an n -bit random string needs n bits. The pigeonhole argument of Chapter 9 was the crude version of this; entropy is the quantitative version.

32.3. Cross-entropy and KL divergence

Definition 32.4 (cross-entropy).

$$H(p, q) = - \sum_x p(x) \log q(x).$$

The average number of bits used if you encode data from p using a code optimised for q .

Definition 32.5 (Kullback–Leibler divergence).

$$\text{KL}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = H(p, q) - H(p).$$

The **extra** bits wasted by using the wrong distribution.

Theorem 32.6 (Gibbs' inequality).

$\text{KL}(p \parallel q) \geq 0$, with equality iff $p = q$.

Proof. Apply Jensen to the concave $\log$:

$$-\text{KL}(p \parallel q) = \sum_x p(x) \log \frac{q(x)}{p(x)} \leq \log \left(\sum_x p(x) \frac{q(x)}{p(x)} \right) = \log \sum_x q(x) = \log 1 = 0.$$

Equality requires $q(x)/p(x)$ constant, and since both sum to 1 that constant is 1. ■

Trap: KL is not a distance.

$\text{KL}(p\|q) \neq \text{KL}(q\|p)$ in general — it is not symmetric — and it does not satisfy the triangle inequality. Calling it a “distance” will mislead you, which is why the literature says **divergence**.

The asymmetry is not a defect; it is meaningful, and choosing the direction is a real modelling decision.

$\text{KL}(p\|q)$ with p the truth is **mean-seeking**: it is enormous wherever p has mass and q does not, so a q minimising it must cover all of p ’s support. Fitting by maximum likelihood does this, and it is why an over-simple model fitted by MLE spreads itself thin to cover everything.

$\text{KL}(q\|p)$ is **mode-seeking**: it punishes q for putting mass where p has none, so q prefers to hug one mode and ignore the others. Variational inference minimises this direction, which is why variational approximations are notoriously over-confident and under-dispersed.

If you need a symmetric measure, use the Jensen–Shannon divergence $\text{JS}(p, q) = \frac{1}{2} \text{KL}(p\|m) + \frac{1}{2} \text{KL}(q\|m)$ with $m = (p + q)/2$, whose square root is a metric.

The idea: why cross-entropy is the loss function.

Here is the four-line derivation promised in the preface.

A classifier outputs a distribution q over labels; the truth is a distribution p (usually one-hot). You want q close to p , so minimise $\text{KL}(p\|q)$.

But $\text{KL}(p\|q) = H(p, q) - H(p)$, and $H(p)$ does not depend on your model at all — it is a property of the data. So minimising KL is **identical** to minimising the cross-entropy $H(p, q)$.

With one-hot p , all but one term vanishes and

$$H(p, q) = -\log q(\text{true label}).$$

That is the cross-entropy loss, exactly as implemented. It is not an arbitrary choice; it is the KL divergence with a constant dropped.

And one more equivalence: minimising $-\log q(\text{true label})$ summed over the dataset is **maximising the log-likelihood** (Chapter 31). Maximum likelihood, minimum cross-entropy, and minimum KL divergence are three names for one optimisation problem.

32.4. Mutual information

Definition 32.7 (mutual information).

$$I(X; Y) = \text{KL}(p(x, y) \| p(x)p(y)) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

Equivalently,

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y).$$

The idea: mutual information is dependence of any kind.

Read the first form: it is the KL divergence between the true joint and what the joint **would** be if X and Y were independent. So it measures exactly how far from independent they are.

By Gibbs, $I(X; Y) \geq 0$ always, and $I(X; Y) = 0$ **if and only if** X and Y are independent.

That “if and only if” is the crucial difference from correlation. Chapter 29’s example — $Y = X^2$ with X symmetric — has zero correlation and **positive** mutual information, correctly reporting that the variables are dependent. Correlation sees only linear structure; mutual information sees all of it.

The second form reads: **the reduction in uncertainty about X from learning Y** . That is the reading used in feature selection and in decision trees.

Example 32.8 (decision trees are greedy entropy reduction).

A decision tree chooses each split to maximise **information gain**:

$$\text{IG} = H(\text{parent}) - \sum_{\text{children}} \frac{n_c}{n} H(\text{child}),$$

which is exactly $I(Y; \text{split})$ — the mutual information between the label and the split decision.

So the tree greedily asks, at each node, “which question tells me the most about the label?” That is a direct application of this chapter, and it is why the algorithm is called ID3/C4.5 information gain.

(Many implementations use Gini impurity instead, which is a computationally cheaper proxy behaving similarly — it is the second-order Taylor expansion of entropy about the uniform distribution.)

32.5. Continuous entropy, briefly

Definition 32.9 (differential entropy).

$$h(X) = - \int f(x) \log f(x) \, dx.$$

Two warnings. Differential entropy can be **negative** (a distribution concentrated in a small interval has density greater than 1 there), and it is **not** invariant under change of variables — rescaling X shifts h by log of the scale factor. So it is not “the amount of information in a continuous variable” in any absolute sense.

What **is** well behaved is KL divergence and mutual information between continuous variables: the offending terms cancel, and both remain invariant under invertible reparameterisation. Use those.

Theorem 32.10 (the normal maximises entropy for a given variance).

Among all distributions on $\mathbb{R}$ with variance σ^2 , the one maximising differential entropy is $\mathcal{N}(0, \sigma^2)$, with $h = \frac{1}{2} \log(2\pi e \sigma^2)$.

This is the second of the three reasons for the normal distribution’s ubiquity, promised in Chapter 28. If all you know is a mean and a variance, the normal is the distribution that assumes nothing further — the **maximum-entropy** choice.

The pattern generalises: fix the mean of a positive variable and you get the exponential; fix the support only and you get the uniform; fix $\mathbb{E}[\log X]$ and $\mathbb{E}[X]$ and you get the gamma. Every distribution in the exponential family is the maximum-entropy distribution subject to some set

of moment constraints, which is a satisfying explanation for why that particular family keeps appearing.

32.6. Where else this shows up

Coding and compression. Huffman, arithmetic coding, and Lempel–Ziv are all attempts to reach the entropy bound. Modern neural compressors are literally language models plus arithmetic coding: a better model means a lower cross-entropy means fewer bits.

Perplexity. Language models report 2^H (or e^H) rather than cross-entropy. It is the same number on a different scale, interpretable as “the model is as confused as if it were choosing uniformly among this many options”. A perplexity of 20 means the model is effectively picking from 20 equally likely words.

Channel capacity. Shannon’s other theorem: a noisy channel has a maximum reliable rate $C = \max_{p(x)} I(X; Y)$, and error-correcting codes can approach it. This is why your modem, your disk, and your phone all work.

The information bottleneck. One theoretical account of representation learning: a good hidden representation T maximises $I(T; Y)$ (predictive of the label) while minimising $I(T; X)$ (compressed away from the input). Whether this explains deep learning is contested; it is at least a clean way to state what a representation is for.

Kolmogorov complexity. An alternative definition of information: the length of the shortest program that outputs the string. It is uncomputable (by a diagonal argument descended from Chapter 1) but it connects to Shannon entropy — for a random string from a known source, the expected Kolmogorov complexity is approximately its Shannon entropy. It is also the formal grounding of Occam’s razor and of the minimum description length principle.

32.7. Exercises

- 32.1** Compute the entropy in bits of: (a) a fair 8-sided die; (b) a distribution with probabilities $(0.5, 0.25, 0.125, 0.125)$; (c) a distribution with probabilities $(0.97, 0.01, 0.01, 0.01)$. Comment on the relationship between entropy and how compressible each source is.
- 32.2** For (b) above, construct a Huffman code and verify its average codeword length equals the entropy exactly. Then determine whether (a) achieves equality, and explain why (c) cannot.
- 32.3** Show that $H(X, Y) = H(X) + H(Y)$ if and only if X and Y are independent.
- 32.4** Compute $KL(p||q)$ and $KL(q||p)$ for $p = (0.5, 0.5)$ and $q = (0.9, 0.1)$, and confirm they differ. Which is larger, and can you say why in terms of the mean-seeking versus mode-seeking discussion?
- 32.5** Prove that cross-entropy loss with a one-hot target reduces to $-\log q_y$, and show that its gradient with respect to the logits is $p - y$ (you may quote the softmax Jacobian from Chapter 26).
- 32.6** A dataset has 60 positive and 40 negative examples. A candidate split sends 50 examples one way (45 positive, 5 negative) and 50 the other. Compute the information gain of this split in bits.
- 32.7** Show that $I(X; Y) = H(X) - H(X|Y)$ follows from the definition, and use it to explain why mutual information is symmetric even though conditional entropy is not.

- 32.8** A language model achieves a cross-entropy of 3.2 nats per token on a test set. What is its perplexity? If a better model reaches 2.9 nats, by what factor has the perplexity improved, and roughly how much smaller would the compressed text be?

PART VI

Computation, Geometry, and Learning

Where the mathematics meets the machine. Floating point and numerical stability, root finding and quadrature, optimization, the geometry behind graphics and simulation, and finally a full derivation of backpropagation and the mathematics of a transformer.

CHAPTER 33.

Numerical Computing

33.1. The gap between mathematics and machines

Every previous chapter assumed exact real arithmetic. Your machine does not have it. This chapter is about the difference, and it is not a footnote: most “the code is right but the answer is wrong” bugs in numerical work live here.

The organising distinction:

Definition 33.1 (conditioning and stability).

A **problem** is **ill-conditioned** if small changes in the input cause large changes in the exact answer. This is a property of the problem, and no algorithm can fix it.

An **algorithm** is **unstable** if it introduces errors much larger than the problem’s conditioning requires. This is the algorithm’s fault, and a better algorithm can fix it.

The idea: assign blame correctly.

When your numerical result is garbage, exactly one of two things happened.

Either the **problem** was hopeless — you asked for the intersection of two nearly parallel lines, and the answer genuinely does depend enormously on the tenth digit of the input. Then no code will save you and you need to reformulate.

Or your **method** was bad — you subtracted two nearly equal numbers, or eliminated with a tiny pivot, or squared a condition number by forming $A^T A$. Then a different method with the same mathematics will work.

The professional habit is to ask which before reaching for higher precision. Doubling precision buys 16 more digits, which is a lot when the algorithm is slightly unstable and nothing at all when $\kappa = 10^{30}$.

33.2. Floating point

Definition 33.2 (IEEE 754 binary64).

A double is $\pm 1.f \times 2^e$: one sign bit, 11 exponent bits (biased), 52 stored fraction bits giving 53 bits of significand (the leading 1 is implicit for normal numbers).

- **Machine epsilon:** $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$, the gap between 1 and the next representable number.
- **Range:** roughly 10^{-308} to 10^{308} , plus subnormals down to 10^{-324} at reduced precision.
- **Special values:** ± 0 , $\pm\infty$, and NaN.

Theorem 33.3 (the fundamental axiom of floating point).

For any basic operation $\circ \in \{+, -, \times, \div\}$ on representable a, b ,

$$\text{fl}(a \circ b) = (a \circ b)(1 + \delta), \quad |\delta| \leq \varepsilon/2.$$

Each individual operation is correctly rounded, with tiny **relative** error.

The word **relative** is the key. Absolute error scales with magnitude: near 1 the spacing is 10^{-16} ; near 10^9 it is 10^{-7} ; beyond 2^{53} consecutive integers are not representable at all.

Trap: the consequences you must internalise.

Never test equality. $0.1 + 0.2 == 0.3$ is false, because none of the three is representable exactly (Chapter 1). Test `abs(a-b) <= max(atol, rtol * max(abs(a), abs(b)))` — absolute tolerance for values near zero, relative tolerance elsewhere.

Addition is not associative. $(a + b) + c \neq a + (b + c)$. Parallel reductions give different answers on different runs and different thread counts. If you need reproducibility you need a fixed reduction order, or compensated summation.

NaN is not equal to itself. `NaN == NaN` is false, by design, so `x != x` is the standard NaN test. NaN also poisons everything it touches, which is useful for debugging: the first NaN is where the bug is, and everything after it is a symptom.

Integers above 2^{53} lose precision as doubles. This is why JSON is not safe for 64-bit IDs, and why they get quoted as strings.

Subnormals are slow. On many CPUs, arithmetic on subnormal numbers takes a hundred times longer, and audio code and physics simulations that decay towards zero hit this. Flush-to-zero mode exists for this reason.

33.3. Catastrophic cancellation

The single most important failure mode.

The idea: subtraction does not lose precision — it reveals loss.

Take $a = 1.0000000000000002$ and $b = 1.0000000000000000$, each with 16 correct digits. Their difference is 2×10^{-16} , computed exactly — the subtraction itself is fine.

The problem is that the **inputs** already had rounding error of order 10^{-16} , which was invisible when it was in the 16th digit of a number near 1, and is now the **entire content** of the result. The relative error went from 10^{-16} to order 1.

Cancellation does not create error. It promotes existing error from irrelevant to dominant. Which is why the cure is never “be more careful with the subtraction” and always “rearrange so the subtraction does not happen”.

Example 33.4 (four cancellations and their fixes).

The quadratic formula. When $b^2 \gg 4ac$, $-b + \sqrt{b^2 - 4ac}$ cancels. Fix (Chapter 2): compute the root with matching signs, then get the other from $x_1 x_2 = c/a$.

Variance. $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$ cancels when the mean is large relative to the spread. Fix (Chapter 28): Welford’s algorithm.

$\log(1 + x)$ for tiny x . Computing $1.0 + x$ rounds away x ’s low bits. Fix: `log1p(x)`, which evaluates the series directly. Likewise `expm1`.

$\sqrt{x+1} - \sqrt{x}$ for large x . Two nearly equal numbers. Fix: multiply by the conjugate,

$$\sqrt{x+1} - \sqrt{x} = \frac{1}{\sqrt{x+1} + \sqrt{x}},$$

which has no subtraction at all. Algebraically identical, numerically entirely different.

Kahan compensated summation

```
sum := 0.0
c   := 0.0          # running compensation for lost low-order bits
for each x:
    y := x - c       # add back what was lost last time
    t := sum + y      # this rounds, losing the low bits of y
    c := (t - sum) - y # recover exactly what was lost
    sum := t
```

Naive summation of n terms has error growing like $n\varepsilon$; Kahan's is bounded by about 2ε regardless of n . The trick is that $(t - \text{sum}) - y$ computes the rounding error of the previous addition exactly — an identity that holds because floating-point subtraction of nearby numbers is exact.

33.4. Condition numbers

Definition 33.5 (condition number of a function).

$$\kappa_f(x) = \left| \frac{x f'(x)}{f(x)} \right|.$$

The factor by which relative input error is amplified into relative output error.

Example 33.6.

$f(x) = \sqrt{x}$: $\kappa = 1/2$. Well conditioned everywhere — square roots are safe.

$f(x) = x_1 - x_2$ for $x_1 \approx x_2$: the condition number blows up as the arguments approach each other. This is cancellation, quantified.

$f(x) = \tan(x)$ near $\pi/2$: enormous. A tiny error in x produces an arbitrary output.

Root-finding for a polynomial with nearly-repeated roots: enormous, which is Wilkinson's polynomial and the reason Chapter 23 warned against the characteristic-polynomial route to eigenvalues.

For linear systems the condition number is $\kappa(A) = \sigma_1/\sigma_n$ (Chapter 25), and the rule of thumb is: with $\kappa \approx 10^k$, expect to lose k decimal digits. Always check it before trusting a solve. `numpy` exposes it as `np.linalg.cond`.

33.5. Root finding

Method	Order	Needs	Notes
Bisection	linear	sign change	Always converges. One bit per step.
Secant	≈ 1.618	two points	No derivative needed. Golden ratio order — genuinely.
Newton	quadratic	f'	Fastest, can diverge (Chapter 15).
Brent	superlinear	sign change	Hybrid. Guaranteed and fast. The default.

The idea: why Brent's method is the right answer.

Bisection is guaranteed and slow. Newton is fast and can fail. Brent's method keeps a bracket at all times — so it can never diverge — but attempts an inverse quadratic interpolation step at each iteration and takes it only if it lands inside the bracket and is making adequate progress, falling back to bisection otherwise.

Guaranteed convergence with near-Newton speed. This **hybridise a safe method with a fast one** pattern recurs throughout numerical computing: trust-region optimisation, adaptive quadrature, and line searches all do it.

Use your library's `brentq`. Do not hand-roll a Newton iteration for production root-finding.

33.6. Numerical differentiation and integration

Chapter 14 established the trade-off for differentiation: truncation error falls with h , roundoff error rises, and the optimum is $h \approx \sqrt{\varepsilon}$ for forward differences (about 8 good digits) or $\varepsilon^{1/3}$ for central differences (about 10). Automatic differentiation avoids the trade-off entirely and is what you should use when you control the code.

For integration, Chapter 16 gave midpoint, trapezoid, and Simpson. Two things to add.

Definition 33.7 (Gaussian quadrature).

Approximate $\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i f(x_i)$, choosing **both** the nodes x_i and the weights w_i optimally. With n nodes this is exact for all polynomials of degree up to $2n - 1$.

The idea: why choosing the nodes doubles the accuracy.

Newton–Cotes rules (trapezoid, Simpson) fix the nodes at equal spacing and choose only the n weights: n free parameters, so exactness for degree $n - 1$.

Gauss chooses the nodes too: $2n$ free parameters, so exactness for degree $2n - 1$. The optimal nodes turn out to be the roots of the Legendre polynomials, and they cluster towards the endpoints.

A 5-point Gauss rule is exact for degree-9 polynomials. Simpson with 5 points is exact for degree 3. The gap is enormous and it comes from nothing more than not insisting on an even grid.

The catch: Gauss nodes are fixed by n , so refining means recomputing everything. Gauss–Kronrod extends a Gauss rule with extra points so the two estimates share evaluations and their difference gives an error estimate. That is what `scipy.integrate.quad` does, inside an adaptive subdivision loop.

Also worth knowing: for high-dimensional integrals nothing grid-based works (the curse of dimensionality), and Monte Carlo's dimension-independent $O(1/\sqrt{n})$ rate wins — Chapter 27.

33.7. Ordinary differential equations

Definition 33.8 (initial value problem).

Find $y(t)$ with $y' = f(t, y)$, $y(t_0) = y_0$.

Method	Step	Error per step / global
Forward Euler	$y_{n+1} = y_n + hf(t_n, y_n)$	$O(h^2)$ / $O(h)$
Backward Euler	$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$	$O(h^2)$ / $O(h)$, implicit
Midpoint (RK2)	evaluate at the half-step	$O(h^3)$ / $O(h^2)$
RK4	four evaluations, weighted	$O(h^5)$ / $O(h^4)$

Classical fourth-order Runge–Kutta

```

k1 := f(t, y)
k2 := f(t + h/2, y + h*k1/2)
k3 := f(t + h/2, y + h*k2/2)
k4 := f(t + h, y + h*k3)
y := y + h*(k1 + 2*k2 + 2*k3 + k4)/6

```

RK4 is four function evaluations per step for fourth-order accuracy, which is an unusually good deal, and it is the default for non-stiff problems.

The idea: stiffness, and why implicit methods exist.

A system is **stiff** when it contains processes on wildly different timescales — a fast transient that decays almost immediately alongside slow dynamics you actually care about.

Explicit methods are **stability-limited**: forward Euler on $y' = -\lambda y$ requires $h < 2/\lambda$ or the numerical solution oscillates and blows up, even though the true solution decays smoothly. With $\lambda = 10^6$ from some irrelevant fast mode, you are forced to take a million steps to integrate one second of interesting behaviour.

Implicit methods have no such limit. Backward Euler on $y' = -\lambda y$ gives $y_{n+1} = y_n/(1 + h\lambda)$, which decays for **every** $h > 0$: unconditionally stable. The price is that each step requires solving an equation for y_{n+1} — typically Newton’s method with a Jacobian, so each step is far more expensive. For stiff problems it is still a massive win.

Diagnosing stiffness: if your explicit solver is taking absurdly small steps and the solution looks smooth, you are stiff. Switch to an implicit solver (BDF, Radau) and the problem usually evaporates.

For **stiff** problems, and for anything you actually care about, use a library: adaptive step size control (estimate the local error, shrink or grow h to hit a tolerance) is what makes solvers usable, and it is fiddly to implement correctly.

33.8. Practical advice

1. **Use the library.** LAPACK, BLAS, SciPy, and Eigen encode decades of hard-won numerical judgement. Your hand-rolled version will be slower and less accurate.
2. **Check the condition number** before trusting any linear solve or fit.
3. **Prefer orthogonal transformations.** QR over normal equations, SVD over eigendecomposition of $A^T A$, Householder over Gram–Schmidt.

4. **Scale your variables.** If one feature is in nanoseconds and another in terabytes, your matrix is ill-conditioned for a purely cosmetic reason. Normalising features is not just an ML convention; it is conditioning.
5. **Work in log space** for products of probabilities (Chapter 4), and use `logsumexp` rather than `log(sum(exp(...)))`.
6. **Test against a slow reference.** Compare your fast method to a naive high-precision implementation on small inputs. Most numerical bugs are visible at $n = 3$.
7. **Do not use `float32` because it is faster, without checking.** It has $\varepsilon \approx 10^{-7}$ — about 7 decimal digits — which is fine for neural network weights and disastrous for accumulating a long sum or inverting an ill-conditioned matrix. Mixed precision (compute in `float32`, accumulate in `float32` or higher) exists for exactly this reason.

33.9. Exercises

- 33.1** Compute $\sqrt{10001} - \sqrt{10000}$ (a) directly and (b) using the conjugate rearrangement, to 4 significant figures at each intermediate step. Compare with the true value 0.0049999.
- 33.2** Explain why $0.1 + 0.2 \neq 0.3$ in binary floating point, referring to the repeating-expansion result of Chapter 1. What is the actual value of $0.1 + 0.2$ to 20 digits?
- 33.3** Estimate the condition number of $f(x) = 1/(1 - x)$ at $x = 0.999$, and state how many digits you would expect to lose.
- 33.4** Implement (on paper) three iterations of Kahan summation on the values 1.0, 10^{-16} , 10^{-16} in a system with $\varepsilon = 2^{-52}$, and compare with naive summation.
- 33.5** Apply two steps of RK4 to $y' = y$, $y(0) = 1$, with $h = 0.5$. Compare with the exact answer e and with two steps of forward Euler.
- 33.6** For the stiff equation $y' = -1000y$, find the largest step size for which forward Euler is stable, and verify that backward Euler is stable for every positive step size.
- 33.7** A 3×3 linear system has condition number 10^7 and your inputs are accurate to 10 significant digits. How many digits of the solution can you trust? What if you switched to `float32` inputs?
- 33.8** You need $\int_0^1 x^7 dx$. How many Gauss–Legendre nodes are needed for an **exact** answer, and how many Simpson subintervals would give comparable accuracy? Explain the gap.

CHAPTER 34.

Optimization

34.1. The shape of the problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad h_j(x) = 0.$$

Training a model, fitting a curve, allocating resources, planning a route, scheduling jobs — all of it is this. What determines whether the problem is easy or hopeless is not the size of n ; it is **convexity**.

34.2. Convexity

Definition 34.1 (convex set and convex function).

A set C is **convex** if for all $x, y \in C$ and $t \in [0, 1]$, $tx + (1 - t)y \in C$: the segment between any two points stays inside.

A function f on a convex set is **convex** if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y),$$

i.e. the chord lies above the graph. **Strictly** convex if the inequality is strict for $x \neq y$.

Theorem 34.2 (characterisations of convexity).

For differentiable f , convexity is equivalent to

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \text{for all } x, y,$$

and for twice-differentiable f , to the Hessian $H(x)$ being positive semidefinite everywhere.

The first-order form says the tangent plane lies **below** the function everywhere — so a local linear model never overpromises. The second-order form is the n -dimensional version of $f'' \geq 0$.

Theorem 34.3 (the reason convexity matters).

For a convex f on a convex set:

- (a) Every local minimum is a global minimum.
- (b) The set of minimisers is convex.
- (c) If f is strictly convex, the minimiser is unique.
- (d) $\nabla f(x) = 0$ is **sufficient**, not merely necessary, for a global minimum.

Proof. (a) Suppose x is a local minimum and $f(y) < f(x)$ for some y . For small $t > 0$,

$$f(x + t(y - x)) \leq (1 - t)f(x) + tf(y) < f(x),$$

giving points arbitrarily close to x with smaller value — contradicting local minimality.

(d) If $\nabla f(x) = 0$ then the first-order condition reads $f(y) \geq f(x)$ for every y . ■

The idea: convex versus non-convex is the real dividing line.

Convex problems are solved. Not “solvable in principle” — solved. Interior-point methods find the global optimum of a convex problem to high accuracy in polynomial time, reliably, with certificates of optimality. If you can formulate your problem as convex, you are done; pick a solver.

Non-convex problems are, in general, hopeless. Finding the global minimum of a general non-convex function is NP-hard. You get local minima, you get no guarantees, and you get heuristics.

So the single most valuable skill in applied optimisation is **recognising or reformulating a problem as convex**. Sometimes a change of variables does it; sometimes a relaxation (replace rank with nuclear norm, replace count-of-nonzeros with ℓ_1) gives a convex problem whose solution is good enough.

Deep learning is spectacularly non-convex and works anyway, which is a genuine puzzle rather than a counterexample. The current partial explanation involves the saddle-point picture of Chapter 18: in very high dimensions, bad local minima are rare, most critical points are saddles, and the many global-ish minima that exist are of similar quality.

Convexity-preserving operations, worth knowing so you can check quickly: nonnegative combinations of convex functions; pointwise maximum; composition with an affine map; and $f(Ax + b)$ whenever f is convex. Norms are convex. $-\log$ is convex. Squared error is convex. Cross-entropy in the **logits of a linear model** is convex; in the weights of a deep network it is not, and that is where the difficulty enters.

34.3. Gradient descent

Gradient descent

```
x := x0
repeat:
    x := x - eta * grad_f(x)
until converged
```

The direction is justified by Chapter 18: $-\nabla f$ is the direction of steepest decrease. The step size η (the **learning rate**) is the entire difficulty.

Theorem 34.4 (convergence for smooth convex functions).

If f is convex with L -Lipschitz gradient, and $\eta \leq 1/L$, then

$$f(x_k) - f^* \leq \frac{\|x_0 - x^*\|^2}{2\eta k}.$$

So $O(1/\varepsilon)$ iterations for accuracy ε .

If additionally f is μ -strongly convex, convergence is **linear**:

$$f(x_k) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^*),$$

needing only $O(\kappa \log(1/\varepsilon))$ iterations, where $\kappa = L/\mu$.

The idea: the condition number is the learning rate problem.

Read those two rates. Everything depends on $\kappa = L/\mu$, the ratio of the largest to the smallest curvature — which for a quadratic is exactly the condition number of the Hessian (Chapter 25).

Geometrically: the level sets are ellipsoids with axis ratio $\sqrt{\kappa}$. If $\kappa = 1$ they are spheres, the gradient points straight at the minimum, and one step of the right size finishes.

If κ is large they are long thin valleys. The gradient points mostly **across** the valley, not along it. You are forced to use a small step size to avoid overshooting the narrow direction, and then you crawl along the long direction. The result is the characteristic zigzag.

This single picture explains most practical optimisation advice:

- **Feature scaling** reduces κ , which is why normalising inputs accelerates training so dramatically. It is not a superstition; it is conditioning.
- **Batch normalisation and layer normalisation** keep the effective conditioning under control during training.
- **Momentum** damps the oscillation across the valley while accumulating speed along it, improving the rate from κ to $\sqrt{\kappa}$.
- **Adam** and its relatives estimate a per-coordinate scale, which is a diagonal approximation to preconditioning — an attempt to make the problem look better-conditioned without computing a Hessian.
- **Newton's method** uses the exact Hessian and is therefore invariant to conditioning altogether. That is why it converges so fast, and why it costs so much.

Definition 34.5 (momentum and Nesterov acceleration).

$$\begin{aligned} v_{k+1} &= \beta v_k + \nabla f(x_k), & x_{k+1} &= x_k - \eta v_{k+1} && \text{(heavy ball)} \\ v_{k+1} &= \beta v_k + \nabla f(x_k - \eta \beta v_k), & x_{k+1} &= x_k - \eta v_{k+1} && \text{(Nesterov)} \end{aligned}$$

Nesterov's version evaluates the gradient at the **look-ahead** point, and that small change is enough to achieve the optimal $O(1/k^2)$ rate for smooth convex functions — provably the best possible for any method using only gradients. The physical reading of momentum: a heavy ball rolling downhill does not reverse direction every time the slope wobbles.

34.4. Stochastic gradient descent

For $f(x) = (1/n) \sum_{i=1}^n f_i(x)$ with n enormous, computing the full gradient costs $O(n)$ per step. SGD uses a random subset:

Minibatch SGD

```
repeat:
    sample a minibatch B of indices
    g := (1/|B|) * sum over i in B of grad f_i(x)
    x := x - eta_k * g
```

g is an **unbiased** estimate of the true gradient: $\mathbb{E}[g] = \nabla f(x)$. The noise does not bias the direction, only adds variance.

Theorem 34.6 (SGD convergence).

With a decreasing step size satisfying $\sum_k \eta_k = \infty$ and $\sum_k \eta_k^2 < \infty$ (for instance $\eta_k = c/k$), SGD converges to the optimum for convex f , at rate $O(1/\sqrt{k})$.

The two conditions have clean readings: $\sum \eta_k = \infty$ means the steps must total enough distance to reach the optimum from anywhere; $\sum \eta_k^2 < \infty$ means the accumulated noise must be finite so you can settle.

The idea: why noisy gradients are not just tolerable but useful.

The obvious framing is that SGD trades accuracy for speed: n times cheaper per step, more steps needed. For large n that trade is overwhelmingly favourable, and that alone justifies it.

But the noise does more:

- It escapes saddle points. A deterministic method can stall exactly at a saddle; noise knocks you off.
- It is implicitly regularising. SGD tends to find flatter minima, which generalise better — a sharp minimum is one where the noise cannot settle.
- It is why the learning rate **schedule** matters so much. A large rate early explores; decaying it later lets you settle into the basin you found. Warmup, cosine decay, and cyclic schedules are all engineering around this exploration/exploitation trade.

34.5. Second-order methods

Definition 34.7 (Newton's method for optimisation).

$$x_{k+1} = x_k - H(x_k)^{-1} \nabla f(x_k).$$

Derived by minimising the second-order Taylor model $f(x) + \nabla f^T d + (1/2)d^T H d$ over the step d : differentiate and set to zero, giving $Hd = -\nabla f$.

Quadratically convergent near the optimum and **affine invariant** — rescaling the variables does not change its trajectory at all, so conditioning is irrelevant to it. In exchange: computing H is $O(n^2)$ entries, solving with it is $O(n^3)$, and H may not be positive definite away from a minimum, in which case the Newton step can point uphill.

Definition 34.8 (quasi-Newton and the practical alternatives).

- **BFGS**: build an approximation to H^{-1} from successive gradients. No Hessian required, super-linear convergence, $O(n^2)$ memory.
- **L-BFGS**: store only the last $m \approx 10$ gradient pairs and reconstruct the action of H^{-1} implicitly. $O(mn)$ memory. The standard choice for smooth medium-scale problems.
- **Gauss–Newton and Levenberg–Marquardt**: for least-squares objectives, approximate $H \approx 2J^T J$ using only first derivatives. Levenberg–Marquardt adds a damping term λI that interpolates between Gauss–Newton (fast, when the model is good) and gradient descent (safe, when it is not).
- **Adam**: a diagonal approximation to preconditioning, using running estimates of the first and second moments of the gradient. Cheap, robust, and the default in deep learning — where the true Hessian has 10^{12} entries and is out of the question.

34.6. Constrained optimization

Definition 34.9 (the Lagrangian).

For $\min f(x)$ subject to $g_i(x) \leq 0$ and $h_j(x) = 0$:

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x), \quad \lambda_i \geq 0.$$

Theorem 34.10 (Karush–Kuhn–Tucker conditions).

Under mild regularity, x^* is optimal only if there exist λ^*, ν^* with:

- | | |
|---|---------------------------|
| (K1) $\nabla_x \mathcal{L} = 0$ | (stationarity) |
| (K2) $g_i(x^*) \leq 0, h_j(x^*) = 0$ | (primal feasibility) |
| (K3) $\lambda_i^* \geq 0$ | (dual feasibility) |
| (K4) $\lambda_i^* g_i(x^*) = 0$ for every i | (complementary slackness) |

For a convex problem these conditions are also **sufficient**.

The idea: complementary slackness is the interesting one.

(K4) says: for each inequality constraint, either it is **active** ($g_i = 0$, you are pressed against the boundary) or its multiplier is zero (the constraint is irrelevant and could be deleted). Never both nonzero.

This is exactly the “shadow price” reading from Chapter 18. A constraint you are not touching has zero value; a constraint you are pressed against has a price equal to how much the objective would improve if you relaxed it by one unit.

In a support vector machine, the constraints are one per training point and the active ones are precisely the **support vectors**. Every other point has $\lambda_i = 0$ and could be deleted from the dataset without changing the solution at all. That fact — which is what makes SVMs sparse in the data — is complementary slackness and nothing else.

Definition 34.11 (duality).

The **dual function** is

$$d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu),$$

and the **dual problem** is to maximise it subject to $\lambda \geq 0$.

Weak duality always holds: $d(\lambda, \nu) \leq f(x^*)$ for any feasible dual point. So any dual point certifies a lower bound on the optimum — which is how solvers prove optimality and how branch-and-bound prunes. **Strong duality** — equality — holds for convex problems under Slater’s condition (a strictly feasible point exists).

The dual is often easier: it may have fewer variables, or nicer constraints. The kernel trick in SVMs works because the dual involves the data only through inner products $x_i^T x_j$, which can then be replaced by a kernel.

34.7. Regularisation

Definition 34.12.

$$\min_x f(x) + \lambda R(x).$$

- $R = \|x\|_2^2$: **ridge**. Shrinks coefficients smoothly, keeps all of them, improves conditioning (it adds λI to the Hessian).
- $R = \|x\|_1$: **lasso**. Produces exact zeros — sparsity — for the geometric reason given in Chapter 19.
- $R = \|x\|_*$ (nuclear norm): low rank, the convex relaxation of rank.

Chapter 31 gave the other reading: regularisation is a prior, and λ is the inverse prior variance. Ridge is a Gaussian prior; lasso is Laplace.

Example 34.13 (ridge regression, three ways).

$$\min_x \|Ax - b\|^2 + \lambda \|x\|^2.$$

By calculus (Chapter 26): the gradient is $2A^T(Ax - b) + 2\lambda x$, so $(A^T A + \lambda I)x = A^T b$.

By conditioning: adding λI shifts every eigenvalue of $A^T A$ up by λ , so the condition number falls from σ_1^2/σ_n^2 to $(\sigma_1^2 + \lambda)/(\sigma_n^2 + \lambda)$. A singular problem becomes solvable.

By SVD (Chapter 25): the solution filters the singular values, $1/\sigma_i \mapsto \sigma_i/(\sigma_i^2 + \lambda)$. Directions with large σ_i are barely touched; directions where the data carries almost no information are suppressed rather than amplified.

Three derivations, one answer, and each explains a different aspect of **why** it works. That is what it looks like when you actually understand something.

34.8. Exercises

- 34.1** Prove that $f(x) = x^4$ is convex on $\mathbb{R}$ but not strongly convex, and identify where the strong convexity fails.
- 34.2** Show that the maximum of two convex functions is convex, but the minimum need not be. Give a counterexample for the minimum.
- 34.3** For $f(x, y) = x^2 + 10y^2$, compute the Hessian, its condition number, and the largest stable learning rate for gradient descent. Then trace two iterations from $(1, 1)$ with η half that maximum and observe the zigzag.
- 34.4** Show that gradient descent on a quadratic $f(x) = (1/2)x^T A x$ with A symmetric positive definite converges iff $0 < \eta < 2/\lambda_{\max}$. (Write the iteration in the eigenbasis.)
- 34.5** Use Lagrange multipliers with the KKT conditions to solve $\min x^2 + y^2$ subject to $x + y \geq 2$. Identify whether the constraint is active and state the multiplier's value.
- 34.6** Derive the Newton step for minimising $f(x) = e^x - 2x$ and carry out two iterations from $x_0 = 0$. Compare with two gradient descent steps at $\eta = 0.5$.
- 34.7** Show that adding $\lambda\|x\|^2$ to a least-squares objective changes the singular values of the effective solution operator from $1/\sigma_i$ to $\sigma_i/(\sigma_i^2 + \lambda)$, and explain what happens to the directions with $\sigma_i \approx 0$.

- 34.8** You are minimising a function of 10 million parameters. Explain in a short paragraph why Newton's method is not an option, why L-BFGS is marginal, and what Adam is actually approximating.

CHAPTER 35.

Geometry for Graphics and Simulation

35.1. Transformations, and the affine problem

Chapter 20 established that linear maps are matrices. There is one thing they cannot do: translation. $T(x) = x + t$ is not linear, because $T(0) \neq 0$.

That is inconvenient, because a graphics pipeline is a long chain of transformations and you would like to compose them all into one matrix.

Definition 35.1 (homogeneous coordinates).

Represent a point $(x, y, z) \in \mathbb{R}^3$ as the 4-vector $(x, y, z, 1)$, and a **direction** as $(x, y, z, 0)$. Then a translation becomes a matrix:

$$T = \begin{pmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x + t_x \\ y + t_y \\ z + t_z \\ 1 \end{pmatrix}.$$

The idea: why the extra coordinate works.

An affine map in n dimensions is a linear map in $n + 1$ dimensions, restricted to the hyperplane $w = 1$. Adding a dimension buys you the translation.

Two immediate benefits, both large:

Composition. Every transformation — translate, rotate, scale, shear, and projection — is now a 4×4 matrix, so a whole chain multiplies into one matrix applied once per vertex. This is why GPUs are built around 4×4 matrix multiplication.

Points versus directions is now type-safe. Points have $w = 1$; directions and normals have $w = 0$. Translating a direction should do nothing, and with $w = 0$ the translation column is multiplied by zero, so it correctly does nothing. The representation enforces a distinction you would otherwise have to remember.

Also, with a general w , the convention is that (x, y, z, w) represents $(x/w, y/w, z/w)$ — and **that** is what makes perspective projection expressible as a matrix, since the divide is deferred to the end.

Definition 35.2 (the standard 3D transforms).

$$\text{Scale : } \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{Rotate about } z : \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and similarly about x and y , each being the 2×2 rotation of Chapter 5 embedded in the appropriate plane.

Trap: transform order matters, and normals do not transform like points.

Order. TR and RT are different: rotating then translating puts the object somewhere different from translating then rotating (the second rotates the translated position about the origin). Matrix multiplication is composition (Chapter 20), and composition is not commutative. Read a chain right to left: in $M = TRS$, the scale is applied first.

Normals. If you transform a surface by M , its normal vectors must be transformed by $(M^{-1})^T$, not by M .

Why: a normal n satisfies $n^T v = 0$ for every tangent v . After the transform the tangent is Mv , and we need the new normal n' to satisfy $n'^T Mv = 0$. Taking $n' = (M^{-1})^T n$ gives $n'^T Mv = n^T M^{-1} Mv = n^T v = 0$. Correct.

For a rotation, $(M^{-1})^T = M$, so nothing goes wrong and people forget the rule. Introduce a non-uniform scale and normals visibly break — they stop being perpendicular to the surface and the lighting goes wrong. This is a classic graphics bug, and it is one line of linear algebra.

35.2. Rotations in three dimensions

Definition 35.3 (rotation matrix).

A 3×3 matrix R is a rotation if $R^T R = I$ and $\det R = +1$. The set of these is the group $\text{SO}(3)$.

The condition $R^T R = I$ makes it orthogonal — preserving lengths and angles (Chapter 20) — and $\det R = +1$ excludes reflections (Chapter 22).

Theorem 35.4 (Euler's rotation theorem).

Every element of $\text{SO}(3)$ is a rotation by some angle about some fixed axis.

Proof. R is a real 3×3 matrix, so its characteristic polynomial is a real cubic and therefore has at least one real root (Chapter 6). Since R is orthogonal, all eigenvalues have modulus 1, and since $\det R = 1$ is the product of the eigenvalues, the real one must be $+1$.

The eigenvector for $\lambda = 1$ is fixed by R : that is the axis. In the plane perpendicular to it, R acts as a 2×2 rotation with eigenvalues $e^{\pm i\theta}$. ■

So a 3D rotation has three degrees of freedom: two for the axis direction, one for the angle.

35.2.1. Euler angles and their failure

Represent a rotation as three successive rotations about coordinate axes — yaw, pitch, roll. Intuitive, compact (three numbers), and defective.

Trap: gimbal lock is a real loss of a degree of freedom.

With the common yaw–pitch–roll convention, set pitch to 90° . Now the yaw axis and the roll axis have been rotated into alignment: changing yaw and changing roll do **the same thing**. You have lost a degree of freedom, and there is no way to rotate out of the configuration smoothly.

It is not a bug in a particular convention — every three-parameter representation of $\text{SO}(3)$ has singularities somewhere. That is a topological fact: $\text{SO}(3)$ is not homeomorphic to any open subset of $\mathbb{R}^3$, so no global three-number chart exists.

Practical consequences: interpolating between two orientations in Euler angles produces wobble and can pass through a singularity; and Apollo 11 really did have to be flown around gimbal lock in its physical IMU.

The fix is to use four numbers.

35.2.2. Quaternions

Definition 35.5 (quaternions).

$$q = w + xi + yj + zk,$$

with $i^2 = j^2 = k^2 = ijk = -1$, from which

$$ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \dots$$

Multiplication is associative but **not commutative**.

Theorem 35.6 (quaternions represent rotations).

A rotation by angle θ about a unit axis u corresponds to the **unit** quaternion

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(u_x i + u_y j + u_z k).$$

A point v is rotated by

$$v' = qvq^{-1},$$

treating v as a quaternion with zero real part. Composition of rotations is quaternion multiplication.

The idea: why the half-angle, and what it costs.

The rotation formula applies q **twice** — once on each side — so each application contributes half the rotation. Hence $\theta/2$.

This has a visible consequence: q and $-q$ give the **same** rotation, since the two sign flips cancel in qvq^{-1} . A full 2π turn takes q to $-q$, and you need 4π to return to q . Quaternions form a **double cover** of $\text{SO}(3)$.

That sounds like a defect and it is the source of the advantages:

- **No gimbal lock.** Four parameters with one constraint ($\|q\| = 1$) covers $\text{SO}(3)$ smoothly with no singularities.
- **Cheap composition.** 16 multiplications, versus 27 for two 3×3 matrices.
- **Cheap renormalisation.** Numerical drift is corrected by dividing by $\|q\|$, one square root. Re-orthogonalising a drifted rotation matrix requires Gram–Schmidt or an SVD.
- **Correct interpolation.** Spherical linear interpolation (**slerp**) walks along the great circle between two unit quaternions, giving constant angular velocity. Interpolating Euler angles or matrix entries does not. (Watch the double cover: negate one quaternion if their dot product is negative, or you interpolate the long way round.)

Quaternions are the reason every game engine, robotics stack, and IMU library represents orientation the way it does. And they are the direct descendant of Chapter 6: complex numbers rotate the plane; quaternions rotate space.

35.3. Projection

Definition 35.7 (orthographic and perspective).

Orthographic: drop a coordinate. Parallel lines stay parallel; no foreshortening. Used for CAD and isometric games.

Perspective: divide by depth. With the eye at the origin looking down $-z$, a point (x, y, z) projects to $(-dx/z, -dy/z)$ on the plane $z = -d$.

The division by z is what homogeneous coordinates were for. The perspective matrix leaves a z -dependent value in the w component, and the pipeline's **perspective divide** — dividing all components by w at the end — does the projection.

Example 35.8 (the graphics pipeline, as matrix algebra).

$$\text{clip} = \underbrace{P}_{\text{projection}} \cdot \underbrace{V}_{\text{view}} \cdot \underbrace{M}_{\text{model}} \cdot \text{vertex}.$$

M places the object in the world, V moves the world into the camera's frame (it is the inverse of the camera's pose), and P applies the projection frustum. The three multiply into one matrix per object, uploaded once and applied to every vertex in parallel.

After the perspective divide you are in normalised device coordinates and a viewport transform maps to pixels.

The whole pipeline is Chapter 20's "composition is matrix multiplication", built into silicon.

35.4. Curves and surfaces

Definition 35.9 (Bézier curve).

For control points $P_0, \dots, P_n$,

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i P_i, \quad t \in [0, 1].$$

The coefficients are the **Bernstein polynomials** — and they are the binomial theorem of Chapter 9 with $x = t$ and $y = 1 - t$, which is why they sum to 1. That fact matters: the curve is a **weighted average** of the control points with weights summing to one, so it always lies in their convex hull.

Properties, each with a one-line reason:

- Passes through P_0 and P_n (all other weights vanish at the endpoints).
- Tangent at P_0 points towards P_1 (differentiate at $t = 0$).
- Lies in the convex hull of the control points (weights are nonnegative and sum to 1).
- Affine invariant: transforming the control points transforms the curve.

The cubic case ($n = 3$) is the one that runs the world: SVG paths, font outlines, animation easing curves, and vector illustration tools are all cubic Béziers.

Definition 35.10 (de Casteljau's algorithm).

Evaluate $B(t)$ by repeated linear interpolation: interpolate between consecutive control points at parameter t , producing n new points; repeat until one point remains. That point is $B(t)$.

Numerically stable (it is a sequence of convex combinations, so nothing can blow up), and it **also** splits the curve into two Bézier curves at t for free — which is what makes adaptive subdivision rendering possible.

Splines chain low-degree pieces together with continuity conditions at the joins: C^0 matching position, C^1 matching tangent, C^2 matching curvature. B-splines and NURBS generalise this with local control, so moving one control point does not perturb the entire curve.

35.5. Distances, intersections, and the primitives

The bread and butter of graphics, collision detection, and computational geometry. All of it is Chapter 19.

Problem	Solution
Point to plane $n \cdot x = d$	$(n \cdot p - d)/\ n\ $; signed, so it also tells you which side
Point to line through a with direction v	$\ (p - a) - \text{proj}_v(p - a)\ $
Ray–sphere	substitute the ray into $\ x - c\ ^2 = r^2$; a quadratic in t , and the discriminant decides hit or miss
Ray–plane	solve $n \cdot (o + td) = D$ for t ; parallel iff $n \cdot d = 0$
Ray–triangle	Möller–Trumbore: solve for barycentric coordinates directly
Which side of a line is p ?	sign of the 2×2 determinant — the cross product of Chapter 19
Do segments intersect?	the two endpoints of each must straddle the other’s line: four orientation tests

The idea: barycentric coordinates.

Any point in a triangle ABC is

$$P = \alpha A + \beta B + \gamma C, \quad \alpha + \beta + \gamma = 1.$$

The weights (α, β, γ) are **barycentric coordinates**, and they are ratios of sub-triangle areas — each of which is a determinant (Chapter 22).

P is inside the triangle exactly when all three are nonnegative, which is the standard point-in-triangle test.

And they are how a rasteriser interpolates: colour, texture coordinates, normals, and depth at an interior pixel are the barycentric-weighted averages of the vertex values. The entire inner loop of a GPU is barycentric interpolation, and it is nothing more than “write the point as a weighted average of the corners”.

35.6. Rotations elsewhere

Two connections worth naming, since the same mathematics keeps reappearing.

Rotary position embeddings. A transformer with RoPE encodes token position by rotating pairs of embedding coordinates by an angle proportional to position. The dot product between a query at position m and a key at position n then depends only on $m - n$, because composing a rotation with the inverse of another leaves a rotation by the difference. That is the angle addition formula of Chapter 5, doing load-bearing work in a language model.

Orthogonal initialisation and constraints. Orthogonal matrices preserve norms (Chapter 20), so a recurrent network whose transition matrix is constrained to be orthogonal cannot suffer exploding or vanishing gradients from that matrix — all its singular values are exactly 1. Orthogonal initialisation is a cheaper approximation to the same idea.

35.7. Exercises

- 35.1** Write the 4×4 homogeneous matrix that translates by $(2, 3, 4)$, and the one that scales by 2 about the origin. Compute both products and describe geometrically how the two results differ.
- 35.2** Verify that the z -axis rotation matrix is orthogonal with determinant 1, and find its eigenvalues. Interpret the complex ones using Chapter 6.
- 35.3** Show that if M is a rotation matrix then $(M^{-1})^T = M$, so normals transform correctly under rotation without the special rule. Then find a non-uniform scale for which they do not.
- 35.4** Convert a rotation of 90° about the y -axis to a unit quaternion, and verify that q and $-q$ rotate the point $(1, 0, 0)$ identically.
- 35.5** Compute the point at $t = 0.5$ on the cubic Bézier with control points $(0, 0)$, $(0, 1)$, $(1, 1)$, $(1, 0)$, using de Casteljau's algorithm. Sketch the curve.
- 35.6** A ray starts at $(0, 0, 0)$ with direction $(1, 1, 1)/\sqrt{3}$. Find where it meets the sphere of radius 2 centred at $(3, 3, 3)$, by substituting into the sphere equation and solving the quadratic.
- 35.7** Find the barycentric coordinates of the point $(2, 1)$ with respect to the triangle $(0, 0)$, $(4, 0)$, $(0, 3)$, and determine whether the point is inside.
- 35.8** Explain in a paragraph why interpolating between two orientations by linearly interpolating their 3×3 rotation matrices entrywise produces a matrix that is not a rotation, and what slerp does instead.

CHAPTER 36.

Backpropagation, Derived

36.1. Everything, in one place

This chapter builds a neural network and derives its training algorithm from first principles. Every ingredient has already appeared:

- the chain rule (Chapter 14) and its matrix form (Chapter 18);
- matrix multiplication as composition (Chapter 20);
- the matrix calculus identities (Chapter 26);
- cross-entropy from information theory (Chapter 32);
- gradient descent and conditioning (Chapter 34);
- floating-point behaviour (Chapter 33).

Nothing new is required. That is the point of the chapter: what looks like a separate subject is an assembly of things you now know.

36.2. The model

Definition 36.1 (a feedforward network).

With input $a^{(0)} = x$, for $\ell = 1, \dots, L$:

$$z^{(\ell)} = W^{(\ell)} a^{(\ell-1)} + b^{(\ell)} \quad (\text{affine})$$

$$a^{(\ell)} = \varphi(z^{(\ell)}) \quad (\text{elementwise nonlinearity})$$

and a scalar loss $\mathcal{L} = \text{loss}(a^{(L)}, y)$.

The idea: why the nonlinearity is not optional.

Compose two affine maps:

$$W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2),$$

which is one affine map. By induction, a network of L affine layers with no nonlinearity is **exactly** a single affine layer — the depth buys literally nothing, and every extra parameter is wasted.

That is Chapter 20: the composition of linear maps is a linear map, and the set of matrices is closed under multiplication.

The elementwise φ is what breaks the closure. It is the only source of expressive power in the architecture, and the universal approximation theorem says that one hidden layer with a non-polynomial φ and enough width can approximate any continuous function on a compact set to any accuracy.

(That theorem is much less useful than it sounds — it says nothing about how **wide**, or whether gradient descent will find the weights. Depth wins in practice for reasons the theorem does not address.)

Definition 36.2 (the usual activations).

$$\begin{aligned}\text{ReLU}(z) &= \max(0, z), & \varphi'(z) &= \begin{cases} 1 & z > 0 \\ 0 & z < 0 \end{cases} \\ \sigma(z) &= \frac{1}{1 + e^{-z}}, & \varphi'(z) &= \sigma(z)(1 - \sigma(z)) \\ \tanh(z) &= \frac{e^{2z} - 1}{e^{2z} + 1}, & \varphi'(z) &= 1 - \tanh^2(z) \\ \text{GELU}(z) &= z\Phi(z), & \Phi & \text{ the normal CDF}\end{aligned}$$

ReLU's derivative at exactly 0 does not exist (Chapter 14: it is a corner). Frameworks pick 0 by convention. This is a **subgradient** choice, it is harmless in practice, and it is worth knowing that it is a choice rather than a derivative.

36.3. Forward and backward

Theorem 36.3 (backpropagation).

Define $\delta^{(\ell)} = \partial\mathcal{L}/\partial z^{(\ell)}$. Then

$$\begin{aligned}\delta^{(L)} &= \nabla_{a^{(L)}} \mathcal{L} \circ \varphi'(z^{(L)}) \\ \delta^{(\ell)} &= (W^{(\ell+1)})^T \delta^{(\ell+1)} \circ \varphi'(z^{(\ell)}) \\ \frac{\partial \mathcal{L}}{\partial W^{(\ell)}} &= \delta^{(\ell)} (a^{(\ell-1)})^T \\ \frac{\partial \mathcal{L}}{\partial b^{(\ell)}} &= \delta^{(\ell)}\end{aligned}$$

where $\circ$ is the elementwise product.

Proof. The last two lines are Chapter 26's linear-layer proposition applied with upstream gradient $\delta^{(\ell)}$.

For the recursion, apply the chain rule through the path $z^{(\ell)} \rightarrow a^{(\ell)} \rightarrow z^{(\ell+1)}$:

$$\frac{\partial \mathcal{L}}{\partial z^{(\ell)}} = \left(\frac{\partial a^{(\ell)}}{\partial z^{(\ell)}} \right)^T \left(\frac{\partial z^{(\ell+1)}}{\partial a^{(\ell)}} \right)^T \frac{\partial \mathcal{L}}{\partial z^{(\ell+1)}}.$$

The second Jacobian is $W^{(\ell+1)}$, so its transpose is $(W^{(\ell+1)})^T$. The first is diagonal with entries $\varphi'(z^{(\ell)})$ (Chapter 26), and multiplying by a diagonal matrix is an elementwise product. Composing gives the stated recursion.

The base case is the same computation at the output layer, where the upstream gradient is the loss gradient directly. ■

One training step

```
# forward
a[0] := x
for l = 1..L:
    z[l] := W[l] @ a[l-1] + b[l]
    a[l] := phi(z[l])
loss := loss_fn(a[L], y)
```

```

# backward
d := grad_loss(a[L], y) * phi_prime(z[L])
for l = L..1:
    gradW[l] := outer(d, a[l-1])
    gradb[l] := d
    if l > 1:
        d := (W[l].T @ d) * phi_prime(z[l-1])

# update
for l = 1..L:
    W[l] -= eta * gradW[l]
    b[l] -= eta * gradb[l]

```

The idea: why this is efficient, stated precisely.

There are P parameters and one scalar loss.

Finite differences would need $P + 1$ forward passes — perturb each parameter, recompute. For $P = 10^9$ that is impossible, and each estimate would carry the numerical error of Chapter 14.

Forward-mode autodiff propagates derivatives forwards alongside the values. Each pass gives the derivative with respect to **one** input direction, so you again need P passes. Forward mode is the right choice when there are few inputs and many outputs.

Reverse mode — backpropagation — propagates derivatives backwards from the single scalar output, and gets the gradient with respect to **all** P parameters in one backward pass costing about the same as one forward pass.

The reason is the associativity observed in Chapter 18. The gradient is a product of Jacobians; the leftmost factor is a $1 \times n$ row vector because the loss is a scalar; so multiplying left to right keeps every intermediate a **vector**, and every operation is a matrix–vector product rather than matrix–matrix.

One scalar output, many inputs: reverse mode. That is the entire argument, and it is a statement about associativity of matrix multiplication.

The cost is **memory**: the backward pass needs every $a^{(\ell)}$ from the forward pass, so activation memory scales with depth times batch size. That is what gradient checkpointing trades against — recompute some activations instead of storing them, spending time to save memory.

36.4. The gradient pathologies

Trap: vanishing and exploding gradients, quantified.

Unroll the recursion:

$$\delta^{(1)} = \left(\prod_{\ell=2}^L (W^{(\ell)})^T D^{(\ell-1)} \right) \delta^{(L)}, \quad D^{(\ell)} = \text{diag}(\varphi'(z^{(\ell)})).$$

The gradient at layer 1 is a **product of $L - 1$ matrices**. Taking norms, its magnitude scales roughly as $\prod_{\ell} \|W^{(\ell)}\| \|D^{(\ell)}\|$.

If those factors average below 1, the product decays exponentially in L : early layers receive essentially no gradient and never learn. If above 1, it grows exponentially and training diverges to NaN.

It is Chapter 14's "gains multiply along a chain", with $L = 100$ factors.

For sigmoid, $\max \varphi' = 0.25$, so even with perfectly scaled weights the gradient shrinks by at least 4^{-L} . That is why deep sigmoid networks were untrainable, and why the field stalled for two decades on exactly this fact.

The fixes, each attacking one factor of the product:

- **ReLU:** $\varphi' = 1$ on the active half, so D contributes no shrinkage.
- **Careful initialisation:** He/Xavier scaling chooses $\text{Var}(W_{ij}) \approx 2/n_{\text{in}}$ precisely so that $\|W^{(\ell)}\|$ is near 1 and the product neither grows nor decays.
- **Residual connections:** $a^{(\ell)} = a^{(\ell-1)} + F(a^{(\ell-1)})$ makes the layer Jacobian $I + \partial F$, so the product is $\prod(I + \partial F_\ell)$ — close to the identity, with a gradient path that reaches every layer without attenuation. This is the single change that made hundred-layer networks trainable.
- **Normalisation layers:** keep the scale of activations, and hence of the Jacobians, under control at every depth.
- **Gradient clipping:** rescale the gradient when its norm exceeds a threshold. A blunt instrument, and effective against the exploding direction.

36.5. Automatic differentiation as a general technique

Backpropagation is a special case. The general framework is worth understanding because it is what your framework implements.

Definition 36.4 (computation graph).

A DAG (Chapter 10) whose nodes are variables and whose edges are primitive operations. Each primitive knows its own local derivative.

Definition 36.5 (the two modes).

Forward mode: propagate $\dot{v} = \partial v / \partial x$ forwards from the inputs. Implementable with **dual numbers**: carry pairs $(v, \dot{v})$ with the arithmetic

$$(a, \dot{a}) \times (b, \dot{b}) = (ab, a\dot{b} + \dot{a}b),$$

which is the product rule built into the number type. One pass gives one column of the Jacobian.

Reverse mode: run forwards recording the graph, then propagate $\bar{v} = \partial \mathcal{L} / \partial v$ backwards from the output. One pass gives one row of the Jacobian — which is the whole gradient when there is one output.

Cost: forward mode is $O(n)$ passes for n inputs; reverse is $O(m)$ passes for m outputs. For neural networks $n \approx 10^9$ and $m = 1$, so reverse wins by nine orders of magnitude. For a function $\mathbb{R} \rightarrow \mathbb{R}^{1000}$ — sensitivity of many outputs to one parameter — forward mode wins instead.

The idea: AD is neither symbolic nor numerical differentiation.

Symbolic differentiation manipulates expressions and produces a formula. It suffers **expression swell**: the derivative of a product of n terms has n terms, and nested products explode combinatorially.

Numerical differentiation uses finite differences and loses half your precision to the trade-off of Chapter 14.

Automatic differentiation applies the chain rule to the **execution trace** of the program, at the level of individual operations, evaluating derivatives numerically as it goes. It gives exact derivatives — to machine precision, with no step size — at a constant factor of the cost of the original function.

It is one of the genuinely great ideas in scientific computing, it is much older than deep learning, and it is the reason modern frameworks let you differentiate through arbitrary code including loops and branches.

36.6. The mathematics of a transformer

To close, the current dominant architecture — entirely in terms of this book.

Definition 36.6 (scaled dot-product attention).

Given queries $Q \in \mathbb{R}^{n \times d}$, keys $K \in \mathbb{R}^{m \times d}$, and values $V \in \mathbb{R}^{m \times d_v}$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V.$$

The idea: reading the formula piece by piece.

QK^T . Every query dotted with every key: an $n \times m$ matrix of similarity scores. This is Chapter 19 — the dot product measures agreement, and if the vectors are normalised it is cosine similarity. Computing all pairs at once is a matrix multiplication, which is why attention maps onto GPUs so well.

$/\sqrt{d}$. Suppose the entries of q and k are independent with mean 0 and variance 1. Then $q \cdot k = \sum_{i=1}^d q_i k_i$ has variance d (Chapter 29: variances of independent terms add), so a typical score has magnitude $\sqrt{d}$.

For $d = 512$ that is around 23. Feeding scores of that size into softmax saturates it: one entry dominates, the output is nearly one-hot, and the Jacobian (Chapter 26) is nearly zero — so no gradient flows.

Dividing by $\sqrt{d}$ restores unit variance and keeps softmax in its responsive regime. It is a variance calculation, and it is the whole justification for that term.

softmax. Chapter 4: turn scores into a probability distribution over positions. Each output is a convex combination of the value vectors, so attention outputs live in the convex hull of the values — the same observation as Bézier curves in Chapter 35.

$\cdot V$. Take the weighted average. Attention is a differentiable, content-addressed lookup: a soft dictionary where every key contributes in proportion to how well it matches.

Definition 36.7 (multi-head attention).

Run h attention operations in parallel with separate learned projections, and concatenate:

$$\text{MHA}(X) = [\text{head}_1; \dots; \text{head}_h]W^O, \quad \text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V).$$

Each head projects into a d/h -dimensional subspace (Chapter 19) and attends within it, so different heads can specialise — one on syntax, one on coreference, and so on. The concatenation and W^O recombine them. In implementation it is one batched matrix multiplication, not h separate ones (Chapter 20's block matrices).

The rest of a transformer block:

- **Residual connections** $x + \text{sublayer}(x)$: the identity-Jacobian trick above, which is what allows the depth.
- **Layer normalisation**: standardise each token's activation vector to zero mean and unit variance, then rescale by learned parameters. Keeps the Jacobian norms controlled at every depth.
- **A feedforward network**: two linear layers with a nonlinearity, applied independently at each position. This is where most of the parameters live.
- **Positional encoding**: attention is permutation-equivariant, so position must be injected. Sinusoidal encodings (Chapter 5) or rotary embeddings (Chapter 35) do it.
- **Training objective**: cross-entropy on the next token, which by Chapter 32 is maximum likelihood, which is minimum KL divergence to the data distribution.
- **Optimiser**: Adam, a diagonal preconditioner (Chapter 34), on minibatch gradients computed by reverse-mode AD.

The idea: what you should take from this.

There is no step in that description that this book has not covered.

A transformer is: dot products for similarity, softmax for normalisation, a variance calculation for the scaling, matrix multiplication for composition and for batching, the chain rule for training, cross-entropy from information theory, and gradient descent with a diagonal preconditioner.

The engineering is formidable — the systems work, the data curation, the scale — and the **mathematics** is a few hundred pages of undergraduate material assembled thoughtfully.

That is the honest picture, and it is the one worth having. Papers that look impenetrable are usually assembling familiar pieces in an unfamiliar order. Having the pieces is what lets you see the order.

36.7. Exercises

- 36.1** For a two-layer network $y = W_2\varphi(W_1x)$ with squared loss $\mathcal{L} = \|y - t\|^2$, derive $\partial\mathcal{L}/\partial W_1$ and $\partial\mathcal{L}/\partial W_2$ explicitly, and state the shape of every intermediate quantity.
- 36.2** Show that a network with L linear layers and no activation function is equivalent to a single linear layer, and count how many parameters are wasted for $L = 5$ layers of width 100.
- 36.3** A 50-layer network uses sigmoid activations and weights initialised with $\|W\| \approx 1$. Estimate the factor by which the gradient at layer 1 is smaller than at layer 50. Repeat for ReLU.
- 36.4** Verify that for a residual block $a = x + F(x)$, the Jacobian is $I + \partial F/\partial x$, and explain in one sentence why a product of such Jacobians resists both vanishing and exploding.

- 36.5** Implement (on paper) dual-number arithmetic for $f(x) = x^2 \sin(x)$ at $x = 2$: carry the pair $(2, 1)$ through each operation and read off $f'(2)$. Check against the analytic derivative.
- 36.6** Show that if q and k have i.i.d. entries with mean 0 and variance 1, then $\text{Var}(q \cdot k) = d$. Then explain what would go wrong in a transformer with $d = 4096$ and no $1/\sqrt{d}$ scaling.
- 36.7** Attention has cost $O(n^2d)$ in sequence length n . Derive that from the matrix shapes in the definition, and state which of the two matrix products dominates for large n .
- 36.8** Explain, in terms of the associativity argument of Chapter 18, why reverse-mode AD is preferred for training neural networks but forward-mode is preferred for computing the sensitivity of a large simulation output to a single scalar input.

PART VII

Appendices

Worked solutions to every exercise, a curated library of free and open-licensed mathematics books, a notation reference, and a study plan that puts the whole thing in order.

CHAPTER A.

Worked Solutions

Solutions to every exercise, worked rather than merely stated.

Trap: read these second.

A solution read before a genuine attempt produces the **feeling** of understanding without any of the substance. Give each problem fifteen real minutes first.

When you do come here and find you were wrong, the useful step is not reading the solution — it is finding the specific sentence in the chapter you failed to use. That is the thing that will not happen again.

Solutions are numbered to match the exercises: solution 14.3 answers exercise 3 of Chapter 14.

A.1. Chapter 1 — Numbers

1.1 Suppose $\sqrt{3} = p/q$ in lowest terms. Then $p^2 = 3q^2$, so $3 \mid p^2$.

Claim: $3 \mid p^2$ implies $3 \mid p$. Check the cases: if $p = 3k + 1$ then $p^2 = 9k^2 + 6k + 1 \equiv 1$; if $p = 3k + 2$ then $p^2 = 9k^2 + 12k + 4 \equiv 1 \pmod{3}$. So an integer not divisible by 3 has a square not divisible by 3.

Hence $p = 3m$, giving $9m^2 = 3q^2$, so $q^2 = 3m^2$ and by the same claim $3 \mid q$. Both divisible by 3, contradicting lowest terms.

Where it breaks for $\sqrt{4}$. The claim fails: $4 \mid p^2$ does **not** imply $4 \mid p$. Take $p = 2$: $p^2 = 4$ is divisible by 4, p is not. The claim needs the divisor to be **prime** (Euclid's lemma, Chapter 12), and 4 is not. This is precisely why $\sqrt{n}$ is irrational exactly when n is not a perfect square.

1.2 Let r be rational and x irrational, and suppose $r + x = s$ were rational. Then $x = s - r$ is a difference of two rationals, hence rational — contradiction. So $r + x$ is irrational.

Two irrationals can sum either way: $\sqrt{2} + (-\sqrt{2}) = 0$ is rational, while $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$ is irrational (if it were rational, halving it would make $\sqrt{2}$ rational).

1.3 Let $x = 0.\overline{142857}$, period 6. Then $10^6 x = 142857.\overline{142857}$, so

$$10^6 x - x = 142857 \Rightarrow x = \frac{142857}{999999}.$$

Both are divisible by 142857: indeed $999999 = 7 \times 142857$, so $x = 1/7$.

1.4 Write $x = (x - y) + y$ and apply the triangle inequality:

$$|x| \leq |x - y| + |y| \Rightarrow |x| - |y| \leq |x - y|.$$

Swapping the roles of x and y gives $|y| - |x| \leq |y - x| = |x - y|$.

So both $|x| - |y|$ and its negative are at most $|x - y|$, which is exactly $||x| - |y|| \leq |x - y|$.

1.5 Finite strings: countable. There are 2^k strings of length k , finitely many. List all length-0 strings, then all length-1, then all length-2, and so on, alphabetically within each block. Every finite string has some finite length, so it appears in some block and therefore at some finite position.

Infinite strings: uncountable. Suppose $s_1, s_2, \dots$ were a complete list. Build t whose n -th character is a if the n -th character of s_n is b , and b otherwise. Then t differs from every s_n in position n , so it is not on the list. This is Cantor's diagonal argument verbatim; indeed infinite binary strings are in bijection with subsets of $\mathbb{N}$, and with $[0, 1]$ up to the $0.\bar{9}$ ambiguity.

1.6 Take $a = 10^{16}$, $b = -10^{16}$, $c = 1$.

$(a + b) + c$: the first sum is exactly 0, so the result is 1.

$a + (b + c)$: near 10^{16} the spacing between doubles is 2 (since $10^{16} > 2^{53} \approx 9.007 \times 10^{15}$). So $-10^{16} + 1$ falls exactly halfway between two representable values and rounds to -10^{16} under round-half-to-even. Then $a + (-10^{16}) = 0$.

So the two groupings give 1 and 0. The mechanism: the addition $b + c$ is **absorbed** — c is smaller than the representation gap at b 's magnitude, so it vanishes entirely.

1.7 $S = \{0, 1/2, 2/3, 3/4, \dots\}$. Every element is less than 1, so 1 is an upper bound. And no smaller number is: given any $b < 1$, choose $n > 1/(1 - b)$, and then $1 - 1/n > b$. So $\sup S = 1$.

But $1 \notin S$, since $1 - 1/n = 1$ would need $1/n = 0$.

This is exactly the difference between a supremum and a maximum: the maximum is a supremum that happens to be **attained**. Every nonempty bounded-above set has a supremum (completeness); not every one has a maximum. The Extreme Value Theorem of Chapter 13 is precisely a statement that in certain circumstances the supremum **is** attained.

A.2. Chapter 2 — Algebra as a Rewriting System

2.1 Suppose $a + z = a$ for every a . Apply it with $a = 0$: $0 + z = 0$.

But by commutativity (F1) and the identity axiom (F3), $0 + z = z + 0 = z$.

So $z = 0$. Note the shape of the argument: a claim about **all** a is pinned down by choosing one convenient a .

2.2 $\sqrt{2x + 3} = x$. Squaring gives $2x + 3 = x^2$, so $x^2 - 2x - 3 = 0$, factoring as $(x - 3)(x + 1) = 0$: candidates $x = 3$ and $x = -1$.

Squaring is the one-way step — it is not injective, so it can gain solutions. Both candidates must be checked in the original equation.

$x = 3$: $\sqrt{9} = 3$. Valid. $x = -1$: $\sqrt{1} = 1 \neq -1$. Rejected.

There is also a structural reason $x = -1$ cannot work: $\sqrt{\cdot}$ is nonnegative by definition, so the equation forces $x \geq 0$ before any algebra happens. Answer: $x = 3$.

2.3 $2x^2 - 12x + 7 = 2(x^2 - 6x) + 7 = 2[(x - 3)^2 - 9] + 7 = 2(x - 3)^2 - 11$.

Since $(x - 3)^2 \geq 0$ with equality only at $x = 3$, the minimum value is -11 , attained at $x = 3$.

No calculus needed: a square is nonnegative, so the expression is minimised when the square vanishes.

2.4 Telescope $\sum_{i=1}^n [(i + 1)^4 - i^4] = (n + 1)^4 - 1$.

Expanding, $(i + 1)^4 - i^4 = 4i^3 + 6i^2 + 4i + 1$, so

$$4 \sum i^3 + 6 \sum i^2 + 4 \sum i + n = (n + 1)^4 - 1.$$

Substituting the known sums $\sum i = n(n+1)/2$ and $\sum i^2 = n(n+1)(2n+1)/6$:

$$4 \sum i^3 = (n+1)^4 - 1 - n - 2n(n+1) - n(n+1)(2n+1).$$

Expanding the right side: $(n+1)^4 - 1 - n = n^4 + 4n^3 + 6n^2 + 3n$, and $2n(n+1) + n(n+1)(2n+1) = 2n^3 + 5n^2 + 3n$. Subtracting, $4 \sum i^3 = n^4 + 2n^3 + n^2 = n^2(n+1)^2$, so

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Check at $n = 3$: $1 + 8 + 27 = 36 = 6^2$. ✓

2.5 $x^2 - x - 6 = (x-3)(x+2)$, so write

$$\frac{5x-4}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}, \quad 5x-4 = A(x+2) + B(x-3).$$

Put $x = 3$: $11 = 5A$, so $A = 11/5$. Put $x = -2$: $-14 = -5B$, so $B = 14/5$.

$$\frac{5x-4}{x^2-x-6} = \frac{11/5}{x-3} + \frac{14/5}{x+2}.$$

2.6 Move everything to one side rather than cross-multiplying:

$$\frac{x-1}{x+2} - 1 \geq 0 \Rightarrow \frac{(x-1) - (x+2)}{x+2} \geq 0 \Rightarrow \frac{-3}{x+2} \geq 0.$$

The numerator is the negative constant -3 , so the fraction is nonnegative exactly when the denominator is negative: $x+2 < 0$, i.e. $x < -2$. (It is never zero, so the inequality is strict in effect.)

Solution set: $(-\infty, -2)$.

What goes wrong with cross-multiplication. Multiplying both sides by $x+2$ gives $x-1 \geq x+2$, i.e. $-1 \geq 2$, which is false — so you would conclude there are no solutions. The error is that $x+2$ may be negative, in which case the inequality must flip. The sign of the multiplier is unknown, so the step is illegal.

2.7 $(x-y)^2 \geq 0$ for all real x, y , since squares of reals are nonnegative. Expanding: $x^2 - 2xy + y^2 \geq 0$, so $x^2 + y^2 \geq 2xy$. Equality holds iff $x-y=0$, i.e. $x=y$.

Now substitute $x = \sqrt{a}$, $y = \sqrt{b}$ for $a, b \geq 0$:

$$a+b \geq 2\sqrt{a}\sqrt{b} = 2\sqrt{ab},$$

which rearranges to $(a+b)/2 \geq \sqrt{ab}$: AM–GM, with equality iff $\sqrt{a} = \sqrt{b}$, i.e. $a=b$.

2.8 p has degree 3 and four distinct roots 0, 1, 2, 3.

By the corollary to the factor theorem, a **nonzero** polynomial of degree n has at most n roots. Four roots exceed the bound for degree 3, so p cannot be a nonzero polynomial of degree 3.

The only resolution is that p is the zero polynomial — which has every number as a root and no well-defined degree. (Strictly, the hypothesis “ p has degree 3” is inconsistent with the four roots; the exercise’s point is that you should notice the inconsistency rather than hunt for a cubic.)

A.3. Chapter 3 — Functions

3.1 (a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 2n$. Injective: $2a = 2b$ gives $a = b$. Not surjective: no integer maps to 1.

(b) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = \lfloor n/2 \rfloor$. Not injective: $f(0) = f(1) = 0$. Surjective: $f(2k) = k$ for every k .

(c) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - x$. Not injective: $f(-1) = f(0) = f(1) = 0$. Surjective: it is a continuous odd-degree polynomial, so it takes arbitrarily large positive and negative values and by the Intermediate Value Theorem hits everything between.

(d) $f : \mathbb{R} \rightarrow (0, \infty)$, $f(x) = e^x$. Injective (strictly increasing) and surjective onto the stated codomain. Bijective.

3.2 Suppose $g \circ f$ is injective and $f(a_1) = f(a_2)$. Applying g , $g(f(a_1)) = g(f(a_2))$, so $(g \circ f)(a_1) = (g \circ f)(a_2)$, and injectivity of the composite gives $a_1 = a_2$. So f is injective.

g need not be. Take $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x$ (whose image is $(0, \infty)$) and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$. Then $(g \circ f)(x) = e^{2x}$, which is injective, while g is not ($g(1) = g(-1)$).

The reason: g only needs to be injective **on the image of f** , and it can misbehave freely outside it.

3.3 $(f \circ g)(x) = f(x^2) = 2x^2 + 3$. $(g \circ f)(x) = g(2x + 3) = (2x + 3)^2 = 4x^2 + 12x + 9$.

These differ, confirming composition is not commutative. They agree when

$$2x^2 + 3 = 4x^2 + 12x + 9 \Rightarrow 2x^2 + 12x + 6 = 0 \Rightarrow x^2 + 6x + 3 = 0,$$

so $x = -3 \pm \sqrt{6}$.

3.4 Rewrite the inside first: $-3f(2x - 4) + 1 = -3f(2(x - 2)) + 1$.

Horizontal (inside the function, so backwards):

1. compress horizontally by a factor of 2 (from the $2x$);
2. shift right by 2 (from the $x - 2$).

Writing it as $2(x - 2)$ makes the order unambiguous. If you instead read $2x - 4$ literally, the equivalent order is **shift right by 4, then compress by 2** — the same final picture reached differently, which is exactly why factoring the inside out first is the safe habit.

Vertical (outside, so as written):

1. stretch vertically by 3;
2. reflect in the x -axis (the minus sign);
3. shift up by 1.

3.5

$$\text{even part} = \frac{e^x + e^{-x}}{2} = \cosh x,$$

$$\text{odd part} = \frac{e^x - e^{-x}}{2} = \sinh x.$$

And indeed $\cosh x + \sinh x = e^x$, as the decomposition requires. Checking the parity: $\cosh(-x) = \cosh(x)$ and $\sinh(-x) = -\sinh(x)$, directly from swapping the two exponentials.

3.6 Total. A function $A \rightarrow B$ is a choice of an output for each of the m elements of A , with n options each and the choices independent. By the product rule that is n^m functions. (This is why the notation B^A is used for the set of such functions.)

Injective. Now the choices are not independent: each element of A must get a **distinct** output. There are n options for the first element, $n - 1$ for the second, and so on:

$$n(n-1)(n-2)\cdots(n-m+1) = \frac{n!}{(n-m)!},$$

which is 0 when $m > n$ — correctly, since you cannot inject a bigger set into a smaller one (pigeonhole, Chapter 9).

3.7 $f(x) = (2x^2 - 3)/(x^2 - 4)$.

Domain. The denominator vanishes at $x = \pm 2$, so the domain is $\mathbb{R} \setminus \{-2, 2\}$.

Vertical asymptotes. At $x = 2$ and $x = -2$: the numerator is nonzero there ($2(4) - 3 = 5$), so the function blows up rather than having a removable hole.

Behaviour at infinity. Numerator and denominator have equal degree, so divide through by x^2 :

$$f(x) = \frac{2 - 3/x^2}{1 - 4/x^2} \rightarrow 2$$

as $x \rightarrow \pm\infty$. Horizontal asymptote $y = 2$.

A.4. Chapter 4 — Exponentials and Logarithms

4.1 (a) $\log_2 32 = 5$, since $2^5 = 32$. (b) $\log_9 3 = 1/2$, since $9^{1/2} = 3$. (c) $\log_5(1/125) = -3$, since $5^{-3} = 1/125$. (d) $\log_a(a^7) = 7$, by the definition of the logarithm as the inverse. (e) $2^{\log_2 11} = 11$, likewise.

4.2 By definition, $2^{\log_2 n} = n$. Take the natural logarithm of both sides:

$$\ln(2^{\log_2 n}) = \ln n.$$

By the power law, the left side is $(\log_2 n)(\ln 2)$. Dividing by $\ln 2 \neq 0$:

$$\log_2 n = \frac{\ln n}{\ln 2}.$$

Note $1/\ln 2 \approx 1.4427$, so $\log_2 n$ and $\ln n$ differ by a constant factor — which is the whole reason bases are invisible inside big-O.

4.3 (a) $3^{2x-1} = 27 = 3^3$. Since $x \mapsto 3^x$ is injective, $2x - 1 = 3$, so $x = 2$.

(b) $\ln x + \ln(x - 3) = \ln 10$ gives $\ln(x(x - 3)) = \ln 10$, so $x^2 - 3x - 10 = 0$, factoring as $(x - 5)(x + 2) = 0$. Candidates 5 and -2 .

Check the domain: both $x > 0$ and $x - 3 > 0$ are required, so $x > 3$. $x = -2$ is rejected (and would have made both logs undefined). Answer $x = 5$.

(c) Substitute $u = e^x > 0$: $u^2 - 5u + 6 = 0$, so $u = 2$ or $u = 3$, both positive. Hence $x = \ln 2$ or $x = \ln 3$.

4.4

$$\sigma(-x) = \frac{1}{1+e^x} = \frac{e^{-x}}{e^{-x}+1} = \frac{(1+e^{-x})-1}{1+e^{-x}} = 1 - \sigma(x).$$

Consequence for classification: a binary classifier must output $\mathbb{P}(\text{class 1})$ and $\mathbb{P}(\text{class 0})$, which sum to 1. The identity says σ applied to the negated logit gives exactly the complementary probability, so one number $\sigma(z)$ determines both. Emitting two logits and applying softmax gives the same thing with a redundant degree of freedom — which is why binary classifiers use a single output and `binary_cross_entropy`.

4.5 **Shift invariance.**

$$\frac{e^{z_i+c}}{\sum_j e^{z_j+c}} = \frac{e^c e^{z_i}}{e^c \sum_j e^{z_j}} = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

The e^c cancels top and bottom.

Log-sum-exp. With $m = \max_j z_j$,

$$\ln \sum_j e^{z_j} = \ln \left(e^m \sum_j e^{z_j-m} \right) = m + \ln \sum_j e^{z_j-m},$$

using $\log(ab) = \log a + \log b$. In exact arithmetic this is an identity. In floating point it is far better behaved: every exponent $z_j - m$ is ≤ 0 , so no term exceeds 1 and nothing overflows, while the largest term is exactly $e^0 = 1$ so the sum cannot underflow to zero either.

4.6 Set $n \lg n = n^2/2$. Dividing by n (valid for $n > 0$):

$$\lg n = \frac{n}{2}.$$

By inspection $n = 2$ works ($1 = 1$) and $n = 4$ works ($2 = 2$). Between them the logarithm is larger (at $n = 3$: $1.585 > 1.5$), and beyond $n = 4$ the line wins and grows away.

So mathematically the two costs cross at $n = 2$ and $n = 4$.

Numerically, and more usefully: the real question is where $c_1 n \lg n = c_2 n^2/2$ with the implementations' actual constants, since $c_2 \ll c_1$ in practice. Measure both at a range of n , plot on log-log axes, and find the intersection — which is empirically around $n = 10$ to 50 for real sort implementations, and is exactly why library sorts switch to insertion sort for small subarrays.

4.7 Assume cost $= Cn^k$. Taking the ratio of the two measurements eliminates C :

$$\frac{16}{2} = \left(\frac{4000}{1000} \right)^k \Rightarrow 8 = 4^k.$$

Taking logs: $k = \ln 8 / \ln 4 = 3/2$. So the routine is $\Theta(n^{1.5})$.

If the second measurement were 8 ms instead: $4 = 4^k$ gives $k = 1$ — linear.

Note how little data this needs: two points and a logarithm. This is the practical content of the log-log plot remark in the chapter.

4.8

$$2\sigma(2x) - 1 = \frac{2}{1+e^{-2x}} - 1 = \frac{2 - (1+e^{-2x})}{1+e^{-2x}} = \frac{1-e^{-2x}}{1+e^{-2x}}.$$

Multiply numerator and denominator by e^{2x} :

$$= \frac{e^{2x} - 1}{e^{2x} + 1} = \tanh x,$$

matching the definition given in the chapter. So $\tanh$ is a sigmoid rescaled to output $(-1, 1)$ and horizontally compressed by 2 — which is why the two activations behave so similarly and why $\tanh$ was generally preferred (its output is centred at zero, which conditions the next layer better).

A.5. Chapter 5 — Trigonometry

5.1 $\sin(7\pi/6)$. $7\pi/6 = \pi + \pi/6$, so it is in the third quadrant with reference angle $\pi/6$. In the third quadrant both coordinates are negative, so $\sin$ is negative: $-1/2$.

$\cos(3\pi/4)$. Second quadrant, reference angle $\pi/4$. There $x < 0$, so $\cos$ is negative: $-\sqrt{2}/2$.

$\tan(5\pi/3)$. Fourth quadrant, reference angle $\pi/3$. There $x > 0$ and $y < 0$, so $\tan$ is negative: $-\sqrt{3}$.

5.2 Write $\sin(\alpha - \beta) = \sin(\alpha + (-\beta))$ and apply the addition formula:

$$= \sin \alpha \cos(-\beta) + \cos \alpha \sin(-\beta).$$

Now $\cos$ is even so $\cos(-\beta) = \cos \beta$, and $\sin$ is odd so $\sin(-\beta) = -\sin \beta$. Substituting:

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

5.3 Matching $3 \cos t + 4 \sin t = R \cos(t - \varphi) = R \cos \varphi \cos t + R \sin \varphi \sin t$ requires $R \cos \varphi = 3$ and $R \sin \varphi = 4$.

Squaring and adding: $R^2 = 9 + 16 = 25$, so $R = 5$.

Dividing: $\tan \varphi = 4/3$, and since both $R \cos \varphi$ and $R \sin \varphi$ are positive, φ is in the first quadrant. Hence $\varphi = \operatorname{atan2}(4, 3) = \arctan(4/3) \approx 0.927$ radians.

$$3 \cos t + 4 \sin t = 5 \cos(t - \arctan(4/3)).$$

5.4

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}.$$

Divide numerator and denominator by $\cos \alpha \cos \beta$:

$$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}.$$

Validity. The division requires $\cos \alpha \neq 0$ and $\cos \beta \neq 0$, and the original quotient requires $\cos(\alpha + \beta) \neq 0$. So the identity fails when any of α , β , $\alpha + \beta$ is an odd multiple of $\pi/2$ — which is where the tangents are undefined anyway.

5.5 Apply the half-angle formula twice.

$$\sin^4 \theta = (\sin^2 \theta)^2 = \left(\frac{1 - \cos 2\theta}{2} \right)^2 = \frac{1 - 2 \cos 2\theta + \cos^2 2\theta}{4}.$$

Now $\cos^2 2\theta = (1 + \cos 4\theta)/2$, so

$$\sin^4 \theta = \frac{1}{4} \left(1 - 2 \cos 2\theta + \frac{1 + \cos 4\theta}{2} \right) = \frac{3}{8} - \frac{1}{2} \cos 2\theta + \frac{1}{8} \cos 4\theta.$$

Check at $\theta = 0$: $3/8 - 1/2 + 1/8 = 0$. ✓ At $\theta = \pi/2$: $3/8 + 1/2 + 1/8 = 1$. ✓

Every term is now a cosine of a multiple angle, so integrating is immediate — which is the point of the manipulation.

5.6 $r = \sqrt{9 + 16} = 5$ — distance from the origin.

The point $(-3, -4)$ is in the **third** quadrant. Its angle is

$$\theta = \text{atan2}(-4, -3) = -(\pi - \arctan(4/3)) \approx -(3.1416 - 0.9273) = -2.214 \text{ rad},$$

equivalently $\approx -126.9^\circ$, which lies in $(-\pi, \pi]$ as required.

What $\arctan(y/x)$ would give. $y/x = (-4)/(-3) = 4/3$, and $\arctan(4/3) \approx 0.927$ rad — a **first-quadrant** angle. The two negatives cancelled in the ratio, so $\arctan$ cannot tell $(-3, -4)$ from $(3, 4)$ and returns the wrong answer by exactly π .

5.7 Consider triangle ABC with the usual labelling, and drop a perpendicular from C to side AB , of length h .

In the right triangle containing angle A , the hypotenuse is b (the side AC), so $h = b \sin A$.

In the right triangle containing angle B , the hypotenuse is a , so $h = a \sin B$.

Equating: $b \sin A = a \sin B$, hence

$$\frac{\sin A}{a} = \frac{\sin B}{b}.$$

Dropping a perpendicular from a different vertex gives the third ratio.

(If the foot of the perpendicular falls outside the segment — an obtuse triangle — the same computation works with the supplementary angle, whose sine is the same.)

A.6. Chapter 6 — Complex Numbers

6.1 (a) $(3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 - 10i + 8 = 11 - 10i$.

(b) Multiply by the conjugate of the denominator:

$$\frac{2 + i}{3 - i} = \frac{(2 + i)(3 + i)}{(3 - i)(3 + i)} = \frac{6 + 2i + 3i + i^2}{9 + 1} = \frac{5 + 5i}{10} = \frac{1}{2} + \frac{1}{2}i.$$

(c) Powers of i cycle with period 4. $2027 = 4 \times 506 + 3$, so $i^{2027} = i^3 = -i$.

(d) $|3 - 4i| = \sqrt{9 + 16} = 5$.

6.2 $z = -1 + i$: $r = |z| = \sqrt{2}$, and the point is in the second quadrant, so $\theta = 3\pi/4$. Hence $z = \sqrt{2} e^{i3\pi/4}$.

By De Moivre,

$$z^8 = (\sqrt{2})^8 e^{i \cdot 8 \cdot 3\pi/4} = 16 e^{i6\pi} = 16,$$

since 6π is a whole number of full turns.

Sanity check: $z^2 = (-1 + i)^2 = 1 - 2i + i^2 = -2i$, so $z^4 = (-2i)^2 = -4$, so $z^8 = 16$. ✓

6.3 $z^3 = 8$. Write $z = re^{i\theta}$: then $r^3 = 8$ so $r = 2$, and 3θ must be a multiple of 2π , so $\theta = 0, 2\pi/3, 4\pi/3$.

$$\begin{aligned}
 z_1 &= 2e^{i0} = 2, \\
 z_2 &= 2e^{i2\pi/3} = 2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -1 + i\sqrt{3}, \\
 z_3 &= 2e^{i4\pi/3} = -1 - i\sqrt{3}.
 \end{aligned}$$

Geometrically they are three points equally spaced around the circle of radius 2, one of them on the positive real axis — the vertices of an equilateral triangle. Note z_2 and z_3 are conjugates, as they must be since $z^3 - 8$ has real coefficients.

6.4 By computation. With $z = a + bi$, $w = c + di$:

$$zw = (ac - bd) + (ad + bc)i,$$

so

$$\begin{aligned}
 |zw|^2 &= (ac - bd)^2 + (ad + bc)^2 \\
 &= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 \\
 &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2.
 \end{aligned}$$

The cross terms cancelling is the entire content.

In polar form. $zw = (re^{i\alpha})(se^{i\beta}) = rs e^{i(\alpha+\beta)}$, whose modulus is $rs = |z||w|$. One line.

6.5 Write each cosine as an average of exponentials:

$$\cos \alpha \cos \beta = \left(\frac{e^{i\alpha} + e^{-i\alpha}}{2} \right) \left(\frac{e^{i\beta} + e^{-i\beta}}{2} \right).$$

Multiplying out gives four terms:

$$= \frac{1}{4} (e^{i(\alpha+\beta)} + e^{i(\alpha-\beta)} + e^{-i(\alpha-\beta)} + e^{-i(\alpha+\beta)}).$$

Pair the first with the last and the middle two:

$$= \frac{1}{4} (2 \cos(\alpha + \beta) + 2 \cos(\alpha - \beta)) = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)].$$

6.6 With $\omega = e^{2\pi i/5}$, the sum $1 + \omega + \omega^2 + \omega^3 + \omega^4$ is geometric with ratio $\omega \neq 1$:

$$\sum_{k=0}^4 \omega^k = \frac{\omega^5 - 1}{\omega - 1} = \frac{1 - 1}{\omega - 1} = 0.$$

Geometrically: the five fifth-roots of unity are five unit vectors equally spaced around the circle at 72° intervals. That configuration is symmetric under rotation by 72° , and the only vector unchanged by a nontrivial rotation is the zero vector — so the sum must be zero.

6.7 Let $U = \{z \in \mathbb{C} : |z| = 1\}$.

Closed under multiplication: if $|z| = |w| = 1$ then $|zw| = |z||w| = 1$ by the previous exercise, so $zw \in U$.

Closed under inverses: if $|z| = 1$ then $z \neq 0$, so z^{-1} exists, and $|z^{-1}| = 1/|z| = 1$. (Indeed $z^{-1} = \bar{z}$ when $|z| = 1$, since $z\bar{z} = |z|^2 = 1$.)

Together with associativity (inherited from $\mathbb{C}$) and the identity $1 \in U$, this makes U a group. Writing $z = e^{i\theta}$, multiplication adds angles, so U is precisely the group of rotations of the plane — which is why it is called the circle group and why it is the same object as $\text{SO}(2)$.

A.7. Chapter 7 — Logic and Proof

7.1 Original: “If a graph is a tree then it has no cycles.” **True** (by definition of a tree).

Contrapositive: “If a graph has a cycle then it is not a tree.” **True** — it is the same statement.

Converse: “If a graph has no cycles then it is a tree.” **False.** Two disjoint edges form an acyclic graph that is not connected, hence not a tree. (Such a thing is a forest.)

Inverse: “If a graph is not a tree then it has a cycle.” **False**, with the same counterexample — and necessarily so, since the inverse is the contrapositive of the converse and therefore has the same truth value.

7.2 Flip each quantifier in turn and negate the innermost statement:

$$\exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N : |a_n - L| \geq \varepsilon.$$

In words: **there is some tolerance ε that the sequence keeps violating** — no matter how far out you go, you can always find a later term at least ε away from L .

Note the structure: convergence says the bad terms eventually stop; the negation says bad terms occur infinitely often. It does **not** say all terms are far from L .

7.3 Contrapositive: if n is odd, then $n^3 + 5$ is even.

Let $n = 2k + 1$. Then n^3 is a product of three odd numbers, hence odd — or explicitly, $n^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1$.

So $n^3 + 5 = (2m + 1) + 5 = 2(m + 3)$, which is even.

Since the contrapositive holds, so does the original.

7.4 **Base case $n = 4$:** $4! = 24 > 16 = 2^4$. ✓

Inductive step. Assume $k! > 2^k$ for some $k \geq 4$. Then

$$(k + 1)! = (k + 1) \cdot k! > (k + 1) \cdot 2^k \geq 5 \cdot 2^k > 2 \cdot 2^k = 2^{k+1},$$

using $k \geq 4$ so $k + 1 \geq 5 > 2$.

Where $n \geq 4$ is needed. The inductive step only requires $k + 1 > 2$, i.e. $k \geq 2$. The real constraint is the **base case**: at $n = 3$, $3! = 6 < 8 = 2^3$, so the statement is false there and no induction can start earlier. The hypothesis $n \geq 4$ is exactly the point at which the factorial overtakes the exponential.

7.5 **Base case $n = 0$:** $0^3 - 0 = 0$, divisible by 6. ✓

Inductive step. Assume $6 \mid (k^3 - k)$. Then

$$\begin{aligned} (k + 1)^3 - (k + 1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3k^2 + 3k = (k^3 - k) + 3k(k + 1). \end{aligned}$$

The first term is divisible by 6 by hypothesis. For the second: k and $k + 1$ are consecutive, so one is even, so $k(k + 1)$ is even and $3k(k + 1)$ is divisible by 6.

A sum of two multiples of 6 is a multiple of 6.

(Alternative without induction: $n^3 - n = (n-1)n(n+1)$, a product of three consecutive integers, which is divisible by 2 and by 3, hence by 6.)

7.6 Base cases. $8 = 3 + 5$, $9 = 3 + 3 + 3$, $10 = 5 + 5$. ✓

Inductive step. Let $n \geq 11$ and assume every integer m with $8 \leq m < n$ is representable. Then $n - 3 \geq 8$ and $n - 3 < n$, so by hypothesis $n - 3 = 3a + 5b$. Hence $n = 3(a + 1) + 5b$.

Why three base cases. The step reduces n by 3, so it lands on $n - 3$; to cover every residue class mod 3 you need the three consecutive values 8, 9, 10 established directly. With only 8 verified, the induction would reach 11, 14, 17, ... and nothing else.

7.7 The step fails exactly at $k = 1$, i.e. going from 1 horse to 2.

With $k + 1 = 2$ horses $\{h_1, h_2\}$: removing h_2 leaves $\{h_1\}$, all one colour (trivially). Removing h_1 leaves $\{h_2\}$, all one colour. But the argument then needs these two conclusions to **chain** — and chaining requires a horse present in both subsets, which transfers the colour from one to the other.

For $k + 1 \geq 3$ the two k -subsets overlap and the chaining works. For $k + 1 = 2$ they are disjoint singletons and there is nothing to chain through.

This is a genuinely useful lesson: an inductive step can be valid for all large k and fail at the very bottom, which is precisely where nobody checks. When an induction proves something false, examine the step at the smallest k where it is applied.

7.8 Invariant: at the top of each iteration, $m = \max(A[0..i - 1])$.

Initialisation. Before the loop, $i = 1$ and $m = A[0]$, and $\max(A[0..0]) = A[0]$. Holds. (This is where nonemptiness is used: $A[0]$ must exist.)

Maintenance. Suppose the invariant holds with values m, i . The body sets $m' = \max(m, A[i])$ — either explicitly or by leaving m alone — and $i' = i + 1$. Then

$$m' = \max(\max(A[0..i - 1]), A[i]) = \max(A[0..i]) = \max(A[0..i' - 1]).$$

The invariant holds again.

Termination. The loop exits when $i = n$, so $m = \max(A[0..n - 1])$, which is the maximum of the whole array. And the loop does terminate: $n - i$ is a nonnegative integer that strictly decreases each iteration (well-ordering).

A.8. Chapter 8 — Sets, Relations, and Structure

8.1 ($\subseteq$) Let $x \in A \setminus (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$. By De Morgan for logic, “not ($x \in B$ or $x \in C$)” means $x \notin B$ **and** $x \notin C$. So $x \in A \setminus B$ and $x \in A \setminus C$, i.e. $x \in (A \setminus B) \cap (A \setminus C)$.

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A$, $x \notin B$, and $x \notin C$. Hence $x \notin B \cup C$, so $x \in A \setminus (B \cup C)$.

Every step is reversible, so a single chain of “iff”s would also have done it; the double inclusion is written out here because it is the form that always works even when steps are not reversible.

8.2 $\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\},$

which is 8 elements, $= 2^3$. ✓

The bijection to 0, ..., 7. Assign bit 0 to a , bit 1 to b , bit 2 to c , setting the bit iff the element is present:

$$\begin{aligned}\emptyset &\rightarrow 000_2 = 0, & \{a\} &\rightarrow 001_2 = 1, & \{b\} &\rightarrow 010_2 = 2, & \{a, b\} &\rightarrow 011_2 = 3, \\ \{c\} &\rightarrow 100_2 = 4, & \{a, c\} &\rightarrow 101_2 = 5, & \{b, c\} &\rightarrow 110_2 = 6, & \{a, b, c\} &\rightarrow 111_2 = 7.\end{aligned}$$

This is exactly the bitmask representation, and it is a bijection because every 3-bit pattern occurs exactly once. Union becomes bitwise OR, intersection becomes AND, and complement becomes NOT — which is the Boolean-algebra correspondence of the chapter, made executable.

8.3

Relation	refl.	symm.	antisymm.	trans.
(a) $a \leq b$	yes	no	yes	yes
(b) $a \mid b$ on $\mathbb{Z}$	yes	no	no	yes
(c) $a + b$ even	yes	yes	no	yes
(d) $a \neq b$	no	yes	no	no

(a) A total order.

(b) The interesting one: divisibility is antisymmetric on $\mathbb{N}$ but **not** on $\mathbb{Z}$, since $2 \mid -2$ and $-2 \mid 2$ while $2 \neq -2$. Sign is the obstruction.

(c) An equivalence relation. Transitivity: $a + c = (a + b) + (b + c) - 2b$, a sum of even numbers. The two classes are the evens and the odds — this is congruence mod 2.

(d) Not reflexive ($a = a$, so aRa fails) and not transitive: $1 \neq 2$ and $2 \neq 1$, but $1 = 1$.

8.4 Write $a \sim b$ for “ a and b have the same remainder mod 5”, i.e. $5 \mid (a - b)$.

Reflexive: $5 \mid 0$. **Symmetric:** if $5 \mid (a - b)$ then $5 \mid (b - a)$. **Transitive:** if $5 \mid (a - b)$ and $5 \mid (b - c)$ then 5 divides their sum $(a - c)$.

The classes are the five residue classes

$$\begin{aligned}[0] &= \{\dots, -5, 0, 5, 10, \dots\}, & [1] &= \{\dots, -4, 1, 6, 11, \dots\}, \\ [2] &= \{\dots, -3, 2, 7, \dots\}, & [3] &= \{\dots, -2, 3, 8, \dots\}, & [4] &= \{\dots, -1, 4, 9, \dots\},\end{aligned}$$

and they partition $\mathbb{Z}$ — every integer is in exactly one, by the division algorithm (Chapter 12).

8.5 **A strict example.** Take $f(x) = x^2$, $S = \{1\}$, $T = \{-1\}$. Then $S \cap T = \emptyset$ so $f(S \cap T) = \emptyset$, while $f(S) = f(T) = \{1\}$ so $f(S) \cap f(T) = \{1\}$. Strict.

If f is injective, equality holds. The inclusion $\subseteq$ is always true: if $y = f(x)$ with $x \in S \cap T$, then $y \in f(S)$ and $y \in f(T)$.

For $\supseteq$: let $y \in f(S) \cap f(T)$. Then $y = f(s)$ for some $s \in S$ and $y = f(t)$ for some $t \in T$. Injectivity forces $s = t$, so this common element lies in $S \cap T$, and $y \in f(S \cap T)$.

The failure without injectivity is exactly the collision: two **different** points, one from each set, landing on the same image.

8.6 Let M be the maths-takers and P the physics-takers, inside a universe of all 100 students.

$$|M \cup P| = |M| + |P| - |M \cap P| = 60 + 45 - 20 = 85.$$

So the number taking neither is

$$|\overline{M \cup P}| = 100 - 85 = 15.$$

The universe being complemented against is the set of all 100 students — which must be stated, because “neither” is meaningless without it.

8.7 $(\mathbb{R}, -)$ is **not** a monoid, and it fails on more than one count.

Associativity fails, which is the decisive one:

$$(1 - 2) - 3 = -4, \quad \text{but} \quad 1 - (2 - 3) = 1 - (-1) = 2.$$

Identity is one-sided. $a - 0 = a$, so 0 is a right identity. But $0 - a = -a \neq a$ in general, so it is not a left identity. A monoid requires a two-sided identity.

Since associativity fails, subtraction cannot be safely parallelised or reassociated — which is why `reduce(-)` over a list is a bug waiting to happen and `foldl` versus `foldr` give different answers.

8.8 An example. On $\{1, 2, 3\}$, let $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.

Symmetric: yes, every pair’s reverse is present. Transitive: yes, check the four compositions among $\{1, 2\}$. Reflexive: **no** — $(3, 3) \notin R$.

(The simplest example of all is $R = \emptyset$ on a nonempty set: vacuously symmetric and transitive, not reflexive.)

Why the “obvious” argument is wrong. The bogus reasoning is: “ aRb implies bRa by symmetry, and then aRb with bRa implies aRa by transitivity.”

The flaw: that derivation assumes there **exists** some b with aRb . If a is related to nothing at all — as 3 is above — the chain never starts, and reflexivity at a is never forced.

The argument does prove something weaker and true: in a symmetric transitive relation, every element that is related to **anything** is related to itself.

A.9. Chapter 9 — Counting

9.1 Build the hand by an ordered sequence of independent decisions.

1. **Choose the two ranks that will be paired:** $\binom{13}{2} = 78$. Using $\binom{13}{2}$ rather than 13×12 is exactly where the double-counting is avoided — “a pair of aces and a pair of kings” is one hand, not two.
2. **Choose two suits for the first paired rank:** $\binom{4}{2} = 6$.
3. **Choose two suits for the second:** $\binom{4}{2} = 6$.
4. **Choose the fifth card,** from a rank not already used: $11 \text{ ranks} \times 4 \text{ suits} = 44$.

$$78 \times 6 \times 6 \times 44 = 123,152.$$

(Out of $\binom{52}{5} = 2,198,160$ hands, about 4.75%.)

9.2 Complementary counting. Total strings of length 10 over a 3-letter alphabet: $3^{10} = 59,1049$. Strings with **no** A — so built from $\{B, C\}$ only: $2^{10} = 1,024$.

$$59,1049 - 1,024 = 58,1025.$$

Doing it directly would mean summing over the number of As, $\sum_{k=1}^{10} \binom{10}{k} 2^{10-k}$ — which is correct, much more work, and by the binomial theorem equals the same thing.

9.3 Count the subsets of an n -element set, two ways.

Way 1. A subset is a yes/no decision per element: 2^n .

Way 2. Classify by size. There are $\binom{n}{k}$ subsets of size k , and every subset has exactly one size, so the classes are disjoint and exhaustive: $\sum_{k=0}^n \binom{n}{k}$.

Two counts of the same set must agree:

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

9.4 Nonnegative. Stars and bars with $k = 15$ stars and $n = 4$ bins:

$$\binom{15+4-1}{15} = \binom{18}{15} = \binom{18}{3} = \frac{18 \cdot 17 \cdot 16}{6} = 816.$$

Positive. Substitute $y_i = x_i - 1 \geq 0$. Then $y_1 + y_2 + y_3 + y_4 = 15 - 4 = 11$, and

$$\binom{11+3}{11} = \binom{14}{3} = \frac{14 \cdot 13 \cdot 12}{6} = 364.$$

The substitution is the standard move: hand out the mandatory minimum first, then distribute the remainder freely.

9.5 Write each of the chosen integers uniquely as $2^a m$ with m odd (repeatedly divide out 2s).

Every such m is an odd number in $\{1, 2, \dots, 2n\}$, and there are exactly n of those: $1, 3, 5, \dots, 2n - 1$.

We have $n + 1$ integers and n possible odd parts, so by pigeonhole two of them share the same m : say $2^a m$ and $2^b m$ with $a < b$.

Then $2^b m = 2^{b-a} \cdot (2^a m)$, so the first divides the second.

The whole difficulty was finding the right pigeonholes; once “odd part” is the classifier, the argument is two lines.

9.6 Let A_d be the multiples of d in $1..1000$, so $|A_d| = \lfloor 1000/d \rfloor$.

$$\begin{aligned} |A_2| &= 500, & |A_3| &= 333, & |A_5| &= 200, \\ |A_6| &= 166, & |A_{10}| &= 100, & |A_{15}| &= 66, & |A_{30}| &= 33. \end{aligned}$$

By inclusion–exclusion,

$$|A_2 \cup A_3 \cup A_5| = 500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

So the count divisible by none of them is $1000 - 734 = 266$.

9.7 Count pairs (S, x) where S is a k -subset and $x \in S$ is a distinguished element.

By choosing S first: $\binom{n}{k}$ subsets, then k choices of x within it: $k \binom{n}{k}$.

By choosing x first: n choices of x , then the remaining $k - 1$ members from the other $n - 1$ elements: $n \binom{n-1}{k-1}$.

Equating and dividing by k :

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}.$$

9.8 Exact. Insert the keys one at a time. The i -th key (0-indexed) avoids collision iff it lands in one of the $m - i$ still-empty buckets, so

$$\mathbb{P}(\text{no collision}) = \prod_{i=0}^{n-1} \frac{m-i}{m} = \prod_{i=0}^{n-1} \left(1 - \frac{i}{m}\right).$$

Approximation. Using $1 - x \approx e^{-x}$ for small x :

$$\mathbb{P} \approx \prod_{i=0}^{n-1} e^{-i/m} = \exp\left(-\frac{1}{m} \sum_{i=0}^{n-1} i\right) = \exp\left(-\frac{n(n-1)}{2m}\right).$$

This drops to $1/2$ when $n(n-1)/(2m) \approx \ln 2$, i.e.

$$n \approx \sqrt{2m \ln 2} \approx 1.18\sqrt{m}.$$

So collisions become likely at around $\sqrt{m}$ insertions, not m — the birthday phenomenon. It is quadratic because collisions are about **pairs**, and there are $\binom{n}{2} \approx n^2/2$ of those.

A.10. Chapter 10 — Graphs

10.1 The sum of the degrees would be $7 \times 3 = 21$, which is odd.

But by the handshake lemma the sum of all degrees equals $2|E|$, which is even. Contradiction, so no such graph exists.

Equivalently by the corollary: all seven vertices have odd degree, and the number of odd-degree vertices must be even.

10.2 Let T be a tree with at least two vertices, and let $P = v_0 v_1 \dots v_k$ be a **longest** path in T (one exists since T is finite). Since T is connected with ≥ 2 vertices, $k \geq 1$.

Claim: v_0 has degree 1. Suppose v_0 had another neighbour $w \neq v_1$. If w is on P , say $w = v_j$ with $j \geq 2$, then $v_0 v_1 \dots v_j v_0$ is a cycle — impossible in a tree. If w is not on P , then $w v_0 v_1 \dots v_k$ is a longer path — contradicting maximality.

So v_0 is a leaf, and by the identical argument so is v_k . They are distinct because $k \geq 1$. Hence at least two leaves.

10.3 Let $P = v_0 v_1 \dots v_k$ be a longest path in G .

v_k has degree at least 2, so it has a neighbour w other than v_{k-1} .

w cannot be off the path: $v_0 \dots v_k w$ would be longer.

So $w = v_i$ for some $i \leq k-2$. Then

$$v_i v_{i+1} \dots v_k v_i$$

is a closed walk of length at least 3 with no repeated vertex except the endpoints — a cycle.

10.4 The graph is the 6-cycle with edges 12, 23, 34, 45, 56, 61.

It is bipartite, with parts $\{1, 3, 5\}$ and $\{2, 4, 6\}$: every edge joins an odd vertex to an even one. (Equivalently, every cycle — there is only one, of length 6 — is even.)

Adding edge 14: still bipartite. Vertex 1 is in the first part and vertex 4 is in the second, so the new edge respects the same 2-colouring. It creates two new cycles, 12341 and 14561, both of length 4 — even, as König’s theorem requires.

Worth noting what **would** break it: adding an edge between two same-parity vertices, say 13, creates the odd cycle 1231 of length 3 and destroys bipartiteness immediately.

10.5 The four landmasses have degrees 3, 3, 3, 5 — all odd.

No circuit. Euler’s theorem requires **every** vertex to have even degree for a closed circuit using each edge once. Four odd vertices, so no.

No open path either. An Euler **path** (using every edge once, allowed to finish somewhere else) exists in a connected graph iff there are exactly 0 or 2 vertices of odd degree — the start and the end are the only places where the enter/leave pairing can fail. Königsberg has four odd vertices, so no such walk exists either.

This is the problem Euler solved in 1736, generally regarded as the beginning of graph theory.

10.6 Minimum degree at most 5. Suppose every vertex had degree ≥ 6 . Then

$$2|E| = \sum_v \deg(v) \geq 6V \Rightarrow |E| \geq 3V > 3V - 6,$$

contradicting $|E| \leq 3V - 6$. So some vertex has degree ≤ 5 .

Six-colourability, by induction on V . For $V \leq 6$, colour every vertex differently.

For larger V : pick a vertex v of degree at most 5 (which exists by the above; note every subgraph of a planar graph is planar, so the bound keeps applying). Delete v ; the rest is planar with fewer vertices, so by hypothesis it is 6-colourable. Restore v : it has at most 5 neighbours, so at most 5 of the 6 colours are excluded, and at least one remains. Assign it.

10.7 $(A^2)_{ii}$ counts walks of length 2 from i back to i : go to a neighbour and come back. There is exactly one such walk per neighbour, so $(A^2)_{ii} = \deg(i)$.

$(A^3)_{ii}$ counts closed walks of length 3 from i . In a simple graph such a walk must visit three distinct vertices — a triangle through i — and each triangle can be traversed in 2 directions. So $(A^3)_{ii} = 2t_i$, where t_i is the number of triangles containing i .

tr(A^3)/6. Summing, $\text{tr}(A^3) = 2 \sum_i t_i$. Each triangle is counted once at each of its 3 vertices, so $\sum_i t_i = 3T$ where T is the total number of triangles. Hence $\text{tr}(A^3) = 6T$, and

$$T = \frac{\text{tr}(A^3)}{6}.$$

A genuinely usable algorithm: cube the adjacency matrix and read off the triangle count.

10.8 Let $u \leq v$ mean “there is a directed path from u to v ” in a DAG.

Reflexive: the empty path from u to itself.

Transitive: concatenate a path $u \rightsquigarrow v$ with $v \rightsquigarrow w$.

Antisymmetric: suppose $u \leq v$ and $v \leq u$ with $u \neq v$. Concatenating the two paths gives a closed directed walk through two distinct vertices, which contains a directed cycle — contradicting acyclicity. So $u = v$.

What fails with a cycle: exactly antisymmetry. On a directed cycle every vertex reaches every other, so all of them are mutually $\leq$ without being equal. Reflexivity and transitivity survive; it becomes a **preorder** rather than a partial order, and the standard repair is to collapse each strongly connected component to a point — giving the condensation, which is a DAG and therefore a genuine partial order again.

A.11. Chapter 11 — Recurrences and Asymptotics

11.1 Upper bound. For $n \geq 1$, each of n and 1 is at most n^2 , so

$$3n^2 + 100n + 7 \leq 3n^2 + 100n^2 + 7n^2 = 110n^2.$$

Lower bound. All terms are positive, so $3n^2 + 100n + 7 \geq 3n^2$.

So with $c = 3$, $C = 110$, $n_0 = 1$: $3n^2 \leq 3n^2 + 100n + 7 \leq 110n^2$ for all $n \geq n_0$, which is the definition of $\Theta(n^2)$.

11.2 Two preliminary observations. $2^{\log n} = n$ when the log is base 2 (and is a fixed power of n for any other base). And $\log(n!) = \Theta(n \log n)$ by Stirling.

In increasing order:

$$\sqrt{n} \prec 2^{\log n} = n \prec n \log n \prec \log(n!) \prec n(\log n)^2 \prec n^{1.5} \prec 2^n \prec n!$$

Justifications for the non-obvious comparisons:

- $n \log n \prec n(\log n)^2$: divide by $n \log n$, leaving $\log n \rightarrow \infty$.
- $n(\log n)^2 \prec n^{1.5}$: divide by n , compare $(\log n)^2$ with $\sqrt{n}$. Logs beat powers, and squaring a log does not change that (substitute $m = \log n$ and compare m^2 with $2^{m/2}$).
- $n^{1.5} \prec 2^n$: every exponential beats every polynomial (Chapter 15).
- $2^n \prec n!$: the ratio $n!/2^n$ has each successive factor $k/2$, which exceeds 1 from $k = 3$ onwards and grows.

11.3 Unrolling,

$$T(n) = T(n-1) + \frac{1}{n} = T(n-2) + \frac{1}{n-1} + \frac{1}{n} = \dots = T(1) + \sum_{k=2}^n \frac{1}{k}.$$

With $T(1) = 1$ this is $\sum_{k=1}^n 1/k = H_n$, the n -th harmonic number.

By Chapter 17, $H_n = \ln n + \gamma + O(1/n)$, so $T(n) = \Theta(\log n)$.

11.4 (a) $T = 4T(n/2) + n$. $c = \log_2 4 = 2$. $f(n) = n = O(n^{2-1})$, so case (i): $\Theta(n^2)$.

(b) $T = 4T(n/2) + n^2$. $c = 2$, $f = \Theta(n^2)$, case (ii): $\Theta(n^2 \log n)$.

(c) $T = 4T(n/2) + n^3$. $c = 2$, $f = \Omega(n^{2+1})$. Regularity: $4f(n/2) = 4n^3/8 = n^3/2 = (1/2)f(n)$, so $k = 1/2 < 1$. ✓ Case (iii): $\Theta(n^3)$.

(d) $T = 2T(n/2) + n/\log n$. $c = 1$. Here $f(n) = n/\log n$ is smaller than $n^c = n$, but not **polynomially** smaller — it is smaller only by a log factor, so $f \neq O(n^{1-\varepsilon})$ for any $\varepsilon > 0$. The Master Theorem does not apply.

Recursion tree. At level i there are 2^i subproblems of size $n/2^i$, contributing

$$2^i \cdot \frac{n/2^i}{\log(n/2^i)} = \frac{n}{\log n - i}.$$

Summing over $i = 0$ to $\log n - 1$:

$$T(n) = n \sum_{j=1}^{\log n} \frac{1}{j} = nH_{\log n} = \Theta(n \log \log n).$$

11.5 Guess: $T(n) \leq Cn \lg n$ for suitable C and all n above a base.

Step. Assume it for all smaller arguments:

$$\begin{aligned} T(n) &\leq C \left(\frac{n}{3}\right) \lg\left(\frac{n}{3}\right) + C \left(2\frac{n}{3}\right) \lg\left(2\frac{n}{3}\right) + n \\ &= C \frac{n}{3} (\lg n - \lg 3) + C \frac{2n}{3} (\lg n - \lg(3/2)) + n \\ &= Cn \lg n - Cn \left[\frac{1}{3} \lg 3 + \frac{2}{3} \lg(3/2) \right] + n. \end{aligned}$$

The bracket evaluates to

$$\frac{1}{3}(1.585) + \frac{2}{3}(0.585) = 0.528 + 0.390 = 0.918.$$

So $T(n) \leq Cn \lg n - 0.918Cn + n$, and this is at most $Cn \lg n$ provided $0.918C \geq 1$, i.e. $C \geq 1.09$.

Choose C at least that and large enough to cover the base cases, and the induction closes. Hence $T(n) = O(n \log n)$.

This recurrence is the one for quicksort with a guaranteed one-third/two-thirds split, and the computation shows why an unbalanced but **constant-ratio** split is still $O(n \log n)$ — only the constant suffers.

11.6 Characteristic equation $x^2 - 5x + 6 = 0$, factoring as $(x - 2)(x - 3) = 0$: roots 2 and 3, distinct.

So $a_n = A \cdot 2^n + B \cdot 3^n$. From the initial conditions:

$$\begin{aligned} n = 0 : \quad A + B &= 1 \\ n = 1 : \quad 2A + 3B &= 4 \end{aligned}$$

Subtracting twice the first from the second: $B = 2$, hence $A = -1$.

$$a_n = 2 \cdot 3^n - 2^n.$$

Check $n = 2$: $18 - 4 = 14$, and the recurrence gives $5(4) - 6(1) = 14$. ✓

11.7 Characteristic equation $x^2 - 4x + 4 = (x - 2)^2$: a **repeated** root $x = 2$.

With multiplicity 2, the general solution is $a_n = (A + Bn)2^n$ — the extra factor of n is what the repeated-root case supplies (without it the solution space would have dimension 1, too small).

$n = 0$: $A = 1$. $n = 1$: $(1 + B) \cdot 2 = 6$, so $B = 2$.

$$a_n = (1 + 2n)2^n.$$

Check $n = 2$: $(5)(4) = 20$, and $4(6) - 4(1) = 20$. ✓

11.8 Accounting method. Charge each push 2 credits:

- 1 pays for the push itself;
- 1 is deposited **with the pushed item**, to pay for its eventual removal.

pop costs 1 real unit, paid by the credit sitting on the item being removed. Amortised cost 0.

multipop(k) removes $j = \min(k, \text{size})$ items at real cost j , each paid by that item's own deposited credit. Amortised cost 0.

The bank never goes negative because every item on the stack carries exactly one unit of credit, and you can only pop items that are on the stack.

So the amortised costs are: push 2, pop 0, multipop 0 — all $O(1)$ — even though a single **multipop** can cost $\Theta(n)$ in real time. The total over any sequence of m operations is at most $2m$.

A.12. Chapter 12 — Number Theory and Modular Arithmetic

12.1 Euclid.

$$1071 = 2 \times 462 + 147$$

$$462 = 3 \times 147 + 21$$

$$147 = 7 \times 21 + 0$$

So $\gcd(1071, 462) = 21$.

Extended, by back-substitution. From the second line, $21 = 462 - 3 \times 147$. From the first, $147 = 1071 - 2 \times 462$. Substituting:

$$21 = 462 - 3(1071 - 2 \times 462) = 7 \times 462 - 3 \times 1071.$$

So $x = -3$, $y = 7$.

Check: $-3(1071) + 7(462) = -3213 + 3234 = 21$. ✓

12.2 Run Euclid on 43 and 17:

$$43 = 2 \times 17 + 9$$

$$17 = 1 \times 9 + 8$$

$$9 = 1 \times 8 + 1$$

Back-substitute:

$$1 = 9 - 8$$

$$= 9 - (17 - 9) = 2 \times 9 - 17$$

$$= 2(43 - 2 \times 17) - 17 = 2 \times 43 - 5 \times 17.$$

So $-5 \times 17 \equiv 1 \pmod{43}$, giving $17^{-1} \equiv -5 \equiv 38 \pmod{43}$.

Check: $17 \times 38 = 646 = 15 \times 43 + 1 = 645 + 1$. ✓

12.3 $N = 3 \times 5 \times 7 = 105$.

$$N_1 = 35, \quad 35 \equiv 2 \pmod{3}, \quad 2 \times 2 = 4 \equiv 1, \quad M_1 = 2$$

$$N_2 = 21, \quad 21 \equiv 1 \pmod{5}, \quad M_2 = 1$$

$$N_3 = 15, \quad 15 \equiv 1 \pmod{7}, \quad M_3 = 1$$

$$x = 2(35)(2) + 3(21)(1) + 2(15)(1) = 140 + 63 + 30 = 233.$$

Reducing mod 105: $233 - 210 = 23$.

Check: $23 = 7 \times 3 + 2 \checkmark$; $23 = 4 \times 5 + 3 \checkmark$; $23 = 3 \times 7 + 2 \checkmark$.

12.4 By Fermat. 11 is prime and $11 \nmid 7$, so $7^{10} \equiv 1$. Writing $222 = 22 \times 10 + 2$:

$$7^{222} = (7^{10})^{22} \cdot 7^2 \equiv 1^{22} \cdot 49 \equiv 49 - 44 = 5 \pmod{11}.$$

By square-and-multiply. $7^2 = 49 \equiv 5$; $7^4 \equiv 5^2 = 25 \equiv 3$; $7^8 \equiv 9$; $7^{16} \equiv 81 \equiv 4$; $7^{32} \equiv 16 \equiv 5$; $7^{64} \equiv 25 \equiv 3$; $7^{128} \equiv 9$. Now $222 = 128 + 64 + 16 + 8 + 4 + 2$, so

$$7^{222} \equiv 9 \times 3 \times 4 \times 9 \times 3 \times 5 \pmod{11}.$$

Reducing as we go: $9 \times 3 = 27 \equiv 5$; $5 \times 4 = 20 \equiv 9$; $9 \times 9 = 81 \equiv 4$; $4 \times 3 = 12 \equiv 1$; $1 \times 5 = 5$.

Both give 5. $\checkmark$ Fermat's route is dramatically shorter, which is the point: reduce the exponent modulo $p - 1$ before doing anything else.

12.5 Suppose $\gcd(a, n) = d > 1$ and that $ax \equiv 1 \pmod{n}$ had a solution. Then $ax - 1 = kn$ for some integer k , so

$$ax - kn = 1.$$

Now $d \mid a$ and $d \mid n$, so d divides the left-hand side — hence $d \mid 1$.

But $d > 1$, and no integer greater than 1 divides 1. Contradiction. So no inverse exists.

Note the structure: this is the Bézout argument run backwards. Bézout says the **smallest** positive value of $ax + ny$ is $\gcd(a, n)$; getting the value 1 therefore requires the gcd to be 1.

12.6 Squares mod 4. If $n = 2k$ then $n^2 = 4k^2 \equiv 0$. If $n = 2k + 1$ then $n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1 \equiv 1$. So every square is 0 or 1 mod 4.

Sums of two squares. Adding two values each in $\{0, 1\}$ gives 0, 1, or 2 modulo 4. Never 3.

An integer of the form $4k + 3$ is congruent to 3 mod 4, so it cannot be a sum of two squares.

This is a small instance of a much more general theorem (a positive integer is a sum of two squares iff every prime $\equiv 3 \pmod{4}$ occurs to an even power in its factorisation), but the mod-4 obstruction is the whole content of this case, and it is two lines.

12.7 Let $n > 2$ and pair each a coprime to n with $n - a$.

The partner is also coprime: $\gcd(n - a, n) = \gcd(-a, n) = \gcd(a, n) = 1$.

The pairing has no fixed point: $a = n - a$ would force $n = 2a$, and then $a \mid n$ with $a > 1$ (since $n > 2$), contradicting $\gcd(a, n) = 1$. The only escape would be $a = 1$ and $n = 2$, excluded by hypothesis.

And it is an involution: $n - (n - a) = a$.

So the $\varphi(n)$ coprime residues split into disjoint pairs, and $\varphi(n)$ is even.

12.8 Truncated division (C, C++, Java, Rust, Go): the quotient is truncated towards zero, $\text{trunc}(-17/5) = \text{trunc}(-3.4) = -3$, so the remainder is

$$-17 - 5(-3) = -17 + 15 = -2.$$

Floored division (Python, Haskell): the quotient is $\lfloor -3.4 \rfloor = -4$, so the remainder is

$$-17 - 5(-4) = -17 + 20 = 3.$$

The mathematically correct residue in $[0, 5)$ is 3.

A portable expression:

```
((x % m) + m) % m
```

Under floored division the inner $\%$ is already in $[0, m)$, so adding m and reducing again is a no-op. Under truncated division the inner $\%$ lies in $(-m, m)$, so adding m makes it positive and the outer $\%$ brings it back into range. Either way the result is correct, at the cost of one extra modulo.

A.13. Chapter 13 — Limits and Continuity

13.1 Scratch. $|(3x - 2) - 1| = |3x - 3| = 3|x - 1|$, which is less than ε when $|x - 1| < \varepsilon/3$.

Proof. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon/3$. Suppose $0 < |x - 1| < \delta$. Then

$$|(3x - 2) - 1| = 3|x - 1| < 3\delta = \varepsilon.$$

Since ε was arbitrary, $\lim_{x \rightarrow 1} (3x - 2) = 1$.

13.2 Scratch. $|x^2 - 4| = |x - 2||x + 2|$. The first factor is what we control; the second must be **bounded**. Restricting to $\delta \leq 1$ gives $1 < x < 3$, hence $|x + 2| < 5$. Then $|x^2 - 4| < 5|x - 2|$, which is under ε when $|x - 2| < \varepsilon/5$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \min(1, \varepsilon/5)$. Suppose $0 < |x - 2| < \delta$. Since $\delta \leq 1$ we have $|x + 2| < 5$, and since $\delta \leq \varepsilon/5$,

$$|x^2 - 4| = |x - 2||x + 2| < \left(\frac{\varepsilon}{5}\right)(5) = \varepsilon.$$

The min with 1 is the standard device: one part of δ bounds the nuisance factor, the other delivers the tolerance.

13.3 (a) $(x^2 - 9)/(x - 3) = x + 3$ for $x \neq 3$, so the limit is 6.

(b) Multiply by the conjugate:

$$\frac{\sqrt{1+x}-1}{x} \cdot \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1} \rightarrow \frac{1}{2}.$$

(c)

$$\frac{\sin 5x}{x} = 5 \cdot \frac{\sin 5x}{5x} \rightarrow 5 \cdot 1 = 5,$$

using the fundamental trigonometric limit with argument $5x$.

(d) Divide top and bottom by x^3 :

$$\frac{4 - 1/x^2}{2 + 7/x^3} \rightarrow \frac{4}{2} = 2.$$

13.4 $|\cos(1/x^2)| \leq 1$ for every $x \neq 0$, so

$$-|x| \leq x \cos\left(\frac{1}{x^2}\right) \leq |x|.$$

Both bounds tend to 0 as $x \rightarrow 0$, so by the squeeze theorem the middle does too:

$$\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x^2}\right) = 0.$$

Note that $\cos(1/x^2)$ itself has no limit at 0 — it oscillates ever faster — so the product law is unavailable and the squeeze is genuinely needed.

13.5 Take $f(x) = \operatorname{sgn}(x)$ and $g(x) = -\operatorname{sgn}(x)$.

Neither has a limit at 0: the one-sided limits are $+1$ and -1 for f , and the reverse for g .

But $f + g$ is identically 0, so $\lim_{x \rightarrow 0}(f + g) = 0$ exists.

No contradiction. The sum law states: **if** both limits exist, **then** the limit of the sum exists and equals their sum. It is an implication in one direction. It says nothing about the case where the hypotheses fail, and in particular does not claim the converse.

Confusing a theorem with its converse — Chapter 7 — in its natural habitat.

13.6 For $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1,$$

so $\lim_{x \rightarrow 1} f(x) = 2$.

Continuity at 1 requires $f(1) = \lim_{x \rightarrow 1} f(x)$, so $c = 2$.

This is a **removable** discontinuity: the function had a hole, and defining the value at the point to be the limit fills it.

13.7 Let $f(x) = x^3 - x - 1$, continuous everywhere (a polynomial).

$f(1) = 1 - 1 - 1 = -1 < 0$ and $f(2) = 8 - 2 - 1 = 5 > 0$.

Since 0 lies between $f(1)$ and $f(2)$, the Intermediate Value Theorem gives $c \in (1, 2)$ with $f(c) = 0$.

Bisection steps. After k bisections the bracket has width $1/2^k$. Requiring $1/2^k < 10^{-6}$ gives $2^k > 10^6$, so $k \geq 20$ (since $2^{20} = 1,048,576$).

13.8 Let $p(x) = a_n x^n + \dots + a_0$ with n odd and $a_n \neq 0$; assume $a_n > 0$ (otherwise apply the argument to $-p$, which has the same roots).

For large $|x|$ the leading term dominates: dividing by x^n ,

$$\frac{p(x)}{x^n} = a_n + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} \rightarrow a_n > 0.$$

So for sufficiently large x , $p(x)$ has the same sign as x^n .

Since n is odd, $x^n \rightarrow +\infty$ as $x \rightarrow +\infty$ and $x^n \rightarrow -\infty$ as $x \rightarrow -\infty$. So there exist a with $p(a) < 0$ and b with $p(b) > 0$.

p is continuous, so by the IVT it takes the value 0 somewhere between a and b .

(Even-degree polynomials can fail: $x^2 + 1$ has no real root, because both ends go the same way.)

A.14. Chapter 14 — Derivatives

14.1

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{\frac{x-(x+h)}{x(x+h)}}{h} \\ &= \frac{-h}{hx(x+h)} = \frac{-1}{x(x+h)}.\end{aligned}$$

The cancellation of h is legal because the limit never evaluates at $h = 0$. Letting $h \rightarrow 0$:

$$f'(x) = -\frac{1}{x^2}.$$

Consistent with the power rule applied to x^{-1} .

14.2 (a) Product rule: $3x^2e^x + x^3e^x = x^2e^x(3+x)$.

(b) Quotient rule:

$$\frac{2(x^2-3) - (2x+1)(2x)}{(x^2-3)^2} = \frac{2x^2-6-4x^2-2x}{(x^2-3)^2} = \frac{-2x^2-2x-6}{(x^2-3)^2}.$$

(c) Chain rule: $\cos(x^2+1) \cdot 2x = 2x \cos(x^2+1)$.(d) Chain rule: $e^{\cos x} \cdot (-\sin x) = -\sin x e^{\cos x}$.

(e) Chain rule:

$$\frac{d}{dx} \ln(\ln x) = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}.$$

(f) Chain rule twice:

$$\frac{d}{dx} (1 + \tan x)^{1/2} = \frac{1}{2} (1 + \tan x)^{-1/2} \cdot \sec^2 x = \frac{\sec^2 x}{2\sqrt{1 + \tan x}}.$$

14.3

$$\left(\frac{f}{g}\right)(x+h) - \left(\frac{f}{g}\right)(x) = \frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} = \frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h)g(x)}.$$

Add and subtract $f(x)g(x)$ in the numerator:

$$\begin{aligned}&= \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{g(x+h)g(x)} \\ &= \frac{g(x)[f(x+h) - f(x)] - f(x)[g(x+h) - g(x)]}{g(x+h)g(x)}.\end{aligned}$$

Divide by h and let $h \rightarrow 0$. The bracketed differences become $hf'(x)$ and $hg'(x)$ in the limit, and $g(x+h) \rightarrow g(x)$ by continuity:

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

The add-and-subtract trick is exactly the one from the product rule — which is not a coincidence, since the quotient rule is the product rule in disguise.

14.4 Differentiate $x^3 + y^3 = 6xy$ with respect to x , treating y as a function of x :

$$3x^2 + 3y^2y' = 6y + 6xy'.$$

Collecting:

$$y'(3y^2 - 6x) = 6y - 3x^2 \Rightarrow y' = \frac{2y - x^2}{y^2 - 2x}.$$

Horizontal tangent requires $y' = 0$, so $2y = x^2$, i.e. $y = x^2/2$ (with the denominator nonzero). Substituting into the curve:

$$x^3 + \frac{x^6}{8} = 6x \cdot \frac{x^2}{2} = 3x^3 \Rightarrow \frac{x^6}{8} = 2x^3 \Rightarrow x^3(x^3 - 16) = 0.$$

$x = 0$ gives $y = 0$, but there the denominator $y^2 - 2x$ also vanishes: the origin is a singular point of the curve (it self-intersects), not a horizontal tangent.

The genuine solution is $x^3 = 16$, so $x = 2 \cdot 2^{1/3} \approx 2.52$ and $y = x^2/2 = 2^{5/3} \approx 3.17$.

14.5 Take logs first:

$$\ln y = 3 \ln(x^2 + 1) + \frac{1}{2} \ln(x - 1) - 5 \ln(x + 4).$$

Differentiate both sides (implicitly on the left):

$$\frac{y'}{y} = \frac{6x}{x^2 + 1} + \frac{1}{2(x - 1)} - \frac{5}{x + 4}.$$

Multiply through by y :

$$y' = \frac{(x^2 + 1)^3 \sqrt{x - 1}}{(x + 4)^5} \left[\frac{6x}{x^2 + 1} + \frac{1}{2(x - 1)} - \frac{5}{x + 4} \right].$$

Doing this with the product and quotient rules directly is possible and substantially worse. Taking logs turned a product of four factors into a sum of four terms — the same reason log-likelihood is what gets differentiated in statistics.

14.6 $\tanh = \sinh / \cosh$, so by the quotient rule and $\sinh' = \cosh$, $\cosh' = \sinh$:

$$\tanh'(x) = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x},$$

using $\cosh^2 - \sinh^2 = 1$. And

$$\frac{1}{\cosh^2 x} = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = 1 - \tanh^2 x.$$

Computational significance. The forward pass already computed $a = \tanh(x)$ and stored it. The backward pass needs $\tanh'(x)$, and the identity says that is $1 - a^2$ — one multiplication and one subtraction, using a value already in memory. The **input** x never has to be retained.

For a network with millions of activations, not having to store the pre-activation values is a real memory saving, and it is why activations whose derivative is expressible in terms of their output were preferred.

14.7 $f'(0)$ exists. From the definition,

$$f'(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h) - 0}{h} = \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0,$$

by the squeeze theorem, since $|h \sin(1/h)| \leq |h|$.

f' away from zero. By the product and chain rules, for $x \neq 0$:

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

f' is not continuous at 0. As $x \rightarrow 0$ the first term tends to 0 (squeeze again), but $\cos(1/x)$ oscillates between -1 and 1 without settling. So $\lim_{x \rightarrow 0} f'(x)$ does not exist, and in particular does not equal $f'(0) = 0$.

So f is differentiable everywhere and f' is not continuous: the class of differentiable functions is strictly larger than the class of continuously differentiable (C^1) ones. This is why theorems often say “continuously differentiable” rather than just “differentiable”.

- 14.8** With $C = e/2 \approx 1.359$ and $\varepsilon_{\text{mach}} = 2.2 \times 10^{-16}$, the error model is $E(h) \approx Ch + \varepsilon_{\text{mach}}|f(x)|/h$; taking $|f(1)| = e \approx 2.718$:

h	truncation Ch	roundoff $\varepsilon e/h$	total
10^{-3}	1.36×10^{-3}	6.0×10^{-13}	1.4×10^{-3}
10^{-8}	1.36×10^{-8}	6.0×10^{-8}	7.4×10^{-8}
10^{-14}	1.36×10^{-14}	6.0×10^{-2}	6.0×10^{-2}

$h = 10^{-8}$ wins by a wide margin.

The theoretical optimum is $h^* = \sqrt{\varepsilon_{\text{mach}}|f|/C} \approx \sqrt{4.4 \times 10^{-16}} \approx 2 \times 10^{-8}$, confirming the rule of thumb $h \approx \sqrt{\varepsilon_{\text{mach}}}$.

Note the last row: making h **smaller** made the answer catastrophically worse. This is the single most counter-intuitive fact in numerical differentiation and the reason automatic differentiation is preferred.

A.15. Chapter 15 — What Derivatives Are For

- 15.1** $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 4]$.

$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$, so the critical points are $x = -1$ and $x = 3$, both interior to the interval. f' exists everywhere, so there are no other candidates besides the endpoints.

x	-2	-1	3	4
$f(x)$	3	10	-22	-15

(For instance $f(-1) = -1 - 3 + 9 + 5 = 10$ and $f(3) = 27 - 27 - 27 + 5 = -22$.)

Absolute maximum 10 at $x = -1$; absolute minimum -22 at $x = 3$.

- 15.2** $f(x) = x^3 - x$ is a polynomial, hence continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. And $f(-1) = -1 + 1 = 0 = 1 - 1 = f(1)$. All hypotheses hold.

$f'(x) = 3x^2 - 1 = 0$ gives $x = \pm 1/\sqrt{3} \approx \pm 0.577$, both inside $(-1, 1)$.

So Rolle's theorem promises at least one such point and there are two.

- 15.3** If $a = b$ both sides are zero. Otherwise, $\sin$ is differentiable everywhere, so by the mean value theorem on the interval with endpoints a and b there is a c between them with

$$\sin a - \sin b = \cos(c)(a - b).$$

Taking absolute values and using $|\cos c| \leq 1$:

$$|\sin a - \sin b| = |\cos c||a - b| \leq |a - b|.$$

In the vocabulary of the chapter: $\sin$ is 1-Lipschitz, because its derivative is bounded by 1.

- 15.4 (a)** $0/0$. Apply twice:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}.$$

(Check with the series: $e^x - 1 - x = x^2/2 + O(x^3)$.)

- (b)** $0 \cdot \infty$ — must be converted first. Write $x \ln x = (\ln x)/(1/x)$, now $-\infty/\infty$:

$$\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

- (c)** 1^∞ . Take logs: $\ln y = x \ln(1 + 1/x) = \ln(1 + 1/x)/(1/x)$, which is $0/0$. Differentiating top and bottom with respect to x :

$$\frac{-1/x^2}{1 + 1/x} \cdot \frac{1}{-1/x^2} = \frac{1}{1 + 1/x} \rightarrow 1.$$

So $\ln y \rightarrow 1$ and $y \rightarrow e$. (Which is the definition of e from Chapter 4, recovered.)

- (d)** $0/0$:

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{\tan^2 x}{3x^2} = \frac{1}{3},$$

using $\sec^2 - 1 = \tan^2$ and $\tan x/x \rightarrow 1$.

- 15.5** With fixed volume $V = \pi r^2 h$, the surface area of a closed cylinder is $A = 2\pi r^2 + 2\pi r h$. Substituting $h = V/(\pi r^2)$:

$$A(r) = 2\pi r^2 + \frac{2V}{r}, \quad r > 0.$$

$A'(r) = 4\pi r - 2V/r^2 = 0$ gives $4\pi r^3 = 2V$, so $V = 2\pi r^3$. Then

$$h = \frac{V}{\pi r^2} = \frac{2\pi r^3}{\pi r^2} = 2r.$$

So $h/r = 2$: **the optimal can is as tall as it is wide** — the height equals the diameter.

$A''(r) = 4\pi + 4V/r^3 > 0$ for $r > 0$, so A is convex and this is the global minimum.

(Real drinks cans are noticeably taller than this, because the top and bottom are thicker than the walls and because people like the shape.)

- 15.6** $f(x) = x^2 - 2$, $f'(x) = 2x$, so $x_{n+1} = x_n - (x_n^2 - 2)/(2x_n) = (x_n + 2/x_n)/2$.

n	x_n	error $ x_n - \sqrt{2} $	correct digits
0	1	4.14×10^{-1}	0

n	x_n	error $ x_n - \sqrt{2} $	correct digits
1	1.5	8.58×10^{-2}	1
2	1.4166666...	2.45×10^{-3}	2.6
3	1.4142156...	2.1×10^{-6}	5.7

($\sqrt{2} = 1.41421356\dots$)

The digit count roughly doubles at each step: $1 \rightarrow 2.6 \rightarrow 5.7$, and a fourth iteration would give about 12. That is quadratic convergence, exactly as the theorem predicts, with the asymptotic constant $|f''|/(2|f'|) = 2/(2 \cdot 2\sqrt{2}) \approx 0.354$.

15.7 $f(x) = x^{1/3}$, so $f'(x) = (1/3)x^{-2/3}$. The Newton step is

$$x_{n+1} = x_n - \frac{x_n^{1/3}}{(1/3)x_n^{-2/3}} = x_n - 3x_n^{1/3}x_n^{2/3} = x_n - 3x_n = -2x_n.$$

So $x_n = (-2)^n x_0$: the iterates alternate in sign and **double** in magnitude every step. From any $x_0 \neq 0$ the method diverges, and it does so spectacularly.

Which hypothesis fails. The convergence theorem requires $f'(r) \neq 0$ at the root $r = 0$. Here $f'(x) = (1/3)x^{-2/3} \rightarrow \infty$ as $x \rightarrow 0$: f is not differentiable at the root at all (it has a vertical tangent, Chapter 14). The tangent line at any x_n is nearly vertical near the root, so its intercept overshoots wildly.

15.8 The pile is a cone with $h = r$ always, so

$$V = \frac{\pi r^2 h}{3} = \frac{\pi h^3}{3}.$$

Differentiate the relation with respect to t before substituting any values:

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}.$$

Now substitute $(dV)/(dt) = 10$ and $h = 5$:

$$10 = \pi(25) \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{10}{25\pi} = \frac{2}{5\pi} \approx 0.127.$$

The height is increasing at about 0.127 units per second.

Substituting $h = 5$ **before** differentiating would have turned h into a constant and given $(dV)/(dt) = 0$ — the classic error in this topic.

A.16. Chapter 16 — Integration

16.1 Partition $[0, 1]$ into n equal pieces, $\Delta x = 1/n$, right endpoints $x_i = i/n$:

$$\sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \left(\frac{1}{n}\right) = \frac{1}{n^3} \sum_{i=1}^n i^2.$$

By Chapter 2, $\sum i^2 = n(n+1)(2n+1)/6$, so the sum is

$$\frac{n(n+1)(2n+1)}{6n^3} = \frac{(1)(1+1/n)(2+1/n)}{6} \rightarrow \frac{2}{6} = \frac{1}{3}.$$

So $\int_0^1 x^2 dx = 1/3$, agreeing with the power rule antiderivative $x^3/3$.

16.2 (a) Substitute $u = x^2 + 1$, $du = 2x dx$:

$$\int x\sqrt{x^2+1} dx = \frac{1}{2} \int \sqrt{u} du = \frac{1}{3} u^{3/2} + C = \frac{(x^2+1)^{3/2}}{3} + C.$$

(b) Substitute $u = \ln x$, $du = dx/x$:

$$\int \frac{\ln x}{x} dx = \int u du = \frac{(\ln x)^2}{2} + C.$$

(c) Substitute $u = \sin x$, $du = \cos x dx$; the limits become 0 and 1:

$$\int_0^{\pi/2} \sin^3 x \cos x dx = \int_0^1 u^3 du = \frac{1}{4}.$$

(d)

$$\int \frac{dx}{x^2+9} = \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C,$$

from the standard form with $u = x/3$.

16.3 Take $u = x^2$, $dv = e^{-x} dx$, so $du = 2x dx$ and $v = -e^{-x}$:

$$\int x^2 e^{-x} dx = -x^2 e^{-x} + 2 \int x e^{-x} dx.$$

Parts again on the new integral, with $u = x$:

$$\int x e^{-x} dx = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x}.$$

Combining:

$$\int x^2 e^{-x} dx = -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + C = -e^{-x}(x^2 + 2x + 2) + C.$$

Each application of parts reduced the power of x by one — which is the reason to choose the polynomial as u .

16.4 Take $u = \ln x$ and $dv = dx$, so $du = dx/x$ and $v = x$:

$$\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C.$$

The trick is recognising that “there is only one factor” is not an obstacle: the second factor is 1, and $dv = dx$ is a legitimate choice. The same device handles $\int \arctan x dx$ and $\int \arcsin x dx$.

16.5 $x^2 + x - 6 = (x+3)(x-2)$, so

$$\frac{x+5}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2}, \quad x+5 = A(x-2) + B(x+3).$$

Put $x = 2$: $7 = 5B$, so $B = 7/5$. Put $x = -3$: $2 = -5A$, so $A = -2/5$.

$$\int \frac{x+5}{x^2+x-6} dx = -\frac{2}{5} \ln|x+3| + \frac{7}{5} \ln|x-2| + C.$$

16.6 (a)

$$\int_1^\infty x^{-3/2} dx = [-2x^{-1/2}]_1^\infty = 0 - (-2) = 2.$$

Converges ($p = 3/2 > 1$).

(b)

$$\int_0^1 \frac{dx}{x} = \lim_{t \rightarrow 0^+} [\ln x]_t^1 = \lim_{t \rightarrow 0^+} (-\ln t) = +\infty.$$

Diverges.

(c)

$$\int_0^\infty e^{-x} dx = [-e^{-x}]_0^\infty = 0 + 1 = 1.$$

Converges.

16.7 Area. The curves $y = x^2$ and $y = x + 2$ meet where $x^2 = x + 2$, i.e. $x^2 - x - 2 = 0$, giving $x = -1$ and $x = 2$. On that interval the line is above the parabola, so

$$A = \int_{-1}^2 (x + 2 - x^2) dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}.$$

Volume. Rotating $y = \sqrt{x}$ on $[0, 4]$ about the x -axis, the disc method gives cross-sectional area $\pi(\sqrt{x})^2 = \pi x$:

$$V = \int_0^4 \pi x dx = \pi [x^2/2]_0^4 = 8\pi.$$

16.8 $h = 0.25$, and $f(x) = e^{-x^2}$:

$$\begin{aligned} f(0) &= 1, & f(0.25) &= 0.939413, & f(0.5) &= 0.778801, \\ f(0.75) &= 0.569783, & f(1) &= 0.367879. \end{aligned}$$

Trapezoid.

$$T = h \left[\frac{f_0}{2} + f_1 + f_2 + f_3 + \frac{f_4}{2} \right] = 0.25[0.5 + 2.287997 + 0.183940] = 0.742984.$$

Error: $|0.746824 - 0.742984| = 3.84 \times 10^{-3}$.

Simpson.

$$S = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + f_4] = \frac{0.25}{3} [1 + 3.757652 + 1.557602 + 2.279132 + 0.367879] = 0.746855.$$

Error: $|0.746824 - 0.746855| = 3.1 \times 10^{-5}$.

Comparison. The errors differ by a factor of about 124. The theory predicts trapezoid error $\propto h^2$ and Simpson error $\propto h^4$, so at $h = 0.25$ the ratio should be of order $1/h^2 = 16$ times

the ratio of the two constants ($1/12$ versus $1/180$, another factor of 15). $16 \times 15 = 240$, the same order of magnitude as observed. The agreement is as good as one should expect from a four-interval estimate.

A.17. Chapter 17 — Sequences, Series, and Taylor Expansion

17.1 (a) $\sum n/(n^2 + 1)$. Limit comparison with $\sum 1/n$: the ratio $(n/(n^2 + 1))/(1/n) = n^2/(n^2 + 1) \rightarrow 1$. Since the harmonic series diverges, so does this. **Diverges.**

(b) $\sum 1/(n^2 + 1)$. Comparison: $1/(n^2 + 1) < 1/n^2$, and $\sum 1/n^2$ converges ($p = 2 > 1$). **Converges.**

(c) $\sum n!/n^n$. Ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{n^n}{(n+1)^n} = \frac{1}{(1 + 1/n)^n} \rightarrow \frac{1}{e} < 1.$$

Converges.

(d) $\sum (-1)^n/\sqrt{n}$. Alternating series test: $1/\sqrt{n}$ is positive, decreasing, and tends to 0. **Converges** — but only conditionally, since $\sum 1/\sqrt{n}$ is a p -series with $p = 1/2 \leq 1$ and diverges.

(e) $\sum 2^n/n!$. Ratio test: $a_{n+1}/a_n = 2/(n+1) \rightarrow 0 < 1$. **Converges** (to $e^2 - 1$ if the sum starts at $n = 1$).

17.2 First. Geometric with first term 3 and ratio $1/4$:

$$\sum_{n=0}^{\infty} \frac{3}{4^n} = 3 \cdot \frac{1}{1 - 1/4} = 3 \cdot \frac{4}{3} = 4.$$

Second. Geometric with ratio $2/3$, starting at $n = 2$ so the first term is $(2/3)^2 = 4/9$:

$$\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n = \frac{4/9}{1 - 2/3} = \frac{4/9}{1/3} = \frac{4}{3}.$$

17.3 Ratio test on $\sum (x-2)^n/(n3^n)$:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x-2|}{3} \cdot \frac{n}{n+1} \rightarrow \frac{|x-2|}{3}.$$

This is less than 1 iff $|x-2| < 3$, so $R = 3$ and the open interval is $(-1, 5)$.

At $x = 5$: the series becomes $\sum 3^n/(n3^n) = \sum 1/n$ — the harmonic series. **Diverges.**

At $x = -1$: it becomes $\sum (-3)^n/(n3^n) = \sum (-1)^n/n$ — alternating harmonic. **Converges** (conditionally).

Interval of convergence: $[-1, 5)$.

17.4 By substitution. In the geometric series $1/(1-u) = \sum u^n$ put $u = -x^2$:

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - \dots, \quad |x| < 1.$$

By repeated differentiation one gets the same coefficients, but the computation is unpleasant — f'' , f''' , $f^{(4)}$ of $(1+x^2)^{-1}$ are already messy. This is the general lesson: build

new series from known ones by substitution, differentiation, and integration rather than from the definition.

Integrating term by term from 0 to x (legal inside the radius):

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

At $x = 1$ (where it converges, by the alternating series test, though the radius argument alone does not cover the endpoint):

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

— the Leibniz formula. It converges so slowly that computing π to 10 digits would need about 10^{10} terms.

17.5 Four terms means the degree-3 Taylor polynomial:

$$P_3(0.5) = 1 + 0.5 + \frac{0.25}{2} + \frac{0.125}{6} = 1 + 0.5 + 0.125 + 0.0208333 = 1.6458333.$$

Error bound. By Taylor's theorem,

$$R_3 = \frac{e^{\xi}}{4!}(0.5)^4 \quad \text{for some } \xi \in (0, 0.5).$$

Since $e^{\xi} \leq e^{0.5} \approx 1.6487$,

$$|R_3| \leq (1.6487) \frac{0.0625}{24} = 4.29 \times 10^{-3}.$$

Actual error. $e^{0.5} = 1.6487213$, so the true error is $1.6487213 - 1.6458333 = 2.89 \times 10^{-3}$.

The bound 4.29×10^{-3} correctly exceeds the actual 2.89×10^{-3} , and is within a factor of 1.5 — Taylor bounds are usually pessimistic by a small constant because they use the worst-case derivative on the interval.

17.6

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

so

$$\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \dots$$

and dividing by x^3 :

$$\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \dots \rightarrow -\frac{1}{6}.$$

Compare the L'Hôpital route: three applications, each requiring you to differentiate correctly and confirm the form is still $0/0$. The series does it by inspection — which is generally true, and is why series are the better tool for limits at a point once you have them.

17.7 Convergence. $1/n$ is positive, decreasing, and tends to 0, so the alternating series test applies. (The sum is $\ln 2$, from the series for $\ln(1+x)$ at $x = 1$.)

Error. The alternating series bound says the error after N terms is at most the first omitted term, $1/(N+1)$. Requiring $1/(N+1) < 10^{-3}$ gives $N \geq 1000$.

Why nobody does this. A thousand terms for three decimal digits — and by the same bound, a million terms for six, and 10^{15} for fifteen. Each extra digit costs a factor of ten in work.

The practical alternative: use

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$$

with $x = 1/3$, which gives $\ln 2$ and converges like 9^{-n} — about one digit per term. Argument reduction plus a fast-converging series is the standard recipe for every transcendental function in a library.

17.8 Integral test with $f(x) = 1/(x(\ln x)^p)$, which is positive and decreasing for $x \geq 2$.

Substitute $u = \ln x$, $du = dx/x$:

$$\int_2^\infty \frac{dx}{x(\ln x)^p} = \int_{\ln 2}^\infty \frac{du}{u^p}.$$

By the p -test of Chapter 16, this converges iff $p > 1$.

So $\sum 1/(n(\ln n)^p)$ converges iff $p > 1$.

Note the case $p = 1$: $\sum 1/(n \ln n)$ **diverges**, but only just — its partial sums grow like $\ln \ln n$. Iterating gives an infinite hierarchy of series each diverging more slowly than the last, and no single “slowest divergent series” exists.

A.18. Chapter 18 — Calculus in Many Variables

18.1 $f(x, y) = x^3y - 4xy^2 + y$.

$$\frac{\partial f}{\partial x} = 3x^2y - 4y^2, \quad \frac{\partial f}{\partial y} = x^3 - 8xy + 1.$$

At $(1, 2)$:

$$\nabla f(1, 2) = \begin{pmatrix} 3(1)(2) - 4(4) \\ 1 - 8(1)(2) + 1 \end{pmatrix} = \begin{pmatrix} -10 \\ -14 \end{pmatrix}.$$

Second partials:

$$f_{xx} = 6xy, \quad f_{xy} = 3x^2 - 8y, \quad f_{yy} = -8x,$$

so

$$H = \begin{pmatrix} 6xy & 3x^2 - 8y \\ 3x^2 - 8y & -8x \end{pmatrix}, \quad H(1, 2) = \begin{pmatrix} 12 & -13 \\ -13 & -8 \end{pmatrix}.$$

Symmetric, as Clairaut requires.

18.2 $f(x, y) = x^2 + 3xy$, so $\nabla f = (2x + 3y, 3x)$ and $\nabla f(1, 2) = (8, 3)$.

Normalise the direction: $\|(3, 4)\| = 5$, so $u = (3/5, 4/5)$.

$$D_u f(1, 2) = \nabla f \cdot u = (8)(3/5) + (3)(4/5) = \frac{24 + 12}{5} = \frac{36}{5} = 7.2.$$

Forgetting to normalise would give 36 — five times too large, and dimensionally meaningless, since a directional derivative is a rate per unit distance.

18.3 $f(x, y) = x^2 e^{xy}.$

$$f_x = 2xe^{xy} + x^2 y e^{xy} = e^{xy}(2x + x^2 y),$$

$$f_{xy} = x e^{xy}(2x + x^2 y) + e^{xy}(x^2) = e^{xy}(2x^2 + x^3 y + x^2) = e^{xy}(3x^2 + x^3 y).$$

Other order:

$$f_y = x^3 e^{xy}, \quad f_{yx} = 3x^2 e^{xy} + x^3 y e^{xy} = e^{xy}(3x^2 + x^3 y).$$

Equal. ✓

18.4 $f(x, y) = x^3 - 3x + y^2.$

$\nabla f = (3x^2 - 3, 2y) = 0$ gives $x = \pm 1, y = 0$: two critical points, $(1, 0)$ and $(-1, 0)$.

$$H = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}.$$

At $(1, 0)$: $H = \text{diag}(6, 2)$, both eigenvalues positive — positive definite. **Local minimum**, value $f(1, 0) = -2$.

At $(-1, 0)$: $H = \text{diag}(-6, 2)$, eigenvalues of both signs — indefinite. **Saddle point**, value $f(-1, 0) = 2$.

Geometrically: along the x -axis f behaves like a cubic (up then down), along the y -axis like a parabola (up). At $(-1, 0)$ those two disagree, which is exactly what a saddle is.

18.5 By the chain rule. With $z = x^2 y, x = \cos t, y = \sin t$:

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= (2xy)(-\sin t) + (x^2)(\cos t) \\ &= -2 \cos t \sin^2 t + \cos^3 t. \end{aligned}$$

By substitution first. $z = \cos^2 t \sin t$, so by the product and chain rules

$$\frac{dz}{dt} = 2 \cos t (-\sin t) \sin t + \cos^2 t \cos t = -2 \cos t \sin^2 t + \cos^3 t.$$

Identical. ✓ The chain rule version generalises to cases where substituting is impossible; the direct version is a good check.

18.6 Minimise $f = x^2 + y^2 + z^2$ subject to $g = 2x + 3y + z - 12 = 0$.

Why squaring is legitimate: $\sqrt{\cdot}$ is strictly increasing on $[0, \infty)$, so it preserves the order of nonnegative quantities and therefore has the same minimiser (Chapter 2's “safe transformation”). And the squared distance is differentiable at the origin while the distance is not.

$\nabla f = \lambda \nabla g$ gives

$$2x = 2\lambda, \quad 2y = 3\lambda, \quad 2z = \lambda,$$

so $x = \lambda$, $y = 3\lambda/2$, $z = \lambda/2$.

Substituting into the constraint:

$$2\lambda + \frac{9\lambda}{2} + \frac{\lambda}{2} = 12 \Rightarrow 7\lambda = 12 \Rightarrow \lambda = \frac{12}{7}.$$

So the closest point is $(12/7, 18/7, 6/7)$, at distance $\sqrt{144 + 324 + 36}/7 = \sqrt{504}/7 = 12/\sqrt{14}$.

Sanity check: the closest point on a plane to the origin lies along the normal direction $(2, 3, 1)$ — and indeed our point is $(6/7)(2, 3, 1)$. ✓

18.7

$$\int_0^1 \int_0^2 (x + y^2) dx dy = \int_0^1 \left[\frac{x^2}{2} + y^2 x \right]_{x=0}^{x=2} dy = \int_0^1 (2 + 2y^2) dy = 2 + \frac{2}{3} = \frac{8}{3}.$$

Other order:

$$\int_0^2 \int_0^1 (x + y^2) dy dx = \int_0^2 \left[xy + \frac{y^3}{3} \right]_{y=0}^{y=1} dx = \int_0^2 \left(x + \frac{1}{3} \right) dx = 2 + \frac{2}{3} = \frac{8}{3}.$$

Equal, as Fubini guarantees for a continuous integrand on a rectangle.

18.8 In polar coordinates $x^2 + y^2 = r^2$ and $dA = r dr d\theta$, and the unit disc is $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$:

$$\iint_R (x^2 + y^2) dA = \int_0^{2\pi} \int_0^1 r^2 \cdot r dr d\theta = 2\pi \left[\frac{r^4}{4} \right]_0^1 = \frac{2\pi}{4} = \frac{\pi}{2}.$$

The extra r from the Jacobian is what makes the inner integral $\int r^3 dr$ rather than $\int r^2 dr$ — getting it wrong gives $2\pi/3$, and the error is invisible unless you know it should be there.

18.9 Write $h = W_1 x \in \mathbb{R}^h$, $a = \sigma(h)$, $y = W_2 a \in \mathbb{R}$, $L = f(y)$.

Chain rule as a product of Jacobians, from the loss backwards:

$$\frac{\partial L}{\partial W_1} \quad \text{requires} \quad \underbrace{\frac{\partial L}{\partial y}}_{1 \times 1} \cdot \underbrace{\frac{\partial y}{\partial a}}_{1 \times h} \cdot \underbrace{\frac{\partial a}{\partial h}}_{h \times h \text{ diagonal}} \cdot \underbrace{\frac{\partial h}{\partial W_1}}.$$

Concretely, with $g = \partial L / \partial y$ (a scalar here):

$$\delta = (W_2^T g) \circ \sigma'(h) \in \mathbb{R}^h, \quad \frac{\partial L}{\partial W_1} = \delta x^T \in \mathbb{R}^{h \times n}.$$

Cost. Multiplying **left to right** (reverse mode): $W_2^T g$ is a matrix–vector product costing $O(h)$, the elementwise product costs $O(h)$, and the outer product δx^T costs $O(hn)$. Total $O(hn)$ — the same order as one forward pass.

Multiplying **right to left** (forward mode) would require materialising $\partial h / \partial W_1$, which has $h \times (hn)$ entries, and then contracting through the $h \times h$ Jacobian: $O(h^2 n)$.

Reverse mode is cheaper by a factor of h , the hidden width — and in a real network, by a factor equal to the parameter count.

A.19. Chapter 19 — Vectors and Geometry

19.1 $u = (1, -2, 3)$, $v = (4, 0, -1)$.

$$u \cdot v = 4 + 0 - 3 = 1. \quad \|u\| = \sqrt{1 + 4 + 9} = \sqrt{14}, \quad \|v\| = \sqrt{16 + 0 + 1} = \sqrt{17}.$$

$$\cos \theta = \frac{1}{\sqrt{14}\sqrt{17}} = \frac{1}{\sqrt{238}} \approx 0.0648, \quad \theta \approx 86.3^\circ.$$

Nearly perpendicular.

$$\text{proj}_v(u) = \left(\frac{u \cdot v}{v \cdot v} \right) v = \frac{1}{17}(4, 0, -1).$$

$$u \times v = ((-2)(-1) - (3)(0), (3)(4) - (1)(-1), (1)(0) - (-2)(4)) = (2, 13, 8).$$

Checks: $u \cdot (u \times v) = 2 - 26 + 24 = 0 \checkmark$ and $v \cdot (u \times v) = 8 + 0 - 8 = 0 \checkmark$.

19.2 (a) $\{(x, y, z) : x + 2y = 0\}$. Subspace. Contains 0; if $x_1 + 2y_1 = 0$ and $x_2 + 2y_2 = 0$ then the sum and any scalar multiple satisfy the same equation (it is linear and homogeneous).

(b) $\{(x, y, z) : x \geq 0\}$. Not a subspace. $(1, 0, 0)$ is in it, $(-1)(1, 0, 0) = (-1, 0, 0)$ is not. Fails under negative scaling.

(c) $\{(x, y, z) : xy = 0\}$. Not a subspace. It is the union of two planes. $(1, 0, 0)$ and $(0, 1, 0)$ are both in it, but their sum $(1, 1, 0)$ has $xy = 1 \neq 0$. Fails under addition.

(d) $\{(x, y, z) : x = y = z\}$. Subspace — it is the line spanned by $(1, 1, 1)$, and a span is always a subspace.

19.3 Look for a, b with $a(1, 2, 3) + b(2, 1, 0) = (4, 5, 6)$:

$$a + 2b = 4$$

$$2a + b = 5$$

$$3a = 6$$

The third gives $a = 2$, then the first gives $b = 1$, and the second checks out: $4 + 1 = 5$. $\checkmark$

So the vectors are **dependent**, with the explicit relation

$$2(1, 2, 3) + 1(2, 1, 0) - 1(4, 5, 6) = (0, 0, 0).$$

19.4 Suppose the set is $\{v_1, \dots, v_k\}$ with $v_1 = 0$. Then

$$1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_k = 0$$

is a linear combination equal to zero in which not all coefficients are zero — the coefficient of v_1 is 1.

That is precisely the definition of linear dependence.

(Intuitively: the zero vector contributes no direction, so it is always redundant.)

19.5 Expand both squared norms using $\|w\|^2 = w \cdot w$ and bilinearity:

$$\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2,$$

$$\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2.$$

Adding, the cross terms cancel:

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$

Geometrically: in a parallelogram with sides u and v , the diagonals are $u + v$ and $u - v$. The law says **the sum of the squares of the diagonals equals the sum of the squares of the four sides**.

It also characterises which norms come from an inner product: a norm satisfies the parallelogram law iff it is Euclidean-like. The ℓ_1 and ℓ_∞ norms do not, which is why there is no “ ℓ_1 dot product” and no notion of ℓ_1 angle.

19.6 $\|v\|_\infty \leq \|v\|_2$. Let j be an index achieving the max. Then

$$\|v\|_\infty^2 = v_j^2 \leq \sum_i v_i^2 = \|v\|_2^2.$$

$\|v\|_2 \leq \|v\|_1$. Square the right side:

$$\|v\|_1^2 = \left(\sum_i |v_i| \right)^2 = \sum_i v_i^2 + \sum_{i \neq j} |v_i| |v_j| \geq \sum_i v_i^2 = \|v\|_2^2,$$

since the cross terms are nonnegative.

Equality. Both become equalities exactly when at most one component is nonzero — e.g. $v = (5, 0, 0)$, where all three norms equal 5. The inequalities are strict as soon as two components are nonzero, and the gap grows with the spread: for $v = (1, 1, \dots, 1) \in \mathbb{R}^n$ the three norms are 1, $\sqrt{n}$, and n .

19.7 Basis. $\{1, x, x^2\}$ spans $\mathcal{P}_2$ by definition of a polynomial of degree at most 2. Independent: if $a + bx + cx^2 = 0$ **as a function**, then the polynomial has infinitely many roots, so by Chapter 2 it is the zero polynomial and $a = b = c = 0$.

Coordinates of $p(x) = 3 - x + 2x^2$: $(3, -1, 2)$.

In the basis $\{1, 1 + x, (1 + x)^2\}$. Substitute $u = 1 + x$, so $x = u - 1$ and $x^2 = u^2 - 2u + 1$:

$$\begin{aligned} p &= 3 - (u - 1) + 2(u^2 - 2u + 1) \\ &= 3 - u + 1 + 2u^2 - 4u + 2 = 2u^2 - 5u + 6. \end{aligned}$$

Coordinates $(6, -5, 2)$ in the order $(1, 1 + x, (1 + x)^2)$.

Same vector, different names — which is exactly the point of a basis.

19.8 $\|u + v\|^2 = (u + v) \cdot (u + v) = \|u\|^2 + 2u \cdot v + \|v\|^2 = \|u\|^2 + \|v\|^2,$

since $u \cdot v = 0$.

This is the **Pythagorean theorem**. In $\mathbb{R}^2$ with u and v along the axes, it is literally $a^2 + b^2 = c^2$ — but the vector proof works in any dimension and, more importantly, in any inner product space, which is why it applies to random variables (Chapter 29) and to functions (Chapter 24).

A.20. Chapter 20 — Matrices Are Functions

20.1 $AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$

Different. ✓

What B does: it sends $e_1 \mapsto (0, 1) = e_2$ and $e_2 \mapsto (1, 0) = e_1$ — it swaps the coordinates. Geometrically that is a **reflection in the line $y = x$** . (It is orthogonal with determinant -1 , confirming a reflection rather than a rotation.)

Reading the two products: AB swaps first then applies A , so its columns are A 's columns in the other order. BA applies A first then swaps, so its **rows** are A 's rows in the other order.

20.2 Reflection in the xy -plane negates z and fixes x, y :

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Rotation by 90° about the z -axis sends $e_1 \rightarrow e_2$, $e_2 \rightarrow -e_1$, $e_3 \rightarrow e_3$:

$$R = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Both products:

$$RM = MR = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

They commute, which is unusual and has a reason: the reflection is in the plane **perpendicular to** the rotation axis, so the two operations act on disjoint sets of coordinates (z for the reflection, xy for the rotation).

The result has determinant $(-1)(1) = -1$, so it is orientation-reversing: a **rotary reflection** (an improper rotation), not a rotation.

20.3 Inverse of a product.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

and symmetrically $(B^{-1}A^{-1})(AB) = I$. So AB is invertible with the stated inverse.

The order reversal is forced: without it, $(AB)(A^{-1}B^{-1})$ has BA^{-1} stuck in the middle and does not simplify.

Sum need not be invertible. Take $A = I$ and $B = -I$. Both are invertible; $A + B = 0$ is not.

More sharply, invertibility is a property of the **map**, and adding two invertible maps can produce one that collapses everything.

20.4

$$\operatorname{tr}(A^T A) = \sum_i (A^T A)_{ii} = \sum_i \sum_k (A^T)_{ik} A_{ki} = \sum_i \sum_k A_{ki} A_{ki} = \sum_{i,k} A_{ki}^2.$$

So $\operatorname{tr}(A^T A)$ is the sum of the squares of all entries — the squared Frobenius norm.

If it is zero, then a sum of squares of real numbers is zero, which forces every term to be zero, so $A = 0$.

This is the standard way to prove a matrix identity of the form $X = Y$: show $\|X - Y\|_F^2 = 0$.

20.5

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}. \text{ Row 2 is exactly twice Row 1, so } \operatorname{rank}(A) = 1.$$

Column space: every column is a multiple of $(1, 2)^T$, so $\text{Col}(A) = \text{span}\{(1, 2)^T\}$, a line in $\mathbb{R}^2$. Dimension 1.

Null space: the single independent equation is $x_1 + 2x_2 + 3x_3 = 0$. With x_2, x_3 free:

$$\text{Null}(A) = \text{span}\{(-2, 1, 0)^T, (-3, 0, 1)^T\}.$$

Dimension 2.

Check: $1 + 2 = 3 = \text{number of columns}$. ✓ (Rank-nullity.)

20.6
$$(Q_1 Q_2)^T (Q_1 Q_2) = Q_2^T Q_1^T Q_1 Q_2 = Q_2^T I Q_2 = Q_2^T Q_2 = I.$$

So the product is orthogonal.

Together with the facts that I is orthogonal and that $Q^{-1} = Q^T$ is orthogonal whenever Q is, this makes the set of $n \times n$ orthogonal matrices a **group** under multiplication — the orthogonal group $O(n)$. Restricting to determinant +1 gives the rotation group $\text{SO}(n)$ of Chapter 35.

20.7 A is 100×5 , B is 5×100 , C is 100×1 . Computing an $(m \times p)(p \times n)$ product costs mpn scalar multiplications.

(AB)C: AB costs $100 \times 5 \times 100 = 50,000$ and gives a 100×100 matrix; then $(100 \times 100)(100 \times 1)$ costs 10,000. Total **60,000**.

A(BC): BC costs $5 \times 100 \times 1 = 500$ and gives a 5×1 vector; then $(100 \times 5)(5 \times 1)$ costs 500. Total **1,000**.

Ratio: **60 to 1**, for the identical mathematical result.

Note **why:** the right-to-left order never materialises a large intermediate matrix. This is exactly the reverse-mode-versus-forward-mode argument of Chapter 18, in miniature.

20.8 Let $S = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}$.

Closed under addition: obvious, entrywise.

Closed under multiplication:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix},$$

which has the same form with $a' = ac - bd$ and $b' = ad + bc$.

This is exactly complex multiplication: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. So the map $a + bi \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ preserves both addition and multiplication — it is an isomorphism of $\mathbb{C}$ onto S .

The matrix for i is $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and

$$J^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

And J is the 90° rotation matrix. So “ $i^2 = -1$ ” reads as “two quarter turns make a half turn” — exactly the geometric account given in Chapter 6, now as a matrix identity.

A.21. Chapter 21 — Solving Linear Systems

21.1 Augmented matrix and elimination:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 3 & 3 & 0 & 5 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & -3 & -3 & -7 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(using $R_2 - 2R_1$, $R_3 - 3R_1$, then $R_3 - R_2$).

Rank 2, three unknowns, so one free variable: z .

From row 2: $-3y - 3z = -7$, so $y = 7/3 - z$. From row 1: $x + 2(7/3 - z) + z = 4$, so $x = z - 2/3$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2/3 \\ 7/3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

A particular solution plus the null space, exactly as Chapter 20 predicted.

21.2

$$A = \begin{pmatrix} 1 & 3 & 1 & 4 \\ 2 & 6 & 3 & 9 \\ 1 & 3 & 2 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivots in columns 1 and 3, so $\text{rank}(A) = 2$. Free variables x_2, x_4 , so the nullity is 2.

Null space. Row 2: $x_3 + x_4 = 0$, so $x_3 = -x_4$. Row 1: $x_1 + 3x_2 + x_3 + 4x_4 = 0$, so $x_1 = -3x_2 - (-x_4) - 4x_4 = -3x_2 - 3x_4$.

Setting $(x_2, x_4) = (1, 0)$ and $(0, 1)$:

$$\text{Null}(A) = \text{span}\{(-3, 1, 0, 0)^T, (-3, 0, -1, 1)^T\}.$$

Rank-nullity: $2 + 2 = 4 = \text{number of columns}$. ✓

21.3 The coefficient matrix is $\begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}$ with determinant $4 - k^2$.

(a) **Unique solution** iff the determinant is nonzero: $k \neq \pm 2$.

(b) and (c). When $k = 2$ the system is $x + 2y = 1$ and $2x + 4y = 2$ — the second equation is exactly twice the first, so they describe the same line: **infinitely many solutions**.

When $k = -2$: $x - 2y = 1$ and $-2x + 4y = 2$. Dividing the second by -2 gives $x - 2y = -1$, contradicting the first: **no solution** (two parallel lines).

So: unique for $k \neq \pm 2$; infinitely many for $k = 2$; none for $k = -2$.

21.4 A is 3×5 , so $\text{rank}(A) \leq 3$ (the rank cannot exceed either dimension).

By rank-nullity,

$$\dim \text{Null}(A) = 5 - \text{rank}(A) \geq 5 - 3 = 2 > 0.$$

A null space of positive dimension contains nonzero vectors, so $Ax = 0$ has nontrivial solutions.

The **smallest possible** nullity is 2, attained when A has full row rank 3.

21.5 $A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$.

Eliminate: multiplier $\ell_{21} = 2$ gives $R_2 - 2R_1 = (0, 1, 1)$; $\ell_{31} = 4$ gives $R_3 - 4R_1 = (0, 3, 5)$; then $\ell_{32} = 3$ gives $R_3 - 3R_2 = (0, 0, 2)$.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Forward substitution, $Ly = b = (4, 10, 24)^T$: $y_1 = 4$; $2(4) + y_2 = 10$ so $y_2 = 2$; $4(4) + 3(2) + y_3 = 24$ so $y_3 = 2$.

Back substitution, $Ux = y = (4, 2, 2)^T$: $2x_3 = 2$ so $x_3 = 1$; $x_2 + x_3 = 2$ so $x_2 = 1$; $2x_1 + x_2 + x_3 = 4$ so $x_1 = 1$.

$x = (1, 1, 1)^T$. Check: the row sums of A are 4, 10, 24. ✓

21.6 System: $10^{-4}x + y = 1$, $x + y = 2$.

Exact solution. Subtracting: $(10^{-4} - 1)x = -1$, so $x = 1/(1 - 10^{-4}) = 1.00010001\dots$ and $y = 2 - x = 0.99989999\dots$

Without pivoting, to 3 significant figures. Multiplier $1/10^{-4} = 10000$. Then

$$R_2 - 10000R_1 : \text{coefficient of } y : 1 - 10000 = -9999 \rightarrow -10000,$$

$$\text{RHS} : 2 - 10000(1) = -9998 \rightarrow -10000.$$

So $y = (-10000)/(-10000) = 1.00$. Substituting back into row 1: $10^{-4}x + 1.00 = 1$, giving $x = 0.00$.

Answer: (0, 1) — and x is wrong by 100%.

With pivoting. Swap the rows first. Multiplier 10^{-4} :

$$R_2 - 10^{-4}R_1 : 1 - 10^{-4} = 0.9999 \rightarrow 1.00, \quad 1 - 2 \times 10^{-4} = 0.9998 \rightarrow 1.00.$$

So $y = 1.00$, and then $x = 2 - 1.00 = 1.00$.

Answer: (1.00, 1.00) — correct to 3 significant figures.

The matrix was perfectly well conditioned ($\kappa \approx 2.6$). The entire failure was the choice of pivot, which forced a subtraction that annihilated the smaller entries.

21.7 $\text{rank}(AB) \leq \text{rank}(A)$. Every column of AB is A times the corresponding column of B , so every column of AB lies in $\text{Col}(A)$. Hence $\text{Col}(AB) \subseteq \text{Col}(A)$, and the dimension of a subspace is at most that of the containing space.

$\text{rank}(AB) \leq \text{rank}(B)$. Apply the first result to transposes:

$$\text{rank}(AB) = \text{rank}((AB)^T) = \text{rank}(B^T A^T) \leq \text{rank}(B^T) = \text{rank}(B),$$

using $\text{rank}(M) = \text{rank}(M^T)$ twice.

Together, $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.

Consequence worth noting: multiplying can only ever **lose** rank. This is why a low-rank factorisation $A = CR$ with inner dimension r forces $\text{rank}(A) \leq r$ — and why LoRA can only make rank- r changes to a weight matrix.

21.8 At most one solution. $\text{rank}(A) = n$ and A has n columns, so by rank-nullity $\dim \text{Null}(A) = 0$. If $Ax_1 = Ax_2 = b$ then $A(x_1 - x_2) = 0$, so $x_1 - x_2 \in \text{Null}(A) = \{0\}$, giving $x_1 = x_2$.

When there is none. $Ax = b$ is solvable iff $b \in \text{Col}(A)$. Here $\text{Col}(A)$ is an n -dimensional subspace of $\mathbb{R}^m$ with $m > n$ — a proper subspace, of measure zero. So for a **generic** b there is no solution at all.

This is the overdetermined case: more equations than unknowns, so the equations generically conflict. The response is not to give up but to change the question — find the x minimising $\|Ax - b\|$, which is exactly the least-squares problem of Chapter 24. And since the columns are independent, $A^T A$ is invertible and that problem has a unique answer.

A.22. Chapter 22 — Determinants

22.1

$$\det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} = (3)(4) - (1)(2) = 10.$$

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 0.$$

Why, in terms of the columns: $c_1 - 2c_2 + c_3 = (1 - 4 + 3, 4 - 10 + 6, 7 - 16 + 9) = (0, 0, 0)$. The columns are linearly dependent, so by property (e) the determinant is zero. Geometrically, the three columns lie in a plane rather than spanning $\mathbb{R}^3$, so the unit cube is flattened and the image has zero volume.

22.2 Row reduce, tracking the pivots (no row swaps are needed, so no sign change):

$$\begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 1 & 2 & 1 \\ 0 & -0.5 & 1 & 2.5 \end{pmatrix}$$

$(R_2 - 0.5R_1, R_4 - 0.5R_1)$. Then $R_3 - 0.4R_2$ and $R_4 + 0.2R_2$:

$$\rightarrow \begin{pmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 0 & 1.6 & 1.2 \\ 0 & 0 & 1.2 & 2.4 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 0 & 1 \\ 0 & 2.5 & 1 & -0.5 \\ 0 & 0 & 1.6 & 1.2 \\ 0 & 0 & 0 & 1.5 \end{pmatrix}$$

$(R_4 - 0.75R_3)$.

Adding a multiple of one row to another does not change the determinant, so

$$\det = 2 \times 2.5 \times 1.6 \times 1.5 = 12.$$

22.3 A is 4×4 with $\det A = 5$.

$$\begin{aligned}\det(2A) &= 2^4 \det A = 16 \times 5 = 80 \quad (\text{not } 10!) \\ \det(A^T) &= \det A = 5 \\ \det(A^{-1}) &= 1/5 \\ \det(A^3) &= 5^3 = 125 \\ \det(A^T A) &= \det(A^T) \det(A) = 25\end{aligned}$$

The first is the one that catches people: scaling a matrix scales **every column**, so the determinant picks up c once per dimension.

22.4 From $Q^T Q = I$, take determinants:

$$\det(Q^T) \det(Q) = \det(I) = 1.$$

Since $\det(Q^T) = \det(Q)$, this is $(\det Q)^2 = 1$, so $\det Q = \pm 1$.

$\det Q = +1$: a rotation. Orientation is preserved — a right-handed frame stays right-handed. These form the special orthogonal group $\text{SO}(n)$.

$\det Q = -1$: a reflection (or, in odd dimensions, a rotation composed with a reflection). Orientation is reversed; a right-handed frame becomes left-handed.

This is the algebraic version of the fact that you cannot continuously deform a rotation into a reflection: the determinant is continuous and cannot jump from $+1$ to -1 without passing through 0, which orthogonal matrices never do.

22.5 Translate so that P_1 is at the origin; the other two vertices become $u = (x_2 - x_1, y_2 - y_1)$ and $v = (x_3 - x_1, y_3 - y_1)$.

The parallelogram spanned by u and v has area $|\det[u \mid v]|$ — the determinant is the area scale factor applied to the unit square. The triangle is half of it:

$$\text{Area} = \frac{1}{2} \left| \det \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} \right|.$$

The sign as an orientation test. Without the absolute value, the determinant is positive iff (P_1, P_2, P_3) are in counterclockwise order and negative iff clockwise (and zero iff collinear).

That single sign is the primitive underneath computational geometry: Graham scan and Andrew's monotone chain build a convex hull by repeatedly asking “does adding this point turn left or right?” and popping the stack while the turn is the wrong way. Segment intersection, point-in-polygon, and Delaunay triangulation all reduce to the same test. It is one 2×2 determinant, and it is exact in integer arithmetic — which is why robust geometry code keeps coordinates as integers where it can.

22.6 From $A^2 = I$, take determinants: $(\det A)^2 = \det I = 1$, so $\det A = \pm 1$.

$\det A = +1$: take $A = I$. Then $A^2 = I$. ✓

$\det A = -1$: take $A = \text{diag}(1, -1, \dots, -1)$ with an odd number of -1 s — or simply $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ in two dimensions, a reflection. Reflecting twice returns you to the start, so $A^2 = I$, and $\det A = -1$. ✓

(Such matrices are called **involutions**, and geometrically they are reflections in some subspace: $A^2 = I$ forces the eigenvalues to be ± 1 , and the determinant is -1 exactly when an odd number of them are -1 .)

22.7 With $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, $z = \rho \cos \varphi$:

$$J = \begin{pmatrix} \sin \varphi \cos \theta & \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \\ \cos \varphi & -\rho \sin \varphi & 0 \end{pmatrix}.$$

Expand along the third row, which has a zero.

The cofactor of $\cos \varphi$ (position 3,1, sign $+$) is

$$\det \begin{pmatrix} \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \end{pmatrix} = \rho^2 \sin \varphi \cos \varphi (\cos^2 \theta + \sin^2 \theta) = \rho^2 \sin \varphi \cos \varphi.$$

The cofactor of $-\rho \sin \varphi$ (position 3,2, sign $-$) is

$$-\det \begin{pmatrix} \sin \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \sin \varphi \cos \theta \end{pmatrix} = -\rho \sin^2 \varphi (\cos^2 \theta + \sin^2 \theta) = -\rho \sin^2 \varphi.$$

So

$$\det J = \cos \varphi (\rho^2 \sin \varphi \cos \varphi) + (-\rho \sin \varphi)(-\rho \sin^2 \varphi) = \rho^2 \sin \varphi (\cos^2 \varphi + \sin^2 \varphi) = \rho^2 \sin \varphi.$$

Hence $dV = \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$, which is the volume element you have seen quoted — now derived rather than memorised. It correctly vanishes at the poles ($\varphi = 0, \pi$), where the coordinate system degenerates.

22.8 Magnitude. For an $n \times n$ matrix with i.i.d. entries of variance σ^2 , the typical magnitude of the determinant is around $\sigma^n \sqrt{n!}$. For $U[0, 1]$ entries, $\sigma^2 = 1/12$ and $n = 200$:

$$\log_{10} |\det| \approx 200 \log_{10} (1/\sqrt{12}) + \frac{1}{2} \log_{10} (200!) \approx -108 + 187 \approx 79.$$

So $|\det|$ is of order 10^{79} — nowhere near 1, and nowhere near 0.

Why this makes `det` a bad singularity test. The point is not that the value is small; it is that the value is **arbitrary**. It is not a normalised quantity, and it moves by hundreds of orders of magnitude for reasons having nothing to do with invertibility:

- Scale every entry by c and the determinant scales by c^{200} . Draw entries from $U[0, 0.01]$ instead — a matrix with **identical** conditioning, since scaling multiplies all singular values equally and leaves κ unchanged — and the determinant becomes $10^{79} \times 10^{-400} = 10^{-321}$, which underflows a double to exactly zero.
- Conversely $10^{-6}I$ in 200 dimensions has determinant 10^{-1200} (zero, in floating point) and is trivially invertible with $\kappa = 1$.

The determinant is a **product** of all the singular values, so it measures volume. Conditioning is a **ratio** of the largest to the smallest. A product tells you nothing about a ratio. Test with the smallest singular value or `rcond`, never with `det`.

A.23. Chapter 23 — Eigenvalues and Eigenvectors

23.1 $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.

$$\det(A - \lambda I) = (3 - \lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4), \text{ so } \lambda = 2, 4.$$

$$\lambda = 4: (A - 4I) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \text{ giving } v_1 = v_2, \text{ so } v = (1, 1)^T.$$

$$\lambda = 2: (A - 2I) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \text{ giving } v_1 = -v_2, \text{ so } v = (1, -1)^T.$$

Orthogonality check: $(1, 1) \cdot (1, -1) = 1 - 1 = 0$. ✓ As the spectral theorem requires, since A is symmetric.

$$\text{Also } \lambda_1 + \lambda_2 = 6 = \text{tr}(A) \text{ and } \lambda_1 \lambda_2 = 8 = \det A. \quad \checkmark$$

23.2 $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$. $\det(A - \lambda I) = (2 - \lambda)^2 + 1 = 0$, so $\lambda = 2 \pm i$.

Geometric interpretation. Factor out the modulus:

$$A = \sqrt{5} \begin{pmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{pmatrix} = \sqrt{5} R_\theta, \quad \cos \theta = \frac{2}{\sqrt{5}}, \sin \theta = \frac{1}{\sqrt{5}}.$$

So A is a **rotation by** $\theta = \arctan(1/2) \approx 26.6^\circ$, **scaled by** $\sqrt{5}$.

The eigenvalues say exactly that: $|\lambda| = \sqrt{4 + 1} = \sqrt{5}$ is the scaling, and $\arg(\lambda) = \arctan(1/2)$ is the rotation angle. Writing $\lambda = \sqrt{5}e^{\pm i\theta}$ makes it explicit.

A real matrix with non-real eigenvalues has **no** real eigenvector — and it cannot, since a rotation leaves no direction fixed. The conjugate pair encodes a rotation in an invariant plane. This is Chapter 6 doing real work.

23.3 $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.

$$\det(A - \lambda I) = (2 - \lambda)^2, \text{ so } \lambda = 2 \text{ with algebraic multiplicity } 2.$$

$$(A - 2I) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \text{ Solving } (A - 2I)v = 0 \text{ gives } v_2 = 0 \text{ with } v_1 \text{ free, so the eigenspace is } \text{span}\{(1, 0)^T\}: \text{geometric multiplicity } 1.$$

Since $1 < 2$, there is no basis of eigenvectors and A is not diagonalisable — it is **defective**.

Geometrically A is a shear (composed with a scaling by 2): it fixes the x -axis and tilts everything else. A shear has exactly one invariant direction, so there is nothing else to find.

23.4 $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I),$

$$\text{using } (A - \lambda I)^T = A^T - \lambda I \text{ and } \det(M^T) = \det(M).$$

Same characteristic polynomial, hence same eigenvalues with the same algebraic multiplicities.

Eigenvectors generally differ. Take $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Its eigenvector for $\lambda = 1$ is $(1, 0)^T$. But $A^T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ has eigenvector $(0, 1)^T$.

The eigenvectors of A^T are called the **left** eigenvectors of A (they satisfy $w^T A = \lambda w^T$), and they coincide with the right ones exactly when A is symmetric — or more generally normal.

23.5 Let $Av = \lambda v$ with $v \neq 0$ and A invertible. Then $\lambda \neq 0$ (otherwise $Av = 0$ with $v \neq 0$ would make A singular).

Inverse. Apply A^{-1} to both sides of $Av = \lambda v$:

$$v = \lambda A^{-1}v \Rightarrow A^{-1}v = \frac{1}{\lambda}v.$$

Same eigenvector, reciprocal eigenvalue.

Square. $A^2v = A(\lambda v) = \lambda Av = \lambda^2 v$.

More generally $p(A)v = p(\lambda)v$ for any polynomial p — which is why $A = PDP^{-1}$ makes every polynomial (and, by extension, the matrix exponential) trivial to evaluate.

23.6 $A = \begin{pmatrix} 5 & -2 \\ -2 & 5 \end{pmatrix}$. Characteristic: $(5 - \lambda)^2 - 4 = 0$, so $5 - \lambda = \pm 2$, giving $\lambda = 3, 7$.

$\lambda = 7$: $(A - 7I) = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}$, so $v = (1, -1)^T$. $\lambda = 3$: $(A - 3I) = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, so $v = (1, 1)^T$.

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 7 & 0 \\ 0 & 3 \end{pmatrix}, \quad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Then $A^{10} = PD^{10}P^{-1}$ with $7^{10} = 282, !475, !249$ and $3^{10} = 59, !049$:

$$A^{10} = \frac{1}{2} \begin{pmatrix} 7^{10} + 3^{10} & 3^{10} - 7^{10} \\ 3^{10} - 7^{10} & 7^{10} + 3^{10} \end{pmatrix} = \begin{pmatrix} 141 & !267 & !149 & -141 & !208 & !100 \\ -141 & !208 & !100 & 141 & !267 & !149 \end{pmatrix}.$$

Two scalar exponentiations rather than ten matrix multiplications — and for A^{10^6} the saving becomes the difference between possible and not.

23.7 $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Characteristic $(2 - \lambda)^2 - 1 = 0$ gives $\lambda = 1, 3$. Both positive, so **positive definite**.

Direct check with $x = (1, -1)^T$: $x^T Ax = 2 - 2 - 2 + 2 = 0$? Let us compute properly: $x^T Ax = 2(1)^2 + 2(1)(-1) + 2(-1)^2 = 2 - 2 + 2 = 2 > 0$. ✓ (The general quadratic form is $2x_1^2 + 2x_1x_2 + 2x_2^2$, which is positive for all nonzero x .)

$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. Characteristic $(1 - \lambda)^2 - 4 = 0$ gives $1 - \lambda = \pm 2$, so $\lambda = 3, -1$. Eigenvalues of both signs: **indefinite**, not positive definite.

Direct check with $x = (1, -1)^T$: $x^T Ax = 1 - 2 - 2 + 1 = -2 < 0$. ✓ Confirms indefiniteness, and shows the vector realising it is the eigenvector for $\lambda = -1$ — as it must be, since that is the direction of most negative curvature.

23.8 $P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}$ (columns sum to 1).

Stationary distribution. Solve $(P - I)\pi = 0$:

$$\begin{pmatrix} -0.1 & 0.4 \\ 0.1 & -0.4 \end{pmatrix} \pi = 0 \quad \Rightarrow \quad \pi_1 = 4\pi_2.$$

Normalising so $\pi_1 + \pi_2 = 1$:

$$\pi = (0.8, 0.2)^T.$$

Second eigenvalue. $\text{tr}(P) = 0.9 + 0.6 = 1.5 = \lambda_1 + \lambda_2 = 1 + \lambda_2$, so $\lambda_2 = 0.5$.

Mixing time. Deviations from stationarity decay like $|\lambda_2|^k = 0.5^k$. Requiring a factor of 100:

$$0.5^k \leq 0.01 \quad \Rightarrow \quad k \geq \frac{\ln 0.01}{\ln 0.5} = 6.64,$$

so **7 steps**.

The spectral gap here is $1 - 0.5 = 0.5$, which is large — this chain mixes very fast. A chain with $\lambda_2 = 0.999$ would need about 4600 steps for the same factor, and that gap is exactly what “slow mixing” means.

A.24. Chapter 24 — Orthogonality, Projection, and Least Squares

24.1 $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$.

Step 1. $u_1 = (1, 1, 0)$, $\|u_1\| = \sqrt{2}$, so $q_1 = (1, 1, 0)/\sqrt{2}$.

Step 2. $v_2 \cdot q_1 = 1/\sqrt{2}$, so

$$u_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right), \quad \|u_2\| = \sqrt{3/2},$$

giving $q_2 = (1, -1, 2)/\sqrt{6}$.

Step 3. $v_3 \cdot q_1 = 1/\sqrt{2}$ and $v_3 \cdot q_2 = (0 - 1 + 2)/\sqrt{6} = 1/\sqrt{6}$, so

$$u_3 = (0, 1, 1) - \frac{1}{2}(1, 1, 0) - \frac{1}{6}(1, -1, 2) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right),$$

with $\|u_3\| = (2/3)\sqrt{3}$, giving $q_3 = (-1, 1, 1)/\sqrt{3}$.

Checks. $q_1 \cdot q_2 = (1 - 1 + 0)/\sqrt{12} = 0 \checkmark$; $q_1 \cdot q_3 = (-1 + 1 + 0)/\sqrt{6} = 0 \checkmark$; $q_2 \cdot q_3 = (-1 - 1 + 2)/\sqrt{18} = 0 \checkmark$. All unit length by construction.

24.2 With $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ (columns $a_1 = (1, 0, 1)$, $a_2 = (0, 1, 1)$) and $b = (1, 2, 3)$:

$$A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad A^T b = \begin{pmatrix} 1 + 3 \\ 2 + 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.$$

Solving $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} x = (4, 5)^T$ (determinant 3): $x_1 = (8 - 5)/3 = 1$, $x_2 = (10 - 4)/3 = 2$.

$$\hat{b} = 1 \cdot (1, 0, 1) + 2 \cdot (0, 1, 1) = (1, 2, 3).$$

The residual is zero. And indeed it should be: $b = a_1 + 2a_2$ exactly, so b was already in the plane and its projection is itself.

This is a legitimate outcome and a useful check on the machinery: when $b \in \text{Col}(A)$, least squares reduces to exact solution, the residual is 0, and 0 is trivially orthogonal to both spanning vectors.

24.3 Data: $(1, 1), (2, 3), (3, 4), (4, 6)$. So $n = 4$,

$$\sum x = 10, \quad \sum y = 14, \quad \sum xy = 1 + 6 + 12 + 24 = 43, \quad \sum x^2 = 30.$$

The normal equations are

$$\begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix} \begin{pmatrix} m \\ c \end{pmatrix} = \begin{pmatrix} 43 \\ 14 \end{pmatrix}.$$

Solving (determinant $120 - 100 = 20$):

$$m = \frac{4(43) - 10(14)}{20} = \frac{172 - 140}{20} = \frac{32}{20} = 1.6,$$

$$c = \bar{y} - m\bar{x} = 3.5 - 1.6(2.5) = -0.5.$$

Fitted line: $y = 1.6x - 0.5$.

Residuals: $-0.1, +0.3, -0.3, +0.1$ — summing to zero, as they must, since the residual is orthogonal to the column of ones.

24.4 Idempotent. $(I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P$, using $P^2 = P$.

Symmetric. $(I - P)^T = I^T - P^T = I - P$.

So $I - P$ is a projection matrix.

Onto what. For $v \in W$: $Pv = v$, so $(I - P)v = 0$. For $v \in W^\perp$: $Pv = 0$, so $(I - P)v = v$.

So $I - P$ projects onto $W^\perp$, the orthogonal complement. Together, $v = Pv + (I - P)v$ is exactly the orthogonal decomposition theorem, realised as a pair of matrices summing to the identity.

24.5 Null(A) $\subseteq$ Null($A^T A$). If $Ax = 0$ then $A^T Ax = A^T 0 = 0$.

Null($A^T A$) $\subseteq$ Null(A). If $A^T Ax = 0$, multiply on the left by x^T :

$$0 = x^T A^T Ax = (Ax)^T (Ax) = \|Ax\|^2,$$

so $Ax = 0$.

Hence the two null spaces are equal.

Rank. Both A and $A^T A$ have n columns, so rank-nullity gives

$$\text{rank}(A^T A) = n - \dim \text{Null}(A^T A) = n - \dim \text{Null}(A) = \text{rank}(A).$$

No hypothesis about independence was needed — the identity holds for every A . (When the columns **are** independent, both ranks are n and $A^T A$ is invertible, which is the lemma the projection formula needed.)

24.6 For orthogonal Q , $Q^T Q = I$, so $Q^T Q$ has every eigenvalue equal to 1, so every singular value of Q is $\sqrt{1} = 1$. Hence

$$\kappa_2(Q) = \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{1}{1} = 1.$$

(Directly: $\|Qx\| = \|x\|$ for all x , so $\|Q\|_2 = 1$, and $Q^{-1} = Q^T$ is also orthogonal so $\|Q^{-1}\|_2 = 1$.)

Why this makes QR the right method. Condition numbers multiply through a chain of operations. An orthogonal transformation contributes a factor of 1, so it cannot amplify error at all. QR factorisation transforms the least squares problem using only orthogonal operations, so the solve inherits $\kappa(A)$.

The normal equations instead form $A^T A$, whose condition number is $\kappa(A)^2$ — squaring the difficulty and halving the number of correct digits, before any arithmetic has been done.

24.7 $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, columns $a_1 = (1, 1, 0)$, $a_2 = (1, 0, 1)$.

Gram-Schmidt (the first two steps of Solution 24.1):

$$q_1 = \frac{1, 1, 0}{\sqrt{2}}, \quad q_2 = \frac{1, -1, 2}{\sqrt{6}}.$$

$$R = \begin{pmatrix} a_1 \cdot q_1 & a_2 \cdot q_1 \\ 0 & a_2 \cdot q_2 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{pmatrix}.$$

Least squares for $b = (1, 2, 3)$:

$$Q^T b = \begin{pmatrix} q_1 \cdot b \\ q_2 \cdot b \end{pmatrix} = \begin{pmatrix} \frac{1+2}{\sqrt{2}} \\ \frac{1-2+6}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} 3/\sqrt{2} \\ 5/\sqrt{6} \end{pmatrix}.$$

Back-substitute in $Rx = Q^T b$:

$$\sqrt{3/2} x_2 = 5/\sqrt{6} \Rightarrow x_2 = \frac{5}{\sqrt{6}\sqrt{3/2}} = \frac{5}{3}.$$

$$\sqrt{2}x_1 + (1/\sqrt{2})(5/3) = 3/\sqrt{2} \Rightarrow 2x_1 + \frac{5}{3} = 3 \Rightarrow x_1 = \frac{2}{3}.$$

$$\hat{x} = (2/3, 5/3)^T.$$

Cross-check with the normal equations: $A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $A^T b = (3, 4)^T$, and solving gives $x_1 = (6 - 4)/3 = 2/3$, $x_2 = (8 - 3)/3 = 5/3$. ✓

24.8 Use the product-to-sum identity (Chapter 5):

$$\sin(2x) \cos(3x) = \frac{1}{2}[\sin(5x) + \sin(-x)] = \frac{1}{2}[\sin 5x - \sin x].$$

Each of $\sin(5x)$ and $\sin(x)$ integrates to a cosine, evaluated over a whole number of periods on $[-\pi, \pi]$:

$$\int_{-\pi}^{\pi} \sin(kx) dx = \left[-\frac{\cos(kx)}{k} \right]_{-\pi}^{\pi} = 0$$

since $\cos$ is even. So the integral is 0.

Faster argument: $\sin(2x)$ is odd and $\cos(3x)$ is even, so the product is odd, and the integral of an odd function over an interval symmetric about the origin vanishes.

The general principle: distinct members of the Fourier basis $\{1, \cos nx, \sin nx\}$ are orthogonal with respect to the inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$. That orthogonality is what makes the Fourier coefficient formula a simple projection, and it is the entire reason the decomposition works.

A.25. Chapter 25 — The Singular Value Decomposition

25.1 $A = \text{diag}(3, -2)$ already scales the two coordinate axes, so the singular values — which must be **nonnegative** — are 3 and 2.

The sign has to go somewhere, and it goes into one of the orthogonal factors. One valid choice:

$$U = I, \quad \Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

since then $U\Sigma V^T = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$. ✓

Equally valid: $U = \text{diag}(1, -1)$, $V = I$. The SVD is not unique when signs or repeated singular values are involved — only Σ is determined.

Geometrically: A reflects the y -axis and stretches by $(3, 2)$. The reflection is absorbed into V (or U), leaving a pure positive scaling in Σ .

25.2 $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, so

$$A^T A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Characteristic polynomial: $(1 - \lambda)(2 - \lambda) - 1 = \lambda^2 - 3\lambda + 1$, with roots

$$\lambda = \frac{3 \pm \sqrt{5}}{2} \approx 2.618 \text{ and } 0.382.$$

Singular values are the square roots:

$$\sigma_1 = \sqrt{2.618} \approx 1.618, \quad \sigma_2 = \sqrt{0.382} \approx 0.618.$$

Those are φ and $1/\varphi$, the golden ratio and its reciprocal — which is not a coincidence, since $\lambda^2 - 3\lambda + 1$ is the characteristic polynomial of φ^2 .

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_2} = \varphi^2 \approx 2.618.$$

So a unit shear is mildly ill-conditioned: it stretches by 1.618 in one direction and squeezes by 0.618 in another.

25.3 Let $A = U\Sigma V^T$ with A of rank n (full column rank), $m > n$.

Then $A^T A = V\Sigma^T \Sigma V^T$, where $\Sigma^T \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ is $n \times n$ and invertible.

$$\begin{aligned} (A^T A)^{-1} A^T &= (V\Sigma^T \Sigma V^T)^{-1} V\Sigma^T U^T \\ &= V(\Sigma^T \Sigma)^{-1} V^T V\Sigma^T U^T \\ &= V(\Sigma^T \Sigma)^{-1} \Sigma^T U^T. \end{aligned}$$

Now $(\Sigma^T \Sigma)^{-1} \Sigma^T$ is the $n \times m$ matrix with $1/\sigma_i$ on the diagonal — which is exactly Σ^+ . So

$$(A^T A)^{-1} A^T = V\Sigma^+ U^T = A^+.$$

The two formulas agree whenever the second one is defined; the pseudoinverse is the version that survives rank deficiency, where $A^T A$ is singular and the other formula does not exist.

25.4 From Solution 20.4, $\text{tr}(A^T A) = \sum_{i,j} A_{ij}^2 = \|A\|_F^2$.

Now substitute the SVD and use cyclicity of the trace (Chapter 20):

$$\text{tr}(A^T A) = \text{tr}(V\Sigma^T U^T U \Sigma V^T) = \text{tr}(V\Sigma^T \Sigma V^T) = \text{tr}(\Sigma^T \Sigma V^T V) = \text{tr}(\Sigma^T \Sigma),$$

using $U^T U = I$ and $V^T V = I$.

And $\Sigma^T \Sigma$ is diagonal with entries σ_i^2 , so its trace is $\sum_i \sigma_i^2$.

$$\|A\|_F = \sqrt{\sum_i \sigma_i^2}.$$

So the Frobenius norm is blind to the orthogonal factors — it depends only on the stretching. This is the precise sense in which it is a rotation-invariant measure of “how big” a matrix is.

25.5 With $\sigma_i = 100/i$ for $i = 1, \dots, 800$:

$$\sum_i \sigma_i^2 = 10^4 \sum_{i=1}^{800} \frac{1}{i^2} \approx 10^4 \times 1.6437 = 16,437.$$

(The sum converges to $\pi^2/6 \approx 1.6449$; truncating at 800 removes about $1/800$.)

Ninety per cent of that is $0.9 \times 1.6437 = 1.4793$ in the normalised units. Accumulating:

k	1	2	3	4	5	6
$\sum_{i \leq k} 1/i^2$	1.000	1.250	1.361	1.424	1.464	1.491

$k = 5$ gives $1.464 < 1.479$; $k = 6$ gives $1.491 \geq 1.479$. So **6 singular values** suffice.

Storage. Full matrix: $1000 \times 800 = 800,000$ numbers. Rank-6 truncation: $6 \times (1000 + 800 + 1) = 10,806$ numbers. A saving of about **74×**.

Note how fast this is: a $1/i$ spectrum is not especially rapid decay, and six components still carry 90% of the energy. Genuine data often decays faster still.

25.6 Suppose $A^T A v = \lambda v$ with $\lambda \neq 0$ and $v \neq 0$.

Consider $w = Av$. First, $w \neq 0$: if $Av = 0$ then $A^T A v = 0 = \lambda v$, and since $\lambda \neq 0$ that would force $v = 0$.

Now

$$AA^T w = AA^T Av = A(\lambda v) = \lambda(Av) = \lambda w.$$

So w is an eigenvector of AA^T with the same eigenvalue λ .

The argument with A and A^T swapped gives the reverse inclusion, so the nonzero eigenvalues of $A^T A$ and AA^T coincide (with multiplicity).

They differ only in their **zero** eigenvalues: $A^T A$ is $n \times n$ and AA^T is $m \times m$, so the larger one simply has $|m - n|$ extra zeros. Which is exactly why Σ in the SVD is rectangular with zero rows or columns padding it out.

25.7 With $A = U\Sigma V^T$,

$$A^T A = V\Sigma^T \Sigma V^T,$$

which is an eigendecomposition of $A^T A$ with eigenvalues σ_i^2 . Since $A^T A$ is symmetric, its singular values **are** its eigenvalues (in absolute value), so the singular values of $A^T A$ are σ_i^2 .

Hence

$$\kappa(A^T A) = \frac{\sigma_1^2}{\sigma_n^2} = \left(\frac{\sigma_1}{\sigma_n} \right)^2 = \kappa(A)^2.$$

Practical consequence. The rule of thumb is that $\kappa \approx 10^k$ costs you k decimal digits. Forming the normal equations **doubles** k , so it doubles the number of digits lost — for free, before any arithmetic has been performed. Use QR (or SVD) instead, which works with $\kappa(A)$ directly.

25.8 With $X = U\Sigma V^T$ (rows of X are samples, already centred), for any unit vector w :

$$\|Xw\|^2 = w^T X^T X w = w^T V \Sigma^T \Sigma V^T w.$$

Substitute $z = V^T w$. Since V is orthogonal, $\|z\| = \|w\| = 1$, and

$$\|Xw\|^2 = z^T \Sigma^T \Sigma z = \sum_i \sigma_i^2 z_i^2.$$

This is a weighted average of the σ_i^2 with weights z_i^2 summing to 1, so it is maximised by putting all the weight on the largest — $z = e_1$, i.e. $w = V e_1 = v_1$ — with maximum value σ_1^2 .

So the first principal component is the first right singular vector of the centred data matrix, and the variance it captures is $\sigma_1^2/(n-1)$.

Continuing the argument on the orthogonal complement gives v_2, v_3 , and so on: PCA is the SVD, and no separate algorithm is needed.

A.26. Chapter 26 — Matrix Calculus

26.1 With $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$:

$$x^T A x = x_1^2 + 2x_1x_2 + 3x_2x_1 + 4x_2^2 = x_1^2 + 5x_1x_2 + 4x_2^2.$$

Differentiating entry by entry:

$$\frac{\partial}{\partial x_1} = 2x_1 + 5x_2, \quad \frac{\partial}{\partial x_2} = 5x_1 + 8x_2.$$

And the formula:

$$(A + A^T)x = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 + 5x_2 \\ 5x_1 + 8x_2 \end{pmatrix}.$$

Agreement. ✓ Note that the answer depends on A only through its symmetric part $(A + A^T)/2$ — the antisymmetric part contributes nothing to the quadratic form, which is why $x^T A x = x^T ((A + A^T)/2)x$ always.

26.2 $\|x\| = (x^T x)^{1/2}$. Taking differentials:

$$d\|x\| = \frac{1}{2}(x^T x)^{-1/2} \cdot d(x^T x) = \frac{1}{2} \frac{1}{\|x\|} \cdot 2x^T dx = \frac{x^T}{\|x\|} dx.$$

By the identification rule,

$$\nabla_x \|x\| = \frac{x}{\|x\|},$$

the unit vector in the direction of x .

At $x = 0$ the expression is $0/0$ and the gradient does not exist — $\|x\|$ has a corner at the origin, exactly like $|x|$ in one dimension. (Its directional derivative in direction u is $\|u\| = 1$ regardless of direction, which cannot be $\nabla f^T u$ for any fixed gradient.)

The standard fix in optimisation code is to smooth it: use $\sqrt{\|x\|^2 + \varepsilon}$, whose gradient is $x/\sqrt{\|x\|^2 + \varepsilon}$ and is well defined everywhere. This is what group-lasso and total-variation implementations do, and it is why they have an ε parameter.

26.3 With respect to x . Holding y fixed,

$$d(x^T A y) = (dx)^T A y = (A y)^T dx,$$

$$\text{so } \nabla_x (x^T A y) = A y.$$

With respect to y . Holding x fixed,

$$d(x^T A y) = x^T A dy = (A^T x)^T dy,$$

$$\text{so } \nabla_y (x^T A y) = A^T x.$$

The transpose appears in exactly one of them, and the reason is shapes: the gradient with respect to $y \in \mathbb{R}^n$ must be in $\mathbb{R}^n$, and only $A^T x$ has that shape.

26.4 $L(x) = \|Ax - b\|^2 + \lambda\|x\|^2.$

By the two identities of the chapter,

$$\nabla L = 2A^T(Ax - b) + 2\lambda x.$$

Setting it to zero:

$$A^T Ax - A^T b + \lambda x = 0 \Rightarrow (A^T A + \lambda I)x = A^T b,$$

so

$$\hat{x} = (A^T A + \lambda I)^{-1} A^T b.$$

Effect on conditioning. $A^T A$ is symmetric positive semidefinite with eigenvalues σ_i^2 . Adding λI shifts every eigenvalue up by λ , so the condition number becomes

$$\frac{\sigma_1^2 + \lambda}{\sigma_n^2 + \lambda},$$

which is strictly smaller than σ_1^2/σ_n^2 and tends to 1 as λ grows.

Crucially, the smallest eigenvalue is now at least $\lambda > 0$, so the matrix is **invertible even when $A^T A$ is singular**. Ridge regression turns an ill-posed problem into a well-posed one, at the cost of bias — exactly the bias–variance trade of Chapter 31.

26.5 Start from the given identity

$$d \det A = \det(A) \operatorname{tr}(A^{-1} dA).$$

By the chain rule for the scalar function $\log$,

$$d(\log \det A) = \frac{d \det A}{\det A} = \frac{\det(A) \operatorname{tr}(A^{-1} dA)}{\det A} = \operatorname{tr}(A^{-1} dA).$$

By the identification rule, since $\operatorname{tr}(A^{-1} dA) = \operatorname{tr}\left(\left((A^{-1})^T\right)^T dA\right)$, the derivative is

$$\frac{\partial}{\partial A} \log \det A = (A^{-1})^T.$$

Two sanity checks. For 1×1 matrices this reads $d \log a = da/a$. ✓ And for symmetric positive definite A (the case that matters for Gaussians) the transpose is redundant and the answer is simply A^{-1} .

26.6 Write $h = W_1 x$, $a = \varphi(h)$, $y = W_2 a$, and let $g = \partial L / \partial y$.

Output layer. By the linear-layer proposition with input a :

$$\frac{\partial L}{\partial W_2} = g a^T.$$

Shapes: g is $(\dim y) \times 1$, a^T is $1 \times h$, so the result is $(\dim y) \times h$ — matching W_2 . ✓

Backpropagate through W_2 and φ .

$$\frac{\partial L}{\partial a} = W_2^T g \quad (h \times 1),$$

$$\delta = \frac{\partial L}{\partial h} = (W_2^T g) \circ \varphi'(h) \quad (h \times 1),$$

using the elementwise-nonlinearity proposition.

Input layer.

$$\frac{\partial L}{\partial W_1} = \delta x^T.$$

Shapes: δ is $h \times 1$, x^T is $1 \times n$, so the result is $h \times n$ — matching W_1 . ✓

Every step is a matrix–vector product or an elementwise product; nothing larger than a weight matrix is ever formed.

26.7 With $n = 2$, write $S = e^{z_1} + e^{z_2}$, so $p_1 = e^{z_1}/S$ and $p_2 = e^{z_2}/S$.

Diagonal entry.

$$\frac{\partial p_1}{\partial z_1} = \frac{e^{z_1} S - e^{z_1} \cdot e^{z_1}}{S^2} = \frac{e^{z_1}}{S} - \left(\frac{e^{z_1}}{S}\right)^2 = p_1 - p_1^2 = p_1(1 - p_1).$$

Matching $p_{i(\delta_{ii} - p_i)} = p_1(1 - p_1)$. ✓

Off-diagonal entry.

$$\frac{\partial p_1}{\partial z_2} = \frac{0 \cdot S - e^{z_1} \cdot e^{z_2}}{S^2} = -p_1 p_2.$$

Matching $p_1(\delta_{12} - p_2) = p_1(0 - p_2) = -p_1 p_2$. ✓

Note the rows sum to zero: $p_1(1 - p_1) + (-p_1 p_2) = p_1(1 - p_1 - p_2) = 0$. That is the next exercise.

26.8 The softmax Jacobian is $J = \text{diag}(p) - pp^T$.

Symmetric. $\text{diag}(p)$ is symmetric and $(pp^T)^T = pp^T$. ✓

Positive semidefinite. For any x ,

$$x^T J x = \sum_i p_i x_i^2 - \left(\sum_i p_i x_i\right)^2.$$

Treating p as a probability distribution over indices and x as a random variable taking value x_i with probability p_i , this is exactly

$$\mathbb{E}[x^2] - (\mathbb{E}[x])^2 = \text{Var}(x) \geq 0.$$

A lovely appearance of Chapter 28 inside Chapter 26.

Singular. Let $\mathbf{1}$ be the all-ones vector. Then

$$J\mathbf{1} = p - p(\mathbf{1}^T p) = p - p \cdot 1 = 0,$$

since the entries of p sum to 1.

What that means. The all-ones direction is in the null space, which says: **moving all logits by the same amount changes nothing**. That is precisely the shift invariance of Chapter 4, now visible in the derivative.

Consequence: the softmax parameterisation is **over-parameterised** by exactly one degree of freedom. For n classes, only $n - 1$ logits are identifiable, which is why binary classification uses a single output rather than two, and why softmax regression coefficients are only determined up to a common shift.

A.27. Chapter 27 — Probability

27.1 Sample space: the 36 equally likely ordered pairs.

(a) **Sum is 7.** Six outcomes $((1, 6) \text{ through } (6, 1))$: $\mathbb{P} = 6/36 = 1/6$.

(b) **Sum is 7 given the first is 3.** Given the first die, only the second matters, and it must be 4: $\mathbb{P} = 1/6$. (Notice this equals the unconditional probability — the sum being 7 is independent of the value of either individual die, which is a special feature of 7.)

(c) **At least one 6.** Complementary counting: $1 - (5/6)^2 = 11/36$.

(d) **Sum 7 and at least one 6.** Only $(1, 6)$ and $(6, 1)$: $\mathbb{P} = 2/36 = 1/18$.

Independence of (a) and (c).

$$\mathbb{P}(A) \mathbb{P}(C) = \frac{1}{6} \cdot \frac{11}{36} = \frac{11}{216}, \quad \mathbb{P}(A \cap C) = \frac{1}{18} = \frac{12}{216}.$$

Not equal, so **not independent** — though only just. Conditioning on “at least one 6” slightly raises the chance of a 7, because a 6 is a particularly useful die to have when aiming for 7.

27.2 A is the disjoint union of $A \cap B$ and $A \cap \overline{B}$, so

$$\mathbb{P}(A \cap \overline{B}) = \mathbb{P}(A) - \mathbb{P}(A \cap B).$$

Using independence of A and B :

$$= \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A) \mathbb{P}(\overline{B}).$$

Which is the definition of independence for A and $\overline{B}$.

Applying the result twice shows $\overline{A}$ and $\overline{B}$ are also independent — as they should be, since “these two events carry no information about each other” ought to be immune to how you phrase them.

27.3 5 red, 3 blue, two drawn without replacement.

(a) **Both red.**

$$\left(\frac{5}{8}\right)\left(\frac{4}{7}\right) = \frac{20}{56} = \frac{5}{14}.$$

(b) **Second is red.** By total probability, conditioning on the first:

$$\left(\frac{5}{8}\right)\left(\frac{4}{7}\right) + \left(\frac{3}{8}\right)\left(\frac{5}{7}\right) = \frac{20}{56} + \frac{15}{56} = \frac{35}{56} = \frac{5}{8}.$$

(c) **First red given second red.**

$$\mathbb{P}(\text{1st red} \mid \text{2nd red}) = \frac{\mathbb{P}(\text{both red})}{\mathbb{P}(\text{2nd red})} = \frac{5/14}{5/8} = \frac{4}{7}.$$

Comment on (b). The answer $5/8$ is exactly the unconditional probability that **any** particular draw is red. The reason is **exchangeability**: before you look at anything, the 8 balls are in a uniformly random order, and the ball in position 2 is as likely to be red as the ball in position 1. Labelling the draws “first” and “second” is a description of the order, not a physical asymmetry.

Note that (c) is $4/7$, not $5/8$ — learning the second is red **does** change the first, because it removes a red ball from the pool. The relation is symmetric in a way that surprises people: each draw informs the other equally.

27.4 At 40% spam.

$$\begin{aligned}\mathbb{P}(\text{flag}) &= \mathbb{P}(\text{flag}|\text{spam})\mathbb{P}(\text{spam}) + \mathbb{P}(\text{flag}|\text{ham})\mathbb{P}(\text{ham}) \\ &= (0.95)(0.4) + (0.02)(0.6) = 0.380 + 0.012 = 0.392.\end{aligned}$$

$$\mathbb{P}(\text{spam} \mid \text{flag}) = \frac{0.380}{0.392} = 0.969.$$

A flagged message is 97% likely to be spam. Good filter.

At 1% spam.

$$\mathbb{P}(\text{flag}) = (0.95)(0.01) + (0.02)(0.99) = 0.0095 + 0.0198 = 0.0293,$$

$$\mathbb{P}(\text{spam}|\text{flag}) = \frac{0.0095}{0.0293} = 0.324.$$

Only 32%. The filter has not changed at all — same sensitivity, same specificity — but **two thirds of its flags are now false positives**, because there are 99 times as many legitimate messages to misflag.

This is the base-rate effect, and the practical lesson is direct: a detector’s usefulness depends on the prevalence of what it detects, so quality metrics quoted without a base rate are meaningless. It is also why the same fraud model performs beautifully in one market and floods the review queue in another.

27.5 Take $\Omega = \{1, 2, 3, 4\}$ with the uniform distribution, and

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

Each has probability $1/2$.

Pairwise independent. Each pairwise intersection is $\{1\}$, of probability $1/4$, and $\mathbb{P}(A)\mathbb{P}(B) = 1/4$. ✓ Likewise for the other two pairs.

Not mutually independent. $A \cap B \cap C = \{1\}$, so

$$\mathbb{P}(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

Interpretation: any **one** of the events tells you nothing about any other, but any **two** of them together determine the third (if two hold, the outcome must be 1, so the third holds too). Mutual independence is strictly stronger than pairwise, and the extra content is exactly this kind of higher-order coupling.

27.6

$$\mathbb{P}(\text{all distinct}) = \prod_{i=0}^{n-1} \frac{365-i}{365}.$$

Evaluating:

n	22	23	56	57
$\mathbb{P}(\text{shared})$	0.476	0.507	0.988	0.990

So the smallest n exceeding $1/2$ is **23**, and the smallest exceeding 0.99 is **57**.

The reason 23 feels too small: there are $\binom{23}{2} = 253$ **pairs** of people, and each pair has about a $1/365$ chance of matching. The expected number of matches is about $253/365 \approx 0.69$, which is already of order 1. Intuition anchors on 23 versus 365 rather than on 253 versus 365, and the pairs are what matter.

27.7 Let F = “picked the fair coin”, D = “picked the double-headed coin”, each with prior $1/2$. Let E = “three heads”.

$$\mathbb{P}(E | F) = (1/2)^3 = 1/8 \text{ and } \mathbb{P}(E | D) = 1.$$

$$\mathbb{P}(D | E) = \frac{\mathbb{P}(E|D) \mathbb{P}(D)}{\mathbb{P}(E|D)\mathbb{P}(D) + \mathbb{P}(E|F)\mathbb{P}(F)} = \frac{1 \cdot 1/2}{1 \cdot 1/2 + (1/8)(1/2)} = \frac{1/2}{9/16} = \frac{8}{9}.$$

About 89%. Note that the fair coin is not ruled out — it can produce three heads — but it is eight times less likely to, and that ratio is exactly the factor by which the odds shift: prior odds 1 : 1 become posterior odds 8 : 1.

Thinking in **odds** rather than probabilities makes Bayes a multiplication: posterior odds = prior odds $\times$ likelihood ratio. Often far easier.

27.8 Start from the two-event form

$$\mathbb{P}(B_j | A) = \frac{\mathbb{P}(A | B_j) \mathbb{P}(B_j)}{\mathbb{P}(A)}.$$

The law of total probability, applied to the partition $\{B_i\}$, says

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

Substituting into the denominator gives the expanded form.

What the denominator represents. It is $\mathbb{P}(A)$ — the total probability of observing the evidence, averaged over every hypothesis weighted by its prior. In Bayesian language it is called the **evidence** or the **marginal likelihood**.

Operationally it is just the normalising constant: it is whatever makes the posterior probabilities sum to 1 across the hypotheses. That is why in practice one often computes the unnormalised posterior $\mathbb{P}(A|B_j)\mathbb{P}(B_j)$ for each j and divides by the total at the end — and why, when only the **ratio** of two posteriors is wanted, the denominator cancels and never has to be computed at all.

A.28. Chapter 28 — Random Variables and Distributions

28.1

$$\mathbb{E}[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

$$\mathbb{E}[X^2] = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} \approx 15.167.$$

$$\text{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12} \approx 2.917.$$

Standard deviation ≈ 1.708 .

Note $\mathbb{E}[X] = 3.5$ is not an attainable value — expectations need not be in the range of the variable.

28.2 $X \sim \text{Binomial}(10, 0.3)$.

$$\mathbb{P}(X = 3) = \binom{10}{3} (0.3)^3 (0.7)^7 = 120 \times 0.027 \times 0.0823543 \approx 0.2668.$$

$$\mathbb{E}[X] = np = 3, \quad \text{Var}(X) = np(1-p) = 10(0.3)(0.7) = 2.1.$$

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - (0.7)^{10} = 1 - 0.02825 = 0.9718.$$

The complementary form is both easier and numerically better — summing ten terms would be more work and would lose precision.

28.3 First, $\mathbb{P}(X > k) = (1-p)^k$: the event “more than k trials are needed” is exactly “the first k trials all failed”, and by independence that has probability $(1-p)^k$.

Then

$$\mathbb{P}(X > s+t \mid X > s) = \frac{\mathbb{P}(X > s+t \cap X > s)}{\mathbb{P}(X > s)} = \frac{\mathbb{P}(X > s+t)}{\mathbb{P}(X > s)} = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t = \mathbb{P}(X > t),$$

where the second step uses $\{X > s+t\} \subseteq \{X > s\}$.

Reading. Having already failed s times tells you **nothing** about how many more attempts you need. Each trial is a fresh start.

This is why “I have missed five times, I am due” is wrong for genuinely independent trials, and why an exponentially-distributed remaining lifetime does not shorten with age. It is also the property that makes the geometric and exponential distributions so analytically convenient — and that makes them the wrong model for anything that genuinely wears out.

28.4 (a) $\lambda = 3$:

$$\mathbb{P}(X = 5) = \frac{3^5 e^{-3}}{5!} = (243) \frac{0.049787}{120} \approx 0.1008.$$

(b)

$$\mathbb{P}(X = 0) = e^{-3} \approx 0.0498.$$

(c) Over half a second the rate is $\lambda = 1.5$:

$$\mathbb{P}(X > 1) = 1 - \mathbb{P}(0) - \mathbb{P}(1) = 1 - e^{-1.5} - 1.5e^{-1.5} = 1 - 2.5e^{-1.5} \approx 1 - 0.5578 = 0.4422.$$

Note the rate scales linearly with the interval — that is part of what “Poisson process” means, and it is why you can freely change the time unit as long as you scale λ with it.

28.5 By CDF. For $y \geq 0$,

$$\begin{aligned}\mathbb{P}(Y \leq y) &= \mathbb{P}\left(-\frac{\ln(X)}{\lambda} \leq y\right) = \mathbb{P}(\ln X \geq -\lambda y) \\ &= \mathbb{P}(X \geq e^{-\lambda y}) = 1 - e^{-\lambda y},\end{aligned}$$

using $\mathbb{P}(X \geq c) = 1 - c$ for $X \sim \text{Uniform}(0, 1)$ and $c \in [0, 1]$.

That is exactly the CDF of $\text{Exp}(\lambda)$.

By the change-of-variable formula. $g(x) = -\ln(x)/\lambda$ has inverse $x = g^{-1}(y) = e^{-\lambda y}$, so $|dx/dy| = \lambda e^{-\lambda y}$. Since $f_X = 1$ on $(0, 1)$:

$$f_Y(y) = 1 \cdot \lambda e^{-\lambda y}.$$

✓

This is inverse transform sampling (Chapter 27) worked out for the exponential, and it is exactly how `numpy.random.exponential` generates variates: one uniform, one logarithm.

28.6 Let $I_i = 1$ if the i -th element is a record — larger than everything before it.

Key observation. Among the first i elements, the largest is equally likely to be in any of the i positions, by symmetry of the random permutation. The i -th element is a record exactly when it is that largest, so

$$\mathbb{P}(I_i = 1) = \frac{1}{i}.$$

By linearity of expectation — and the I_i are emphatically **not** independent, which does not matter:

$$\mathbb{E}[\text{records}] = \sum_{i=1}^n \frac{1}{i} = H_n \approx \ln n + \gamma.$$

So a list of a million random numbers has only about 14 records. This is also the expected number of times a “running maximum” variable gets updated, and the reason randomised incremental algorithms (like randomised linear programming or incremental convex hulls) are fast: the expensive update only happens $O(\log n)$ times.

28.7 ($\Leftarrow$) If $X = c$ with probability 1 then $\mathbb{E}[X] = c$ and $\text{Var}(X) = \mathbb{E}[(c - c)^2] = 0$.

($\Rightarrow$) Suppose $\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = 0$. The random variable $(X - \mu)^2$ is nonnegative with expectation zero.

A nonnegative random variable with zero expectation must be zero almost surely: if $\mathbb{P}((X - \mu)^2 > \varepsilon) = \delta > 0$ for some $\varepsilon > 0$, then by Markov’s inequality reasoning $\mathbb{E}[(X - \mu)^2] \geq \varepsilon\delta > 0$, a contradiction.

So $(X - \mu)^2 = 0$ with probability 1, i.e. $X = \mu$ with probability 1.

28.8 Let $I_j = 1$ if bucket j is empty. Each of the n keys independently misses bucket j with probability $1 - 1/m$, so

$$\mathbb{P}(I_j = 1) = \left(1 - \frac{1}{m}\right)^n.$$

By linearity,

$$\mathbb{E}[\text{empty buckets}] = m \left(1 - \frac{1}{m}\right)^n.$$

At $n = m$: using $(1 - 1/m)^m \rightarrow e^{-1}$,

$$\mathbb{E}[\text{empty}] \approx \frac{m}{e} \approx 0.368m.$$

So loading a hash table to exactly 100% occupancy leaves **37% of buckets empty** — and correspondingly, a great many buckets hold two or more keys. Randomness is much lumpier than intuition suggests, which is why chaining needs to handle collisions gracefully and why “load factor 1” does not mean “one key per bucket”.

A.29. Chapter 29 — Several Random Variables

29.1 Marginals.

$$\begin{aligned} p_X(0) &= 0.1 + 0.2 = 0.3, & p_X(1) &= 0.3 + 0.4 = 0.7, \\ p_Y(0) &= 0.1 + 0.3 = 0.4, & p_Y(1) &= 0.2 + 0.4 = 0.6. \end{aligned}$$

$$\mathbb{E}[X] = 0.7, \mathbb{E}[Y] = 0.6.$$

$$\mathbb{E}[XY] = (1)(1)(0.4) = 0.4 \text{ — only the } (1, 1) \text{ cell contributes.}$$

$$\text{Cov}(X, Y) = 0.4 - (0.7)(0.6) = 0.4 - 0.42 = -0.02.$$

Independent? No. Check one cell: $p(0, 0) = 0.1$ but $p_X(0)p_Y(0) = (0.3)(0.4) = 0.12$. The joint does not factor.

Consistent with the nonzero covariance — though note that a zero covariance would **not** have established independence (next exercise).

29.2 $X \sim \text{Uniform}(-1, 1)$, $Y = X^2$.

$\mathbb{E}[X] = 0$ by symmetry. And

$$\mathbb{E}[XY] = \mathbb{E}[X^3] = \int_{-1}^1 x^3 \cdot \frac{1}{2} dx = 0,$$

since x^3 is odd and the density is symmetric.

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 - 0 = 0.$$

No contradiction: covariance measures only the **linear** component of association, and here the dependence is purely quadratic and symmetric — Y increases as X moves away from zero in **either** direction, so the positive and negative contributions cancel exactly.

Y is a deterministic function of X , so they are about as dependent as two variables can be. Mutual information (Chapter 32) would report this correctly; correlation cannot.

29.3 Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$. Then $\mathbb{E}[aX + b] = a\mu_X + b$, so the centred version of $aX + b$ is $a(X - \mu_X)$. Likewise for Y . Hence

$$\text{Cov}(aX + b, cY + d) = \mathbb{E}[a(X - \mu_X) \cdot c(Y - \mu_Y)] = ac \text{Cov}(X, Y).$$

Correlation. The standard deviations scale as $\sigma_{aX+b} = |a|\sigma_X$, so

$$\text{Corr}(aX + b, cY + d) = \frac{ac \text{Cov}(X, Y)}{|a||c|\sigma_X\sigma_Y} = \text{sgn}(ac) \text{Corr}(X, Y).$$

For $a, c > 0$ the correlation is unchanged. So correlation is invariant under shifting and positive rescaling — which is exactly why it is unitless and comparable across variables measured in different units. A negative scaling flips its sign, as it should.

29.4 Two assets, $\sigma^2 = 0.04$, $\rho = 0.6$, equal weights $1/2$:

$$\begin{aligned} \text{Var}(\bar{R}) &= \frac{1}{4}[\text{Var}(R_1) + \text{Var}(R_2) + 2\text{Cov}(R_1, R_2)] \\ &= \frac{1}{4}[0.04 + 0.04 + 2(0.6)(0.04)] = \frac{1}{4}[0.128] = 0.032. \end{aligned}$$

With $\rho = 0$:

$$\frac{1}{4}[0.08] = 0.02.$$

So correlation of 0.6 costs you 60% more variance than independence would have given.

As $n \rightarrow \infty$ with all pairwise correlations ρ :

$$\text{Var}(\bar{R}) = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2 \rightarrow \rho\sigma^2 = 0.6 \times 0.04 = 0.024.$$

You can never do better than 0.024, no matter how many such assets you hold. The idiosyncratic risk averages away; the **common** risk does not. That floor is what “systematic risk” means, and it is why a portfolio of a thousand correlated tech stocks is not diversified.

29.5 $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$.

$$\text{Var}(X_1 + X_2) = \Sigma_{11} + \Sigma_{22} + 2\Sigma_{12} = 4 + 3 + 4 = 11.$$

$$\text{Var}(X_1 - X_2) = \Sigma_{11} + \Sigma_{22} - 2\Sigma_{12} = 4 + 3 - 4 = 3.$$

(Both are $a^T \Sigma a$ with $a = (1, 1)$ and $a = (1, -1)$.)

Direction of maximum variance. Characteristic polynomial $(4 - \lambda)(3 - \lambda) - 4 = \lambda^2 - 7\lambda + 8$, so

$$\lambda = \frac{7 \pm \sqrt{17}}{2} \approx 5.562 \text{ and } 1.438.$$

For $\lambda_1 \approx 5.562$: $(4 - 5.562)v_1 + 2v_2 = 0$ gives $v_2 \approx 0.781v_1$, so the direction is

$$v_1 \approx (0.788, 0.616).$$

The variance in that direction is 5.562 — larger than either coordinate variance, and larger than the naive guess that the maximum must be along an axis. That is the whole point of PCA: the informative directions are generally not the ones you happened to measure.

29.6 Track states by the progress made towards HTH.

- S_0 : no useful prefix.

- S_1 : last flip was H.
- S_2 : last two were HT.

Let E_i be the expected additional flips from state i .

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0 \quad (\text{H} \rightarrow S_1, \text{T} \rightarrow S_0)$$

$$E_1 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_2 \quad (\text{H} \rightarrow S_1 \text{ again}, \text{T} \rightarrow S_2)$$

$$E_2 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_0 \quad (\text{H} \rightarrow \text{done}, \text{T} \rightarrow S_0)$$

The second line is the subtle one: from HT, another H completes the pattern; but from H, another H leaves you still with a single trailing H.

Solving: the first gives $E_0 = 2 + E_1$; the second gives $E_1 = 2 + E_2$; the third gives $E_2 = 1 + E_0/2$.

Substituting upwards: $E_1 = 3 + E_0/2$, then $E_0 = 5 + E_0/2$, so $E_0 = 10$.

Ten flips on average.

Worth knowing: the answer depends on the **pattern**, not just its length. HTT takes 8 on average, HHH takes 14. Patterns that can overlap themselves take longer, because a near-miss throws away more progress — the Conway leading-number algorithm quantifies this exactly.

29.7 $N \sim \text{Poisson}(\lambda)$, weights W_i i.i.d. with mean μ and variance τ^2 , and $S = \sum_{i=1}^N W_i$.

Mean, by the tower property.

$$\mathbb{E}[S] = \mathbb{E}[\mathbb{E}[S \mid N]] = \mathbb{E}[N\mu] = \mu \mathbb{E}[N] = \lambda\mu.$$

Variance, by the law of total variance. Given $N = n$, S is a sum of n i.i.d. terms, so $\mathbb{E}[S \mid N] = N\mu$ and $\text{Var}(S \mid N) = N\tau^2$. Hence

$$\begin{aligned} \text{Var}(S) &= \mathbb{E}[\text{Var}(S \mid N)] + \text{Var}(\mathbb{E}[S \mid N]) \\ &= \mathbb{E}[N\tau^2] + \text{Var}(N\mu) \\ &= \lambda\tau^2 + \mu^2\lambda = \lambda(\tau^2 + \mu^2). \end{aligned}$$

(Using $\mathbb{E}[N] = \text{Var}(N) = \lambda$ for a Poisson.)

This is the **compound Poisson** model, and it is the standard model for aggregate insurance claims, total bytes transferred in a session, and total cost of a random number of randomized operations. Note the variance has two sources — randomness in **how many** and randomness in **how big** — and the decomposition separates them cleanly.

29.8

$$M(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{-(\lambda-t)x} dx.$$

The integral converges iff $\lambda - t > 0$, i.e. $t < \lambda$, and then equals $1/(\lambda - t)$. So

$$M(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Moments.

$$M'(t) = \frac{\lambda}{(\lambda - t)^2}, \quad M'(0) = \frac{1}{\lambda} = \mathbb{E}[X].$$

$$M''(t) = \frac{2\lambda}{(\lambda - t)^3}, \quad M''(0) = \frac{2}{\lambda^2} = \mathbb{E}[X^2].$$

$$\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Matching the values quoted in Chapter 28. The restriction $t < \lambda$ is not a technicality: it says the exponential's tail is exactly at the boundary of what an MGF can handle, and distributions with heavier tails (like the Cauchy, or a power law) have no MGF at all for any $t > 0$ — which is precisely why the CLT machinery fails for them.

A.30. Chapter 30 — The Limit Theorems

30.1 $\mu = 10$, $\sigma = 2$. The event $X \geq 20$ implies $|X - 10| \geq 10$, which is 5σ . By Chebyshev,

$$\mathbb{P}(X \geq 20) \leq \mathbb{P}(|X - \mu| \geq 5\sigma) \leq \frac{1}{25} = 0.04.$$

If X were normal: $\mathbb{P}(Z \geq 5) \approx 2.87 \times 10^{-7}$.

The gap is a factor of about 140,000.

Chebyshev is loose because it uses only the variance and must hold for **every** distribution with that variance — including ones with all their excess mass sitting exactly at $\pm 5\sigma$. The normal has extremely thin tails, so it is nowhere near the worst case.

The lesson is not that Chebyshev is bad; it is that distribution-free bounds buy universality at the cost of tightness. If you know your distribution, use it. If you do not, Chebyshev still gives you something.

30.2 Markov's inequality with $a = 100$:

$$\mathbb{P}(X \geq 100) \leq \frac{\mathbb{E}[X]}{100} = \frac{1}{100} = 0.01.$$

A distribution achieving equality. Let

$$X = \begin{cases} 100 & \text{with probability } 0.01 \\ 0 & \text{with probability } 0.99 \end{cases}$$

Then $\mathbb{E}[X] = 100 \times 0.01 = 1$ ✓ and $\mathbb{P}(X \geq 100) = 0.01$ — exactly the bound.

So Markov cannot be improved without further assumptions. The extremal distribution puts all its mass at 0 and at the threshold, which is precisely the case in which the proof's two inequalities are both tight.

30.3 $\bar{x} = 250$ ms, $s = 40$ ms, $n = 100$.

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{40}{10} = 4 \text{ ms}.$$

A 95% confidence interval:

$$250 \pm 1.96 \times 4 = 250 \pm 7.84 = [242.2, 257.8] \text{ ms}.$$

To halve the width you need to halve the standard error, and since $\text{SE} \propto 1/\sqrt{n}$, you need **four times** the data: $n = 400$.

Worth noting for benchmarking: reducing s is usually cheaper than quadrupling n . Pinning CPU affinity, disabling frequency scaling, warming caches, and discarding the first few runs all attack s directly, and a $2\times$ reduction in s is worth a $4\times$ increase in n .

30.4 $X \sim \text{Binomial}(10000, 0.5)$, so

$$\mu = 5000, \quad \sigma = \sqrt{np(1-p)} = \sqrt{2500} = 50.$$

By the CLT, X is approximately $\mathcal{N}(5000, 50^2)$, so

$$\mathbb{P}(4900 \leq X \leq 5100) \approx \mathbb{P}(-2 \leq Z \leq 2) \approx 0.9545.$$

About 95.5%.

Note how tight that is in **relative** terms: the interval is 5000 ± 100 , which is $\pm 2\%$. With ten thousand tosses the proportion of heads is pinned to within about 1% with high confidence. With a hundred tosses, $\sigma = 5$, and the same $\pm 2\sigma$ is $\pm 10\%$. The $1/\sqrt{n}$ shrinkage, made concrete.

30.5 Independent case. $\text{Var}(\sum X_i) = \sum \text{Var}(X_i) = n\sigma^2$, and $\bar{X} = (1/n) \sum X_i$, so

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Correlated case. Using the general formula,

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) = n\sigma^2 + n(n-1)\rho\sigma^2.$$

Dividing by n^2 :

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2.$$

Limit. As $n \rightarrow \infty$ the first term vanishes and $(n-1)/n \rightarrow 1$, so

$$\text{Var}(\bar{X}) \rightarrow \rho\sigma^2.$$

Averaging removes the independent part of the noise entirely and the correlated part not at all. If $\rho = 0.01$ — a correlation you would barely notice — then no amount of data reduces the variance below $0.01\sigma^2$, which caps the achievable precision at one tenth of a single measurement's standard deviation.

30.6 The CLT requires **finite variance**. The Cauchy distribution has density $f(x) = 1/(\pi(1+x^2))$, whose tails decay like $1/x^2$. Then

$$\int |x|f(x) dx \approx \int \frac{dx}{\pi|x|}$$

diverges logarithmically, so the Cauchy has **no mean**, let alone a finite variance. The hypothesis fails at the first step.

What actually happens. The average of n i.i.d. standard Cauchy variables is **again standard Cauchy** — exactly, for every n . (This follows from the characteristic function $e^{-|t|}$: the average has characteristic function $(e^{-|t|/n})^n = e^{-|t|}$.)

So $\bar{X}_n$ does not concentrate at all. Its distribution is identical for $n = 1$ and $n = 10^9$. Collecting more data buys you literally nothing, which is a startling and genuinely important failure mode.

The correct limit theory here uses **stable distributions**; the Cauchy is itself stable, which is why it is its own limit. And the practical warning is real: heavy-tailed data (network latencies, financial returns, file sizes) can be much closer to this regime than to the Gaussian one, and sample means of such data are far less trustworthy than their standard errors suggest.

- 30.7** Baseline $p = 0.05$; a 2% **relative** lift means the treatment rate is $0.05 \times 1.02 = 0.051$, so the difference to detect is $\Delta = 0.001$.

The standard error of the difference of two proportions, with n per arm, is

$$\text{SE} = \sqrt{\frac{2p(1-p)}{n}} = \sqrt{\frac{2 \times 0.05 \times 0.95}{n}} = \sqrt{\frac{0.095}{n}}.$$

Requiring $\Delta = 2 \cdot \text{SE}$:

$$0.001 = 2\sqrt{\frac{0.095}{n}} \Rightarrow \frac{0.095}{n} = 2.5 \times 10^{-7} \Rightarrow n = 380,000.$$

About 380,000 users per arm.

Two lessons. First, small relative effects on small base rates are **extremely** expensive to measure — and note that the required n scales as $1/\Delta^2$, so detecting a 1% lift instead of 2% would cost four times as much. Second, this calculation must be done **before** running the experiment; discovering afterwards that you were underpowered means the experiment could not have answered the question, whatever it returned.

- 30.8 Why the standard formula does not apply.** The $\bar{X} \pm 1.96 \text{ SE}$ interval rests on the CLT for the **sample mean**, and specifically on $\text{SE} = \sigma/\sqrt{n}$. The sample median has a different asymptotic distribution: it is asymptotically normal with variance $1/(4nf(m)^2)$, where f is the **density at the true median**.

That is the problem: $f(m)$ is a property of the unknown distribution, and estimating a density well is much harder than estimating a mean. There is no formula you can evaluate from the data alone in the way $s/\sqrt{n}$ can be.

Bootstrap procedure.

```
given the sample x[1..200]:
  for b = 1 .. 10000:
    resample 200 values from x, WITH replacement
    m[b] := median of that resample
  sort m[]
  CI := [ m[250], m[9750] ]      # 2.5th and 97.5th percentiles
```

This needs no density estimate and no formula. It works because the empirical distribution converges to the true one, so the simulated variability of the median converges to its real variability.

One caveat: the bootstrap is less reliable for statistics that depend on the extreme tails (the maximum, say) or for very small samples, where the empirical distribution is a poor stand-in. For the median at $n = 200$ it is entirely trustworthy.

A.31. Chapter 31 — Estimation and Inference

31.1 For $X_i \sim \text{Poisson}(\lambda)$,

$$\ell(\lambda) = \sum_i [x_i \log \lambda - \lambda - \log(x_i!)] .$$

Differentiating:

$$\ell'(\lambda) = \frac{\sum_i x_i}{\lambda} - n = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{1}{n} \sum_i x_i = \bar{x} .$$

Maximum? $\ell''(\lambda) = -(\sum x_i)/\lambda^2 < 0$ (assuming not all observations are zero), so ℓ is strictly concave and the critical point is the global maximum.

The sample mean — reassuring, since λ is the mean of a Poisson.

31.2 For $X_i \sim \text{Exp}(\lambda)$,

$$\ell(\lambda) = n \log \lambda - \lambda \sum_i x_i .$$

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_i x_i = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}} .$$

Unbiased? No. $\sum X_i \sim \text{Gamma}(n, \lambda)$, and for that distribution $\mathbb{E}[1/\sum X_i] = \lambda/(n-1)$. So

$$\mathbb{E}[\hat{\lambda}] = n \mathbb{E}\left[\frac{1}{\sum X_i}\right] = \frac{n\lambda}{n-1} > \lambda .$$

The MLE **overestimates** the rate, by a factor $n/(n-1)$. Multiplying by $(n-1)/n$ gives an unbiased estimator.

This is the general situation: MLEs are **consistent** (the bias vanishes as $n \rightarrow \infty$) but not generally unbiased at finite n , because maximum likelihood does not commute with nonlinear transformations. Here the parameter is $1/\text{mean}$, and the reciprocal of an unbiased estimate of the mean is not an unbiased estimate of the reciprocal — Jensen's inequality (Chapter 28), in fact.

31.3 Expand the sum of squared deviations:

$$\sum_i (X_i - \bar{X})^2 = \sum_i X_i^2 - n\bar{X}^2 .$$

Taking expectations, and using $\mathbb{E}[X_i^2] = \sigma^2 + \mu^2$ and $\mathbb{E}[\bar{X}^2] = \text{Var}(\bar{X}) + (\mathbb{E}[\bar{X}])^2 = \sigma^2/n + \mu^2$:

$$\begin{aligned}\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] &= n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2 = (n-1)\sigma^2.\end{aligned}$$

Dividing by $n-1$:

$$\mathbb{E}\left[\frac{1}{n-1} \sum_i (X_i - \bar{X})^2\right] = \sigma^2.$$

Where the missing degree of freedom went. The deviations are measured from $\bar{X}$, not from the true μ — and $\bar{X}$ is the value that **minimises** the sum of squared deviations, so it necessarily sits closer to the data than μ does. Dividing by n would systematically underestimate. The one degree of freedom consumed estimating the mean is exactly the correction.

31.4 10 flips, 7 heads.

MLE: $\hat{p} = 7/10 = 0.700$.

Beta(1, 1) prior (uniform, equivalent to 2 pseudo-observations, one of each). Posterior is $\text{Beta}(1+7, 1+3) = \text{Beta}(8, 4)$, with mean

$$\frac{8}{8+4} = \frac{2}{3} \approx 0.667.$$

Beta(10, 10) prior (equivalent to 20 pseudo-observations, 10 heads and 10 tails). Posterior is $\text{Beta}(17, 13)$, with mean

$$\frac{17}{30} \approx 0.567.$$

The difference. The $\text{Beta}(10, 10)$ prior encodes a fairly strong belief that the coin is near-fair, carrying twice the weight of the actual data (20 pseudo-observations against 10 real ones). So the posterior is pulled most of the way back towards 0.5.

The $\text{Beta}(1, 1)$ prior is nearly uninformative — it shifts the estimate only from 0.700 to 0.667, which is Laplace’s rule of succession $(k+1)/(n+2)$.

The right question is always “how many observations is my prior worth?”, and for the Beta family the answer is literally $\alpha + \beta$.

31.5 With Gaussian likelihood (noise variance σ^2) and prior $x \sim \mathcal{N}(0, \tau^2 I)$, the log posterior is

$$\log p(x | b) = \text{const} - \frac{\|Ax - b\|^2}{2\sigma^2} - \frac{\|x\|^2}{2\tau^2}.$$

Maximising this is minimising

$$\frac{\|Ax - b\|^2}{2\sigma^2} + \frac{\|x\|^2}{2\tau^2},$$

and multiplying through by $2\sigma^2$ (which does not move the minimiser):

$$\|Ax - b\|^2 + \frac{\sigma^2}{\tau^2} \|x\|^2.$$

That is exactly ridge regression with

$$\lambda = \frac{\sigma^2}{\tau^2}.$$

Reading it. λ is the ratio of noise variance to prior variance — the ratio of “how uncertain the data is” to “how uncertain I was beforehand”. Noisy data (large σ^2) or a confident prior (small τ^2) both mean strong regularisation. A vague prior ($\tau^2 \rightarrow \infty$) gives $\lambda \rightarrow 0$ and recovers ordinary least squares.

This is why tuning λ by cross-validation is, in Bayesian terms, empirically choosing how much you believe your prior.

31.6 40 independent tests, no true effects, $\alpha = 0.05$.

Expected number of “significant” results: by linearity of expectation, $40 \times 0.05 = 2$.

Probability of at least one:

$$1 - (0.95)^{40} = 1 - 0.1285 = 0.8715.$$

So you would be **unlucky not to find something**, and whatever you found would be noise.

Bonferroni threshold: to keep the family-wise error rate at 5%, use

$$\alpha' = \frac{\alpha}{m} = \frac{0.05}{40} = 0.00125.$$

Bonferroni is conservative — it controls the probability of **any** false positive, which is a strong requirement, and with many tests it becomes very hard to detect anything real. When you have hundreds of tests and expect some genuine effects, Benjamini–Hochberg (controlling the **false discovery rate**, the expected proportion of discoveries that are false) is usually the better trade.

And the count m must include every test you ran, including the ones you decided not to report. That is not a technicality; it is the whole content of the multiple comparisons problem.

31.7 For a Bernoulli, $\log p(x; p) = x \log p + (1 - x) \log(1 - p)$.

$$\frac{\partial}{\partial p} \log p(x; p) = \frac{x}{p} - \frac{1 - x}{1 - p}.$$

$$\frac{\partial^2}{\partial p^2} \log p(x; p) = -\frac{x}{p^2} - \frac{1 - x}{(1 - p)^2}.$$

Taking $-\mathbb{E}[\cdot]$ with $\mathbb{E}[X] = p$:

$$I(p) = \frac{p}{p^2} + \frac{1 - p}{(1 - p)^2} = \frac{1}{p} + \frac{1}{1 - p} = \frac{1}{p(1 - p)}.$$

Cramér–Rao bound:

$$\text{Var}(\hat{p}) \geq \frac{1}{nI(p)} = \frac{p(1 - p)}{n}.$$

The sample proportion. $\hat{p} = k/n$ with $k \sim \text{Binomial}(n, p)$, so

$$\text{Var}(\hat{p}) = \frac{np(1 - p)}{n^2} = \frac{p(1 - p)}{n}.$$

It attains the bound exactly. The sample proportion is an **efficient** estimator: no unbiased estimator of p can do better, at any n . There is no cleverer statistic waiting to be found.

Note also that $I(p) = 1/(p(1-p))$ is smallest at $p = 1/2$ and blows up near 0 and 1 — the data is **most** informative about extreme rates and least informative about a fair coin, which matches the intuition that distinguishing $p = 0.5$ from $p = 0.51$ is much harder than distinguishing $p = 0.001$ from $p = 0.011$.

31.8 Baseline $p_1 = 0.02$, 10% relative lift means $p_2 = 0.022$, so $\Delta = 0.002$.

For 80% power at two-sided $\alpha = 0.05$, the standard formula for the sample size per arm is

$$n = \frac{(z_{\alpha/2} + z_\beta)^2 [p_1(1-p_1) + p_2(1-p_2)]}{\Delta^2},$$

with $z_{0.025} = 1.96$ and $z_{0.20} = 0.84$.

$$(1.96 + 0.84)^2 = 7.84.$$

$$p_1(1-p_1) = 0.0196, \quad p_2(1-p_2) = 0.021516, \quad \text{sum} = 0.041116.$$

$$n = \frac{7.84 \times 0.041116}{0.002^2} = \frac{0.32235}{4 \times 10^{-6}} \approx 80,566.$$

About 80,000 users per arm, so 160,000 in total.

Two things to note. The requirement scales as $1/\Delta^2$, so halving the effect you want to detect quadruples the cost. And it scales roughly as $1/p$ for small p , so low-conversion funnels are much more expensive to experiment on — which is why teams working on rare events measure upstream proxies instead.

A.32. Chapter 32 — Information Theory

32.1 (a) Fair 8-sided die. Uniform over 8 outcomes: $H = \log_2 8 = 3$ bits.

(b) (0.5, 0.25, 0.125, 0.125).

$$H = 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3) = 0.5 + 0.5 + 0.375 + 0.375 = 1.75 \text{ bits.}$$

(c) (0.97, 0.01, 0.01, 0.01).

$$H = -0.97 \log_2 0.97 - 3(0.01 \log_2 0.01) = 0.0426 + 0.1993 = 0.242 \text{ bits.}$$

Compressibility. Naively each source needs 3, 2, and 2 bits per symbol respectively. The entropies say the true limits are 3, 1.75, and 0.242.

So (a) is **incompressible** — it is already at its limit. (b) can be compressed by 12.5%. And (c) can be compressed by a factor of more than 8, because it is almost always the same symbol and a good coder spends almost nothing saying so.

32.2 Huffman code for (b). Merge the two smallest repeatedly: $0.125 + 0.125 = 0.25$; then $0.25 + 0.25 = 0.5$; then $0.5 + 0.5 = 1$. Reading off the tree:

$$a \rightarrow 0 \quad (1 \text{ bit}), \quad b \rightarrow 10 \quad (2), \quad c \rightarrow 110 \quad (3), \quad d \rightarrow 111 \quad (3).$$

Average length:

$$0.5(1) + 0.25(2) + 0.125(3) + 0.125(3) = 1.75 \text{ bits} = H.$$

✓

Does (a) achieve equality? Yes. Eight equally likely symbols get 3-bit codes, average 3 bits, which equals the entropy.

Why these two work. In both cases every probability is a power of $1/2$, so the ideal codeword length $-\log_2 p_i$ is an **integer** and Huffman can hit it exactly. This is the condition for a **dyadic** distribution.

Why (c) cannot. Here $-\log_2 0.97 = 0.044$ bits — but a Huffman codeword must be at least 1 bit long. So the common symbol is charged 1 bit when it deserves 0.044, and the average length is at least 1 bit against an entropy of 0.242. Huffman is off by a factor of four.

The fix is arithmetic coding, which encodes the whole message as a single number and therefore charges fractional bits per symbol, approaching the entropy arbitrarily closely. This is exactly why modern compressors (and neural compressors) use arithmetic or range coding rather than Huffman.

32.3 Start from the chain rule for entropy,

$$H(X, Y) = H(X) + H(Y|X),$$

which follows from $p(x, y) = p(x)p(y|x)$ and expanding the logarithm.

So $H(X, Y) = H(X) + H(Y)$ holds iff $H(Y|X) = H(Y)$, i.e. iff

$$I(X; Y) = H(Y) - H(Y|X) = 0.$$

And by Gibbs' inequality, $I(X; Y) = \text{KL}(p(x, y) \parallel p(x)p(y)) = 0$ iff the two distributions are equal, i.e. iff $p(x, y) = p(x)p(y)$ — which is exactly independence.

In words: the information in the pair equals the sum of the individual informations precisely when neither tells you anything about the other. Any dependence means some of the information is shared, hence counted twice in the sum.

32.4 $p = (0.5, 0.5)$, $q = (0.9, 0.1)$, in bits.

$$\begin{aligned} \text{KL}(p\|q) &= 0.5 \log_2 \left(\frac{0.5}{0.9} \right) + 0.5 \log_2 \left(\frac{0.5}{0.1} \right) \\ &= 0.5(-0.848) + 0.5(2.322) = -0.424 + 1.161 = 0.737 \text{ bits.} \end{aligned}$$

$$\begin{aligned} \text{KL}(q\|p) &= 0.9 \log_2 \left(\frac{0.9}{0.5} \right) + 0.1 \log_2 \left(\frac{0.1}{0.5} \right) \\ &= 0.9(0.848) + 0.1(-2.322) = 0.763 - 0.232 = 0.531 \text{ bits.} \end{aligned}$$

KL($p\|q$) is larger.

Why, in the language of the chapter. $\text{KL}(p\|q)$ weights the log-ratio by p . Here p puts a full half of its mass on the outcome where q assigns only 0.1, and the ratio there is 5 — so the term $0.5 \log_2 5 = 1.16$ dominates. The first argument punishes q for **under-covering** p 's support, and it does so severely.

$\text{KL}(q\|p)$ weights by q , which is concentrated on the outcome where p is perfectly adequate (0.5 versus 0.9, a ratio of only 1.8), so the penalty is milder.

This is the mean-seeking/mode-seeking asymmetry made numerical: the first direction hates missing mass, the second tolerates it.

32.5 Reduction to $-\log q_y$. With one-hot p , the vector y has $y_c = 1$ for the true class c and 0 elsewhere. So

$$H(p, q) = -\sum_i y_i \log q_i = -\log q_c,$$

because every term but one is multiplied by zero.

Gradient with respect to the logits. Let $p = \text{softmax}(z)$ (writing p now for the model's output). Then

$$L = -\sum_i y_i \log p_i, \quad \frac{\partial L}{\partial z_j} = -\sum_i \frac{y_i}{p_i} \frac{\partial p_i}{\partial z_j}.$$

Substituting the softmax Jacobian $\partial p_i / \partial z_j = p_i(\delta_{ij} - p_j)$:

$$\frac{\partial L}{\partial z_j} = -\sum_i \frac{y_i}{p_i} p_i(\delta_{ij} - p_j) = -\sum_i y_i(\delta_{ij} - p_j) = -y_j + p_j \sum_i y_i = p_j - y_j,$$

using $\sum_i y_i = 1$.

$$\nabla_z L = p - y.$$

Note the cancellation: the $1/p_i$ from the logarithm met the p_i from the softmax and vanished. That is why the fused `softmax_cross_entropy` op exists — computing the two separately performs a division that mathematically was never needed and numerically loses precision.

32.6 Parent entropy. 60 positive, 40 negative, so $p = 0.6$:

$$H(\text{parent}) = -0.6 \log_2 0.6 - 0.4 \log_2 0.4 = 0.442 + 0.529 = 0.971 \text{ bits.}$$

Children. Left: 45 positive, 5 negative, so $p = 0.9$:

$$H(\text{left}) = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 = 0.137 + 0.332 = 0.469.$$

Right: the remaining 15 positive and 35 negative, so $p = 0.3$:

$$H(\text{right}) = -0.3 \log_2 0.3 - 0.7 \log_2 0.7 = 0.521 + 0.360 = 0.881.$$

Weighted child entropy (both children have 50 of the 100 examples):

$$0.5(0.469) + 0.5(0.881) = 0.675.$$

Information gain:

$$IG = 0.971 - 0.675 = 0.296 \text{ bits.}$$

So this split resolves about 0.3 of the 0.97 bits of uncertainty in the label — roughly 30%. A decision tree would compare this against every other candidate split and take the largest.

32.7 From the definition,

$$\begin{aligned}
I(X; Y) &= \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \\
&= \sum_{x,y} p(x, y) \log \frac{p(x|y)}{p(x)} \\
&= \sum_{x,y} p(x, y) \log p(x|y) - \sum_{x,y} p(x, y) \log p(x) \\
&= -H(X|Y) + H(X),
\end{aligned}$$

using $p(x, y)/p(y) = p(x|y)$ in the second line and marginalising over y in the last term.

Why symmetry survives. The **defining** expression $p(x, y)/(p(x)p(y))$ is manifestly symmetric in x and y , so $I(X; Y) = I(Y; X)$ immediately.

But $H(X|Y) \neq H(Y|X)$ in general — they are not even measured against the same total. What is symmetric is the **reduction**:

$$H(X) - H(X|Y) = H(Y) - H(Y|X).$$

Both sides equal $H(X) + H(Y) - H(X, Y)$, which is visibly symmetric. The shared information is symmetric even though the amounts of private information are not — exactly like the overlap of two sets of different sizes.

32.8 Perplexity is e^H when the cross-entropy is in nats.

$$e^{3.2} = 24.53, \quad e^{2.9} = 18.17.$$

Improvement factor: $24.53/18.17 = 1.35$. The better model behaves as though it were choosing among 18 equally likely options rather than 25.

Compressed size. The number of bits needed per token is proportional to the cross-entropy, so the ratio of compressed sizes is

$$\frac{2.9}{3.2} = 0.906.$$

The compressed text would be about **9.4% smaller**.

Worth noticing the scale mismatch: a 35% improvement in perplexity buys only a 9% reduction in file size. Perplexity is exponential in the thing that actually costs bits, so it exaggerates progress — which is one reason cross-entropy (or bits-per-byte) is the more honest metric to report.

A.33. Chapter 33 — Numerical Computing

33.1 (a) Directly, to 4 significant figures. $\sqrt{10001} = 100.0$ and $\sqrt{10000} = 100.0$, so the difference is 0.000.

Every significant digit has been destroyed. The two operands agreed to more digits than we carried, so the difference carried none.

(b) By the conjugate.

$$\sqrt{10001} - \sqrt{10000} = \frac{1}{\sqrt{10001} + \sqrt{10000}} = \frac{1}{100.0 + 100.0} = \frac{1}{200.0} = 0.005000.$$

The true value is 0.00499999..., so this is correct to all four figures carried.

The lesson. The two expressions are algebraically identical. One is a subtraction of nearly equal numbers; the other has no subtraction at all. The rearrangement did not add precision — it avoided throwing precision away.

- 33.2** $1/10 = 1/(2 \times 5)$, and 5 is not a power of 2. By the proposition of Chapter 1 — a fraction has a terminating expansion in base b iff its denominator's prime factors all divide b — $1/10$ has a **repeating** binary expansion:

$$0.1_{10} = 0.\overline{00011}_2.$$

A double stores 53 significant bits, so the expansion must be truncated and rounded, giving a value very slightly **above** $1/10$. The same happens for 0.2.

Adding the two rounded values and rounding again gives a result that is not the nearest double to 0.3:

$$0.1 + 0.2 = 0.3000000000000000444089209850062616169452667236328125.$$

To 20 digits, 0.30000000000000004441.

Meanwhile the double nearest 0.3 is 0.299999999999999988898.... The two differ by one unit in the last place, so `==` returns false.

Nothing went wrong: three exact roundings produced a result one ulp away from the rounding of the exact answer. That is the best that can be guaranteed.

- 33.3** $f(x) = 1/(1-x)$, so $f'(x) = 1/(1-x)^2$.

$$\kappa_f(x) = \left| \frac{xf'(x)}{f(x)} \right| = \left| \frac{x/(1-x)^2}{1/(1-x)} \right| = \left| \frac{x}{1-x} \right|.$$

At $x = 0.999$:

$$\kappa = \frac{0.999}{0.001} = 999 \approx 10^3.$$

Expect to lose about 3 decimal digits. With double precision (about 16 digits) you would retain roughly 13 — acceptable. But the condition number grows without bound as $x \rightarrow 1$: at $x = 1 - 10^{-15}$ it is 10^{15} and essentially nothing survives.

This is a property of the **function near the pole**, not of any algorithm. No implementation of $1/(1-x)$ can do better, because the true answer genuinely does depend that sensitively on the input.

- 33.4** Values $1.0, 10^{-16}, 10^{-16}$, with $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$ (the gap just above 1).

Naive summation. $1.0 + 10^{-16}$: the addend is smaller than half an ulp (1.11×10^{-16}), so it rounds away entirely and the sum stays 1.0. The same happens again. **Result: 1.0.**

Kahan.

step	$y = x - c$	$t = \text{sum} + y$	$c = (t - \text{sum}) - y$	sum
$x = 1.0$	1.0	1.0	0	1.0
$x = 10^{-16}$	10^{-16}	1.0 (rounded)	$0 - 10^{-16} = -10^{-16}$	1.0

step	$y = x - c$	$t = \text{sum} + y$	$c = (t - \text{sum}) - y$	sum
$x = 10^{-16}$	2×10^{-16}	$1 + \varepsilon$	$\varepsilon - 2 \times 10^{-16} = 2.2 \times 10^{-17}$	$1 + \varepsilon$

On the second step the addend is lost — but the compensation c **records exactly how much was lost**. On the third step that recorded loss is added back, making the effective addend 2×10^{-16} , which now exceeds half an ulp and rounds **up**.

Result: **1.0000000000000002**, the correctly rounded answer.

Kahan recovered a quantity that naive summation discarded twice. Over millions of terms this is the difference between a usable answer and drift.

33.5 $y' = y$, $y(0) = 1$, $h = 0.5$. Exact: $y(1) = e = 2.718282$.

RK4, step 1.

$$k_1 = 1, \quad k_2 = 1 + 0.25 = 1.25, \quad k_3 = 1 + 0.25(1.25) = 1.3125, \quad k_4 = 1 + 0.5(1.3125) = 1.65625.$$

$$y_1 = 1 + \frac{0.5}{6}(1 + 2.5 + 2.625 + 1.65625) = 1 + 0.083333(7.78125) = 1.648438.$$

(Exact $e^{0.5} = 1.648721$.)

RK4, step 2. With $y = 1.648438$:

$$k_1 = 1.648438, \quad k_2 = 2.060547, \quad k_3 = 2.163574, \quad k_4 = 2.730225,$$

$$y_2 = 1.648438 + 0.083333(12.826904) = 2.717346.$$

Forward Euler. $y_1 = 1 + 0.5(1) = 1.5$; $y_2 = 1.5 + 0.5(1.5) = 2.25$.

Method	$y(1)$	error
Exact	2.718282	—
RK4, 2 steps	2.717346	9.4×10^{-4}
Euler, 2 steps	2.250000	4.7×10^{-1}

RK4 is **500 times** more accurate for the same number of steps, at 4 times the function evaluations. Even per evaluation it wins by two orders of magnitude, and the gap widens as h shrinks — h^4 against h .

33.6 $y' = -1000y$.

Forward Euler. $y_{n+1} = y_n + h(-1000y_n) = (1 - 1000h)y_n$.

The iteration decays iff the amplification factor has modulus below 1:

$$|1 - 1000h| \leq 1 \quad \Rightarrow \quad 0 \leq 1000h \leq 2 \quad \Rightarrow \quad h \leq 0.002.$$

Any larger step and the numerical solution oscillates with **growing** amplitude, while the true solution decays smoothly to zero. The instability is entirely an artefact of the method.

Backward Euler. $y_{n+1} = y_n + h(-1000y_{n+1})$, so

$$y_{n+1} = \frac{y_n}{1 + 1000h}.$$

For every $h > 0$ the factor $1/(1 + 1000h)$ lies strictly between 0 and 1, so the iteration decays.
Unconditionally stable.

This is the whole argument for implicit methods on stiff problems: the explicit method's step size is dictated by the **fastest** timescale even when that mode has long since decayed and you do not care about it. The implicit method's step is dictated only by the accuracy you want.

33.7 Double precision inputs, 10 accurate digits. With $\kappa = 10^7$, the relative error in the solution can be 10^7 times the relative error in the input. Losing 7 digits from 10 leaves **about 3 trustworthy digits**.

float32 inputs. Single precision carries about 7 decimal digits. Losing 7 leaves **none**. The computed solution would be numerical noise — it would have the right shape and no correct digits.

The general rule: you need input precision **exceeding** $\log_{10} \kappa$ plus however many digits you actually want out. For $\kappa = 10^7$ and a demand of 6 good digits you need 13 digits in, which rules out **float32** entirely and leaves double precision with little margin.

This is why “just use **float32**, it is twice as fast” is a decision that requires knowing κ first.

33.8 Gauss–Legendre. An n -node rule is exact for polynomials of degree up to $2n - 1$. For x^7 we need $2n - 1 \geq 7$, so $n \geq 4$.

Four nodes suffice, and give the exact answer $1/8$.

Simpson. Simpson is exact for cubics only. Its composite error is

$$E \approx \frac{(b-a)h^4}{180} f^{(4)}(\xi), \quad f^{(4)}(x) = 840x^3, \max = 840.$$

To reach machine precision, 10^{-16} :

$$\frac{h^4 \times 840}{180} \leq 10^{-16} \Rightarrow h^4 \leq 2.1 \times 10^{-17} \Rightarrow h \leq 2.1 \times 10^{-5},$$

needing about 48,000 subintervals, hence roughly 10^5 function evaluations.

The gap: 4 evaluations against about 100,000.

Why. Simpson fixes the nodes at even spacing and optimises only the n weights, buying exactness to degree $n - 1$. Gauss optimises the nodes **and** the weights — $2n$ free parameters — buying exactness to degree $2n - 1$. For a smooth integrand that doubling of the exactness degree is worth an enormous amount, and it costs nothing but giving up the even grid.

A.34. Chapter 34 — Optimization

34.1 Convex. $f(x) = x^4$ has $f''(x) = 12x^2 \geq 0$ for all x , so f is convex. (It is even **strictly** convex: $f'' > 0$ except at the single point $x = 0$, which is not enough to create a flat segment.)

Not strongly convex. μ -strong convexity requires $f''(x) \geq \mu > 0$ **everywhere**. Here $f''(0) = 0$, so no positive μ works.

Where it fails, and why it matters. The failure is exactly at the minimiser. Near $x = 0$ the function is extraordinarily flat — it looks like x^4 , not like a parabola — so the gradient carries almost no information about how far you are from the optimum.

Consequently gradient descent on x^4 converges **sublinearly**: the error decays like $1/\sqrt{k}$ rather than geometrically. Strong convexity is precisely the hypothesis that buys the linear rate of Chapter 34's theorem, and this is what its absence costs.

34.2 Maximum of convex functions is convex. Let $h = \max(f, g)$ with f, g convex. For $t \in [0, 1]$:

$$\begin{aligned} h(tx + (1-t)y) &= \max(f(tx + (1-t)y), g(tx + (1-t)y)) \\ &\leq \max(tf(x) + (1-t)f(y), tg(x) + (1-t)g(y)) \\ &\leq t \max(f(x), g(x)) + (1-t) \max(f(y), g(y)) = th(x) + (1-t)h(y). \end{aligned}$$

(The last step uses $tf(x) \leq th(x)$ and similarly for the other three terms.)

Minimum need not be. Take $f(x) = x^2$ and $g(x) = (x-2)^2$, both convex. Then $h = \min(f, g)$ equals x^2 for $x \leq 1$ and $(x-2)^2$ for $x \geq 1$, with a **downward** kink at $x = 1$.

Check convexity at the midpoint of 0 and 2:

$$h(1) = 1, \quad \text{but} \quad \frac{h(0) + h(2)}{2} = \frac{0 + 0}{2} = 0.$$

Since $h(1) > 0$, the chord lies **below** the function. Not convex.

This generalises: an arbitrary maximum of convex functions is convex (which is why the pointwise supremum of linear functions — the support function — is always convex), while minima are not. It is also why “the best of several models” is a hard object to optimise over.

34.3 $f(x, y) = x^2 + 10y^2$.

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 20 \end{pmatrix}, \quad \lambda_{\min} = 2, \lambda_{\max} = 20, \quad \kappa = 10.$$

Largest stable learning rate: $\eta < 2/\lambda_{\max} = 0.1$.

At $\eta = 0.05$. The updates decouple:

$$x_{k+1} = (1 - 0.05 \times 2)x_k = 0.9x_k, \quad y_{k+1} = (1 - 0.05 \times 20)y_k = 0 \cdot y_k.$$

From $(1, 1)$: $(0.9, 0)$, then $(0.81, 0)$.

Here $\eta = 0.05$ happens to be **exactly** $1/\lambda_{\max}$, which annihilates the y component in a single step — a lucky accident of the numbers. The point that remains: the x component decays by only 0.9 per step, needing about 44 iterations to reach 10^{-2} . **The steep direction is solved instantly and the shallow one crawls.**

The classic zigzag appears at a rate closer to the maximum. At $\eta = 0.09$:

$$x \text{ factor} = 1 - 0.18 = 0.82, \quad y \text{ factor} = 1 - 1.8 = -0.8,$$

so from $(1, 1)$ the iterates are $(0.82, -0.8)$, $(0.672, 0.64)$, $(0.551, -0.512)$ — oscillating across the valley in y while creeping along it in x . Exactly the picture in the chapter, and it is caused entirely by $\kappa = 10$.

34.4 $f(x) = (1/2)x^T Ax$ has $\nabla f = Ax$, so the iteration is

$$x_{k+1} = x_k - \eta Ax_k = (I - \eta A)x_k.$$

Diagonalise: $A = Q\Lambda Q^T$ with Q orthogonal (spectral theorem, Chapter 23). Writing $z_k = Q^T x_k$, the iteration decouples completely:

$$z_{k+1,i} = (1 - \eta\lambda_i)z_{k,i},$$

so $z_{k,i} = (1 - \eta\lambda_i)^k z_{0,i}$.

This tends to zero for every component iff

$$|1 - \eta\lambda_i| < 1 \quad \text{for all } i \quad \Leftrightarrow \quad 0 < \eta\lambda_i < 2 \quad \text{for all } i.$$

Since all $\lambda_i > 0$ (positive definite), the binding constraint is the largest:

$$0 < \eta < \frac{2}{\lambda_{\max}}.$$

And the **rate** is governed by the worst factor, $\max_i |1 - \eta\lambda_i|$, which is minimised at $\eta^* = 2/(\lambda_{\min} + \lambda_{\max})$, giving the contraction factor $(\kappa - 1)/(\kappa + 1)$. Large κ pushes that towards 1 — the quantitative version of “ill-conditioning makes gradient descent slow”.

34.5 Write the constraint in standard form $g(x, y) = 2 - x - y \leq 0$, and form

$$\mathcal{L} = x^2 + y^2 + \lambda(2 - x - y), \quad \lambda \geq 0.$$

Stationarity: $2x - \lambda = 0$ and $2y - \lambda = 0$, so $x = y = \lambda/2$.

Complementary slackness: $\lambda(2 - x - y) = 0$.

Case $\lambda = 0$. Then $x = y = 0$, but $g(0, 0) = 2 > 0$ — infeasible. Rejected.

Case constraint active. Then $x + y = 2$, so with $x = y$ we get $x = y = 1$ and $\lambda = 2$.

Check the remaining conditions: primal feasibility $\checkmark$ ($1 + 1 = 2$), dual feasibility $\checkmark$ ($\lambda = 2 \geq 0$). Since the problem is convex, KKT is sufficient, so this is the global minimum.

Answer: $(1, 1)$ with value 2; the constraint is **active**, with multiplier $\lambda = 2$.

Reading the multiplier. Relaxing the constraint to $x + y \geq 2 - \varepsilon$ would reduce the optimal value by about 2ε . Indeed the optimum for a right-hand side c is $c^2/2$, whose derivative at $c = 2$ is 2. $\checkmark$ The shadow price is exact.

34.6 $f(x) = e^x - 2x$, $f'(x) = e^x - 2$, $f''(x) = e^x$. The minimiser is $x^* = \ln 2 = 0.693147$.

Newton, $x_{n+1} = x_n - (e^{x_n} - 2)/e^{x_n}$:

$$x_1 = 0 - \frac{1 - 2}{1} = 1,$$

$$x_2 = 1 - \frac{e - 2}{e} = 1 - 0.264 = 0.736.$$

Gradient descent at $\eta = 0.5$, $x_{n+1} = x_n - 0.5(e^{x_n} - 2)$:

$$x_1 = 0 - 0.5(-1) = 0.5,$$

$$x_2 = 0.5 - 0.5(1.6487 - 2) = 0.5 + 0.176 = 0.676.$$

After 2 steps	x_2	error
Newton	0.736	0.043
Gradient descent	0.676	0.017

Gradient descent is closer — which is not what the theory advertises, and is worth understanding.

Two reasons. First, Newton's first step **overshot** badly: from $x_0 = 0$ the quadratic model is a poor fit and the step lands at 1, well past the minimum. Newton's guarantees are **local**.

Second, and more amusing: $f''(x^*) = e^{\ln 2} = 2$, so the ideal Newton scaling near the optimum is $1/f'' = 0.5$ — exactly the learning rate chosen. For this problem, gradient descent at $\eta = 0.5$ is Newton's method near the optimum.

A third Newton step gives 0.694 (error 9×10^{-4}) and a fourth essentially machine precision; gradient descent continues to halve its error each step. The quadratic advantage appears once you are close, and not before.

34.7 With $A = U\Sigma V^T$, the ridge solution is

$$\begin{aligned}\hat{x} &= (A^T A + \lambda I)^{-1} A^T b \\ &= (V \Sigma^T \Sigma V^T + \lambda V V^T)^{-1} V \Sigma^T U^T b \\ &= V (\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T U^T b.\end{aligned}$$

The middle factor is diagonal with entries

$$\frac{\sigma_i}{\sigma_i^2 + \lambda},$$

against $1/\sigma_i$ for ordinary least squares ($\lambda = 0$).

Behaviour of the filter.

- For $\sigma_i^2 \gg \lambda$: the factor is $\approx 1/\sigma_i$. Directions where the data is informative are essentially untouched.
- For $\sigma_i^2 \ll \lambda$: the factor is $\approx \sigma_i/\lambda \rightarrow 0$. Directions where the data carries almost nothing are **suppressed** rather than amplified.

The directions with $\sigma_i \approx 0$ are exactly where plain least squares divides by nearly zero and produces enormous, noise-driven coefficients. Ridge replaces that blow-up with a graceful decay to zero: it declines to make claims about directions the data cannot see.

Compare the truncated pseudoinverse (Chapter 25), which does the same thing with a hard cutoff. Ridge is the smooth version, and the smoothness is why it is differentiable and easy to tune.

34.8 Newton is impossible. The Hessian of a function of 10^7 parameters has 10^{14} entries. In `float64` that is 8×10^{14} bytes — **800 terabytes**, for one matrix, at one point. Solving with it would cost $O(n^3) = 10^{21}$ operations. Both numbers are off by many orders of magnitude from feasible, and no amount of hardware closes the gap.

L-BFGS is marginal. It stores $m \approx 10$ pairs of n -vectors, so $20 \times 10^7 \times 8$ bytes = 1.6 GB — large but not absurd, and each iteration is $O(mn)$. The real obstacles are elsewhere: L-BFGS assumes a **deterministic, smooth** objective, and it uses a line search. Minibatch gradients are stochastic, so the curvature estimates built from successive noisy gradients are unreliable, and a line search on a noisy objective is meaningless. L-BFGS is excellent for full-batch smooth problems and poorly suited to SGD-style training.

What Adam approximates. Adam maintains two running averages — m_t of the gradient and v_t of the **squared** gradient, both n -vectors, so 160 MB total — and steps with

$$x_{t+1} = x_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \varepsilon}}.$$

Dividing each coordinate by the root-mean-square of its recent gradients is a **diagonal preconditioner**: it approximates multiplying by $\text{diag}(H)^{-1/2}$, rescaling each coordinate so that they all have comparable effective curvature.

In the language of the chapter, it is trying to reduce the **effective condition number** — the same goal as Newton’s method, achieved with $O(n)$ memory instead of $O(n^2)$ by throwing away every off-diagonal coupling. That approximation is crude, and for problems where the important curvature is off-diagonal it helps little. It is nonetheless the best cost–benefit trade anyone has found at this scale.

A.35. Chapter 35 — Geometry for Graphics and Simulation

35.1

$$T = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$TS = \begin{pmatrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad ST = \begin{pmatrix} 2 & 0 & 0 & 4 \\ 0 & 2 & 0 & 6 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Reading them. Remember the rightmost factor acts first.

TS : scale, then translate. The object doubles in size about the origin and then moves by $(2, 3, 4)$.

ST : translate, then scale. The object moves by $(2, 3, 4)$ and then everything — including that displacement — doubles, so the net translation is $(4, 6, 8)$.

The difference is the whole reason transform order matters in a scene graph. Scaling after translating scales the object’s **position** as well as its size, which is almost never what an artist means.

35.2

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Orthogonal. The columns are $(\cos \theta, \sin \theta, 0)$, $(-\sin \theta, \cos \theta, 0)$, $(0, 0, 1)$. Each has norm 1 (Pythagorean identity) and the pairwise dot products are $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$ and 0. ✓

Determinant. Expanding along the last row: $\det = 1 \cdot (\cos^2 \theta + \sin^2 \theta) = 1$. ✓

Eigenvalues. The characteristic polynomial factors as

$$(1 - \lambda)[(\cos \theta - \lambda)^2 + \sin^2 \theta],$$

giving $\lambda = 1$ and $\lambda = \cos \theta \pm i \sin \theta = e^{\pm i\theta}$.

Interpretation. The eigenvalue 1 has eigenvector $(0, 0, 1)$ — the rotation **axis**, the one direction left fixed (Euler's theorem).

The complex pair $e^{\pm i\theta}$ describes the action in the xy -plane, where there is no real invariant direction. By Chapter 6, multiplication by $e^{i\theta}$ is rotation by θ — so the rotation angle is sitting in the argument of the eigenvalue, and the modulus being 1 says no stretching occurs.

35.3 For a rotation. M orthogonal means $M^{-1} = M^T$, so

$$(M^{-1})^T = (M^T)^T = M.$$

The normal transform rule and the point transform rule coincide, and no special handling is needed. This is why the bug stays hidden in rotation-only code.

A non-uniform scale where it fails. Take $S = \text{diag}(2, 1, 1)$ and the plane $x + y = 0$, whose normal is $n = (1, 1, 0)$.

Points on the plane have the form $(t, -t, z)$. Applying S gives $(2t, -t, z)$, which satisfies $x/2 + y = 0$ — a plane with normal proportional to $(1, 2, 0)$.

Now compare:

$$Sn = (2, 1, 0) \quad (\text{wrong}), \quad (S^{-1})^T n = \text{diag}\left(\frac{1}{2}, 1, 1\right)(1, 1, 0) = \left(\frac{1}{2}, 1, 0\right) \propto (1, 2, 0) \quad (\text{right}).$$

Transforming the normal by S gives $(2, 1, 0)$, which is **not** perpendicular to the transformed plane. Visually this shows up as lighting that is subtly wrong on any non-uniformly scaled object — highlights in the wrong place, surfaces that look bent.

35.4 Rotation by 90° about the y -axis, so $u = (0, 1, 0)$ and $\theta = \pi/2$:

$$q = \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)j = \frac{\sqrt{2}}{2}(1 + j) \approx 0.7071 + 0.7071j.$$

Rotating $(1, 0, 0)$. Treating $v = i$ and computing qvq^{-1} (with $q^{-1} = \bar{q}$ since $\|q\| = 1$):

$$qi\bar{q} = -k,$$

i.e. the point $(0, 0, -1)$. ✓ Correct: a right-handed 90° rotation about $+y$ sends $\hat{x}$ to $-\hat{z}$.

q and $-q$ agree. Substituting $-q$:

$$(-q)v(-q)^{-1} = (-q)v(-q^{-1}) = (-1)(-1)qvq^{-1} = qvq^{-1}.$$

The two sign flips cancel exactly, because q appears once on each side.

So the map from unit quaternions to rotations is **two-to-one**. Practically this matters in one place: when interpolating between two orientations, take the dot product of the quaternions first and negate one of them if it is negative — otherwise *slerp* takes the long way round the sphere, and the object spins through nearly 360° to reach a nearby orientation.

35.5 Control points $P_0 = (0, 0)$, $P_1 = (0, 1)$, $P_2 = (1, 1)$, $P_3 = (1, 0)$, at $t = 0.5$ — so every interpolation is a midpoint.

Level 1.

$$Q_0 = (0, 0.5), \quad Q_1 = (0.5, 1), \quad Q_2 = (1, 0.5).$$

Level 2.

$$R_0 = (0.25, 0.75), \quad R_1 = (0.75, 0.75).$$

Level 3.

$$B(0.5) = (0.5, 0.75).$$

The curve is a symmetric arch: it starts at $(0, 0)$ heading straight up (towards P_1), turns over, and lands at $(1, 0)$ heading straight down (away from P_2). Its apex is the computed point $(0.5, 0.75)$ — note it does **not** reach $y = 1$, since the curve is pulled towards but not through its interior control points, and always stays inside their convex hull.

The two halves produced by the algorithm — control points $P_0, Q_0, R_0, B(0.5)$ and $B(0.5), R_1, Q_2, P_3$ — are exactly the two sub-curves, which is the subdivision property that makes adaptive rendering work.

35.6 Ray $p(t) = td$ with $d = (1, 1, 1)/\sqrt{3}$, sphere centre $c = (3, 3, 3)$, radius 2.

Substituting into $\|p - c\|^2 = 4$:

$$t^2 - 2t(d \cdot c) + \|c\|^2 - 4 = 0.$$

Here $d \cdot c = 9/\sqrt{3} = 3\sqrt{3} \approx 5.196$ and $\|c\|^2 = 27$, so

$$t^2 - 6\sqrt{3}t + 23 = 0.$$

Discriminant: $108 - 92 = 16 > 0$, so two intersections.

$$t = \frac{6\sqrt{3} \pm 4}{2} = 3\sqrt{3} \pm 2 \approx 3.196 \text{ and } 7.196.$$

The points are td :

$$\approx (1.845, 1.845, 1.845) \quad \text{and} \quad (4.155, 4.155, 4.155).$$

Check: the distance from $(3, 3, 3)$ to the first point is $\sqrt{3} \times 1.155 = 2.00$. ✓

In a renderer you take the **smaller positive** root (the near hit), and the sign of the discriminant is the entire hit test — negative means miss, zero means tangent.

35.7 Triangle $A = (0, 0)$, $B = (4, 0)$, $C = (0, 3)$; point $P = (2, 1)$.

Since A is the origin, $P = \alpha A + \beta B + \gamma C$ reduces to

$$(2, 1) = \beta(4, 0) + \gamma(0, 3),$$

so $4\beta = 2$ giving $\beta = 1/2$, and $3\gamma = 1$ giving $\gamma = 1/3$. Then

$$\alpha = 1 - \beta - \gamma = 1 - \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

$$(\alpha, \beta, \gamma) = \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{3}\right).$$

All three are **nonnegative** (and they sum to 1), so **P is inside** the triangle.

As a check by areas: the whole triangle has area 6, and the sub-triangle opposite A (namely PBC) should have area $\alpha \times 6 = 1$. Indeed PBC has vertices $(2, 1)$, $(4, 0)$, $(0, 3)$ and area $(1/2) \left| \det \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \right| = (1/2)(2) = 1$. ✓

35.8 Why entrywise interpolation fails. The rotation matrices form $\text{SO}(3)$, which is a **curved manifold** — a three-dimensional surface sitting inside the nine-dimensional space of 3×3 matrices, cut out by the six equations $Q^T Q = I$ plus $\det Q = 1$.

A straight-line interpolation $(1-t)R_1 + tR_2$ is a **chord** through that ambient space, and a chord of a curved surface leaves the surface. The result generally has columns that are neither unit length nor mutually orthogonal: it is not a rotation, and applying it will shear and shrink the object.

The pathological case makes it vivid: in two dimensions interpolate $R(0) = I$ and $R(\pi) = -I$ at $t = 0.5$. You get the **zero matrix** — the object collapses to a point halfway through the animation.

What slerp does instead. Represent each orientation as a unit quaternion and interpolate along the **great circle** joining them on the unit sphere in $\mathbb{R}^4$:

$$\text{slerp}(q_1, q_2, t) = \frac{\sin((1-t)\Omega) q_1 + \sin(t\Omega) q_2}{\sin \Omega}, \quad \cos \Omega = q_1 \cdot q_2.$$

Every intermediate value is a unit quaternion, hence a genuine rotation. And because the path is a geodesic traversed at constant speed, the **angular velocity is constant** — the object turns smoothly rather than speeding up and slowing down.

(Two practical notes: negate q_2 if $q_1 \cdot q_2 < 0$, to take the short way round the double cover; and fall back to linear interpolation when Ω is tiny, since $\sin \Omega$ in the denominator is then a division by nearly zero — a Chapter 33 problem inside a Chapter 35 formula.)

A.36. Chapter 36 — Backpropagation, Derived

36.1 Let $h = W_1 x \in \mathbb{R}^p$, $a = \varphi(h)$, $y = W_2 a \in \mathbb{R}^m$, and $\mathcal{L} = \|y - t\|^2$.

Loss gradient. $g = \partial \mathcal{L} / \partial y = 2(y - t)$, shape $m \times 1$.

Output weights. By the linear-layer rule with input a :

$$\frac{\partial \mathcal{L}}{\partial W_2} = g a^T, \quad (m \times 1)(1 \times p) = m \times p. \checkmark$$

Backpropagate.

$$\frac{\partial \mathcal{L}}{\partial a} = W_2^T g, \quad (p \times m)(m \times 1) = p \times 1,$$

$$\delta = (W_2^T g) \circ \varphi'(h), \quad p \times 1.$$

Input weights.

$$\frac{\partial \mathcal{L}}{\partial W_1} = \delta x^T, \quad (p \times 1)(1 \times n) = p \times n. \checkmark$$

Every intermediate is a vector; the only matrices produced are the two gradients themselves, each the same shape as its weight matrix. That is the signature of reverse mode done correctly, and checking the shapes is the fastest way to catch an error.

36.2 Collapse. Composing L affine maps:

$$W_L(W_{L-1}(\cdots(W_1 x) \cdots)) = (W_L W_{L-1} \cdots W_1)x = W_{\text{eff}}x,$$

since matrix multiplication is associative. The product W_{eff} is a single matrix of the same shape as the composite map, so the network is exactly equivalent to one linear layer.

(With biases the same holds for affine maps: the composition is affine, with a single effective weight and a single effective bias.)

Parameter count for $L = 5$, width 100. The five layers hold $5 \times 100 \times 100 = 50,000$ parameters. The equivalent single layer needs $100 \times 100 = 10,000$.

40,000 parameters are wasted — they add memory, compute, and optimisation difficulty while adding exactly zero expressive power.

Worth noting they are not entirely inert: the **parameterisation** differs, so gradient descent takes a different path through a different landscape (this is the subject of “implicit regularisation of deep linear networks”). But the set of functions representable is identical.

36.3 Sigmoid. $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ is maximised at $z = 0$, where it equals 0.25. So each layer’s diagonal Jacobian contributes a factor of at most 0.25.

With $\|W\| \approx 1$, the gradient at layer 1 relative to layer 50 is at most

$$0.25^{49} = 10^{49 \log_{10} 0.25} = 10^{-29.5} \approx 3 \times 10^{-30}.$$

That is not “small”; it is **below the smallest normal double** (10^{-308} is the limit, so it survives, but in **float32** — minimum around 10^{-38} — it is within a factor of 10^8 of flushing to zero). In practice the early layers receive no usable signal at all and simply do not train.

ReLU. $\varphi'(z) = 1$ on the active half and 0 on the inactive half. On the active path there is **no** systematic shrinkage: with $\|W\| \approx 1$ the product stays $O(1)$ across depth.

The caveat is that roughly half the units are inactive at initialisation, so the effective variance is halved per layer — which is precisely why He initialisation uses $\text{Var}(W_{ij}) = 2/n_{\text{in}}$ rather than $1/n_{\text{in}}$. The factor of 2 exactly compensates for ReLU discarding half its input.

That one factor of 2 is the difference between a 50-layer network that trains and one that does not.

36.4 For a residual block $a = x + F(x)$, differentiate termwise:

$$\frac{\partial a}{\partial x} = I + \frac{\partial F}{\partial x}.$$

Why the product resists both pathologies. Over L blocks the gradient carries a factor

$$\prod_{\ell=1}^L (I + J_{\ell}), \quad J_{\ell} = \frac{\partial F_{\ell}}{\partial x}.$$

Expanding, this is

$$I + \sum_{\ell} J_{\ell} + \sum_{\ell < m} J_{\ell} J_m + \dots$$

The leading term is the **identity**. So there is always a path from the loss back to every layer that passes through no multiplication at all — the gradient cannot be attenuated to nothing by a long product, because one term of the expansion never multiplies anything.

And if the J_{ℓ} are individually small (as they are at initialisation, when F is initialised near zero), the whole product stays close to I rather than exploding.

This is why residual connections were the change that made hundred-layer and thousand-layer networks trainable, and why essentially every deep architecture since 2015 — including every transformer — has them. It is one line of calculus with an enormous consequence.

- 36.5** Dual numbers for $f(x) = x^2 \sin x$ at $x = 2$. Carry the pair $(2, 1)$ — the value and its derivative with respect to x .

Step 1: $u = x \cdot x$. By the product rule built into the arithmetic,

$$u = (2 \times 2, 2 \times 1 + 1 \times 2) = (4, 4).$$

Step 2: $v = \sin x$. By the chain rule for the primitive $\sin$,

$$v = (\sin 2, \cos 2 \times 1) = (0.909297, -0.416147).$$

Step 3: $f = u \cdot v$.

$$\begin{aligned} f &= (4 \times 0.909297, 4 \times (-0.416147) + 4 \times 0.909297) \\ &= (3.637190, -1.664588 + 3.637190) = (3.637190, 1.972602). \end{aligned}$$

Check analytically. $f'(x) = 2x \sin x + x^2 \cos x$, so

$$f'(2) = 4(0.909297) + 4(-0.416147) = 3.637190 - 1.664588 = 1.972602.$$

✓

Exact to every digit carried, with no step size and no cancellation. That is forward-mode automatic differentiation: overload the arithmetic, run the program once, read the derivative out of the second component.

- 36.6** With $q, k \in \mathbb{R}^d$ having i.i.d. entries of mean 0 and variance 1:

$$q \cdot k = \sum_{i=1}^d q_i k_i.$$

The terms are independent, so their variances add:

$$\text{Var}(q \cdot k) = \sum_{i=1}^d \text{Var}(q_i k_i).$$

For each term, $\mathbb{E}[q_i k_i] = \mathbb{E}[q_i] \mathbb{E}[k_i] = 0$, so

$$\text{Var}(q_i k_i) = \mathbb{E}[q_i^2 k_i^2] = \mathbb{E}[q_i^2] \mathbb{E}[k_i^2] = 1 \times 1 = 1.$$

Hence $\text{Var}(q \cdot k) = d$, and the typical magnitude of a score is $\sqrt{d}$.

At $d = 4096$ with no scaling. Typical scores are of order $\sqrt{4096} = 64$, with the largest a few multiples of that.

Feeding scores separated by tens of units into softmax makes it saturate completely: the exponentials differ by factors of $e^{64} \approx 10^{28}$, so the output is numerically one-hot.

Then the softmax Jacobian $\text{diag}(p) - pp^T$ is essentially the zero matrix (every $p_{i(1-p_i)} \approx 0$), and **no gradient flows back through the attention weights at all**. The model cannot learn what to attend to; it is stuck with whatever its initialisation happened to pick.

Dividing by $\sqrt{d}$ restores unit-variance scores, keeping softmax in the regime where its derivative is non-negligible. One line of variance arithmetic, and the architecture works.

36.7 With $Q \in \mathbb{R}^{n \times d}$, $K \in \mathbb{R}^{m \times d}$, $V \in \mathbb{R}^{m \times d_v}$:

1. QK^T : an $(n \times d)(d \times m)$ product, costing $O(nmd)$ and producing an $n \times m$ matrix.
2. **Softmax over each row**: $O(nm)$.
3. $\times V$: an $(n \times m)(m \times d_v)$ product, costing $O(nmd_v)$.

In self-attention $m = n$ and $d_v \approx d$, so the total is

$$O(n^2d).$$

Which dominates. Both matrix products are $O(n^2d)$ — they are the same order, and neither dominates asymptotically. What differs is that the first **materialises** an $n \times n$ matrix, so the memory cost is $O(n^2)$, and for long sequences that is the binding constraint rather than the arithmetic.

This is exactly what FlashAttention addresses: it computes the same result by tiling, never storing the full $n \times n$ matrix, trading a little recomputation for a large reduction in memory traffic. The FLOP count is unchanged; the memory-bandwidth cost is not, and on modern hardware that is what determines the wall clock.

36.8 The gradient of a composition is a product of Jacobians, and matrix multiplication is associative — so you may bracket the product any way you like, and the cost depends entirely on the bracketing (Chapter 20, Exercise 7).

Training a network: many inputs, one output. The chain is

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\underbrace{\partial h_L}_{1 \times n_L}} \cdot \frac{\partial h_L}{\partial h_{L-1}} \cdots \frac{\partial h_1}{\partial \theta}.$$

The **leftmost** factor is a row vector, because the loss is a scalar. So multiplying **left to right** keeps every intermediate a row vector, and every operation is a cheap vector–matrix product. One backward pass yields the gradient with respect to all 10^9 parameters.

Forward mode would need one pass per input direction — 10^9 passes. Reverse mode wins by that factor.

A simulation: one input, many outputs. Now the **rightmost** factor is a column vector, because there is a single scalar parameter. Multiplying **right to left** keeps every intermediate a column vector, and one forward pass gives the sensitivity of **every** output to that one parameter.

Reverse mode would need one pass per output.

The rule. Start from the narrow end. Few outputs, many inputs: reverse. Few inputs, many outputs: forward. Comparable numbers: either, and the optimal bracketing of a general chain is a dynamic programming problem — which is why some autodiff systems perform “optimal Jacobian accumulation” and why the general problem is NP-hard.

CHAPTER B.

A Library of Free Mathematics Books

B.1. How this list is organised and what the labels mean

Everything here is legally free to read. That covers two different things and the distinction matters if you want to redistribute, remix, or print:

Label	Meaning
OPEN	Released under an open licence (Creative Commons, GFDL, or public domain). Free to copy, print, and often to modify.
FREE	The author or publisher provides it at no cost, but the copyright is conventional. Read it, do not redistribute it.
BUY	Not free. Listed because it is the best book on its subject and you should know it exists.

Trap: verify before you rely on a link.

URLs move and licences change. Everything below was accurate as far as I know, but I have not re-checked each one. If a link is dead, search the title and the author's name — these are all well-known works and are usually one search away, frequently mirrored on the author's university page.

Where a book is hosted on an author's personal or departmental page, that page is the canonical source. Prefer it to any aggregator.

A note on how to use a list like this: **do not collect books**. Pick one per subject, work through it, and only then look at a second. The failure mode of having a good list is treating acquisition as progress.

B.2. Proof and mathematical maturity

Start here if Chapter 7 was the hardest part of this book. These are the gentlest genuine on-ramps to proof that exist.

Title	Why	Lic.
Book of Proof — Richard Hammack	The best free introduction to proof, full stop. Logic, sets, induction, relations, functions, cardinality. Clear, patient, well-exercised. Start here.	OPEN
Mathematical Reasoning: Writing and Proof — Ted Sundstrom	Similar territory, more emphasis on the writing of proofs. Good second opinion if Hammack does not land.	OPEN
An Infinite Descent into	More ambitious: proof plus a tour of number theory, combinatorics, and analysis. Good if you want momentum.	OPEN

Title	Why	Lic.
Pure Mathematics — Clive Newstead		
Mathematics for Computer Science — Lehman, Leighton, Meyer	The MIT 6.042 text. Proof, induction, number theory, graphs, counting, probability, all aimed squarely at computer scientists. Enormous and excellent. Arguably the single best companion to this book.	OPEN

B.3. Foundations: algebra, precalculus, trigonometry

If Part I moved too fast, these fill in at a slower pace with far more practice problems.

Title	Why	Lic.
OpenStax Algebra and Trigonometry	Professionally produced, thorough, enormous exercise sets with answers. Unexciting and completely reliable.	OPEN
OpenStax Precalculus	The same, aimed at the transition into calculus.	OPEN
Precalculus — Stitz and Zeager	A well-regarded free alternative with a slightly drier, more rigorous tone.	OPEN
Khan Academy	Not a book, but the best resource in existence for drilling a specific gap. Use it surgically: identify the thing you cannot do, drill that, return.	FREE

B.4. Calculus

Title	Why	Lic.
OpenStax Calculus, Volumes 1–3	Complete single- and multivariable calculus. Volume 1: limits and derivatives. Volume 2: integration and series. Volume 3: multivariable. Solid, conventional, thousands of exercises.	OPEN
Active Calculus — Matt Boelkins	Built around you doing the work rather than reading worked examples. Genuinely different pedagogy, and it suits a self-learner who has time.	OPEN
APEX Calculus — Hartman et al.	Clean, well-illustrated, good on the geometric side.	OPEN
Calculus — Gilbert Strang	Strang’s own calculus text, free from MIT and OpenStax. Idiosyncratic, opinionated, full of insight about why . Best as a second read.	OPEN

Title	Why	Lic.
Paul’s Online Math Notes — Paul Dawkins	Not a textbook: a reference. When you need to remember how to do a specific integral at 2am, this is where you go. Complete worked examples for everything in calculus I–III and differential equations.	FREE
MIT OCW 18.01, 18.02	Full lecture videos, problem sets, and exams with solutions.	OPEN
Essence of Calculus — 3Blue1Brown	Twelve videos. Will not teach you to compute anything, will completely rewire how you see the subject. Watch before or alongside a real text, never instead of one.	FREE

B.5. Real analysis

The rigorous version of calculus — where the ε - δ arguments of Chapter 13 get taken seriously. Optional for most programmers; essential if you want to read theory papers.

Title	Why	Lic.
Basic Analysis — Jiří Lebl	Two volumes, free, well-paced, widely used as a course text. The most approachable free option.	OPEN
Introduction to Real Analysis — William Trench	More traditional and more demanding. Complete and careful.	OPEN
Mathematical Analysis I — Elias Zakon	Free from the Trillia Group. Terse, thorough, old-school.	OPEN
Principles of Mathematical Analysis — Walter Rudin	“Baby Rudin”. The canonical text, famously austere. Not a first book, and not for everyone — but if you can work through it, you can read anything.	BUY

B.6. Linear algebra

The most important part of this book, and correspondingly well served.

Title	Why	Lic.
Introduction to Applied Linear Algebra (VMLS) — Boyd and Vandenberghe	Free PDF from Stanford. Written for exactly this audience: applications-first, no determinants until late, everything motivated by data fitting and least squares. Companion Julia and Python material. If you read one, read this.	FREE

Title	Why	Lic.
Linear Algebra Done Wrong — Sergei Treil	The title is a joke about Axler’s Linear Algebra Done Right . Rigorous, coordinate-light, free. Excellent second book.	FREE
A First Course in Linear Algebra — Rob Beezer	Very complete, very free, with an unusual emphasis on getting every proof written out.	OPEN
Linear Algebra — Jim Hefferon	Free, with a full solutions manual — rare and valuable for self-study.	OPEN
MIT 18.06 — Gilbert Strang	The famous course. Video lectures, problem sets, exams. Strang’s four-fundamental-subspaces framing is the one used in Chapter 24 of this book. The accompanying textbook is BUY, the course is free.	FREE
Essence of Linear Algebra — 3Blue1Brown	Fifteen short videos that make the geometric content of Part IV visible. The single highest-value hour of video on this list.	FREE
Immersive Linear Algebra — Ström, Åström, Akenine-Möller	A linear algebra book with fully interactive figures. Worth an afternoon for the visualisations alone.	FREE
The Matrix Cookbook — Petersen and Pedersen	Not a textbook: a 70-page reference of matrix identities and derivatives. Keep it open when reading ML papers. Chapter 26 of this book is a curated subset.	FREE
Numerical Linear Algebra — Trefethen and Bau	The best book on how linear algebra behaves on actual computers. Twenty-page “lectures”, beautifully written. If Chapter 33 interested you, this is the follow-up.	BUY

B.7. Discrete mathematics and combinatorics

Title	Why	Lic.
Discrete Mathematics: An Open Introduction — Oscar Levin	Friendly, well-scoped, good exercises. The standard free choice.	OPEN
Mathematics for Computer Science —	Listed above and worth listing twice. The definitive free discrete text for programmers.	OPEN

Title	Why	Lic.
Lehman, Leighton, Meyer		
Applied Combinatorics — Keller and Trotter	Deeper on counting, generating functions, and graph theory.	OPEN
generatingfunctionology — Herbert Wilf	Free from the author. The book on generating functions — the technique gestured at in Chapter 17. Short, witty, brilliant.	FREE
A = B — Petkovšek, Wilf, Zeilberger	Free. How to make a computer prove combinatorial identities. Delightful, and genuinely useful if you ever need one proved.	FREE
Concrete Mathematics — Graham, Knuth, Patashnik	Not free. The best book on the mathematics of algorithm analysis, written partly by Knuth for exactly that purpose. Sums, recurrences, generating functions, asymptotics. Buy it eventually.	BUY

B.8. Number theory and cryptography

Title	Why	Lic.
A Computational Introduction to Number Theory and Algebra — Victor Shoup	Free from the author. Rigorous, algorithm-centred, and the right book if you want to actually implement cryptographic primitives.	FREE
Elementary Number Theory: Primes, Congruences, and Secrets — William Stein	Free, computational, uses Sage throughout.	FREE
An Introductory Course in Elementary Number Theory — Wissam Raji	Shorter and gentler than the above.	OPEN
A Graduate Course in Applied Cryptography — Boneh and Shoup	Free draft. The modern reference on the cryptography side. Demanding.	FREE

B.9. Probability and statistics

Title	Why	Lic.
Introduction to Probability — Grinstead and Snell	Free, classic, thorough. Released by the AMS under an open licence. A safe default.	OPEN
Introduction to Probability, Statistics, and Random Processes — Hossein Pishro-Nik	Free online with an excellent set of worked solutions and interactive tools. Covers more ground than Grinstead and Snell.	FREE
Harvard Stat 110 — Joe Blitzstein	Free lecture videos, problem sets, and a superb “cheat sheet”. Blitzstein is one of the best probability teachers alive. The accompanying book (Introduction to Probability) is BUY and is worth it.	FREE
Think Stats and Think Bayes — Allen Downey	Free, CC-licensed, Python-first. Statistics taught by writing code rather than by manipulating formulas. Excellent for a programmer, and a useful counterweight to formula-heavy treatments.	OPEN
Seeing Theory — Brown University	Interactive visual introduction to probability. Half an hour well spent before reading anything else.	FREE
All of Statistics — Larry Wasserman	Not free. A fast, dense survey for people who want the whole subject in one volume and can handle the pace.	BUY
Statistical Rethinking — Richard McElreath	Not free, but the lecture series is on YouTube and is outstanding. The best introduction to Bayesian thinking for people who learn by doing.	BUY

B.10. Information theory

Title	Why	Lic.
Information Theory, Inference, and Learning Algorithms — David MacKay	Free PDF, legally, from the author’s page. One of the great textbooks of the last thirty years: information theory, Bayesian inference, coding theory, and neural networks in one coherent story. Idiosyncratic and wonderful. If Chapter 32 caught your interest, read this.	FREE
Elements of Information Theory — Cover and Thomas	The standard reference. Not free.	BUY

B.11. Numerical computing

Title	Why	Lic.
Fundamentals of Numerical Computation — Driscoll and Braun	Free online (fncbook). Modern, Julia-based, covers everything in Chapter 33 properly.	FREE
What Every Computer Scientist Should Know About Floating-Point Arithmetic — David Goldberg	Free, 70 pages, and the title is not an exaggeration. Read it once properly.	FREE
Numerical Recipes — Press et al.	Older editions readable free online. Encyclopaedic and opinionated; the code is dated and the explanations are still good.	FREE
Accuracy and Stability of Numerical Algorithms — Nicholas Higham	Not free. The definitive treatment of the conditioning-versus-stability distinction of Chapter 33.	BUY

B.12. Optimization

Title	Why	Lic.
Convex Optimization — Boyd and Vandenberghe	Free PDF from Stanford, and the canonical text. Chapter 34 is a compressed sketch of its first half. The accompanying EE364a lectures are also free.	FREE
Algorithms for Decision Making — Kochenderfer, Wheeler, Wray	Free from MIT Press. Decision theory, MDPs, and planning under uncertainty. Well written, Julia-based.	OPEN
Numerical Optimization — Nocedal and Wright	Not free. The reference for what optimisation solvers actually do.	BUY

B.13. Mathematics specifically for machine learning

Title	Why	Lic.
Mathematics for Machine Learning — Deisenroth, Faisal, Ong	Free PDF. The closest published book to what this one is trying to do, aimed slightly higher. Linear algebra, analytic geometry, matrix decompositions, vector calculus, probability, optimisation, then four ML applications. Read it after this book.	FREE
Probabilistic Machine Learning: An Introduction and Advanced Topics — Kevin Murphy	Free draft PDFs. Enormous, current, and the best single reference on the probabilistic view of ML.	FREE
An Introduction to Statistical Learning — James, Witten, Hastie, Tibshirani	Free PDF. The gentlest good introduction to statistical learning, with R and Python editions.	FREE
The Elements of Statistical Learning — Hastie, Tibshirani, Friedman	Free PDF. The harder, deeper sibling of the above. A classic.	FREE
Understanding Deep Learning — Simon Prince	Free PDF from MIT Press. Current, unusually clear, excellent figures. The best modern deep learning text.	FREE
Dive into Deep Learning — Zhang, Lipton, Li, Smola	Free, interactive, runnable. Every concept comes with working code in multiple frameworks.	OPEN
Deep Learning — Goodfellow, Bengio, Courville	Free HTML. Now somewhat dated on architectures, and Part I remains a good compact treatment of the mathematical background.	FREE

B.14. Graphics, geometry, and simulation

Title	Why	Lic.
Physically Based Rendering — Pharr, Jakob, Humphreys	Free online. A complete renderer explained line by line, with the mathematics done properly. Won an Academy Award, which is not a sentence you write about many textbooks.	FREE

Title	Why	Lic.
Ray Tracing in One Weekend — Peter Shirley	Free three-book series. The fastest possible path from nothing to a working ray tracer. Start here if graphics interests you at all.	OPEN
Scratchapixel	Free online course covering the mathematics of rendering from scratch. Uneven but often exactly what you need.	FREE
3D Math Primer for Graphics and Game Development — Dunn and Parberry	Not free. The friendliest treatment of quaternions and 3D transforms.	BUY

B.15. Reference and tools

Resource	Why	Lic.
The Matrix Cookbook	Matrix identities and derivatives. Keep it open.	FREE
OEIS — Online Encyclopedia of Integer Sequences	Type in the first six terms of a sequence you found; it tells you what it is, with formulas and references. Astonishingly useful.	FREE
SymPy	Python computer algebra. Check your symbolic work, expand messy expressions, compute derivatives you do not trust yourself with.	OPEN
Wolfram Alpha	Fast checking for integrals, series, and equations. Free tier shows answers without steps.	FREE
Desmos and GeoGebra	Instant graphing. Plot the function before reasoning about it — half of Chapter 15 is easier with a picture.	FREE
Math Stack Exchange	The archive is the value, not asking. Almost every confusion you will have has been asked and answered well.	FREE

B.16. A suggested route through the list

If you want a single path rather than a menu:

1. **Hammack, Book of Proof.** Cover to cover, doing the exercises. This is the highest-leverage thing on the list.
2. **Boyd and Vandenberghe, VMLS.** Linear algebra as you will actually use it.
3. **OpenStax Calculus Volume 1**, plus 3Blue1Brown's **Essence of Calculus** alongside it.
4. **Lehman, Leighton and Meyer, Mathematics for Computer Science.** Discrete mathematics and a great deal of proof practice.
5. **Blitzstein, Stat 110.** Probability, properly.

6. **Deisenroth, Faisal and Ong, Mathematics for Machine Learning.** The consolidation pass.
7. Then specialise: MacKay for information theory, Boyd for optimisation, Trefethen and Bau for numerical linear algebra, Prince for deep learning.

That is perhaps two years of part-time work, and it is a genuine undergraduate mathematics education for a working programmer. Nothing on it costs money.

CHAPTER C.

Notation Reference

Keep this page open. Nearly every “I cannot read this” is one unfamiliar symbol, not a hard idea.

C.1. Sets and logic

Symbol	Meaning	Ch.
$\in, \notin$	is / is not an element of	8
$\subseteq, \subsetneq$	subset; proper subset	8
$\cup, \cap$	union; intersection	8
$\setminus$	set difference	8
$\overline{A}, A^c$	complement	8
$\emptyset$	the empty set	8
$\mathcal{P}(S)$	power set: all subsets of S	8
$ S $	cardinality (number of elements)	8
$A \times B$	Cartesian product: set of ordered pairs	8
$\{x : P(x)\}$	set-builder: all x such that $P(x)$	8
$\forall, \exists$	for all; there exists	7
$\neg, \wedge, \vee$	not; and; or (inclusive)	7
$\Rightarrow$	implies	7
$\Leftrightarrow$	if and only if	7
$\sim$	is distributed as; or an equivalence relation	8, 28

C.2. Number systems

Symbol	Meaning	Ch.
$\mathbb{N}$	natural numbers $\{0, 1, 2, \dots\}$	1
$\mathbb{Z}$	integers	1
$\mathbb{Q}$	rationals	1
$\mathbb{R}$	real numbers	1
$\mathbb{C}$	complex numbers	6
$\mathbb{R}^n$	n -tuples of reals; n -dimensional space	19
$\mathbb{R}^{m \times n}$	m -by- n real matrices	20
$\mathbb{Z}/n\mathbb{Z}, \mathbb{F}_p$	integers mod n ; the field of p elements	12
$[a, b], (a, b)$	closed / open interval	1
$ x $	absolute value	1
$\lfloor x \rfloor, \lceil x \rceil$	floor; ceiling	3

Symbol	Meaning	Ch.
i	the imaginary unit, $i^2 = -1$	6
$\bar{z}$	complex conjugate	6

C.3. Sums, products, and operators

Symbol	Meaning	Ch.
$\sum_{i=m}^n$	sum from $i = m$ to n ; empty sum is 0	2
$\prod_{i=m}^n$	product; empty product is 1	2
$n!$	factorial; $0! = 1$	9
$\binom{n}{k}$	“ n choose k ”	9
$\gcd(a, b)$	greatest common divisor	12
$a \mid b$	a divides b exactly	12
$a \equiv b \pmod{n}$	congruent modulo n	12
$\varphi(n)$	Euler’s totient	12

C.4. Functions and calculus

Symbol	Meaning	Ch.
$f : A \rightarrow B$	function from A to B	3
$x \mapsto f(x)$	maps to (defines the rule)	3
$g \circ f$	composition; f applied first	3
f^{-1}	inverse function, or preimage of a set	3, 8
$\exp, \ln, \log_a$	exponential; natural log; log base a	4
$\lg$	$\log_2$	4
$\sigma(x)$	logistic / sigmoid function	4
$\lim_{x \rightarrow a}$	limit as x approaches a	13
ε, δ	tolerance and neighbourhood in a limit proof	13
$f', f'', f^{(n)}$	first, second, n -th derivative	14
$\frac{dy}{dx}$	Leibniz notation for the derivative	14
$\frac{\partial f}{\partial x}$	partial derivative	18
dx	differential; the variable of integration	16
$\int_a^b f(x) dx$	definite integral	16
∇f	gradient (column vector of partials)	18
$\nabla^2 f$	Laplacian (or sometimes the Hessian — check)	18
J	Jacobian matrix	18
H	Hessian matrix	18
O, Ω, Θ, o	asymptotic upper / lower / tight / strict bounds	11

Symbol	Meaning	Ch.
$f \sim g$	$f/g \rightarrow 1$ (asymptotic equality)	11

C.5. Linear algebra

Symbol	Meaning	Ch.
$u \cdot v, u^T v$	dot product / inner product	19
$\ v\ $	Euclidean (ℓ_2) norm	19
$\ v\ _1, \ v\ _\infty$	taxicab norm; max norm	19
$u \times v$	cross product (three dimensions only)	19
$\text{span}\{\dots\}$	set of all linear combinations	19
$\dim V$	dimension	19
A^T	transpose	20
A^{-1}	inverse	20
A^+	Moore–Penrose pseudoinverse	25
I, I_n	identity matrix	20
$\text{tr}(A)$	trace: sum of the diagonal	20
$\det A, A $	determinant	22
$\text{rank}(A)$	rank	21
$\text{Col}(A), \text{Null}(A)$	column space; null space (kernel)	20
$W^\perp$	orthogonal complement	24
$\text{proj}_v(u)$	projection of u onto v	19
λ, v	eigenvalue and eigenvector	23
$\rho(A)$	spectral radius: largest $ \lambda $	23
σ_i	singular values	25
$\kappa(A)$	condition number	21, 25
$\ A\ _2, \ A\ _F$	spectral norm; Frobenius norm	25
$\circ$	elementwise (Hadamard) product	26
$(\cdot)$	a matrix	20

C.6. Probability and statistics

Symbol	Meaning	Ch.
Ω	sample space	27
$\mathbb{P}(A)$	probability of event A	27
$\mathbb{P}(A \mid B)$	conditional probability	27
X, Y	random variables (capital letters, by convention)	28

Symbol	Meaning	Ch.
$p(x), f(x)$	probability mass function; probability density	28
$F(x)$	cumulative distribution function	28
$\mathbb{E}[X]$	expectation	28
$\text{Var}(X), \sigma^2$	variance	28
σ	standard deviation	28
$\text{Cov}(X, Y)$	covariance	29
ρ, Corr	correlation	29
Σ	covariance matrix	29
$X \sim \mathcal{N}(\mu, \sigma^2)$	X is normally distributed	28
Binomial, Poisson, Uniform	binomial, Poisson, uniform distributions	28
$\bar{X}$	sample mean	30
$\hat{\theta}$	an estimator of θ (hats mean estimates)	31
$\mathcal{L}(\theta)$	likelihood	31
$\ell(\theta)$	log-likelihood	31
arg max, arg min	the argument achieving the max / min	31
$H(X)$	entropy	32
$\text{KL}(p\ q)$	Kullback–Leibler divergence	32
$I(X; Y)$	mutual information	32
i.i.d.	independent and identically distributed	30

C.7. Conventions that trip people up

Capitalisation. Random variables are capital (X), their realised values lowercase (x). Matrices are capital (A), vectors lowercase bold or plain (v), scalars lowercase. Not universal, but usual.

Hats and bars. $\hat{\theta}$ is an estimate; $\bar{x}$ is an average; $\tilde{x}$ is usually “modified”; x^* is usually “optimal” (or, in complex analysis, a conjugate — context decides).

Vectors are columns. Unless stated otherwise, $v \in \mathbb{R}^n$ is $n \times 1$. So $u^T v$ is a scalar (inner product) and uv^T is an $n \times n$ matrix (outer product). Getting these two backwards is the most common notational error in matrix calculus.

Subscripts versus superscripts in ML. $W^{(\ell)}$ with a parenthesised superscript is layer ℓ , not a power. x_i is usually the i -th sample or the i -th component — and which one is rarely stated. Infer from context and from dimensions.

log without a base. Base e in mathematics, base 2 in computer science and information theory, base 10 in engineering. Inside a big-O it does not matter (Chapter 4). Elsewhere it does; check.

“Iff”. Standard mathematical abbreviation for “if and only if”. Not a typo.

“WLOG”. “Without loss of generality” — meaning a case has been assumed that can be reduced to from the others by symmetry. A legitimate move that is sometimes used to paper over a case the author did not check. When you see it, verify the reduction actually works.

The colon and the vertical bar. In set-builder notation $\{x : P(x)\}$ and $\{x \mid P(x)\}$ mean the same thing. In probability, $\mid$ means “given”. In number theory, $\mid$ means “divides”. Three unrelated meanings for one glyph; context is all you have.

CHAPTER D.

A Study Plan

D.1. The dependency graph

Not everything depends on everything. Here is what actually needs what, so you can skip intelligently rather than randomly.

To read	You genuinely need
Part I (1–7)	Nothing. This is the floor.
Part II (8–12)	Chapter 7 (proof), and Chapter 2 for Chapter 11.
Part III (13–18)	Chapters 1–5 solidly. Chapter 7 to follow the proofs.
Part IV (19–26)	Chapters 1–3. Chapter 5–6 for the geometry. Chapter 18 for Chapter 26 only.
Part V (27–32)	Chapter 9 (counting), Chapter 16 (integration), Chapter 4 (logs).
Ch. 33 (numerical)	Chapters 1, 14, 16, 21, 25.
Ch. 34 (optimization)	Chapters 15, 18, 23, 26.
Ch. 35 (geometry)	Chapters 5, 6, 19, 20, 22.
Ch. 36 (backprop)	Chapters 14, 18, 20, 26, 32, 34. This is the convergence point of the whole book.

Two things worth noting from that table. **Part IV does not depend on Part III**, apart from Chapter 26 — you can do linear algebra before calculus, and many people find it easier. And **Chapter 7 is a prerequisite for everything**; if you skip it you will be able to follow computations and not arguments.

D.2. Four routes

Pick the one matching why you are here. Each assumes a few hours a week.

D.2.1. Route A — “I want to read ML papers” (about 6 months)

The shortest path to the stated goal, skipping what does not serve it.

1. **Weeks 1–4.** Chapters 1–4. Foundations, and get logarithms genuinely automatic — they are everywhere in ML.
2. **Weeks 5–6.** Chapter 7. Do not skip it. You will not be **writing** proofs, but every paper’s appendix is one.
3. **Weeks 7–14.** Part IV, Chapters 19–26. This is the core. Spend the most time here. Chapter 26 is the notation papers are written in.
4. **Weeks 15–20.** Chapters 13–18. Calculus, aiming at the gradient. You can be lighter on Chapter 16 (integration techniques) than the others.
5. **Weeks 21–24.** Chapters 27–32. Probability through information theory. Chapter 32 explains your loss function.

6. **Weeks 25–26.** Chapters 34 and 36. Optimization and backpropagation.

Skippable for this route: Chapters 10 and 12, most of Chapter 16’s techniques, Chapter 35.

D.2.2. Route B — “I want general mathematical maturity” (about 12 months)

Read it in order. All of it. Do every exercise.

The difference from Route A is not the content, it is the **depth**: for maturity, the proofs are the point and the applications are the decoration. Read every proof twice per the preface’s instruction, and after each chapter, close the book and try to reconstruct the two or three central arguments from memory.

Add **Hammack’s Book of Proof** (Appendix B) alongside Part I, and do its exercises too. Proof is a motor skill and this book does not have enough repetitions in it by itself.

D.2.3. Route C — “I want to reason rigorously about my own code” (about 4 months)

1. **Weeks 1–3.** Chapters 1–2, then Chapter 7 in full.
2. **Weeks 4–10.** Part II entire: Chapters 8–12. This is your part. Sets, counting, graphs, recurrences, number theory.
3. **Weeks 11–14.** Chapters 27–30. Probability through the limit theorems — enough to reason about randomised algorithms, hash collisions, load balancing, and A/B tests.
4. **Weeks 15–16.** Chapters 1 and 33 together. Floating point and conditioning.

Skippable for this route: most of Part III and Part IV, though Chapters 19 and 20 are worth an afternoon regardless.

D.2.4. Route D — “I want graphics, geometry, and simulation” (about 6 months)

1. **Weeks 1–6.** Chapters 1–6. Trigonometry and complex numbers are load bearing here in a way they are not for the other routes.
2. **Weeks 7–16.** Part IV, Chapters 19–25. Transformations, determinants, eigenvalues.
3. **Weeks 17–22.** Chapters 13–18. Calculus, with real attention to Chapter 18 (vector calculus, multiple integrals).
4. **Weeks 23–26.** Chapters 33 and 35. Numerical stability and the geometry chapter.

D.3. How to study, mechanically

The instructions from the preface, restated as a procedure you can follow without thinking about it.

Working through one chapter

1. Read the intuition boxes only. Ten minutes. Get the shape.
2. Read the chapter properly, slowly. For each definition:
 - construct one example that satisfies it
 - construct one near-miss that does not
 If you cannot do both, reread.
3. For each proof:
 - pass 1: is every step true?
 - pass 2: why would anyone have thought of this?
4. Close the book. Write down, from memory:

- the two or three main results
- what each is for

Compare with the chapter. The gaps are what you actually missed.

5. Do the exercises. Fifteen real minutes each before looking.
6. Check against Appendix A. For anything you got wrong, find the specific sentence in the chapter you failed to use.

Step 4 is the one people skip and the one that does the work. Recall is what builds memory; rereading builds only the feeling of familiarity, which is worse than useless because it is indistinguishable from knowing.

D.4. On pace and on getting stuck

A chapter a week is a good pace. Faster than that and it will not stick. Much slower and you will lose the thread between chapters.

Twenty minutes a day beats three hours on Sunday. The consolidation happens between sessions, not during them. This is not motivational advice, it is how memory works.

When you are stuck, the problem is upstream. Nine times out of ten, confusion in Chapter 23 is an unresolved gap in Chapter 20. Go back. Every time you go back you will find the gap was real.

Being stuck for twenty minutes is normal and productive. Being stuck for two hours is not; at that point you are missing a prerequisite, not thinking insufficiently hard. Identify what you would need to know, go and learn that, come back.

Write things down by hand. Not for any mystical reason: it is slow enough to force you to actually process each symbol, and typing is not.

Compute things. You are a programmer. When a theorem says the eigenvalues of a symmetric matrix are real, generate a hundred random symmetric matrices and check. When Chapter 30 says the sample mean is approximately normal, simulate it and plot the histogram. This is a form of understanding available to you that is not available to most students, and it is worth using.

D.5. What to do after

You will finish this book able to read most technical mathematics slowly. The next step depends on where you want depth:

- **Analysis** (Lebl, then Rudin) if you want to actually understand why calculus works.
- **Linear algebra again, harder** (Axler, or Trefethen and Bau for the numerical side). Nobody learns linear algebra properly the first time.
- **Probability again, harder** (Blitzstein, then Durrett for measure-theoretic probability) if you go deep into statistics or stochastic processes.
- **Abstract algebra** (Judson's free **Abstract Algebra: Theory and Applications**) if the group and field remarks in Chapter 8 interested you. It is the right next step for cryptography and coding theory.
- **Convex optimization** (Boyd) if you do anything with fitting or resource allocation.
- **Information theory** (MacKay) if Chapter 32 was the best chapter.

All of those are free, and all of them are in Appendix B.

D.6. A last word

The reason to learn this material is not that it makes you better at any particular task, although it does. It is that a large amount of what currently looks to you like magic — why this optimiser works, why that estimate is biased, why the numbers went wrong at scale — turns out to be a small number of ideas applied repeatedly.

There are fewer ideas in mathematics than there appear to be. The dot product turns up as similarity, as projection, as correlation, and as covariance. The chain rule turns up as sensitivity, as backpropagation, and as change of variables. Orthogonality turns up as independence, as uncorrelatedness, and as the optimality condition for every least-squares problem in existence.

Once you have seen an idea in four costumes you stop being surprised by the fifth. That is what mathematical maturity actually is, and it is entirely achievable in a couple of years of unhurried work.